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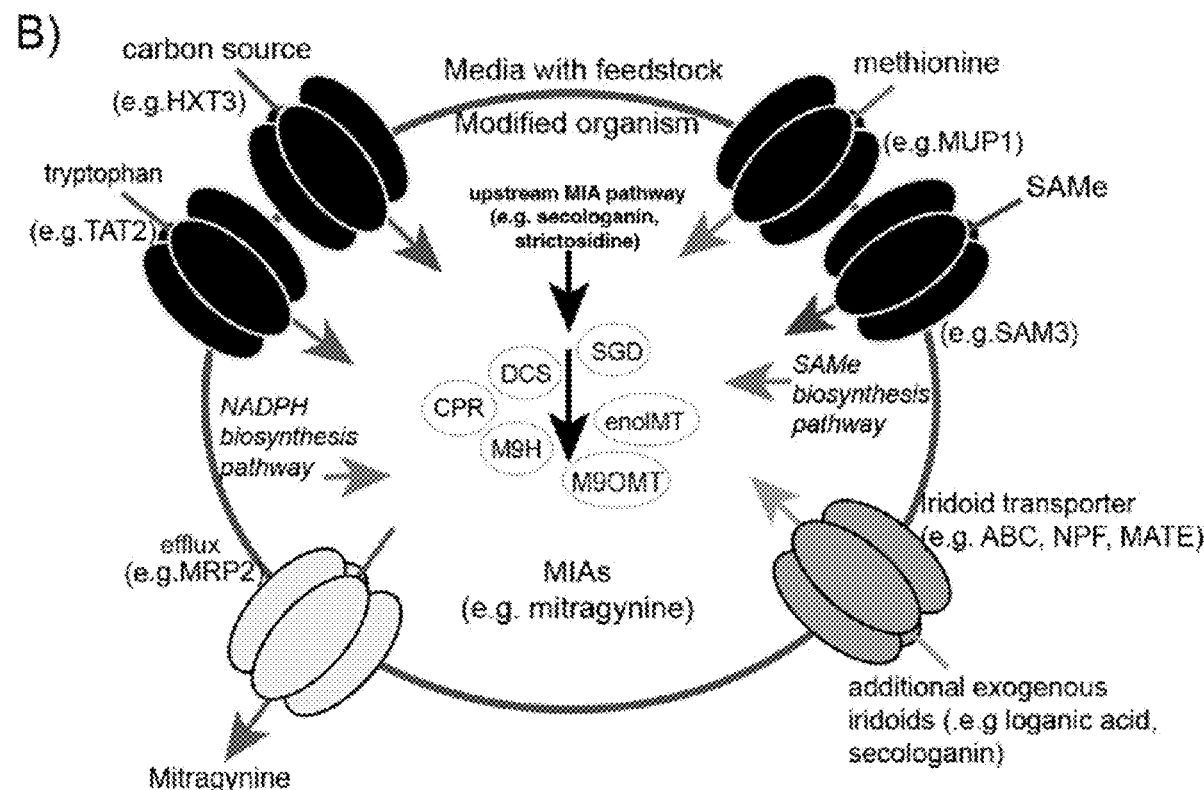
Nwe et al.

(10) **Pub. No.: US 2025/0179544 A1**(43) **Pub. Date: Jun. 5, 2025**(54) **PRODUCTION OF CORYNANTHE-TYPE
MONOTERPENE INDOLE ALKALOID
COMPOUNDS IN A HETEROLOGOUS HOST**(71) Applicant: **CB Therapeutics**, Carlsbad, CA (US)(72) Inventors: **Phu Khat Nwe**, San Diego, CA (US);
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(2013.01); **C12N 1/20** (2013.01); **C12N 15/52**
(2013.01); **C12N 2500/30** (2013.01); **C12N**
2500/76 (2013.01); **C12N 2510/02** (2013.01)

(57)

ABSTRACT

Provided are non-naturally occurring nucleic acids comprising a sequence encoding an enzyme, regulatory protein, or other functional protein that provides for biosynthesis of monoterpene indole alkaloids (e.g., mitragynine), as well as modified enzyme and protein sequences encoded by the nucleic acid sequences. Also provided are recombinant microorganisms, including novel strains, that comprise the nucleic acid sequences and/or express the enzyme or regulatory protein that provides for the biosynthesis of monoterpene indole alkaloids (e.g., mitragynine). Methods of expressing the enzyme or regulatory protein are provided, as are methods of generating monoterpene indole alkaloid mitragynine-type compounds, as well as precursors, intermediates, and analogs thereof.

Specification includes a Sequence Listing.

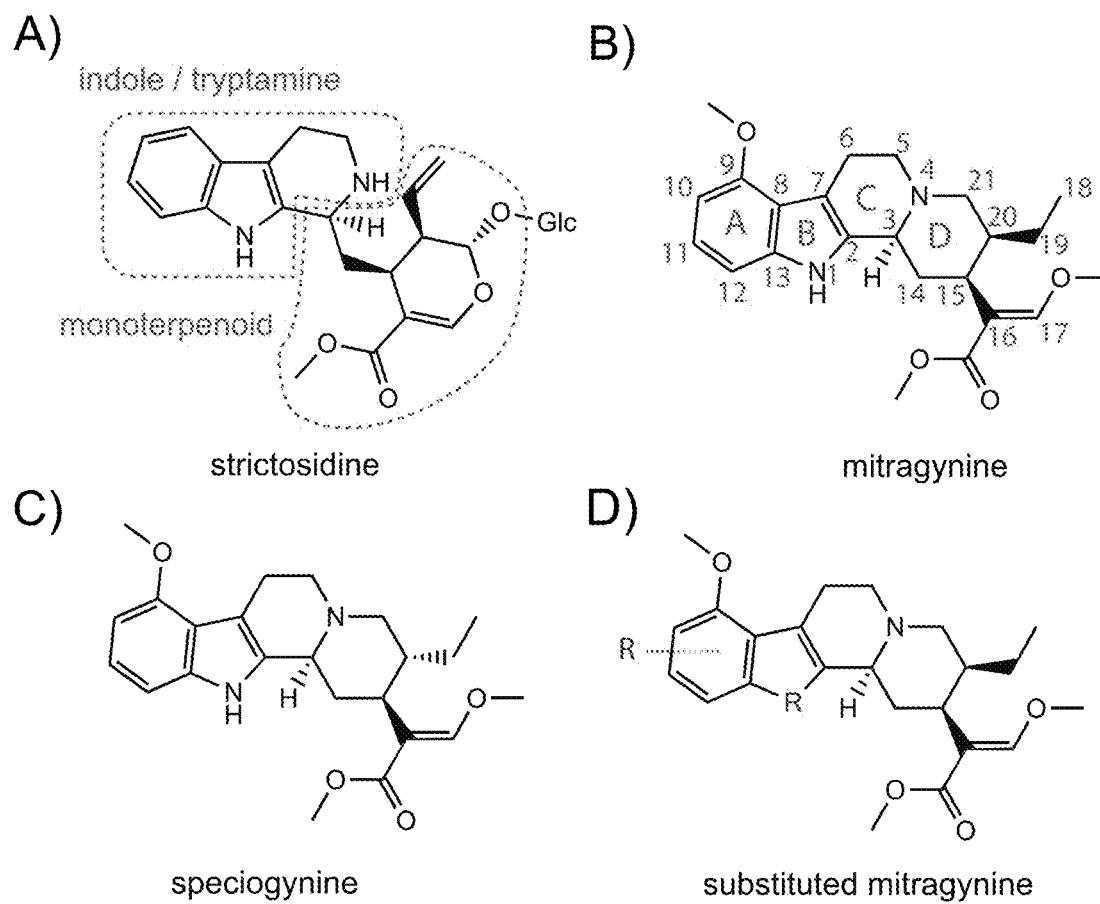


FIG. 1

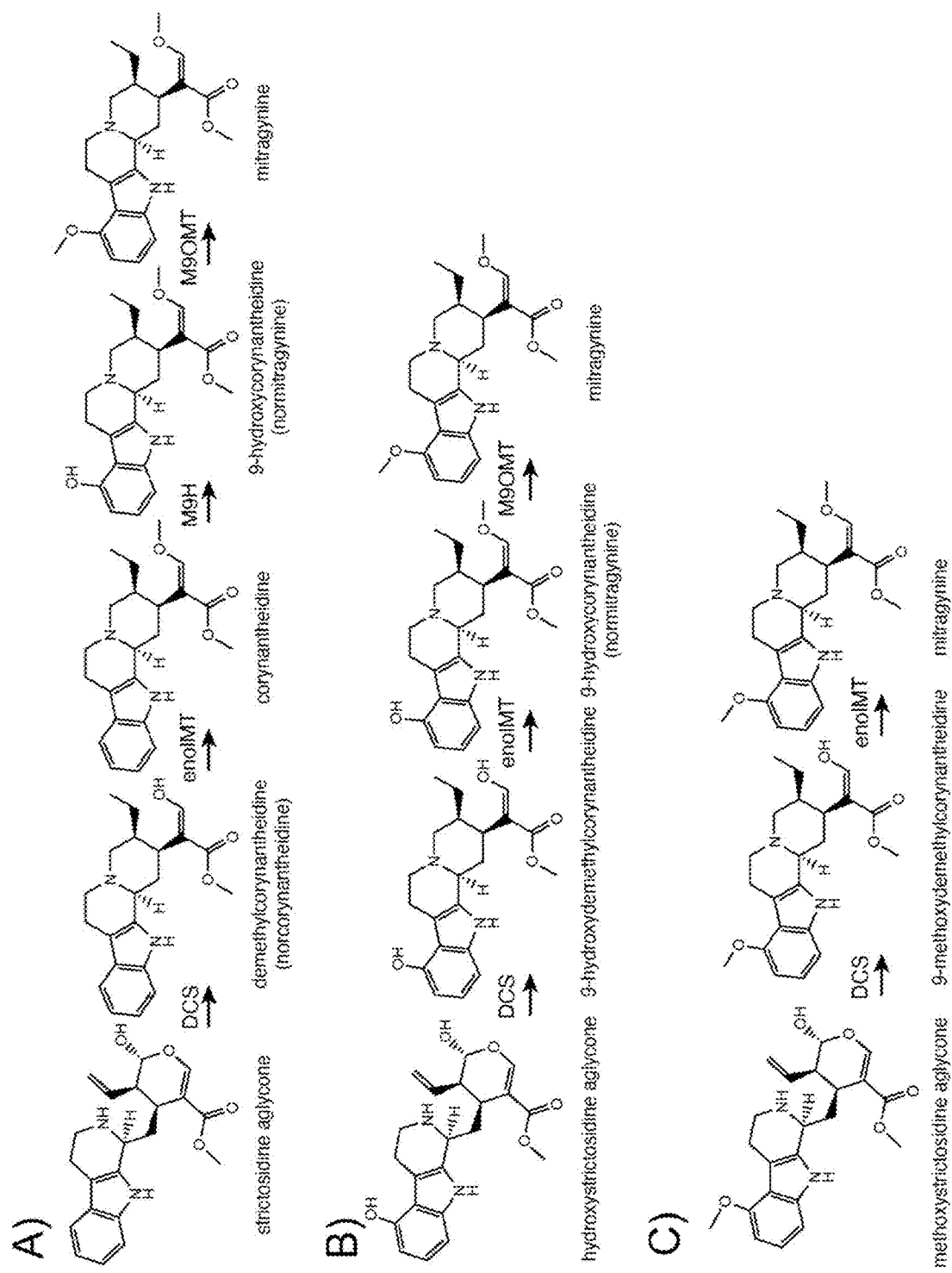


FIG 2

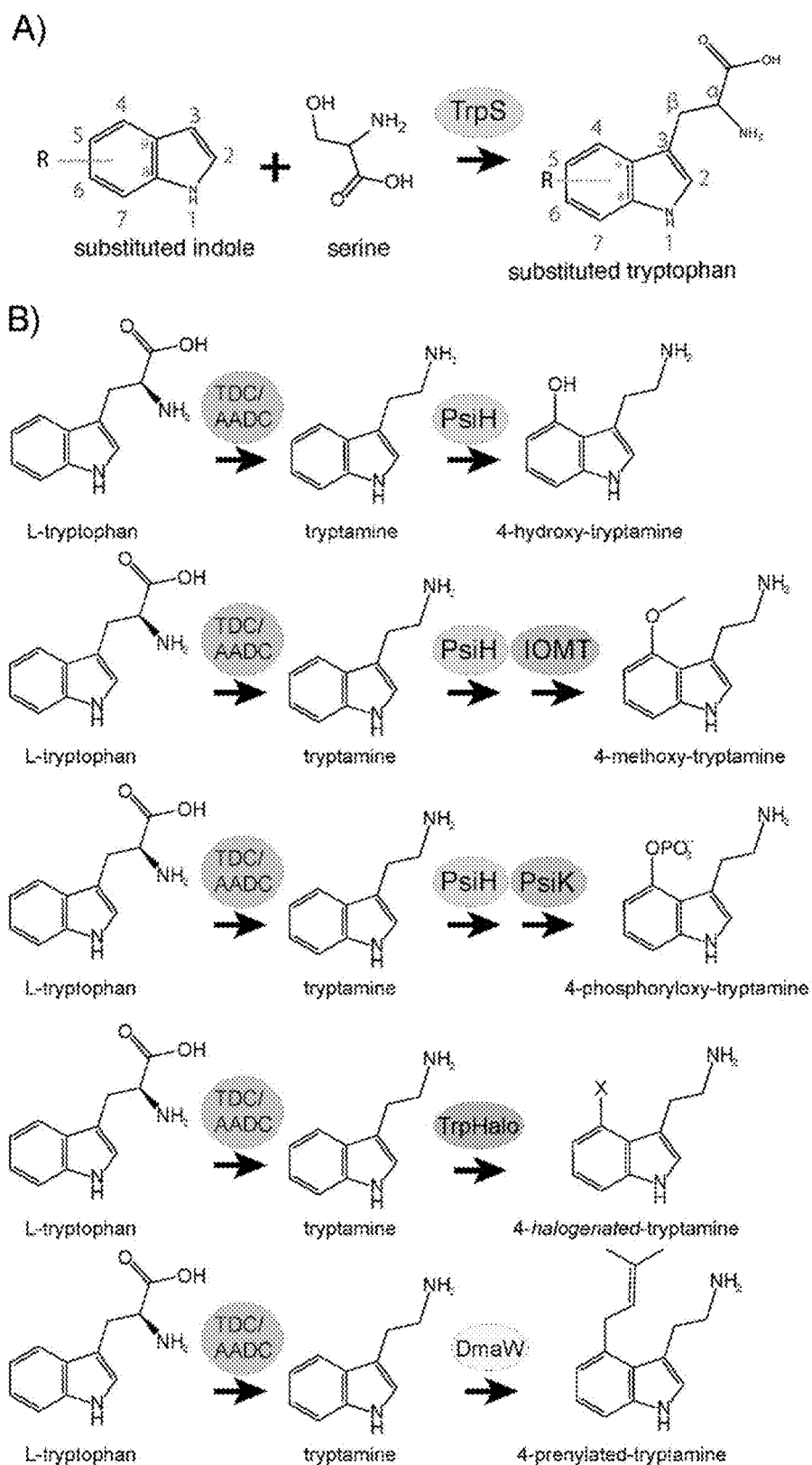


FIG 3



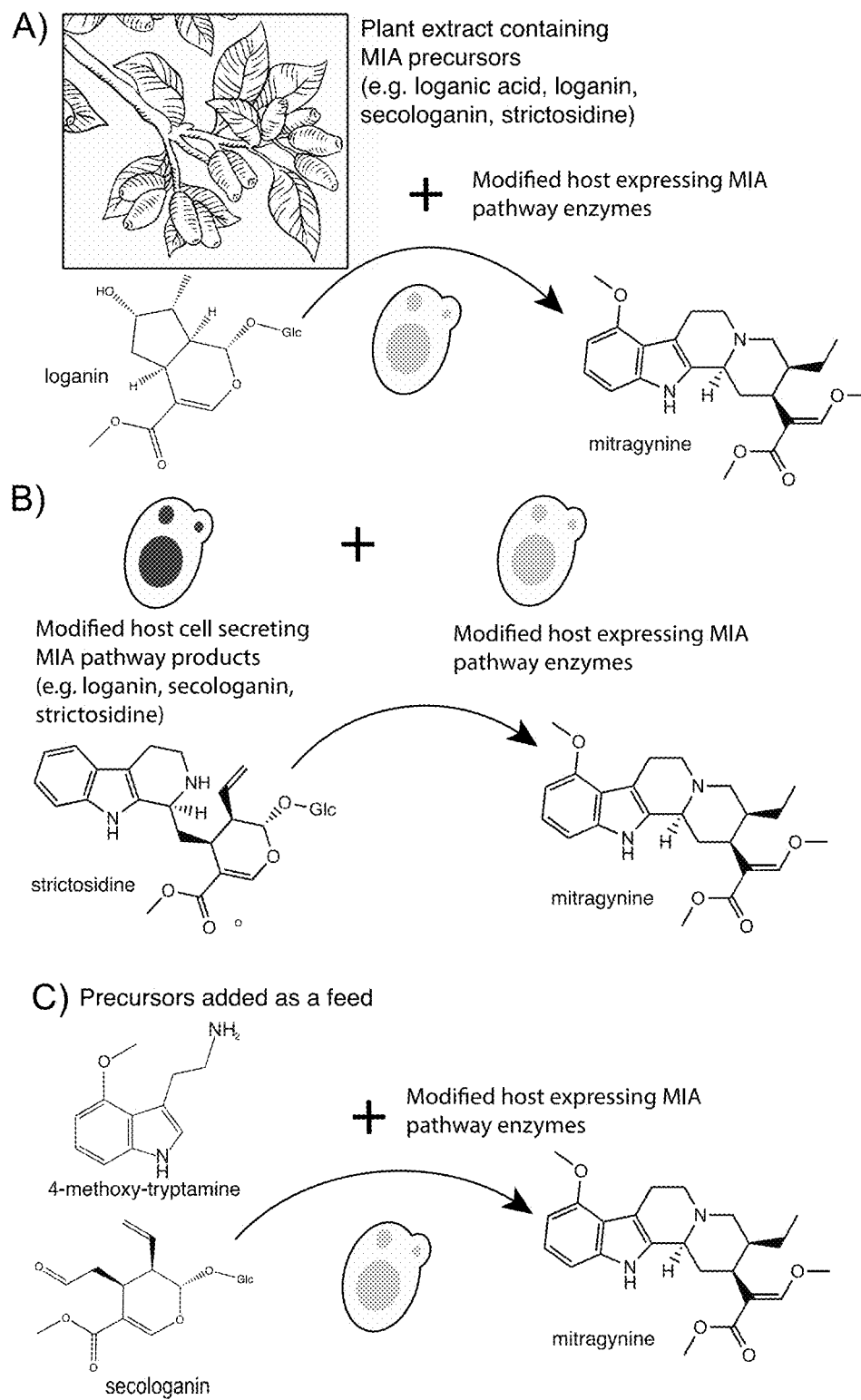


FIG 5

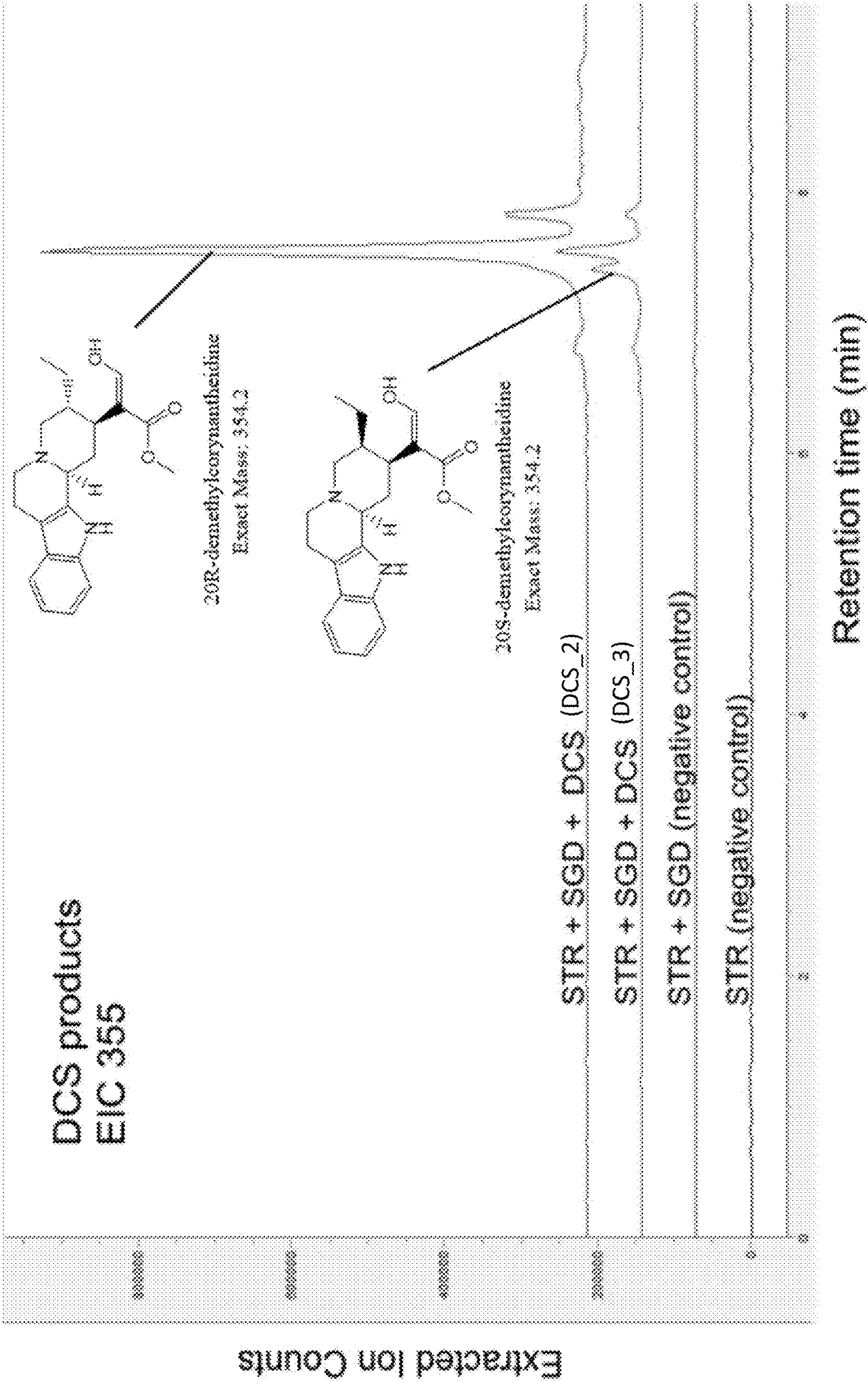


FIG 6

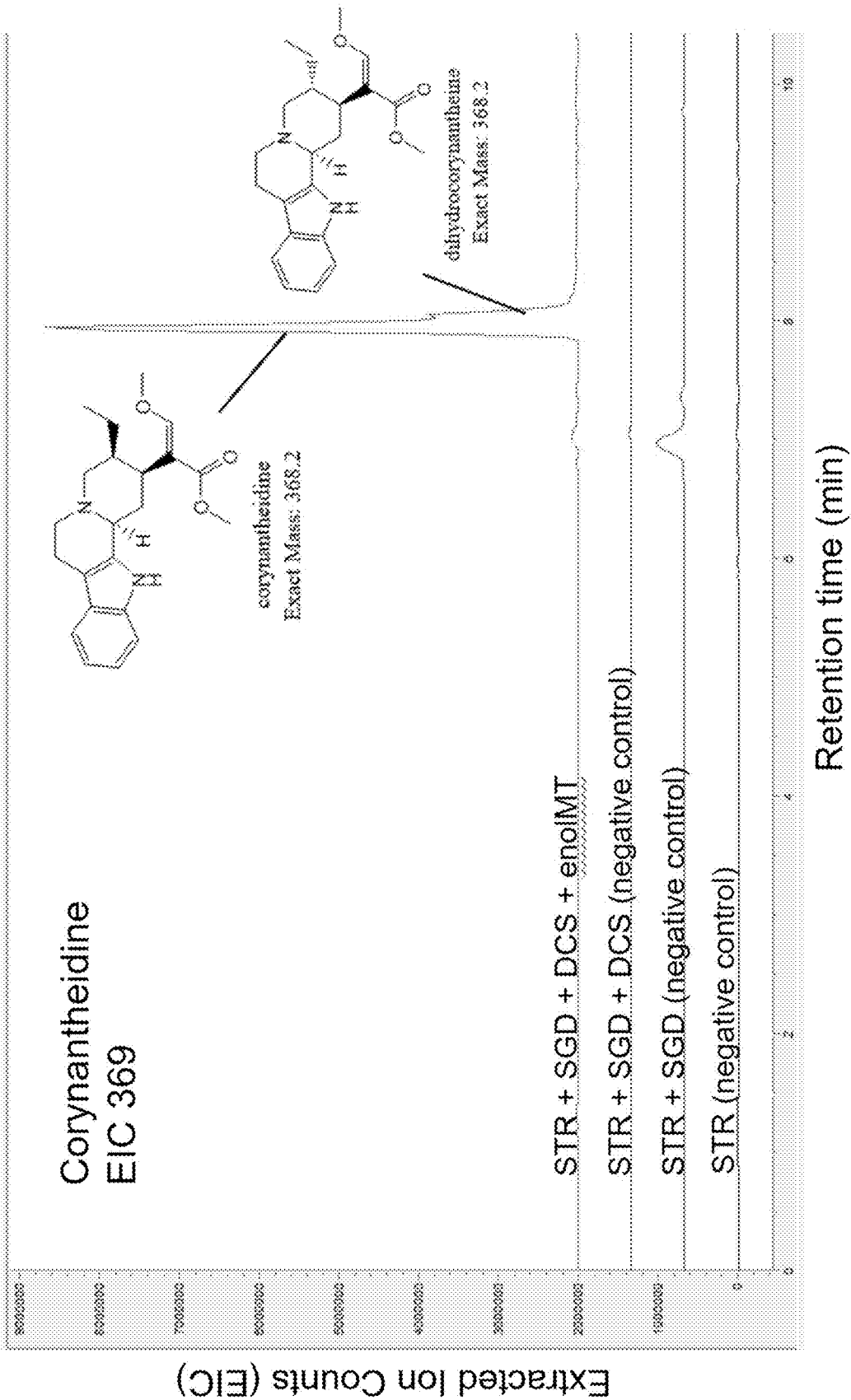


FIG 7

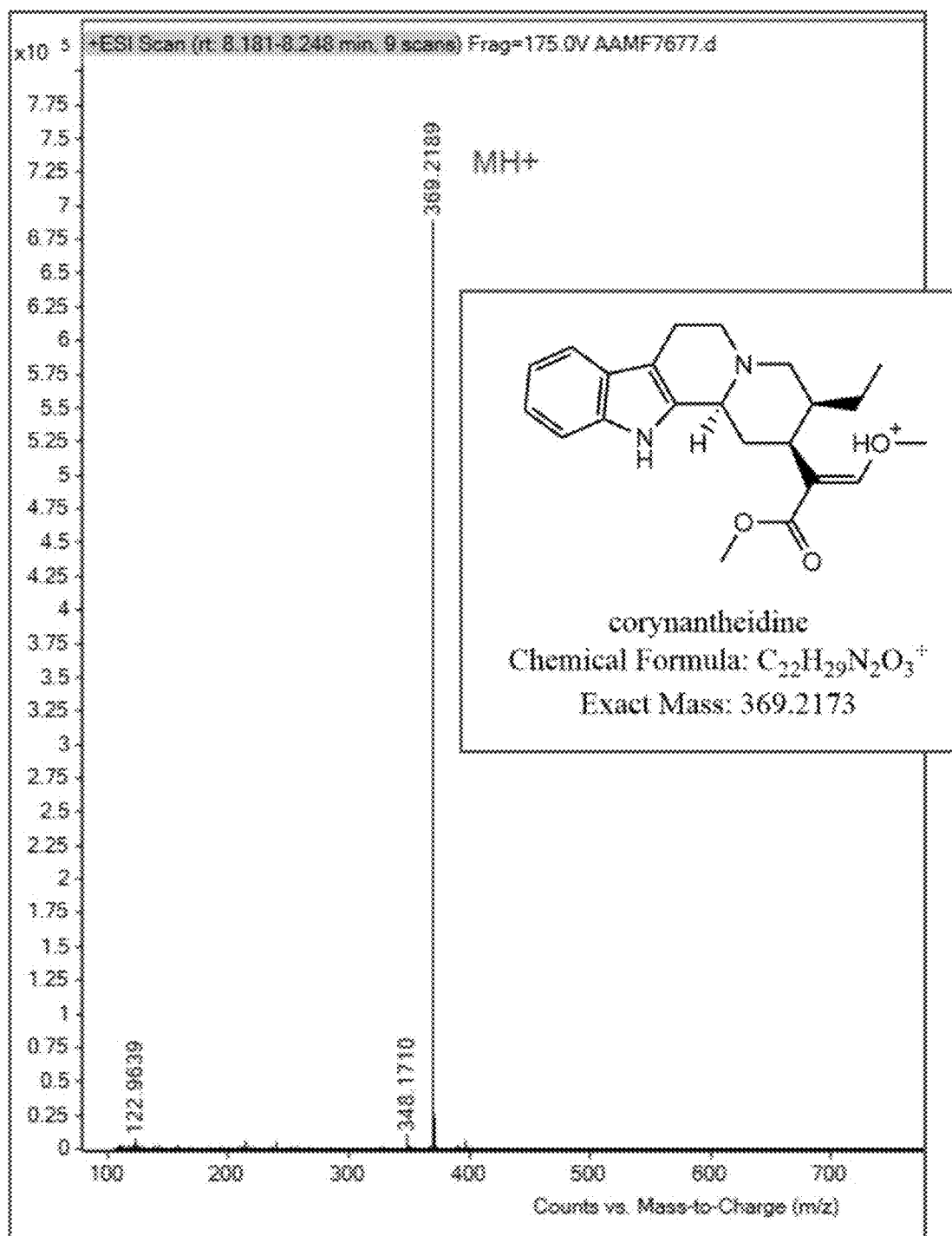


FIG 8

Fragmentation of Corynantheidine

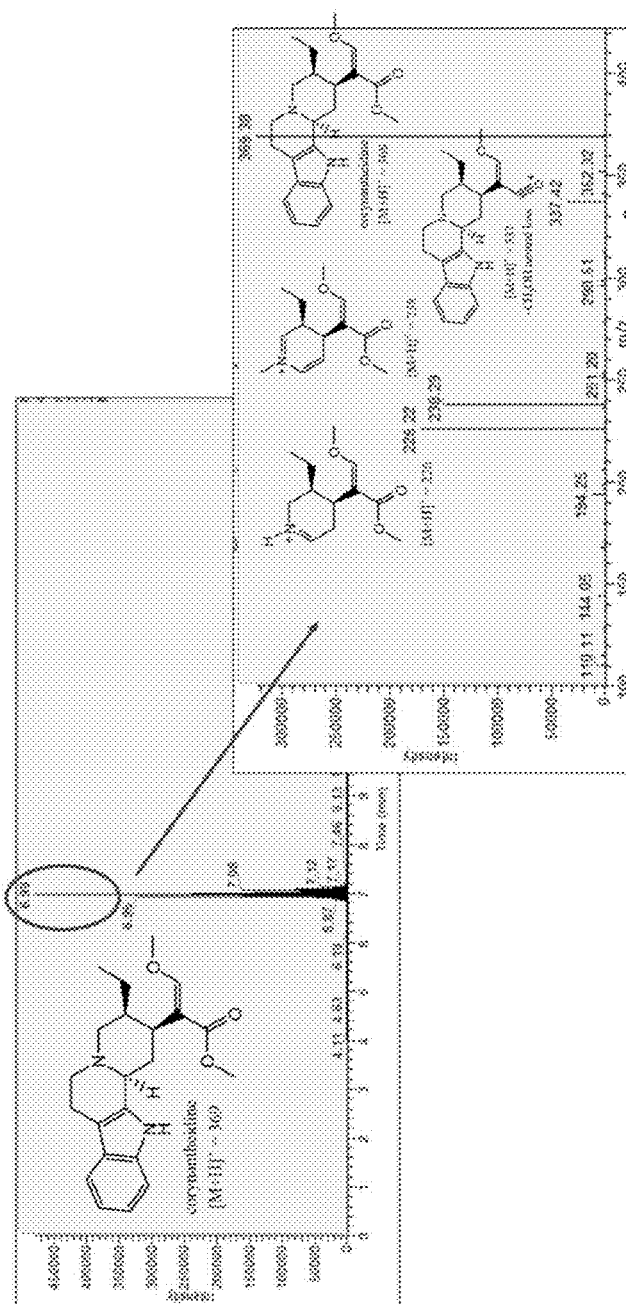
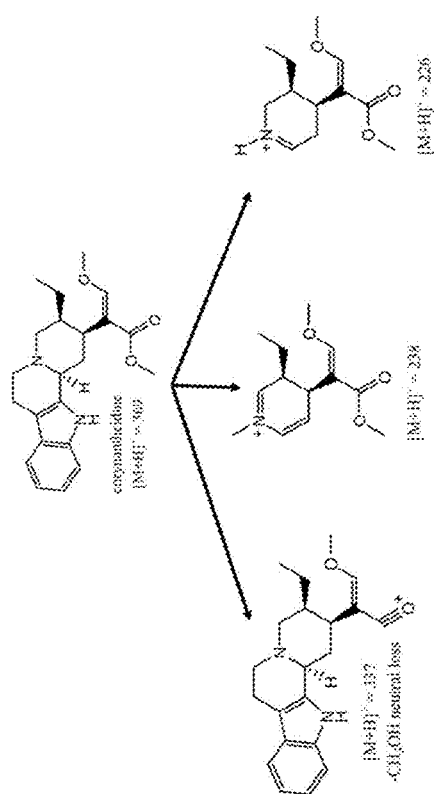
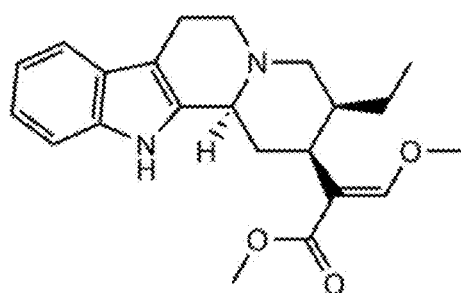
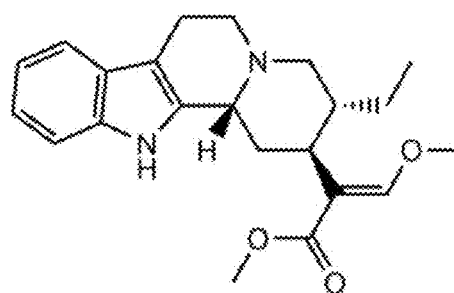


FIG 9

Samples and compounds	RT (min)	MS1	MS2
		m/z	m/z
Recombinant host			
corynantheidine	6.99	369	337
			238
			226
kratom extract			
corynantheidine	6.96	369	337
			238
			226
Analytical standard			
corynantheidine	6.98	369	337
			238
			226
Analytical standard			
hirsutine	7.32	369	337
			238
			226



corynantheidine
[M+H]⁺= 369



hirsutine
[M+H]⁺= 369

FIG 10

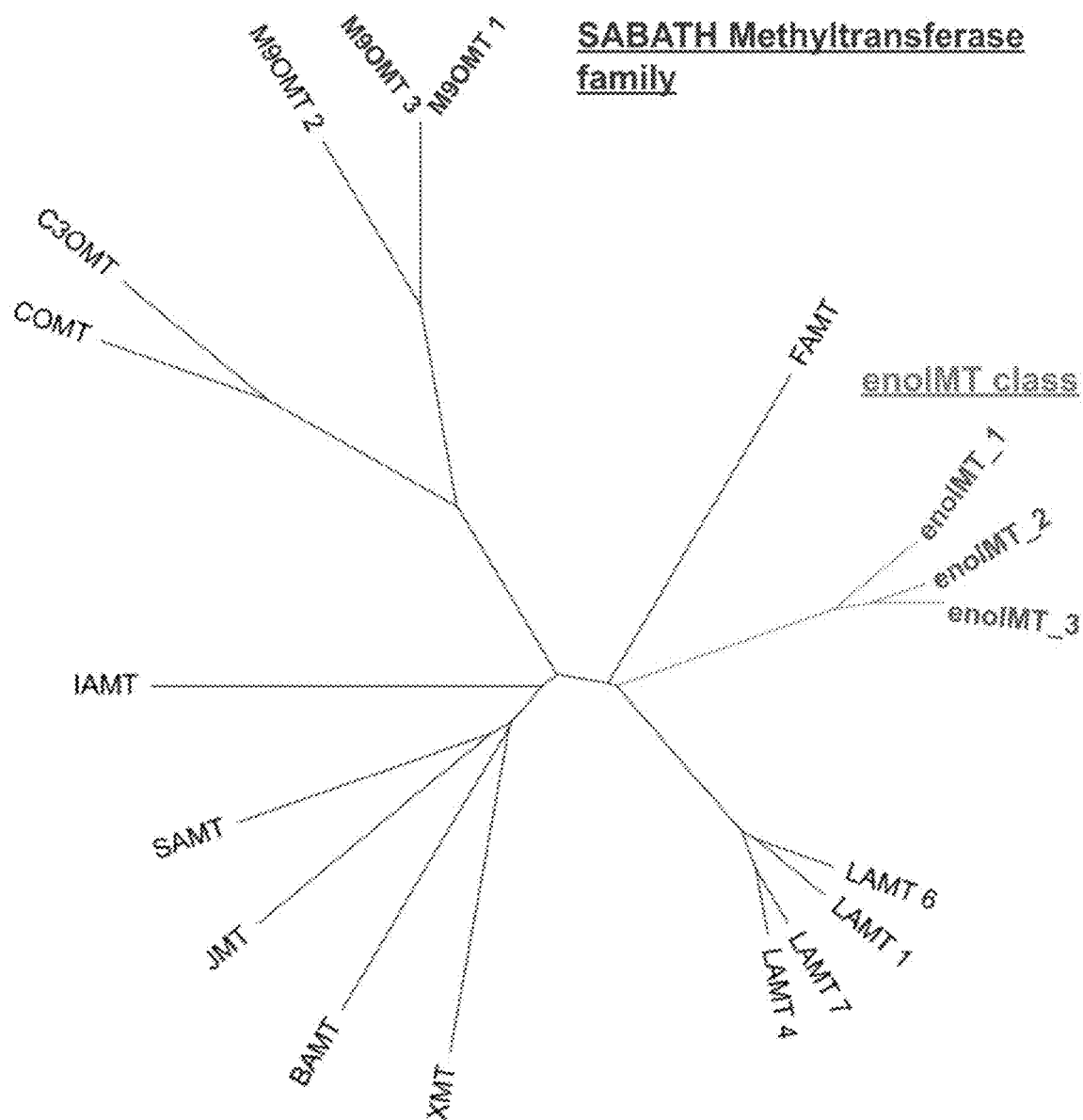


FIG 11

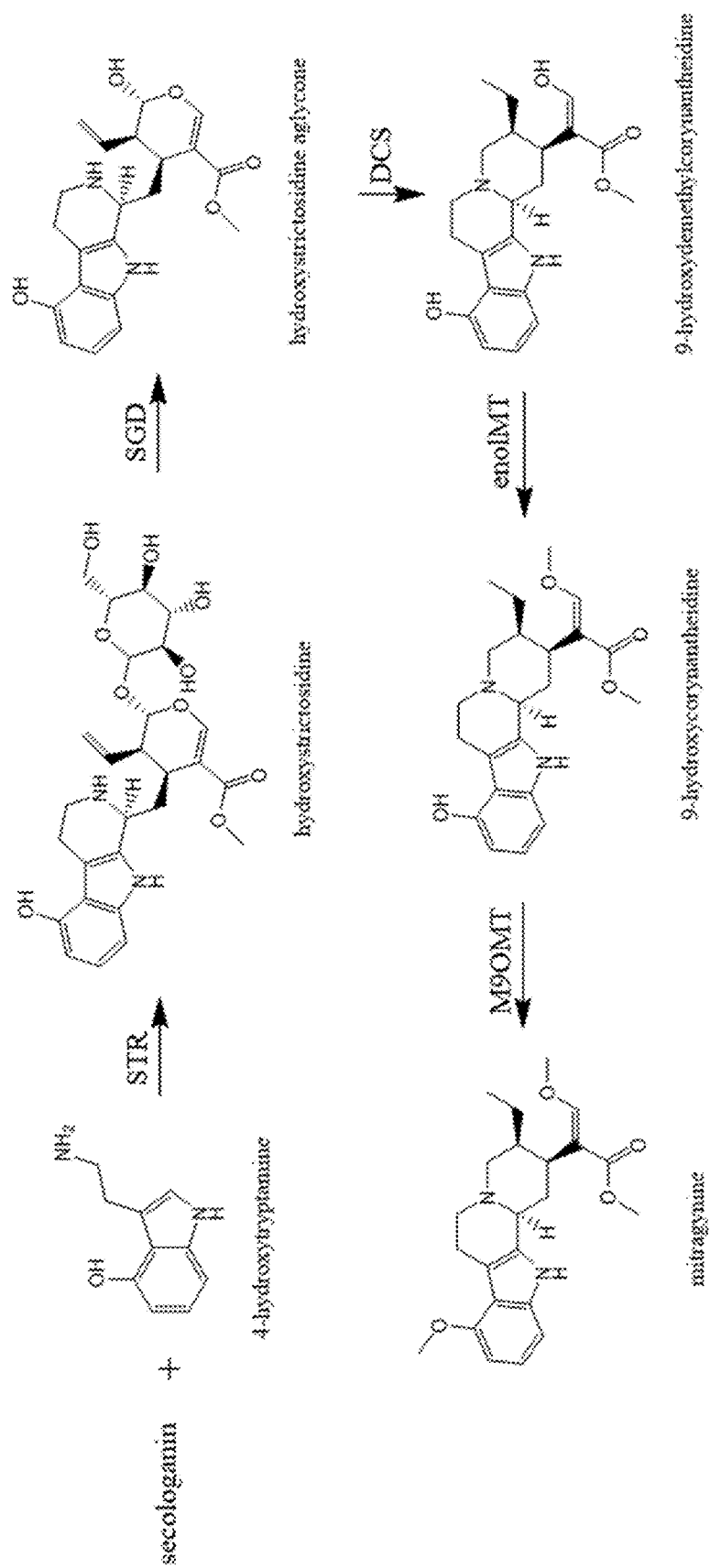


FIG 12

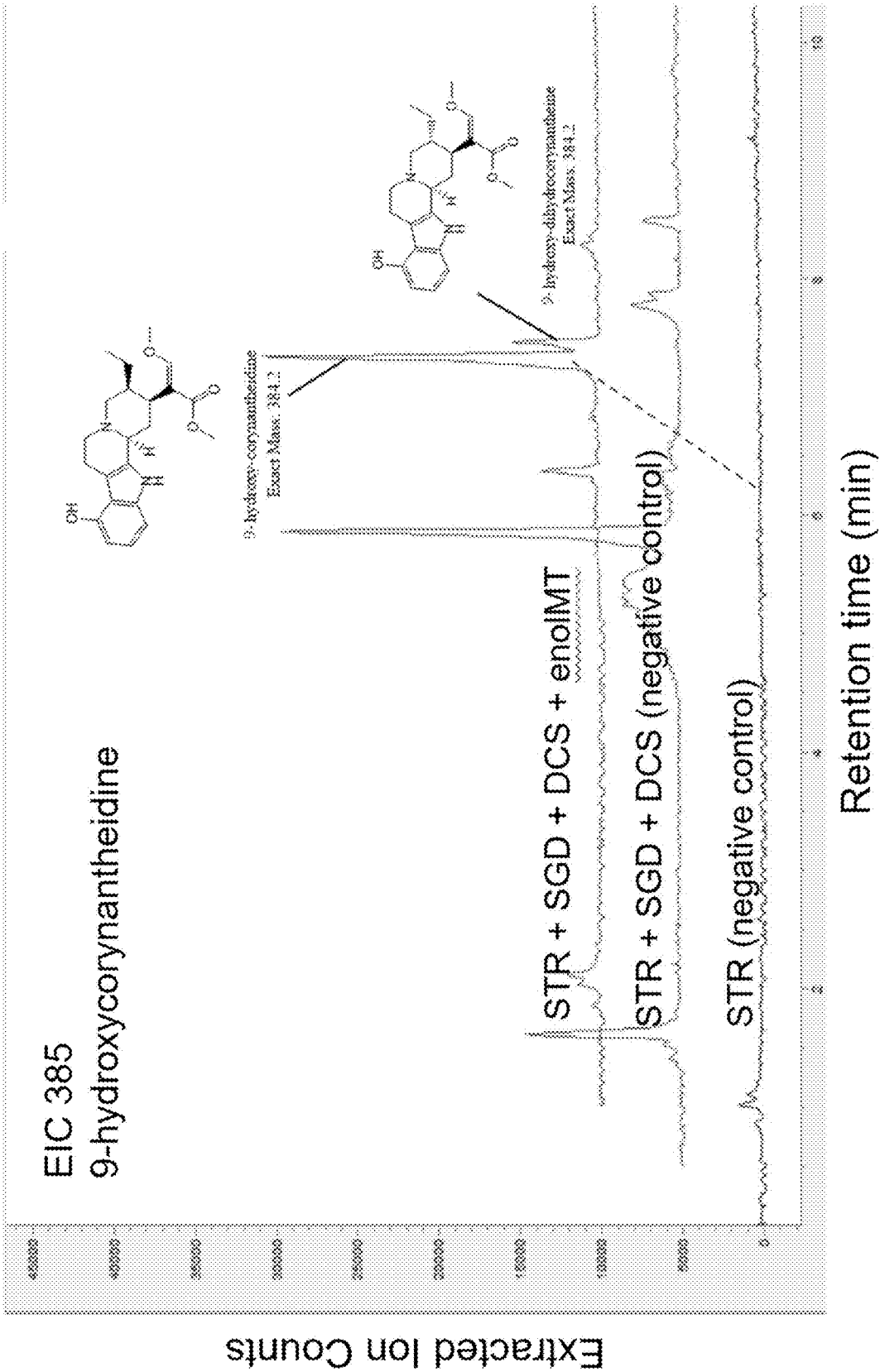


FIG 13

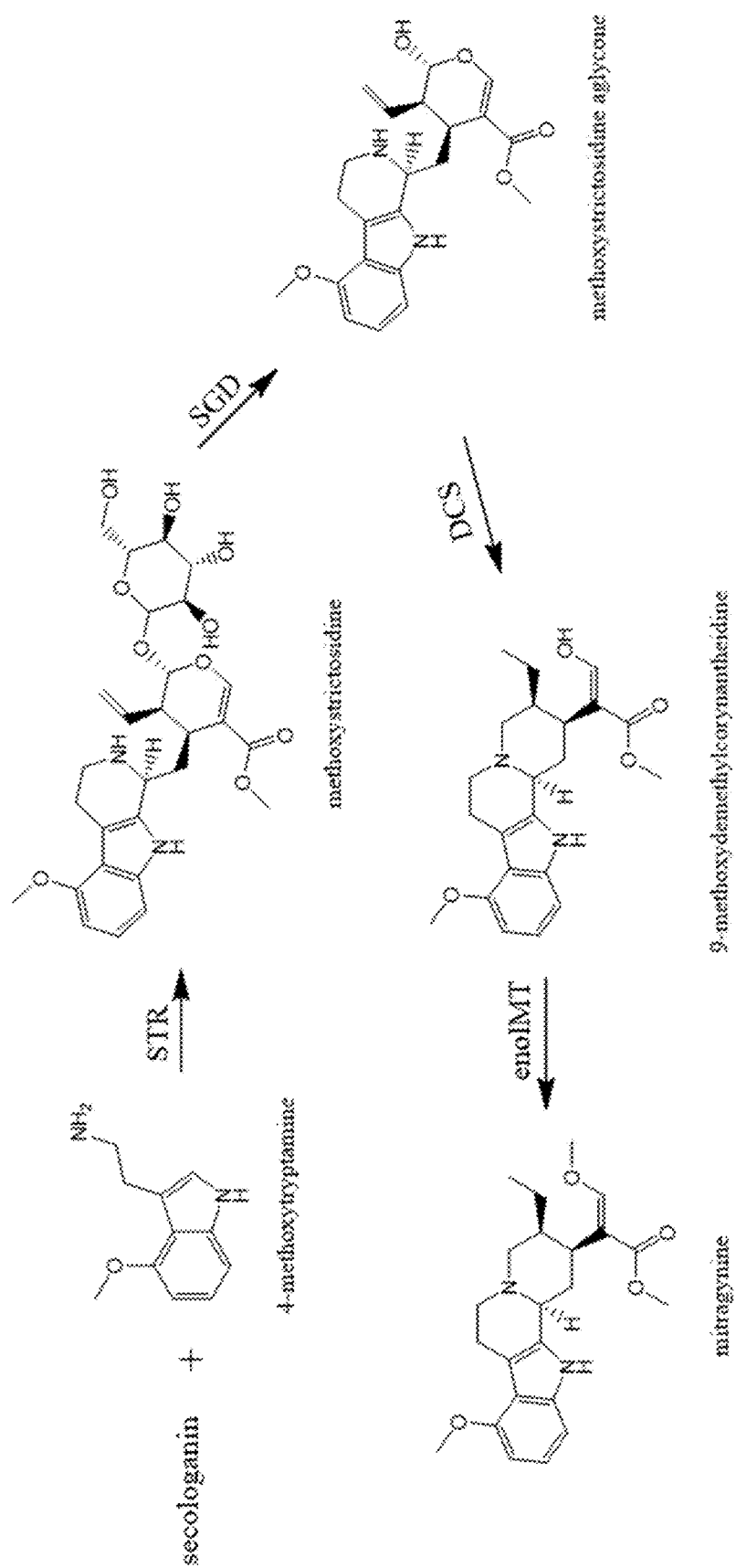


FIG 14

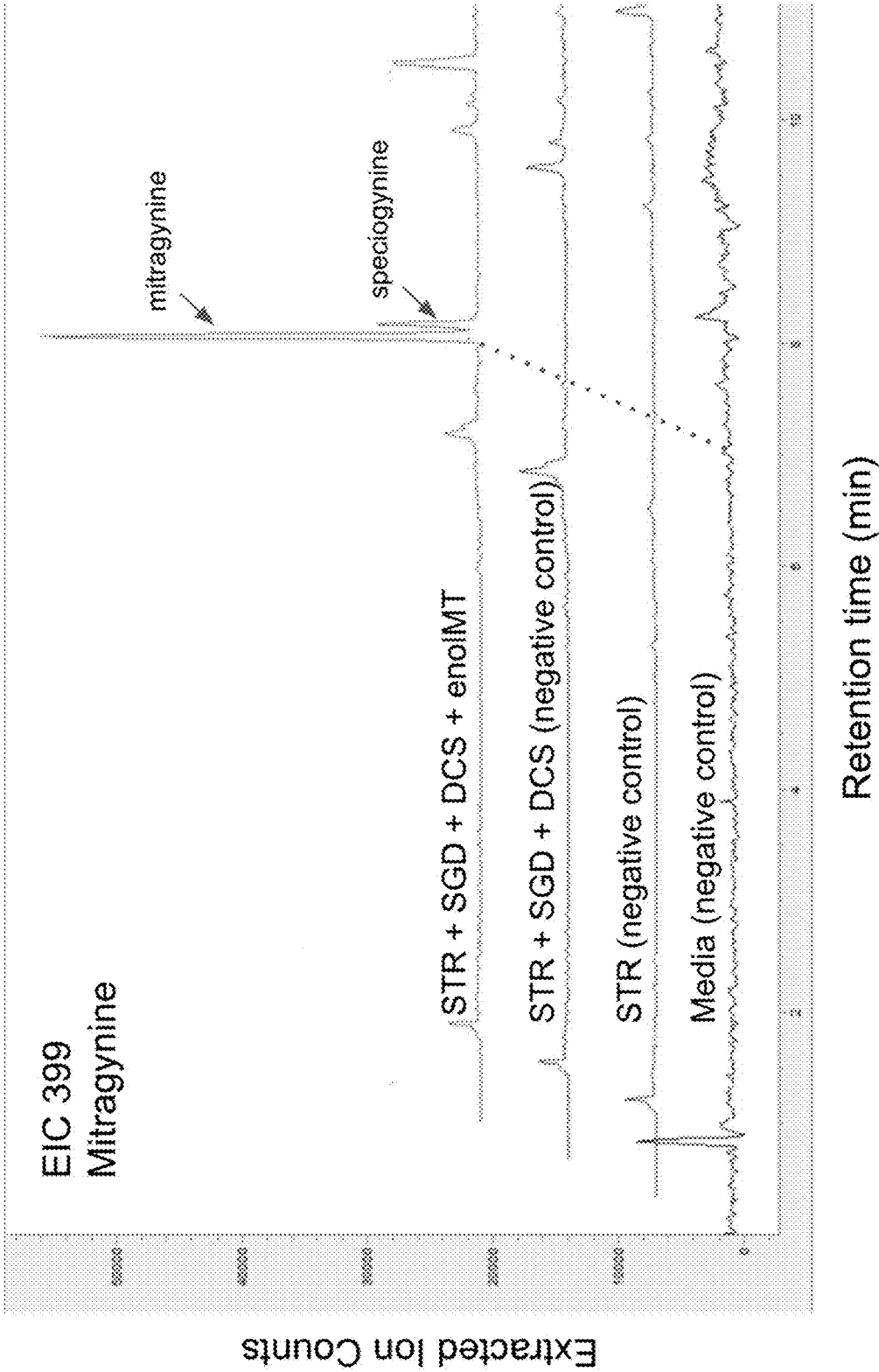


FIG 15.

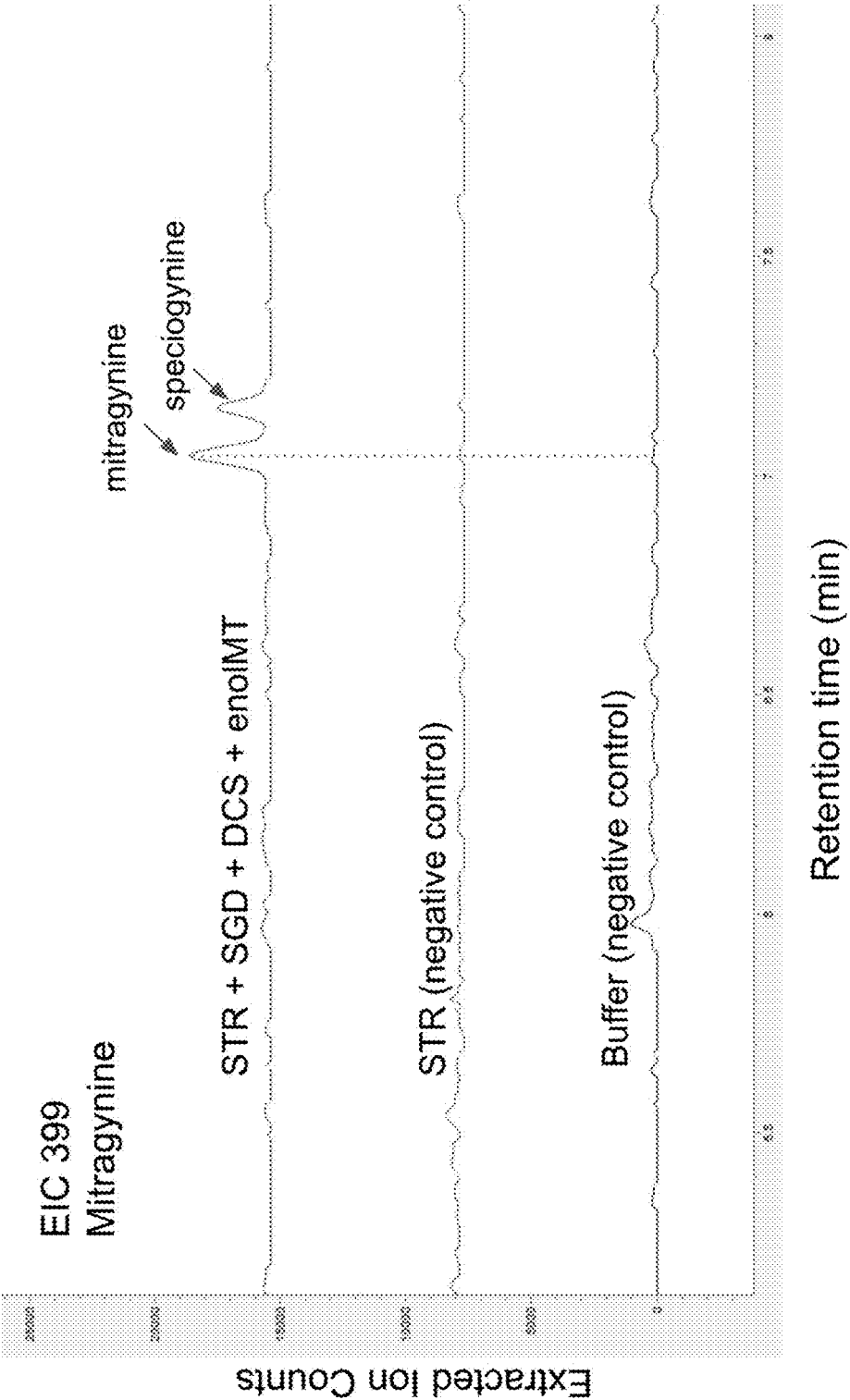


FIG 16.

**PRODUCTION OF CORYNANTHE-TYPE
MONOTERPENE INDOLE ALKALOID
COMPOUNDS IN A HETEROLOGOUS HOST**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application claims the benefit of U.S. Provisional Application No. 63/605,289, filed Dec. 1, 2023, and incorporated by reference herein in its entirety.

SEQUENCE LISTING

[0002] The contents of the electronic sequence listing (CBTH-14-US.xml; Size: 584 kilobytes; and Date of Creation: Nov. 20, 2024) is herein incorporated by reference in its entirety.

FIELD

[0003] The disclosure relates to polynucleotides, enzymes, biosynthetic methods, and recombinant microorganisms useful in the production of downstream monoterpene indole alkaloid compounds. The monoterpene indole alkaloids that can be synthesized according to the disclosure include plant-derived and/or fungal-derived compounds or metabolites that can exert a physiological effect, and are inclusive of compounds such as strictosidine, hydroxystrictosidine, methoxystrictosidine, strictosidine aglycone, hydroxystrictosidine aglycone, methoxystrictosidine aglycone, demethylcorynantheidine (20R, 20S), 9-hydroxydemethylcorynantheidine (20R, 20S), 9-methoxydemethylcorynantheidine (20R, 20S), corynantheidine, dihydrocorynantheine, 9-hydroxycorynantheidine, 9-hydroxydihydrocorynantheine, speciogynine, and/or mitragynine, as well as precursors, intermediates, metabolites, stereoisomers, and derivatives thereof. The monoterpene indole alkaloids herein can be generated from precursor compounds that are produced by upstream monoterpene indole alkaloid biosynthetic pathways, which can feed into the biosynthetic pathways (e.g., mitragynine pathways) useful in the production the target compounds as described herein.

BACKGROUND

[0004] Plants, fungi, and bacteria are biochemical factories that can make complex alkaloids molecules, often to defend against herbivores and protect against pathogens. Many of these plant and fungal natural products have medicinal properties. There are multiple examples of these alkaloid compounds used to treat a wide range of conditions including, for example, drug addiction, depression, pain, cancer, and diabetes.

[0005] Indole alkaloid compounds occur in nature, typically in very small quantities in certain plants, fungi, and/or bacteria. Isolating such compounds from natural sources typically requires large amounts of starting material (e.g., biomass) as well as complicated separation techniques, and only provides the target compound(s) at low yield. Chemical synthesis strategies that target monoterpene indole alkaloids, particularly from readily available and low-cost precursors, are often highly complicated, involve toxic solvents, and also are associated with low yields.

[0006] Accordingly, there is a need for compositions and methods that provide for the production of indole alkaloid molecules from a renewable source such as recombinant microorganism(s). The disclosure provides polynucleotides,

amino acid sequences (i.e., enzymes), recombinant cells and strains, and biosynthetic methods that produce indole alkaloids from low-cost feedstock or precursor compounds, as well as from cell lysates and/or in cell-free reactions. The methods and recombinant cells disclosed herein provide for synthesis of mitragyna- or corynanthe-type indole alkaloid compounds in good quantity and yield, providing advantages in scale, reliability, reproducibility and environmentally-friendly production, particularly when compared to existing production and isolation and/or purification methods.

SUMMARY OF THE DISCLOSURE

[0007] In an aspect the disclosure provides for polynucleotide sequences that encode amino acid sequences (e.g., enzymes, transporters, etc.) that are useful in the biosynthesis of a downstream monoterpene indole alkaloid (e.g., from strictosidine or strictosidine aglycone to mitragynine, and intermediates, derivatives, and metabolites thereof). In embodiments of this aspect the polynucleotide sequence comprises a gene that encodes a polypeptide that is associated with monoterpene indole alkaloid synthesis.

[0008] In one aspect, the disclosure relates to a polynucleotide comprising a sequence as disclosed in the Sequence Listing provided herewith, or a sequence having about 90% sequence identity to any one of the disclosed sequences. In some embodiments, the disclosure relates to a polynucleotide sequence having about 95%, 96%, 97%, 98%, or 99% sequence identity to any one of the polynucleotide sequences disclosed in the Sequence Listing filed herewith.

[0009] In an aspect, the disclosure relates to an expression vector comprising a polynucleotide in accordance with disclosure and an operatively-linked promoter sequence that allows for expression of the one or more polynucleotide sequences in a cell.

[0010] In one aspect, the disclosure relates to a recombinant cell comprising a polynucleotide and/or an expression vector in accordance with disclosure. In some embodiments, the recombinant cell comprises a bacterial cell, a fungal cell, or a yeast cell. In some embodiments, the yeast cell comprises *Saccharomyces*, *Candida*, *Pichia*, *Schizosaccharomyces*, *Scheffersomyces*, *Blakeslea*, *Rhodotorula*, or *Yarrowia*. In some embodiments, the filamentous fungus cell comprises *Aspergillus* or *Penicillium*. In some embodiments, the bacterial cell comprises *Escherichia*, *Corynebacterium*, *Caulobacter*, *Pseudomonas*, *Streptomyces*, *Bacillus*, or *Lactobacillus*.

[0011] In yet further embodiments, the recombinant cell comprises at least one copy of the one or more polynucleotide sequences stably integrated into the genome of the cell. In further embodiments, a plurality of copies of the one or more polynucleotide sequences are stably integrated into the genome of the cell. In some further embodiments, the recombinant cell comprises a polynucleotide sequence or polynucleotide sequences in accordance with the disclosure that comprises the same sequence. In some further embodiments, the recombinant cell comprises a polynucleotide sequence or polynucleotide sequences in accordance with the disclosure that comprises different sequences. In yet further embodiments, the recombinant cell comprises a polynucleotide sequence or polynucleotide sequences in accordance with the disclosure that encode the same class of enzyme or amino acid sequence. In some further embodiments, the recombinant cell comprises a polynucleotide

sequence or polynucleotide sequences in accordance with the disclosure that encode a different class of enzyme or amino acid sequence.

[0012] In another aspect, the disclosure provides a method for the biosynthesis of a downstream mitragyna-type or corynanthe-type monoterpene indole alkaloid (MIA) compound, or a precursor or metabolite thereof, wherein the method comprises culturing a recombinant cell in accordance with any of the aspects and embodiments of the disclosure under conditions that allow for the biosynthesis of the compound. In some embodiments, the method further comprises contacting the recombinant cell with a feedstock comprising one or more exogenous substrate compounds useful in the biosynthesis of one or more MIAs. In some further embodiments, the one or more exogenous substrate compounds may comprise an upstream precursor MIA compound such as, for example, nepetalactol, loganic acid, loganin, secologanin, L-tryptophan, tryptamine, serotonin, 5-methoxytryptamine, 4-methoxytryptamine, or 4-hydroxytryptamine, or the feed can comprise a plant extract that includes the one or more exogenous substrate compounds. In some further embodiments, the method can comprise one or more exogenous substrate compounds that are produced by a recombinant cell that can produce one or more upstream precursor MIA compounds, and which are fed into a recombinant cell according to any one of the aspects and embodiments disclosed herein.

[0013] In some embodiments, the biosynthetic method generates one or more of the following compounds: strictosidine, hydroxystictosidine, methoxystictosidine, strictosidine aglycone, hydroxystictosidine aglycone, methoxystictosidine aglycone, demethylcorynantheidine (20R, 20S), 9-hydroxydemethylcorynantheidine (20R, 20S), 9-methoxydemethylcorynantheidine (20R, 20S), corynantheidine, dihydrocorynantheine, 9-hydroxycorynantheidine, 9-hydroxydihydrocorynantheine, speciogynine, and/or mitragynine, among others, including halogenated analogs.

[0014] In a further aspect, the disclosure provides an amino acid sequence comprising a sequence as disclosed in the Sequence Listing provided herewith, or a sequence having about 90% sequence identity to any one of the disclosed sequences. In some embodiments, the disclosure relates to an amino acid sequence having about 95%, 96%, 97%, 98%, or 99% sequence identity to any one of the amino acid sequences disclosed in the Sequence Listing provided herewith, or a sequence having about 90% sequence identity to any one of the sequences. The disclosure also provides a polynucleotide encoding the amino acid sequences disclosed in the Sequence Listing provided herewith or those with 90%, 95%, 96%, 97%, 98% or 99% sequence identity to those amino acid sequences.

[0015] In some further embodiments, the polynucleotide that encodes a polypeptide associated with monoterpene indole alkaloid synthesis (e.g., arising from MIA precursors such as strictosidine, hydroxystictosidine, methoxystictosidine, and/or strictosidine aglycone, etc.) comprises a sequence that is a modified gene. In some embodiments, the polynucleotide sequence comprises a NADPH-dependent reductase (DCS); an enol-O-methyltransferase (enolMT); an indole ring hydroxylase (M9H); indole ring O-methyltransferase (M9OMT) or a combination of any two or more the genes.

[0016] In some further embodiments, the polynucleotide encodes a polypeptide that is useful in the biosynthesis of a

monoterpene indole alkaloid that encodes for a polypeptide having a wide variety of biological activity including, for example, a polypeptide associated with electron transport, a polypeptide associated with metabolic processes, transporter protein (importer protein, efflux protein, etc.), an oxidase, a reductase, a dehydrogenase, a peptide synthetase, a halogenase, a transferase (e.g., glycosyltransferase), a synthetase (e.g., tryptophan synthetase) or other polypeptides having functional utility in the biosynthesis of a monoterpene indole alkaloid. In some embodiments the polynucleotide sequence encodes a cytochrome P450 (CYP), a cytochrome reductase (CPR), a cytochrome B (CYB), a tryptophan synthetase (TRPS), or combinations thereof. In such further embodiments, one or more of the polynucleotides can be used in combination with one or more polynucleotides, as described herein.

[0017] In some embodiments, the method comprises an NADPH-dependent reductase (DCS) that catalyzes the iminium reduction of an aglycone from compounds such as strictosidine aglycone to produce dihydrocorynantheine aldehyde. In some embodiments, the method comprises an NADPH-dependent reductase (DCS) gene in the Sequence Listing provided herewith, including DCS_In (SEQ ID NO:1), DCS_2n (SEQ ID NO:2), DCS_3n (SEQ ID NO:3), DCS_4n (SEQ ID NO:4), DCS_5n (SEQ ID NO:5), DCS_6n (SEQ ID NO:6), DCS_7n (SEQ ID NO:7), DCS_8n (SEQ ID NO:8), DCS_9n (SEQ ID NO:9), DCS_10n (SEQ ID NO:10), DCS_11n (SEQ ID NO:11), DCS_12n (SEQ ID NO:12), or DCS_13n (SEQ ID NO:13), DCS_14n (SEQ ID NO:14), DCS_15n (SEQ ID NO:15), DCS_16n (SEQ ID NO:16), DCS_17n (SEQ ID NO:17), DCS_18n (SEQ ID NO:18), DCS_19n (SEQ ID NO:19), DCS_20n (SEQ ID NO:20), DCS_21n (SEQ ID NO:21), DCS_22n (SEQ ID NO:22), DCS_23n (SEQ ID NO:23), DCS_24n (SEQ ID NO:24), DCS_25n (SEQ ID NO:25), or DCS_26n (SEQ ID NO:26), or a (DCS) amino acid sequence of the Sequence Listing provided herewith, including DCS_Ip (SEQ ID NO:53), DCS_2p (SEQ ID NO:54), DCS_3p (SEQ ID NO:55), DCS_4p (SEQ ID NO:56), DCS_5p (SEQ ID NO:57), DCS_6p (SEQ ID NO:58), DCS_7p (SEQ ID NO:59), DCS_8p (SEQ ID NO:60), DCS_9p (SEQ ID NO:61), DCS_10p (SEQ ID NO:62), DCS_11p (SEQ ID NO:63), DCS_12p (SEQ ID NO:64), or DCS_13p (SEQ ID NO:65), DCS_14p (SEQ ID NO:66), DCS_15p (SEQ ID NO:67), DCS_16p (SEQ ID NO:68), DCS_17p (SEQ ID NO:69), DCS_18p (SEQ ID NO:70), DCS_19p (SEQ ID NO:71), DCS_20p (SEQ ID NO:72), DCS_21p (SEQ ID NO:73), DCS_22p (SEQ ID NO:74), DCS_23p (SEQ ID NO:75), DCS_24p (SEQ ID NO:76), DCS_25p (SEQ ID NO:77), or DCS_26p (SEQ ID NO:78) or a sequence having about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any of the DCS sequences.

[0018] In some embodiments, the method comprises an indole ring O-methyltransferase (M9OMT) that catalyzes the methylation of oxygen bound to an indole ring, leading to the conversion of compounds such as 9-hydroxycorynantheidine to produce mitragynine. In some embodiments, the method comprises an indole-ring O-methyltransferase (M9OMT) gene in the Sequence Listing provided herewith, including M9OMT_In (SEQ ID NO:27), M9OMT_2n (SEQ ID NO:28), M9OMT_3n (SEQ ID NO:29), M9OMT_4n (SEQ ID NO:30), M9OMT_5n (SEQ ID NO:31), M9OMT_6n (SEQ ID NO:32), M9OMT_7n (SEQ ID NO:33), M9OMT_8n (SEQ ID NO:34), M9OMT_9n (SEQ ID

NO:35), M9OMT_10n (SEQ ID NO:36), M9OMT_11n (SEQ ID NO:37), M9OMT_12n (SEQ ID NO:38), M9OMT_13n (SEQ ID NO:39), M9OMT_14n (SEQ ID NO:40), M9OMT_15n (SEQ ID NO:41), M9OMT_16n (SEQ ID NO: 42), M9OMT_17n (SEQ ID NO:43), M9OMT_18n (SEQ ID NO:44), M9OMT_19n (SEQ ID NO: 45), M9OMT_20n (SEQ ID NO:46), M9OMT_21n (SEQ ID NO:47), M9OMT_22n (SEQ ID NO:48), M9OMT_23n (SEQ ID NO:49), M9OMT_24n (SEQ ID NO:50), M9OMT_25n (SEQ ID NO:51), M9OMT_26n (SEQ ID NO:52), M9OMT_27n (SEQ ID NO: 233), M9OMT_28n (SEQ ID NO:234), M9OMT_29n (SEQ ID NO:235), M9OMT_30n (SEQ ID NO:236), M9OMT_31n (SEQ ID NO:237), M9OMT_32n (SEQ ID NO:238), M9OMT_33n (SEQ ID NO:239), M9OMT_34n (SEQ ID NO:240), M9OMT_35n (SEQ ID NO: 241), M9OMT_36n (SEQ ID NO:242), M9OMT_37n (SEQ ID NO:243), M9OMT_38n (SEQ ID NO:244), M9OMT_39n (SEQ ID NO:245), M9OMT_40n (SEQ ID NO:246), M9OMT_41n (SEQ ID NO:247), M9OMT_42n (SEQ ID NO:248), M9OMT_43n (SEQ ID NO: 249), M9OMT_44n (SEQ ID NO:250), M9OMT_45n (SEQ ID NO:251), M9OMT_46n (SEQ ID NO:252), M9OMT_47n (SEQ ID NO:253), M9OMT_48n (SEQ ID NO:254), M9OMT_49n (SEQ ID NO:255), M9OMT_50n (SEQ ID NO:256), M9OMT_51n (SEQ ID NO: 257), M9OMT_52n (SEQ ID NO:258), M9OMT_53n (SEQ ID NO:259), M9OMT_54n (SEQ ID NO:260), M9OMT_55n (SEQ ID NO:261), M9OMT_56n (SEQ ID NO:262), M9OMT_57n (SEQ ID NO:263), M9OMT_58n (SEQ ID NO:264), M9OMT_59n (SEQ ID NO: 265), M9OMT_60n (SEQ ID NO:266), M9OMT_61n (SEQ ID NO:267), M9OMT_62n (SEQ ID NO:268), M9OMT_63n (SEQ ID NO:269), M9OMT_64n (SEQ ID NO:270), or an indole-ring O-methyltransferase (M9OMT) amino acid sequence in the Sequence Listing provided herewith, including M9OMT_1p (SEQ ID NO:79), M9OMT_2p (SEQ ID NO:80), M9OMT_3p (SEQ ID NO:81), M9OMT_4p (SEQ ID NO:82), M9OMT_5p (SEQ ID NO:83), M9OMT_6p (SEQ ID NO:84), M9OMT_7p (SEQ ID NO:85), M9OMT_8p (SEQ ID NO:86), M9OMT_9p (SEQ ID NO:87), M9OMT_10p (SEQ ID NO:88), M9OMT_11p (SEQ ID NO:89), M9OMT_12p (SEQ ID NO:90), M9OMT_13p (SEQ ID NO:91), M9OMT_14p (SEQ ID NO: 92), M9OMT_15p (SEQ ID NO:93), M9OMT_16p (SEQ ID NO:94), M9OMT_17p (SEQ ID NO: 95), M9OMT_18p (SEQ ID NO:96), M9OMT_19p (SEQ ID NO:97), M9OMT_20p (SEQ ID NO:98), M9OMT_21p (SEQ ID NO:99), M9OMT_22p (SEQ ID NO:100), M9OMT_23p (SEQ ID NO:101), M9OMT_24p (SEQ ID NO: 102), M9OMT_25p (SEQ ID NO: 103), M9OMT_26p (SEQ ID NO: 104), M9OMT_27p (SEQ ID NO: 105), M9OMT_28p (SEQ ID NO:106), M9OMT_29p (SEQ ID NO:107), M9OMT_30p (SEQ ID NO:108), M9OMT_31p (SEQ ID NO:109), M9OMT_32p (SEQ ID NO: 110), M9OMT_33p (SEQ ID NO: 111), M9OMT_34p (SEQ ID NO:112), M9OMT_35p (SEQ ID NO:113), M9OMT_36p (SEQ ID NO:114), M9OMT_37p (SEQ ID NO:115), M9OMT_38p (SEQ ID NO:116), M9OMT_39p (SEQ ID NO:117), M9OMT_40p (SEQ ID NO:118), M9OMT_41p (SEQ ID NO: 119), M9OMT_42p (SEQ ID NO: 120), M9OMT_43p (SEQ ID NO: 121), M9OMT_44p (SEQ ID NO: 122), M9OMT_45p (SEQ ID NO:123), M9OMT_46p (SEQ ID NO:124), M9OMT_47p (SEQ ID NO: 125), M9OMT_48p (SEQ ID NO:126), M9OMT_49p

(SEQ ID NO: 127), M9OMT_50p (SEQ ID NO:128), M9OMT_51p (SEQ ID NO:129), M9OMT_52p (SEQ ID NO:130), M9OMT_53p (SEQ ID NO:131), M9OMT_54p (SEQ ID NO:132), M9OMT_55p (SEQ ID NO:133), M9OMT_56p (SEQ ID NO:134), M9OMT_57p (SEQ ID NO: 135), M9OMT_58p (SEQ ID NO: 136), M9OMT_59p (SEQ ID NO:137), M9OMT_60p (SEQ ID NO:138), M9OMT_61p (SEQ ID NO:139), M9OMT_62p (SEQ ID NO:140), M9OMT_63p (SEQ ID NO:141), M9OMT_64p (SEQ ID NO:142), or a sequence having about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any of the M9OMT sequences.

[0019] In some embodiments, the method comprises an enol O-methyltransferase (enolMT) that catalyzes the methylation of enol oxygens, leading to the conversion of compounds such as demethylcorynantheidine to produce corynantheidine. In some embodiments, the method comprises an enol-O-methyltransferase (enolMT) gene in the Sequence Listing provided herewith, including enolMT_1n (SEQ ID NO:271), enolMT_2n (SEQ ID NO:272), enolMT_3n (SEQ ID NO:273), enolMT_4n (SEQ ID NO:274), enolMT_5n (SEQ ID NO:275), enolMT_6n (SEQ ID NO:276), or enolMT_7n (SEQ ID NO:277), or an enol-O-methyltransferase (enolMT) amino acid sequence in the Sequence Listing provided herewith, including enolMT_1p (SEQ ID NO: 143), enolMT_2p (SEQ ID NO:144), enolMT_3p (SEQ ID NO: 145), enolMT_4p (SEQ ID NO: 146), enolMT_5p (SEQ ID NO: 147), enolMT_6p (SEQ ID NO:148), or enolMT_7p (SEQ ID NO: 149), or a sequence having about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any of the enolMT sequences.

[0020] In some embodiments, the method comprises an indole ring or aromatic hydroxylase (M9H) that catalyzes the hydroxylation of compounds at the 4' position of the indole ring in conjunction with the cofactors NAD (P) H, FMN, and FAD+ such as in corynantheidine to produce 9-hydroxycorynantheidine. Other P450 CYPs enzymes can also catalyze the hydroxylation of compounds at the 7' position of the indole ring such as in mitragynine to produce 7-hydroxymitragynine. In some embodiments, the method comprises an M9H indole ring or aromatic hydroxylase gene of the Sequence Listing provided herewith, including M9H_1n (SEQ ID NO:278), M9H_2n (SEQ ID NO:279), M9H_3n (SEQ ID NO:280), M9H_4n (SEQ ID NO:281), M9H_5n (SEQ ID NO:282), M9H_6n (SEQ ID NO:283), M9H_7n (SEQ ID NO: 284), M9H_8n (SEQ ID NO:285), M9H_9n (SEQ ID NO:286), M9H_10n (SEQ ID NO: 287), M9H_11n (SEQ ID NO:288), M9H_12n (SEQ ID NO:289), M9H_13n (SEQ ID NO: 290), M9H_14n (SEQ ID NO:291), M9H_15n (SEQ ID NO:292), M9H_16n (SEQ ID NO: 293), M9H_17n (SEQ ID NO:294), M9H_18n (SEQ ID NO:295), M9H_19n (SEQ ID NO: 296), M9H_20n (SEQ ID NO:297), M9H_21n (SEQ ID NO:298), M9H_22n (SEQ ID NO: 299), M9H_23n (SEQ ID NO:300), M9H_24n (SEQ ID NO:301), M9H_25n (SEQ ID NO: 302), M9H_26n (SEQ ID NO:303), M9H_27n (SEQ ID NO:304), M9H_28n (SEQ ID NO: 305), M9H_29n (SEQ ID NO:306), M9H_30n (SEQ ID NO:307), M9H_31n (SEQ ID NO: 308), M9H_32n (SEQ ID NO:309), M9H_33n (SEQ ID NO:310), M9H_34n (SEQ ID NO: 311), M9H_35n (SEQ ID NO:312), or M9H_36n (SEQ ID NO:313), or a (M9H) amino acid sequence of the Sequence Listing provided herewith, including M9H_1p (SEQ ID NO: 150), M9H_2p (SEQ ID NO:151), M9H_3p (SEQ ID

NO:152), M9H_4p (SEQ ID NO:153), M9H_5p (SEQ ID NO:154), M9H_6p (SEQ ID NO: 155), M9H_7p (SEQ ID NO:156), M9H_8p (SEQ ID NO: 157), M9H_9p (SEQ ID NO: 158), M9H_10p (SEQ ID NO:159), M9H_11p (SEQ ID NO: 160), M9H_12p (SEQ ID NO:161), M9H_13p (SEQ ID NO:162), M9H_14p (SEQ ID NO: 163), M9H_15p (SEQ ID NO: 164), M9H_16p (SEQ ID NO: 165), M9H_17p (SEQ ID NO: 166), M9H_18p (SEQ ID NO:167), M9H_19p (SEQ ID NO:168), M9H_20p (SEQ ID NO: 169), M9H_21p (SEQ ID NO:170), M9H_22p (SEQ ID NO: 171), M9H_23p (SEQ ID NO: 172), M9H_24p (SEQ ID NO:173), M9H_25p (SEQ ID NO:174), M9H_26p (SEQ ID NO: 175), M9H_27p (SEQ ID NO:176), M9H_28p (SEQ ID NO:177), M9H_29p (SEQ ID NO: 178), M9H_30p (SEQ ID NO:179), M9H_31p (SEQ ID NO:180), M9H_32p (SEQ ID NO: 181), M9H_33p (SEQ ID NO:182), M9H_34p (SEQ ID NO:183), M9H_35p (SEQ ID NO: 184), or M9H_36p (SEQ ID NO:185), or a sequence having about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any of the M9H sequences.

[0021] In embodiments, the polynucleotide sequence comprises a CYB5 gene comprising any one or more of the CYB5 sequences disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences CYB5_In (SEQ ID NO:315) or CYB5_2n (SEQ ID NO: 316).

[0022] In embodiments, the polynucleotide sequence comprises a CPR gene comprising any one or more of the CPR sequences disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences CPR_In (SEQ ID NO:317), CPR_2n (SEQ ID NO: 318), CPR_3n (SEQ ID NO:319), or CPR_4n (SEQ ID NO:320).

[0023] In embodiments, the polynucleotide sequence comprises a transporter gene comprising any one or more of the sequences disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences trxporter_In (SEQ ID NO:321), trxporter_2n (SEQ ID NO: 322), trxporter_3n (SEQ ID NO:323), trxporter_4n (SEQ ID NO:324), trxporter_5n (SEQ ID NO: 325), or trxporter_6n (SEQ ID NO:326).

[0024] In some alternative embodiments, the polynucleotide sequence comprises a tryptamine hydroxylase which can hydroxylate an indole, tryptamine, or tryptophan molecule, such as PsiH-type and other hydroxylase genes, such as a tryptamine-4-hydroxylase (T4H), disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM) and U.S. Patent Application No.: 11,441,164 (BIOSYNTHETIC PRODUCTION OF PSILOCYBIN AND RELATED INTERMEDIATES IN RECOMBINANT ORGANISMS).

[0025] In some alternative embodiments, the polynucleotide sequence comprises an O-methyltransferase which can methylate hydroxyl groups on an indole or tryptamine molecule, such as the indole (or tryptamine)O-methyltransferases (IOMTs) disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM).

[0026] In some alternative embodiments, the polynucleotide sequence comprises a kinase which can phosphorylate

a hydroxylated indole, tryptamine, or tryptophan such as PsiK-type and other kinases genes disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM) and U.S. Patent Application No.: 11,441,164 (BIOSYNTHETIC PRODUCTION OF PSILOCYBIN AND RELATED INTERMEDIATES IN RECOMBINANT ORGANISMS).

[0027] In some alternative embodiments, the polynucleotide sequence comprises a tryptophan or tryptamine halogenase (TrpHalo) which can halogenate indole, tryptamine, and/or tryptophan molecules, such as the halogenases disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM).

[0028] In another aspect the disclosure provides for amino acid sequences (e.g., enzymes, transporters, etc.) that are useful in the biosynthesis of a downstream monoterpene indole alkaloid. In some embodiments, amino acid sequence comprises a modified sequence of a polypeptide (e.g., protein, enzyme, etc.) sequence that comprises a NADPH-dependent reductase (DCS); an enol-O-methyltransferase (enolMT); an indole ring hydroxylase (M9H); or an indole ring O-methyltransferase (M9OMT); or a combination of any two or more of such polypeptides.

[0029] In some further embodiments the amino acid sequence useful in the biosynthesis comprising a polypeptide having a wide variety of biological activity including, for example, a polypeptide associated with electron transport, a polypeptide associated with metabolic processes, transporter protein (importer protein, efflux protein, etc.), an oxidase, a reductase, a dehydrogenase, a peptide synthetase, a halogenase, a transferase (e.g., glycosyltransferase), a synthetase (e.g., tryptophan synthetase) or other polypeptides having functional utility in the biosynthesis of a monoterpene indole alkaloid. In some embodiments the polypeptide comprises a sequence of a cytochrome P450 (CYP), a cytochrome reductase (CPR), a cytochrome B (CYB), a tryptophan synthetase (TRPS), or combinations thereof. In such further embodiments, one or more of the polypeptides can be used in combination.

[0030] In embodiments, the amino acid sequence comprises a DCS sequence disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences DCS_1p-26p.

[0031] In embodiments, the amino acid sequence comprises a M9OMT sequence disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences M9OMT_1p-64p.

[0032] In embodiments, the amino acid sequence comprises an enolMT sequence disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences enolMT_1p-7p.

[0033] In embodiments, the amino acid sequence comprises a M9H sequence disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences M9H_1p-36p.

[0034] In embodiments, the amino acid sequence comprises a CYB5 sequence disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%,

91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences CYB5_Ip (SEQ ID NO: 187) or CYB5_2p (SEQ ID NO:188).

[0035] In embodiments, the amino acid sequence comprises a CPR sequence disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences CPR_Ip (SEQ ID NO: 189), CPR_2p (SEQ ID NO:190), CPR_3p (SEQ ID NO:191), or CPR_4p (SEQ ID NO: 192).

[0036] In embodiments, the amino acid sequence comprises a transporter protein comprising any one or more of the sequences disclosed in the Sequence Listing provided herewith, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any one of sequences trxporter_Ip (SEQ ID NO:193), trxporter_2p (SEQ ID NO: 194), trxporter_3p (SEQ ID NO:195), trxporter_4p (SEQ ID NO:196), trxporter_5p (SEQ ID NO: 197), or trxporter_6p (SEQ ID NO:198).

[0037] In some alternative embodiments, the amino acid sequence comprises a tryptamine hydroxylase which can hydroxylate an indole, tryptamine, or tryptophan molecule, such as PsiH-type and other hydroxylase genes, such as a tryptamine-4-hydroxylase (T4H), disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM) and U.S. Patent Application No.: 11,441,164 (BIOSYNTHETIC PRODUCTION OF PSILOCYBIN AND RELATED INTERMEDIATES IN RECOMBINANT ORGANISMS).

[0038] In some alternative embodiments, the amino acid sequence comprises an O-methyltransferase which can methylate hydroxyl groups on an indole or tryptamine molecule, such as the indole (or tryptamine)O-methyltransferases (IOMTs) disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM).

[0039] In some alternative embodiments, the polynucleotide sequence comprises a kinase which can phosphorylate a hydroxylated indole, tryptamine, or tryptophan molecule, such as PsiK-type and other kinases genes disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM) and U.S. Patent Application No.: 11,441,164 (BIOSYNTHETIC PRODUCTION OF PSILOCYBIN AND RELATED INTERMEDIATES IN RECOMBINANT ORGANISMS).

[0040] In some alternative embodiments, the amino acid sequence comprises a tryptophan or tryptamine halogenase (TrpHalo) which can halogenate indole, tryptamine, and/or tryptophan molecules, such as the halogenases disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM).

[0041] In another aspect the disclosure provides for vectors (e.g., expression cassettes, episomes) that comprise one or more of the polynucleotide sequences, including multiple copies of the same sequence or a sequence encoding the same or similar type of functional amino acid sequence, as described above.

[0042] In another aspect the disclosure provides recombinant cells (or “host cells”) that are engineered to produce an upstream monoterpene indole alkaloid, as discussed herein.

In some embodiments, the recombinant cell comprises one or a plurality of the polynucleotide sequences of the disclosure. In some embodiments, the recombinant cell comprises one or a plurality of the polypeptide sequences of the disclosure. In some embodiments, the recombinant cell is transformed with a vector, or otherwise genetically manipulated, to express one or a plurality of the polynucleotide sequences of the disclosure. In some embodiments, the recombinant cell comprises a modification to one or more naturally-occurring genes in the cell. In some further embodiments, the genetic modification comprises a deletion of an endogenous gene. In some embodiments, the genetic modification comprises an addition of an exogenous gene. In yet further embodiments, the exogenous gene encodes a polypeptide that is different from the polypeptides of the disclosure. In some embodiments, the genetic modification comprises a combination of exogenous genes comprising a sequence that encodes a polypeptide of the disclosure and a sequence that encodes a polypeptide that is different from the polypeptides of the disclosure.

[0043] In any of the above aspects and embodiments relating to a recombinant cell, the cell can be fungal cell or a microbial cell. In some embodiments, a microbial cell can be a bacterial cell (e.g., *E. coli*, or the like) or a yeast cell (e.g., *Saccharomyces* or *Pichia*, or the like). In some embodiments, the recombinant cell comprises a further genetic modification that improves the cells ability to produce a monoterpene indole alkaloid, or a precursor or metabolite thereof.

[0044] In another aspect, the disclosure provides a method for the biosynthesis of a downstream monoterpene indole alkaloid, or a precursor or metabolite thereof, the method comprising culturing a recombinant cell in accordance with the aspects and embodiments of the disclosure under conditions that allow for the biosynthesis.

[0045] In some embodiments, the method provides for the biosynthesis of one or more compounds including strictosidine, hydroxystictosidine, methoxystictosidine, strictosidine aglycone, hydroxystictosidine aglycone, methoxystictosidine aglycone, demethylcorynantheidine (20R, 20S), 9-hydroxydemethylcorynantheidine (20R, 20S), 9-methoxydemethylcorynantheidine (20R, 20S), corynantheidine, dihydrocorynantheine, 9-hydroxycorynantheidine, 9-hydroxydihydrocorynantheine, speciogynine, and/or mitragynine, among others. In some embodiments, the method provides for the biosynthesis of mitragynine.

[0046] In some embodiments, the method provides for the biosynthesis of demethylcorynantheidine.

[0047] In some embodiments, the method provides for the biosynthesis of corynantheidine. In some embodiments, the method provides for the biosynthesis of 9-hydroxycorynantheidine.

[0048] In some embodiments, the method provides for the biosynthesis of a downstream monoterpene indole alkaloid that is fed precursors generated by an upstream MIA biosynthetic pathway providing compounds such as loganin, loganic acid, strictosidine, or strictosidine aglycone, etc.

[0049] Other aspects and embodiments falling within the scope and spirit of the disclosure will be apparent to those of skill in the art in light of the following description and illustrative examples.

BRIEF DESCRIPTION OF DRAWINGS

[0050] FIG. 1 depicts the chemical structures of example monoterpene indole alkaloid (MIA) compounds, including (A) the orientation of the monoterpene and indole structures in an illustrative MIA compound, strictosidine, (B) mitragynine, (C) the mitragynine stereoisomer, speciogynine, and (D) R-group variable substitutions of mitragynine resulting in analogs and derivatives of mitragynine.

[0051] FIG. 2. depicts the biosynthesis pathway to form the 4-ring structure of mitragynine from a monoterpene-derived iridoid glucoside such as secologanin combined with (A) tryptamine to form the strictosidine aglycone precursor for mitragynine, (B) 4-hydroxytryptamine to form the hydroxystictosidine aglycone precursor for mitragynine, and (C) 4-methoxytryptamine to form the methoxystictosidine aglycone precursor for mitragynine.

[0052] FIG. 3 depicts routes to generate mitragynine-type analogs from substituted indoles and tryptamines, such as from (A) a modified indole that leads to a substituted tryptophan, and (B) enzymatic modifications of tryptamine which can lead to downstream analogs of alkaloids including substituted strictosidine through mitragynine analogs.

[0053] FIG. 4 depicts an illustrative diagram of a recombinant host cell expressing combinations of genes for downstream MIA synthesis in accordance with embodiments of the disclosure that can produce MIAs such as (A) a recombinant cell expressing an upstream pathway that leads to precursors and intermediates for mitragynine biosynthesis and (B) a recombinant cell which can express a partial or full pathway to produce mitragynine, upstream mitragynine precursors, and related molecules.

[0054] FIG. 5 depicts the bioconversion of MIA precursors into successive later (downstream) target MIAs, such as (A) MIA precursors, such as loganin, from a plant source fed to a cell to generate mitragynine, a (B) a co-culture, mixed lysate, or cell-free conversion of MIA precursor compounds (e.g., strictosidine, etc.) generated by or sourced from an exogenous recombinant cell, and reacted with (i.e., fed to) a recombinant cell or cell lysate from a cell expressing downstream pathway enzymes to generate mitragynine, and/or (C) a media feed supplemented with any source of precursors, such as modified tryptamines and secologanin to be utilized for mitragynine production.

[0055] FIG. 6 depicts LC-MS extracted ion counts (EIC) and detection of fermentation-derived demethylcorynantheidine from a recombinant host expressing DCS in accordance with example embodiments of the disclosure.

[0056] FIG. 7 depicts LC-MS extracted ion counts (EIC) and detection of fermentation-derived MIA pathway product, corynantheidine, from a recombinant host in accordance with example embodiments of the disclosure.

[0057] FIG. 8 depicts high resolution mass (from HRMS) and detection of fermentation-derived MIA pathway product, corynantheidine, from a recombinant host in accordance with example embodiments of the disclosure.

[0058] FIG. 9 depicts LC-MS/MS fragmentation of yeast fermentation-derived corynantheidine in accordance with example embodiments of the disclosure.

[0059] FIG. 10 depicts the characterization of corynantheidine by comparing retention time, LC-MS, LC-MS/MS fragmentation of yeast fermentation-derived corynantheidine in accordance with example embodiments of the dis-

closure, corynantheidine from kratom extract, hirsutine commercial standard, and corynantheidine commercial standard

[0060] FIG. 11 depicts a phylogenetic analysis of enolMT class in comparison to the SABATH family methyltransferases showing distinct grouping of the enolMT class from IAMT, SAMT, JMT, BAMT, XMT, LAMT, FAMT, M9OMT, C3OMT, and COMT methyltransferase classes.

[0061] FIG. 12 provides an illustrative scheme showing 4-hydroxytryptamine feeding to yeast fermentation cultures to biosynthesize 9-hydroxycorynantheidine (normitragynine).

[0062] FIG. 13 depicts LC-MS extracted ion counts (EIC) and detection of fermentation-derived MIA pathway products 9-hydroxycorynantheidine and its stereoisomer, 9-hydroxy-dihydrocorynantheidine, from a recombinant host in accordance with example embodiments of the disclosure.

[0063] FIG. 14 provides an illustrative scheme showing 4-methoxytryptamine feeding to yeast fermentation cultures to biosynthesize mitragynine in accordance with example embodiments of the disclosure.

[0064] FIG. 15 depicts LC-MS extracted ion counts (EIC) and detection of fermentation-derived MIA pathway products mitragynine and speciogynine, from a recombinant host in accordance with example embodiments of the disclosure.

[0065] FIG. 16 depicts LC-MS extracted ion counts (EIC) and detection of biosynthetic-derived MIA pathway products mitragynine and speciogynine, from a cell-free reaction in accordance with example embodiments of the disclosure.

DETAILED DESCRIPTION

[0066] In a general sense, the disclosure provides for novel sequences (polynucleotide, proteins, and enzymes) that are useful in the production of recombinant cells and biosynthetic processes for the synthesis of indole alkaloids. In various aspects and embodiments, the disclosure provides a biosynthetic method that is sufficient for the production of at least one monoterpene indole alkaloid (MIA) (e.g., a downstream MIA in the mitragynine pathway), synthetic intermediates, metabolites, or precursor species thereof. In various aspects and embodiments, the disclosure provides sequences, recombinant cells, and methods comprising the same for the biosynthesis of monoterpene indole alkaloids, synthetic intermediates, metabolites, or precursor species thereof. In some aspects and embodiments, the disclosure provides a sequence comprising one or more polynucleotide sequences, including polynucleotide sequences that encode for enzymes, transporter proteins, localization proteins, and/or regulatory protein sequences, associated with the biosynthetic pathway for the production of indole alkaloids including mitragynine, that comprise any one or combination of a NADPH-dependent reductase (DCS); an enol-O-methyltransferase (enolMT); an indole ring hydroxylase (M9H); an indole ring O-methyltransferase (M9OMT), a cytochrome b5 (CYB5), a cytochrome P450 reductase (CPR), a tryptophan importer (TAT2), a SAMe importer (SAM3), a transporter (trxporter), a methionine importer (MUP1), a mRNA stabilizer (IME4), a P450/CYP activity enhancer (ICE2 and/or INO2), a SAMe pathway enhancer (ADK1 and/or SAM2), and/or a tryptamine 4-hydroxylase (PsiH or T4H), a tryptamine O-methyltransferase (IOMT), and/or a halogenase (TrpHalo).

Abbreviations and Definitions

[0067] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure belongs. A number of terms and abbreviations appear throughout the disclosure and, unless otherwise indicated, should be understood to have the definitions that follow.

[0068] A “nucleic acid” or “polynucleotide” sequence are used herein to refer to deoxyribonucleotides or ribonucleotides and polymers thereof in either single- or double-stranded form. The terms encompass nucleic acids containing known analogues of natural nucleotides that have similar binding properties as the reference nucleic acid and are metabolized in a manner similar to naturally-occurring nucleotides. Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified or degenerate variants thereof (e.g., degenerate codon substitutions) and complementary sequences, as well as the sequence explicitly indicated.

[0069] The terms “nucleic acid primer,” “nucleic acid probe,” and “oligonucleotide” are all used herein to refer to a short nucleic acid sequence, which may comprise or consist of a fragment of a longer polynucleotide sequence. Oligonucleotides, nucleic acid primers, and/or nucleic acid probes can be DNA, RNA, or a hybrid thereof, or chemically modified analogs or derivatives thereof and are typically single-stranded. However, they can also be double-stranded having two complementing strands that can be separated (e.g., melted) under denaturing conditions (e.g., stringent, moderately stringent, or highly stringent conditions). In some embodiments an oligonucleotide, primer, and/or probe has a length of from about 8 nucleotides to about 200 nucleotides, or from about 12 nucleotides to about 100 nucleotides, or from about 18 to about 50 nucleotides. In embodiments, oligonucleotides, primers, and/or probes can be labeled with detectable markers or modified in any conventional manners for various molecular biological applications (e.g., to inhibit or prevent degradation).

[0070] The terms “amino acid sequence,” “polypeptide,” “protein,” and “peptide” as used herein all refer to a sequence of amino acid residues linked by peptide bonds or modified peptide bonds. The amino acid sequence can be of any length of greater than two amino acids. Polypeptides can include modified forms of the sequence, such as naturally occurring or synthetically generated post-translational modifications, or modifications to the chemical structure of one or more amino acid residues. Non-limiting examples of modified forms include glycosylated sequences, phosphorylated sequences, myristoylated sequences, palmitoylated sequences, ribosylated sequences, acetylated sequences, and the like. Modifications can also include intra- or inter-molecular crosslinking or covalent attachments to moieties such as lipids, flavin, biotin, polyethylene glycol or derivatives thereof, and the like. In addition, modifications may also include protein cyclization, branching of the amino acid chain, and cross-linking of the protein. Further, amino acids other than the naturally-encoded twenty amino acids may also be included in a polypeptide.

[0071] The polypeptide sequences or polynucleotide sequences can be isolated and/or purified, both of which refer to a polypeptide (or a polynucleotide) that is substantially separated from other cellular components (i.e., proteins, DNA, RNA, lipids, membranes, cell debris) of the

organism in which the sequence is produced (e.g., 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99%, 100% free of contaminants).

[0072] As used herein, a “conservative amino acid substitution” refers to an amino acid substitution in a polypeptide sequence wherein the substituted amino acid(s) has similar characteristics to the amino acid in the native sequence, for example charge, hydrophobicity and/or hydrophilicity profile, polarity, size, and the like. Non-limiting examples of conservative amino acid substitutions are Ser for Ala, Thr, or Cys; Lys for Arg; Gln for Asn, His, or Lys; His for Asn; Glu for Asp or Lys; Asn for His or Gln; Asp for Glu; Pro for Gly; Leu for Ile, Phe, Met, or Val; Val for Ile or Leu; Ile for Leu, Met, or Val; Arg for Lys; Met for Phe; Tyr for Phe or Trp; Thr for Ser; Trp for Tyr; and Phe for Tyr. Non-natural amino acids can also serve as a conservative amino acid substitution for a naturally occurring amino acid.

[0073] The term “functional variant” refers to a recombinant biological sequence that is structurally different from a naturally occurring sequence and capable of performing the same function as the naturally occurring sequence. For example, a functional variant of a pathway gene or enzyme comprises a nucleotide and/or amino acid sequence that is altered by one or more nucleotides and/or amino acids compared to the native indole alkaloid pathway gene or enzyme sequences, and is capable of performing the function of the native or parent indole alkaloid pathway gene or enzyme (e.g., an enolMT enzyme capable of generating corynantheidine). In embodiments, a modification to the native or parent sequence may provide for the same or improved functional activity and reaction parameters without altering the underlying function of the native biological sequence. Functional variants may comprise conservative sequence substitutions, sequence additions, and sequence deletions. The sequence modifications can be introduced by standard techniques known in the art, such as site-directed mutagenesis and random PCR-mediated mutagenesis, and may comprise natural as well as non-natural nucleotides and amino acids, and/or analogs thereof. In some embodiments, recombinant biological sequences, including functional variants, comprise amino acid analogs (e.g. amino acids other than the 20 amino acids encoded by DNA or RNA) and/or labeled amino acids and amino acid analogs comprising, for example, fluorescent dyes, radioisotopes, electron dense agents, and the like.

[0074] A “recombinant” nucleic acid or amino acid sequence is a nucleic acid or polypeptide produced by recombinant DNA technology, e.g., as described in Green and Sambrook (2012). The terms “recombinant,” “heterologous,” and “exogenous,” can be used interchangeably herein and, when referring to polynucleotides, mean a polynucleotide (e.g., a DNA or RNA sequence or a gene) that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, a heterologous gene in a host cell includes a gene that is endogenous to the particular host cell but has been modified through, for example, the use of site-directed mutagenesis or other recombinant techniques. The terms also include non-naturally occurring multiple copies of a naturally occurring DNA sequence. Thus, the terms refer to a DNA segment that is foreign or heterologous to the cell, or homologous to the cell but in a position or form within the host cell in which the element is not ordinarily found. Similarly, the terms when referring to polypeptides, means

a polypeptide or amino acid sequence that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. As such, recombinant DNA molecules can be expressed in a host cell to produce a recombinant polypeptide.

[0075] The terms “transformed,” “transgenic,” and “recombinant,” when used with reference to host cells typically refer to an isolated cell or a cell in culture, such as a plant, fungal, or microbial (e.g. bacterial or yeast) cell, into which a heterologous polynucleotide has been introduced or a heterologous amino acid sequence is expressed. The polynucleotide can be integrated into the genome of the host cell, or it can be present as an extrachromosomal molecule. Such an extrachromosomal molecule can be auto-replicating, as discussed herein.

[0076] Transformed cells, tissues, or subjects are understood to encompass not only the end-product of a transformation process, but also transgenic progeny thereof.

[0077] The terms “plasmid,” “vector,” and “cassette” (e.g., transformation cassette, expression vector, or expression cassette) generally refer to an extra-chromosomal element that comprises nucleic acid sequences (e.g., genes, promoters, regulatory elements (inducers, repressors, etc.) and the like) which are not part of the central metabolism of the cell, and can be circular, double-stranded DNA molecules. Such elements may be autonomously replicating sequences, genome integrating sequences, phage or nucleotide sequences (linear or circular) of a single- or double-stranded DNA or RNA, derived from any source, in which a number of nucleotide sequences have been joined or recombined into a unique construction which is capable of introducing a promoter fragment and DNA sequence for a selected gene product along with appropriate 3' untranslated sequence into a cell. Merely for purposes of clarity, a “transformation cassette” refers to a vector containing a foreign gene and having elements in addition to the foreign gene that facilitate transformation of a particular host cell. An “expression cassette” or “expression vector” refers to a vector containing a foreign gene and having elements in addition to the foreign gene that allow for expression and/or enhanced expression of that gene in a foreign host. Thus, the terms can refer to a nucleic acid molecule capable of transporting another nucleic acid molecule to which it has been linked. A non-limiting example of an expression vector includes a gene encoding an enzyme with a promoter that is functional in yeast, where the promoter and gene are oriented such that the promoter drives expression of the enzyme in the yeast cell. A non-limiting example of a vector capable of extra-chromosomal replication is an episome.

[0078] A “linker” refers to a short amino acid sequence that separates multiple domains of a polypeptide. In some embodiments, the linker prohibits energetically or structurally unfavorable interactions between the discrete domains.

[0079] A recombinant gene can be “codon optimized” when its nucleotide sequence is modified to accommodate codon bias of the host organism, typically to improve gene expression and increase translational efficiency of the gene.

[0080] As used herein a “coding sequence” generally refers to a DNA sequence that encodes for a specific amino acid sequence.

[0081] A “regulatory sequence” is generally used to refer to a polynucleotide sequence located upstream (5' non-coding sequences), within, or downstream (3' non-coding sequences) of a coding sequence, and which influence the

transcription, RNA processing or stability, or translation of the associated coding sequence. Regulatory sequences may include promoters, translation leader sequences, introns, and polyadenylation recognition sequences.

[0082] A “promoter” refers to a DNA sequence capable of controlling the expression of a coding sequence or functional RNA. Commonly, a coding sequence is oriented or located 3' to a promoter sequence. Promoters may be derived in their entirety from a native gene, or be composed of different elements derived from different natural promoters, or comprise synthetic DNA segments. Different promoters may direct the expression of a gene in different cell types, or at different stages of development or cell growth/cycle, or in response to different environmental conditions. Promoters that cause a gene to be expressed in most cell types at most times are commonly referred to as “constitutive promoters.”

[0083] The term “operably linked” refers to the association of nucleic acid sequences on a single nucleic acid fragment so that the function of one is affected by the other. For example, a promoter is operably linked with a coding sequence when it is capable of affecting the expression of that coding sequence (i.e., that the coding sequence is under the transcriptional control of the promoter). Coding sequences can be operably linked to regulatory sequences in sense or antisense orientation.

[0084] The term “expression” generally refers to the transcription and stable accumulation of sense (mRNA) or antisense RNA derived from a nucleic acid sequence. “Over-expression” refers to the production of a gene product in transgenic or recombinant organisms that exceeds levels of production in normal or non-transformed organisms.

[0085] “Transformation” is used according to its ordinary and customary meaning as understood by a person of ordinary skill in the art, and is used without limitation to refer to the transfer of a polynucleotide into a target cell. The transferred polynucleotide can be incorporated into the genome or chromosomal DNA of a target cell, resulting in genetically stable inheritance, or it can replicate independent of the host chromosomal. Host organisms containing the transformed nucleic acid fragments are referred to as “transgenic” or “recombinant” or “transformed” organisms.

[0086] Compounds that fall within the scope of the disclosure comprise compounds that can be used in or generated by at least one step or part of the downstream indole alkaloid biosynthetic pathway described herein, including indole alkaloids, synthetic intermediates, metabolites, or precursor species thereof (which may also be referred to as “target” compounds), as generally described herein. Non-limiting examples of such compounds include strictosidine, hydroxystictosidine, methoxystictosidine, strictosidine aglycone, hydroxystictosidine aglycone, methoxystictosidine aglycone, demethylcorynantheidine (20R, 20S), 9-hydroxydemethylcorynantheidine (20R, 20S), 9-methoxydemethylcorynantheidine (20R, 20S), corynantheidine, dihydrocorynantheine, 9-hydroxycorynantheidine, 9-hydroxydihydrocorynantheine, speciogynine, and/or mitragynine, among others.

[0087] Sequences, including polynucleotide and amino acid sequences that fall within the scope of the disclosure include sequences that comprise a protein or enzyme (or sequences encoding the same) involved in at least a portion of the indole alkaloid biosynthetic pathway for mitragynine, including sequences that may facilitate the expression or activity of one or more other sequences of the biosynthetic

pathway. Non-limiting examples of such sequences include enzymes and transport and regulatory proteins, as well as polynucleotides that encode for such enzymes and proteins comprising: a NADPH-dependent reductase (DCS); an enol-O-methyltransferase (enolMT); an indole ring hydroxylase (M9H); or an indole ring O-methyltransferase (M9OMT), among others.

[0088] Other non-limiting examples of such sequences associated with the biosynthetic pathways include enzymes and regulatory proteins, as well as polynucleotides that encode for such enzymes and regulatory proteins comprising: a cytochrome b5 (CYB5), a cytochrome P450 reductase (CPR), a tryptophan importer (TAT2), a SAME importer (SAM3), a transporter (trxporter), a methionine importer (MUP1), a mRNA stabilizer (IME4), a P450/CYP activity enhancer (ICE2 and/or INO2), a SAME pathway enhancer (ADK1 and/or SAM2)), a tryptamine 4-hydroxylase (PsiH or T4H), an indole O-methyltransferase (IOMT), a kinase (PsiK), and/or a halogenase (TrpHalo), among others.

Polynucleotides and Related Enzymes and Proteins

[0089] In various aspects the disclosure provides for polynucleotide sequences that encode amino acid sequences (e.g., enzymes, regulatory proteins, transporters, etc.) that are useful in the biosynthesis of an indole alkaloid. In various aspects the disclosure provides for polynucleotide sequences that encode amino acid sequences (e.g., enzymes, transporters, etc.) that are useful in the biosynthesis of an indole alkaloid such as, for example, a compound in the biosynthetic pathway for mitragynine production. In embodiments of this aspect, the polynucleotide sequence comprises a gene that encodes a polypeptide that is associated with downstream indole alkaloid synthesis (i.e., compounds in a biosynthetic pathway from strictosidine or strictosidine aglycone through to mitragynine, as well as precursors and metabolites thereof).

[0090] Thus, in embodiments, the polynucleotide sequence comprises a modified gene sequence that encodes an enzyme or regulatory protein associated with the indole alkaloid biosynthetic pathway (i.e., from strictosidine or strictosidine aglycone to mitragynine), comprising one or more of a NADPH-dependent reductase (DCS); an enol-O-methyltransferase (enolMT); an indole ring hydroxylase (M9H); or an indole ring O-methyltransferase (M9OMT), a cytochrome b5 (CYB5), a cytochrome P450 reductase (CPR), a tryptophan importer (TAT2), a SAME importer (SAM3), a transporter (trxporter), a methionine importer (MUP1), a mRNA stabilizer (IME4), a P450/CYP activity enhancer (ICE2 and/or INO2), a SAME pathway enhancer (ADK1 and/or SAM2), a tryptamine 4-hydroxylase (PsiH or T4H), an indole O-methyltransferase (IOMT), a kinase (PsiK), and/or a halogenase (TrpHalo).

[0091] In various embodiments of the biosynthetic methods and recombinant organisms disclosed herein the disclosure provides a combination of one or more of the polynucleotide sequences each encoding a functional amino acid sequence (e.g., enzyme, regulatory protein, transport protein, etc.), as described herein, that participates in the biosynthetic process.

[0092] In embodiments, the sequences comprise a DCS polynucleotide sequence disclosed as DCS_In-26n, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any such DCS polynucleotide sequence. In some embodiments, the

DCS polynucleotide encodes a DCS amino acid sequence of DCS_Ip-26p, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any such DCS amino acid sequence.

[0093] In embodiments, the sequences comprise a M9OMT polynucleotide sequence disclosed as M9OMT_In-64n, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any such M9OMT polynucleotide sequence. In some embodiments, the M9OMT polynucleotide encodes a M9OMT amino acid sequence of M9OMT_Ip-64p, or a sequence having about 90% sequence identity to any such M9OMT amino acid sequence.

[0094] In embodiments, the sequences comprise an enolMT polynucleotide sequence disclosed as enolMT_In-7n, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any such enolMT polynucleotide sequence. In some embodiments, the enolMT polynucleotide encodes a enolMT amino acid sequence of enolMT_Ip-7p, or a sequence having about 90% sequence identity to any such enolMT amino acid sequence.

[0095] In embodiments, the sequences comprise a M9H polynucleotide sequence disclosed as M9H_In-36n, or a sequence having at least 90% sequence identity to any such M9H polynucleotide sequence. In some embodiments, the M9H polynucleotide encodes a M9H amino acid sequence of M9H_Ip-36p, or a sequence having about 90% sequence identity to any such M9H amino acid sequence.

[0096] In some embodiments, the polynucleotide comprises a cytochrome P450 reductase (CPR) gene comprising any of the sequences in the Sequence Listing provided herewith, including CPR_In, CPR_2n, CPR_3n, or CPR_4n, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to the CPR sequences. In some embodiments, the polynucleotide comprises a cytochrome P450 reductase (CPR) gene sequence that encodes a cytochrome P450 reductase (CPR) amino acid sequence in the Sequence Listing provided herewith, including CPR_Ip, CPR_2p, CPR_3p, or CPR_4p, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any of the CPR amino acid sequences.

[0097] In some embodiments, the sequences comprise trxporter genes that are in the Nitrate/Peptide Family (NPF) transporters that uptake exogenous iridoid glucosides including loganic acid and loganin, increasing the intracellular concentrations of these iridoid glucosides. In some embodiments, the polynucleotide comprises a transporter (trxporter) gene comprising any of the sequences in the Sequence Listing provided herewith, including trxporter_In, trxporter_2n, trxporter_3n, trxporter_4n, trxporter_5n, or trxporter_6n, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to the trxporter sequences.

[0098] In some embodiments, the polynucleotide comprises a transporter (trxporter) gene sequence that encodes a transporter (trxporter) amino acid sequence in the Sequence Listing provided herewith, including trxporter_Ip, trxporter_2p, trxporter_3p, trxporter_4p, trxporter_5p, or trxporter_6p, or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any of the trxporter amino acid sequences. In some embodiments, the polynucleotide comprises a cytochrome

b5 (CYB5) gene, which together with CPR can relay electrons from NADPH to a cytochrome p450 enzyme using FAD and FMN as cofactors. Coexpression of CPR and CYB5 proteins can enhance enzymatic oxidations, such as increasing the activity of the P450 enzymes described herein. In some embodiments, the polynucleotide comprises a CYB5 gene comprising any of the sequences in the Sequence Listing provided herewith, including CYB5_In (SEQ ID NO:315), or CYB5_2n (SEQ ID NO:216), or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to the CYB5 sequences. In some embodiments, the polynucleotide comprises a cytochrome b5 (CYB5) gene sequence that encodes a cytochrome b5 (CYB5) amino acid sequence in the Sequence Listing provided herewith, including CYB5_1p (SEQ ID NO:187) or CYB5_2p (SEQ ID NO:188), or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to any of the CYB5 amino acid sequences.

[0099] In some embodiments, the polynucleotide comprises a tryptophan importer (TAT2) gene comprising any of the sequences in the Sequence Listing provided herewith, including TAT2_In (SEQ ID NO:366) or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to the TAT2 sequences. In some embodiments, the polynucleotide comprises a tryptophan importer (TAT2) gene sequence that encodes a tryptophan importer (TAT2) amino acid sequence.

[0100] In some embodiments, the polynucleotide comprises a SAMe importer (SAM3) gene comprising any of the sequences in the Sequence Listing provided herewith, including SAM3_In (SEQ ID NO:364) or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to the SAM3 sequences. In some embodiments, the polynucleotide comprises a SAMe importer (SAM3) gene sequence that encodes a SAMe importer (SAM3) amino acid sequence.

[0101] In some embodiments, the polynucleotide comprises a methionine importer (MUP1) gene comprising any of the sequences in the Sequence Listing provided herewith, including MUP1_In (SEQ ID NO:365) or a sequence having at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to the MUP1 sequences. In some embodiments, the polynucleotide comprises a methionine importer (MUP1) gene sequence that encodes a methionine importer (MUP1) amino acid sequence.

[0102] In some embodiments, the polynucleotide comprises a mRNA stabilizer (IME4) gene, a P450/CYP activity enhancer (ICE2 and/or INO2) gene and INO2_In (SEQ ID NO:361), a SAMe pathway enhancer (ADK1 and/or SAM2) gene, comprising any of the sequences identified in the Sequence Listing provided herewith, including IME4_In (SEQ ID NO:367), ICE2_In (SEQ ID NO:360), INO2_In (SEQ ID NO:361), ADK1_In (SEQ ID NO:363), and/or SAM2_In (SEQ ID NO:362), SAM3_In (SEQ ID NO:364), or a sequence having about 90% sequence identity to these sequences. In some embodiments, the polynucleotide comprises a mRNA stabilizer (IME4) gene, e.g., IME4_In (SEQ ID NO:367), a P450/CYP activity enhancer (ICE2 and/or INO2) gene, a SAMe pathway enhancer (ADK1 and/or SAM2) gene that encodes the corresponding amino acid sequence, i.e., (IME4), (ICE2), (INO2), (ADK1), a (SAM2) amino acid sequence, a tryptamine 4-hydroxylase (PsiH or

T4H), an indole O-methyltransferase (IOMT), a kinase (PsiK), and/or a halogenase (TrpHalo).

[0103] In some embodiments, N terminal peptide sequences for membrane localization are fused to the N terminus of P450s such as the M9H and CPR proteins. In some embodiments, membrane localization signals LclTag_1p-28p (SEQ ID NO:202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, and 229, respectively), encoded by LclTag_In-28n (SEQ ID NO:330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356 and 357, respectively), are fused to any of the sequences described herein. Such fusion polypeptides when expressed may lead to increased P450/CYP and CPR activity due to enhanced membrane localization.

[0104] Hmx1 is a heme oxygenase involved in the degradation of heme, and can be incorporated in accordance with some aspects and example embodiments of the disclosure. For example, in some embodiments, the HMX1 gene of the host is deleted to reduce heme degradation. Heme depletion is a major source of cellular stress and cells are particularly susceptible to this source of stress when several p450 monooxygenase enzymes are expressed in the cell. In other embodiments, enzymes to increase cellular heme are expressed, including Hem3, Hem2 and Hem12. Iron (II) citrate and 8-aminolevulinic acid can be fed into fermentation media to contribute to heme production in a recombinant host cell.

[0105] The tryptophan importer (TAT2) is a high affinity tryptophan and tyrosine permease. Sec, e.g., TAT2_In (SEQ ID NO:366). Heterologous pathway enzymes that are expressed to produce compounds with an indole core such as tryptamine and serotonin use L-tryptophan as a directing molecule. Tryptophan production in cells is normally tightly regulated. Tryptophan accumulation in a recombinant host is increased by overexpressing a recombinant L-tryptophan transporter (TAT2). This allows for exogenous tryptophan to be fed to the cells and transported in the recombinant host. See also, WO/2021/248087.

[0106] SAMe importer (SAM3) is a high-affinity S-adenosylmethionine permease. In some embodiments, a recombinant host is modified to increase the accumulation of the methyl donor, SAMe, which is used in the upstream pathway by enzymes such as the recombinant LAMT enzymes to methylate loganic acid. SAMe accumulation in the recombinant host cell can be increased by overexpressing this permease and feeding exogenous SAMe into the media. See also PCT Patent Application Publication WO/2021/248087.

[0107] The methionine importer (MUP1) is a high affinity methionine permease. An example is MUP_In (SEQ ID NO:365). SAMe is a robust methyl donor synthesized from methionine and ATP. In some embodiments, to increase the uptake of exogenous L-methionine fed into the SAMe pathway, recombinant Mup1 is overexpressed. See also PCT Patent Application Publication WO/2021/248087.

[0108] In some embodiments, the nucleic acids described herewith encode a polypeptide or oligopeptide having an amino acid sequence that is naturally occurring. In other embodiments, the nucleic acids encode a polypeptide or oligopeptide having an amino acid sequence that is not naturally occurring. The encoded polypeptides or oligopeptides that are not naturally occurring can vary from a naturally occurring polypeptide or oligopeptide, or portion

thereof, by a small amount (e.g., one conservative amino acid substitution or a histidine tag) or extensively (e.g., further comprising a fusion peptide, a substituted or added domain from another protein, a scaffold, etc.).

[0109] In some of the above embodiments, the nucleic acid provided herein comprises the sequence of any one of the nucleic acid sequences PCT Patent Application Publication.

[0110] Nucleic acid sequences in accordance with the disclosure are synthesized and cloned using techniques known in the art. Gene expression can be controlled by inducible or constitutive promoter systems using the appropriate expression vectors. Genes are transformed into an organism using standard yeast or fungus transformation methods to generate modified host strains (i.e., the recombinant host organism). The modified strains express the genes for biosynthetic pathways that generate indole alkaloid products in the mitragynine biosynthetic pathway, which can be fed with compounds produced in an upstream biosynthetic pathway (e.g., loganin, loganic acid, strictosidine, strictosidine aglycone, etc.). The indole alkaloid pathway genes herein can be integrated into the genome of the cell or maintained as an episomal plasmid. Recombinant host fermentation samples are: (i) prepared and extracted using a combination of fermentation, dissolution, and purification steps; and (ii) analyzed by HPLC for the presence of directing molecules, precursor molecules, intermediate molecules, and target molecules such as those illustrated in the Examples and otherwise disclosed herein.

[0111] The polynucleotides can be used in, or used to generate, modified strains of host cells, which produce target compounds including mitragynine alkaloids, precursors, intermediates, or metabolites thereof.

[0112] In some embodiments, the nucleic acid sequence (i.e., polynucleotide) encoding a gene or a complementary nucleic acid sequence to such a coding sequence can be codon optimized for production in a selected microorganism. A number of factors can be used in determining a codon-optimized sequence (see, e.g., U.S. Pat. No. 10,435,727). Factors can include, for example, (1) selecting a codon for each amino acid residue in the recombinant polypeptide based on the usage frequency of each codon in the heterologous host cell (e.g., *Saccharomyces cerevisiae*) genome; (2) removing sequences that provide for restriction sites for enzymes to prevent DNA cleavage; (3) modifying long repeats (e.g., consecutive sequences of 5 or more nucleotide) to prevent low-complexity regions; (4) adding a ribosome binding site to the N-terminus; (5) adding a stop codon; (6) changing nucleotides that encode amino acids susceptible to undesirable post-translational modifications (e.g., changing codons for a surface exposed LYS to an ARG codon to avoid ubiquitination); (7) removing or replacing a localization signal sequence.

[0113] In various embodiments, the nucleic acid sequences can further comprise additional sequence encoding amino acids that are not part of the included enzymes or regulatory proteins herein. In some of these embodiments, the additional sequences encode additional amino acids present when the nucleic acid is translated, encoding, for example, a co-folding peptide, as disclosed herein, or an additional protein domain, with or without a linker sequence, creating a fusion protein. Other examples are localization sequences, i.e., signals directing the localization of the folded protein to a specific subcellular compartment

or membrane. Additional non-limiting examples are an affibody tag, a localization scaffold, a vacuolar localization tag, a secretion signal, and a histidine tag (e.g., 6xhis tag). Additional examples include cleavage sites, such as a TEV protease recognition sequence or 2A-self-cleaving peptides encoded between two or more genes for cleavage of individual proteins post-translation and enabling polycistronic gene expression.

[0114] In some embodiments, two or more recombinant polynucleotide sequences can be fused together (i.e., co-expressed) by including a nucleotide linker sequence between a polynucleotide sequence and an additional polynucleotide sequence or polynucleotide sequences, such as the sequences disclosed herein. The resulting chimeric genes encode for fusions (i.e., a polypeptide fused to another polypeptide or polypeptides), such as the recombinant amino acid sequences disclosed herein. In embodiments, the linker region(s) can be the same or different, and can comprise from 3 to 50 amino acids. In such embodiments the fused polynucleotide sequences encoding fused polypeptides, such as fusions between the sequences disclosed herein (DCS with SGD, DCS with enoMT, among other combinations), can lead to enhanced yields by increasing the local concentration of substrate and active enzyme due to the increased co-localization of the fused enzymes. These embodiments relating to fusions of one or more sequences in accordance with the disclosure also confer a benefit that can allow for a single promoter to control expression of multiple recombinant proteins, which can decrease the size of expression cassettes.

[0115] In some embodiments, the nucleic acid comprises additional nucleotide sequences that are not translated. Non-limiting examples include promoters, terminators, barcodes, Kozak sequences, targeting sequences, and enhancer elements. In some particular embodiments the polynucleotide sequences comprise a promoter that is functional in yeast, fungi, and bacteria. In embodiments, expression of a gene encoding an enzyme or regulatory protein is controlled by the promoter operably linked to the gene sequence. For a gene to be expressed, a promoter must be present within 1,000 nucleotides upstream of the gene. A gene is generally cloned under the control of a desired promoter. The promoter is placed upstream of the gene in the genome or on an episomal plasmid. The promoter regulates the amount of enzyme expressed in the cell and the timing of expression, or expression in response to external factors such as carbon source.

[0116] Any promoter can be utilized to drive the expression of the enzymes and regulatory proteins described herein. Listings of various promoters in organisms such as yeast are readily available (See, e.g., the registry of Standard Biological Parts for yeast at the website: parts.igem.org/Yeast). Several of the exemplary promoters listed in Table 1 below drive strong expression, constant gene expression, medium or weak gene expression, or provide inducible gene expression. Inducible or repressible gene expression is dependent on the presence or absence of a certain molecule (inducer/repressor). For example, the GAL1, GAL7, and GAL10 promoters are activated by the presence of galactose and are repressed by the presence of glucose. The HO promoter is active and drives gene expression only in the presence of the alpha factor peptide. The HXT1 promoter is activated by the presence of glucose while the ADH2 promoter is repressed by the presence of glucose.

TABLE 1

Exemplary promoters		
Constitutive promoters (Strong)	Constitutive promoters (Medium and weak)	Inducible/repressible promoters
TEF1	STE2	GAL1
PGK1	TP11	GAL7
PGI1	PYK1	GALI0
TDH3	CYC1	HO
T7	ADH1	HXT1
		ADH2
		AOX1
		T5-lac

[0117] As discussed herein, various embodiments of the disclosure provide the nucleic acid sequence as an expression cassette, e.g., a yeast expression cassette. Any yeast expression cassette capable of expressing the enzyme in a yeast cell can be utilized. Additional regulatory elements can also be present in the expression cassette, including restriction enzyme cleavage sites, antibiotic resistance genes, integration sites, auxotrophic selection markers, origins of replication, and degrons.

[0118] The expression cassette can be present in a vector that, when transformed into a host cell, either integrates into chromosomal DNA or remains episomal in the host cell. Such vectors are well-known in the art. One non-limiting example of a yeast vector is a yeast episomal plasmid (YE_p) that contains the pBluescript II SK (+) phagemid backbone, an auxotrophic selectable marker, yeast and bacterial origins of replication and multiple cloning sites enabling gene cloning under a suitable promoter (see Table 1). Other non-limiting vectors include pRS series plasmids.

[0119] Mutations introduced into the DNA can provide enzyme variations that can prevent or promote post-translational modifications of the protein. Non-limiting examples of post-translational modifications include phosphorylation, acetylation, methylation, SUMOylation, ubiquitination, proteolytic cleavage, lipidation, prenylation such as farnesylation or myristoylation, glycosylation, nitrosylation and biotinylation.

[0120] The nucleic acid sequences can be modified from a gene from any source, e.g., any microorganism, protist, virus, plant, or animal. In some embodiments, the gene encoding an enzyme or regulatory protein is derived from a bacterium. For example, the bacterium can be from phylum Abditibacteriota, including class Abditibacteria, including order Abditibacteriales; phylum Abyssobacteriia or Acidobacteriia, including class Acidobacteriia, Blastocatellia, Holophagae, Thermoanaerobaculia, or Vicinamibacteria, including order Acidobacteriales, Bryobacteriales, Blastocatellales, Acanthopleuribacteriales, Holophagales, Thermotomaculales, Thermoanaerobaculales, or Vicinamibacteraceae; phylum Actinobacteria, including class Acidimicrobiia, Actinobacteria, Actinomarinidae, Coriobacteriia, Nitriliruptoria, Rubrobacteria, or Thermoleophilia, including orders Acidimicrobiales, Acidothermales, Actinomycetales, Actinopolysporales, Bifidobacteriales, Nanopezagicales, Catenulisporales, Cornebacteriales, Cryptosporangiales, Frankiales, Geodermatophilales, Glycomycetales, Jiangellales, Micrococcales, Micromonosporales, Nakamurellales, Propionibacteriales, Pseudonocardiales, Sporichthyales, Streptomycetales, Streptosporangiales, Actinomarinales, Coriobacteriales, Eggerthellales, Egi-

bacteriales, Egicoccales, Euzebyales, Nitriliruptorales, Gaiellales, Rubrobacteriales, Solirubrobacteriales, or Thermoleophilales; phylum Aquificae, including class Aquificae, including order Aquificales or Desulfurobacteriales; phylum Armatimonadetes, including class Armatimonadia, including order Armatimonadales, Capsulimonadales, Chthonomonadetes, Chthonomonadales, Fimbriimonadia, or Fimbriimonadales; phylum Aureobacteria or Bacteroidetes, including class Armatimonadia, Bacteroidia, Chitinophagia, Cytophagia, Flavobacteria, Saprospira or Sphingobacteriia, including order Bacteroidales, Marinilabiales, Chitinophagales, Cytophagales, Flavobacteriales, Saprospirales, or Sphingopacteriales; phylum Balneolaeota, Caldiserica, Calditrichaeota, or Chlamydiae, including class Balneolia, Caldiserica, Calditrichae, or Chlamydia, including order Balneolales, Caldisericales, Calditrichales, Anoxychlamydiales, Chlamydiales, or Parachlamydiales; phylum Chlorobi or Chloroflexi, including class Chlorobia, Anaerolineae, Ardentcatenia, Caldilineae, Thermofonsia, Chloroflexia, Dehalococcoidia, Ktedonobacteria, Tepidiformia, Thermoflexia, Thermomicrobia, or Sphaerobacteridae, including order Chlorobiales, Anaerolineales, Ardentcatenales, Caldilineales, Chloroflexales, Herpetosiphonales, Kallotenuales, Dehalococcoidales, Dehalogenimonas, Ktedonobacteriales, Thermogemmatisporales, Tepidiformales, Thermoflexales, Thermomicrobiales, or Sphaerobacteriales; phylum Chrysiogenetes, Cloacimonetes, Coprothermobacterota, Cryoseri-cota, or Cyanobacteria, including class Chrysiogenetes, Coprothermobacteria, Gloeobacteria, or Oscillatorio-phycideae, including order Chrysiogenales, Coprothermobacte- rales, Chroococcidiopsidales, Gloeomargaritales, Nosto- cales, Pleurocapsales, Spirulinales, Synechococcales, Gloeobacteriales, Chroococcales, or Oscillatoriales; phyla: Efferibacteres, Deinococcus-thermus, Dictyoglomi, Dormi- bacteriaeota, Elusimicrobia, Eremiobacteraeota, Fermenti- bacteria, or Fibrobacteres, including class Deferribacteres, Deinococci, Dictyoglomia, Elusimicrobia, Endomicrobia, Chitinispirillia, Chitinivibrionia, or Fibrobacteria, including order Deferribacterales, Deinococcales, Thermales, Dictyo- glomales, Elusimicrobiales, Endomicrobiales, Chitinispiril- lales, Chitinivibrionales, Fibrobacteriales, or Fibro- monadales;

[0121] phylum Firmicutes, Fusobacteria, Gemmatimonada- detes, or Hydrogenedentes, including class Bacilli, Clostridia, Erysipelotrichia, Limnochordia, Negativicutes, Thermolithobacteria, Tissierellia, Fusobacteriia, Gemmati- monadetes, Longimicrobia, including order Bacillales, Lac- tobacillus, Borkfalkiales, Clostridiales, Halanaerobiales, Natranaerobiales, Thermoanaerobacteriales, Erysipelot- richales, Limnochordales, Acidaminococcales, Selenom- onadales, Veillonellales, Thermolithobacteriales, Tissierella- les, Fusobacteriales, Gemmatimonadales, or Longimicrobia; phylum Hydrogenedentes, Ignavibacteriae, Kapabacteria, Kiritimatiellaeota, Krumholzibacteriota, Kryptonia, Latesci- bacteria, LCP-89, Lentisphaerae, Margulisbacteria, Marini- microbia, Melainabacteria, Nitrospinae, or Omnitrophica, including class Ignavibacteria, Kiritimatiellae, Krumholzi- bacteria, Lentisphaeria, Oligosphaeria, or Nitrospinae, including order Ignavibacteriales, Kiritimatiellales, Krum- holzibacteriales, Lentisphaerales, Victivallales, Oligosphae- rales, or Nitrospina; phylum Omnitrophica or Planctomy- cetes, including class Brocadiae, Phycisphaerae, Planctomycetia, or Phycisphaerales, including order Sedi- mentisphaerales, Tepidisphaerales, Gemmatales, Iso-sphaer-

ales, Pirellulales, or Planctomycetales; phylum Proteobacteria including class Acidithiobacillia, Alphaproteobacteria, Betaproteobacteria, Lambdaproteobacteria, Muproteobacteria, Deltaproteobacteria, Epsilonproteobacteria, Gammaproteobacteria, Hydrogenophilales, Oligoflexia, or Zetaproteobacteria, including order Acidithiobacillales, Caulobacteriales, Emericellales, Holosporales, Iodimonadales, Kiloniellales, Koppriimonadales, Kordiimonadales, Magnetococcales, Micropepsales, Minwuiiales, Parvularculales, Pelagibacterales, Rhizobiales, Rhodobacteriales, Rhodospirillales, Rhodothalassiales, Rickettsiales, Sneathiiellales, Sphingomonadales, Burkholderiales, Ferri-trophicales, Ferrocales, Neisseriales, Nitrosomonadales, Procabacteriales, Rhodocyclales, Bradymonadales, Acidulodesulfobacteriales, Desulfarculales, Desulfobacteriales, Desulfobacteriales, Desulfurellales, Desulfuromonadales, Myxococcales, Syntrophobacteriales, Campylobacteriales, Nautiliales, Acidiferrobacteriales, Aeromonadales, Alteromonadales, Arenicellales, Cardiobacteriales, Cellvibrionales, Chromatiales, Enterobacteriales, Immundisolibacteriales, Legionellales, Methylococcales, Nevskiales, Oceanospirillales, Orbales, Pasteurellales, Pseudomonadales, Salinisphaerales, Thiotrichales, Vibrionales, Xanthomonadales, Hydrogenophilales, Bacteriovorales, Bdellovibrionales, Oligoflexiales, Silvanigrellales, or Mariprofundales; phylum Rhodothermaeota, Saganbacteria, Sericytochromatia, Spirochaetes, Synergistetes, Tectomicrobia, or Tenericutes, including class Rhodothermia, Spirochaetia, Synergistia, Izimaplasma, or Mollicutes, including order Rhodothermales, Brachyspirales, Brevinematales, Leptospirales, Spirochaetales, Synergistales, Achleplasmatiales, Anaeroplasmatales, Entomoplasmatales, or Mycoplasmatiales; phylum Thermodesulfobacteria, Thermotogae, Verrucomicrobia, or Zixibacteria, including class Thermodesulfobacteria, Thermotogae, Methylacidiphilae, Opitutae, Spartobacteria, or Verrucomicrobiae, including order Thermodesulfobacteriales, Kosmotogales, Mesoaciditogales, Petrotogales, Thermotogales, Methylacidiphilales, Opitutales, Puniceococcales, Xiphinematobacter, Chthoniobacteriales, Terrimicrobium, or Verrucomicrobiales.

[0122] In other embodiments, the gene encoding the enzyme or regulatory protein is modified from an archaeon. For example, the archaeon can be from: phylum Euryarchaeota, including class Archaeoglobi, Hadesarchaea, Halobacteria, Methanobacteria, Methanococci, Methanofastidiosia, Methanomicrobia, Methanopyri, Nanoarchaea, Theionarchaea, Thermococci, or Thermoplasmata, including order Archaeoglobales, Hadesarchaeales, Halobacteriales, Methanobacteriales, Methanococcales, Methanocellales, Methanomicrobiales, Methanophagales, Methanosarcinales, Methanopyrales, Thermococcales, Methanomassiliicoccales, Thermoplasmatales, or Nanoarchaeales; DPANN superphylum, including subphyla Aenigmarchaeota, Altiarchaeota, Diapherotrites, Micrarchaeota, Nanoarchaeota, Pacearchaeota, Parvarchaeota, or Woesearchaeota; TACK superphylum, including subphylum Korarchaeota, Crenarchaeota, Aigarchaeota, Geoarchaeota, Thaumarchaeota, or Bathyarchaeota; Asgard superphylum including subphylum Odinarchaeota, Thorarchaeota, Lokiarchaeota, Helarchaeota, or Heimdallarchaeota.

[0123] In additional embodiments, the gene encoding the enzyme or regulatory protein is modified from a fungus. For example, the fungus can be from: phyla Chytridiomycota, Basidiomycota, Ascomycota, Blastocladiomycota, Ascomy-

cota, Microsporidia, Basidiomycota, Glomeromycota, Symbiomycota, and Neocallimastigomycota; phylum Ascomycota, including classes and orders Pezizomycotina, Arthoniomycetes, Coniocybomycetes, Dothideomycetes, Eurotiomycetes, Geoglossomycetes, Laboulbeniomycetes, Lecanoromycetes, Leotiomycetes, Lichinomycetes, Orbiliomycetes, Pezizomycetes, Sordariomycetes, Xylonomycetes, Lahmiales, Itchiclahmadion, Triblidiales, Saccharomycotina, Saccharomyces, Taphrinomycotina, Archaeorhizomycetes, Neoelectromycetes, Pneumocystidomycetes, Schizosaccharomycetes, Taphrinomycetes; phylum Basidiomycota including subphyla or classes Pucciniomycotina, Ustilaginomycotina, Wallemiomycetes, and Entorrhizomycetes; subphylum Agaricomycotina including classes Tremellomycetes, Dacrymycetes, and Agaricomycetes; phylum Symbiomycota, including class Entorrhizomycota; subphylum Ustilaginomycotina including classes Ustilaginomycetes and Exobasidiomycetes; phylum Glomeromycota including classes Archaeosporomycetes, Glomeromycetes, and Paraglomeromycetes; subphylum Pucciniomycotina including orders and classes: Pucciniomycotina, Cystobasidiomycetes, Agaricostilbomycetes, Microbotryomycetes, Atractiellomycetes, Classiculomycetes, Mixiomycetes, and Cryptomycocolacomycetes; subphylum incertae sedis Mucoromycota including orders Calcarisporiellomycota and Mucoromycota; phylum Mortierellomycota including class Mortierellomycota; subphylum incertae sedis Entomophthoromycotina including order Entomophthorales; phylum Zoopagomycota including classes Basidiobolomycota, Entomophthoromycota, Kickxellomycota, and Zoopagomycotina; subphylum incertae sedis Mucoromycotina including orders Mucorales, Endogonales, and Mortierellales; phylum Neocallimastigomycota including class Neocallimastigomycetes; phylum Blastocladiomycota including classes Physodermatomycetes and Blastocladiomycetes; phylum Rozellomycota including classes Rozellomycota and Microsporidia; phylum Aphelidiomycota including class Aphelidiomycota; phylum Chytridiomycota including classes Chytridiomycetes and Monoblepharidomycetes; phylum Oomycota including classes or orders Leptomitales, Myzocytiopsidales, Olpidiopsidales, Peronosporales, Pythiales, Rhipidiales, Salilagenidiales, Saprolegniales, Sclerosporales, Anisopidiales, Lagenismatales, Rozellopsidales, and Haptoglossales.

[0124] In additional embodiments, the gene for the enzyme or regulatory protein is derived from the organism below. This includes but is not limited to: *Acanthurus tractus*, *Aplysina aerophoba*, *Stevia rebaudiana*, *Bos Taurus*, *Bufo bufo*, *Bufo viridis*, *Chrysochloris asiatica*, *Fukomys damarensis*, *Streptomyces reticuliscabie*, *Homo sapiens*, *Rattus norvegicus*, *Rhinella marina*, *Rhinella spinulosa*, *Schistosoma mansoni*, *Xenopus laevis*, *Xenopus tropicalis*, *Acacia koa*, *Arabidopsis thaliana*, *Psilocybe cubensis*, *Brassica oleracea*, *Citrus sinensis*, *Hordeum vulgare*, *Juglans cinerea*, *Juglans regia*, *Lophophora williamsii*, *Nymphaea colorata*, *Oryza sativa*, *Ipomoea violacea*, *Rivea corymbosa*, *Argyrea nervosa*, *Gentiana rigescens*, *Nyssa sinensis*, *Camptotheca acuminata*, *Merremia tuberosa*, *Mitragyna speciosa*, *Tabernanthe iboga*, *Tabernaemontana elegans*, *Voacanga africana*, *Solanum lycopersicum*, *Sorghum bicolor*, *Theobroma cacao*, and *Triticum aestivum*, *Catharanthus roseus*, *Actinidia polygama*, *Corydalis cheilanthesifolia*, *Lonicera japonica*, *Olea europaea*, *Antirrhinum*

majus, *Salix suchowensis*, *Sesamum indicum*, *Salvia splendens*, *Nepeta racemosa*, *Nothapodytes nimmonian*, *Vinca minor*, *Rhazya stricta*, *Corydalis yanhusuo*, *Linum usitatissimum*, *Rauvolfia serpentina*, *Ophiorrhiza pumila*, *Kalopanax septemlobus*, *Amsonia hubrichtii*, *Phaedon cochleariae*, *Aphis glycines*, *Pogonomyrmex barbatus*, *Thermobifida fusca* YX, *Nocardiopsis kunsanensis*, *Actinorugispora endophytica*, *Swertia mussotii*, *Aedes aegypti*, *Arabidopsis thaliana*, *Helianthus tuberosus*, *Phratora vitellinae*, *Macadamia integrifolia*, *Mentha canadensis*, *Coptis chinensis*, *Durio zibethinus*, *Hibiscus syriacus*, *Populus alba*, *Capsicum pubescens*, *Uncaria rhynchophylla*, *Coffea canephora* and *Coffea eugenioides*.

Enzymes and Amino Acid Sequences

[0125] The disclosure provides for non-naturally occurring amino acid sequences (i.e., enzymes or regulatory proteins) comprising a sequence encoded by any of the nucleic acids described above. In some embodiments, the amino acid sequence is 85%, 90%, 95%, 98%, or 100% identical to any one of the sequences in the amino acid sequences disclosed herein (e.g., listed in the Sequence Listing provided herewith). In these embodiments, the enzyme or regulatory protein can be isolated in vitro and used in vitro to provide enzyme activity. Alternatively, the enzyme can be expressed in a recombinant organism as described herein. In some embodiments, the recombinant microorganism for the recombinant production of an amino acid sequence is a bacterium, for example an *E. coli*. In other embodiments, the recombinant microorganism is a yeast or fungal cell, e.g., a species of *Saccharomyces* (for example *S. cerevisiae*), *Candida*, *Pichia*, *Schizosaccharomyces*, *Scheffersomyces*, *Blakeslea*, *Rhodotorula*, *Aspergillus* or *Yarrowia*.

[0126] In some embodiments relating to production and purification of the recombinant amino acid sequences, the gene encoding the enzyme and/or regulatory protein is cloned into an expression vector such as the pET expression vectors from Novagen, transformed into a protease deficient strain of *E. coli* such as BL21 and expressed by induction with IPTG. The protein of interest may be tagged with an affinity tag to facilitate purification, e.g. hexahistidine, GST, calmodulin, TAP, AP, CAT, HA, FLAG, MBP, etc. Coexpression of a bacterial chaperone such as dnak, GroES/GroEL or SecY may help facilitate protein folding. See Green and Sambrook (2012).

[0127] In some aspects relating to sequences comprising amino acids (e.g., enzymes) and nucleotides (e.g., polynucleotides/genes), the disclosure provides sequence variants of the sequences disclosed herein. With regard to amino acid sequences, variants (e.g., substitution, deletion, addition) can be considered as “similar” to the original polypeptide or to have a certain “percent identity” to the original polypeptide, and include those polypeptides wherein one or more amino acid residues of a polypeptide are removed and replaced with alternative residues. In embodiments, the substitutions in amino acid sequences are conservative in nature, however, the disclosure embraces substitutions that are also non-conservative.

[0128] For amino acid sequences, sequence identity and/or similarity can be determined by using standard techniques known in the art such as, for example, the local sequence identity algorithm of Smith and Waterman, 1981, Adv. Appl. Math. 2:482, the sequence identity alignment algorithm of

Needleman and Wunsch, 1970, J. Mol. Biol. 48:443, the search for similarity method of Pearson and Lipman, 1988, Proc. Nat. Acad. Sci. U.S.A. 85:2444, computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Drive, Madison, Wis.), the Best Fit sequence program described by Devereux et al., 1984, Nucl. Acid Res. 12:387-395, using the default settings, or by inspection. Various alignment parameters can be set according to known methods (e.g., “Current Methods in Sequence Comparison and Analysis,” Macromolecule Sequencing and Synthesis, Selected Methods and Applications, pp 127-149 (1988), Alan R. Liss, Inc.). Additional useful algorithms include PILEUP, which can align multiple sequences from a group of related sequences using progressive, pairwise alignments; BLAST, including gapped BLAST, WU-BLAST-2 (see, e.g., Altschul et al., 1990, J. Mol. Biol. 215:403-410; Altschul et al., 1993, Nucl. Acids Res. 25:3389-3402; Altschul et al., 1997, Nucleic Acids Res. 25:3389-3402; Altschul et al., 1996, Methods in Enzymology 266:460-480; and Karin et al., 1993, Proc. Natl. Acad. Sci. U.S.A. 90:5873-5787).

[0129] Generally, the amino acid homology, similarity, or identity between sequences are at least 80%, including at least 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, and from 99% to almost 100% identity. Similarly, the “percent (%) nucleic acid sequence identity” with respect to the nucleic acid sequences described herein refers to the percentage of nucleotide residues in a candidate sequence that are identical with the nucleotides disclosed herein, in the gene or coding sequence of the related polypeptide. Specific methods can include the default parameters of algorithms such as, for example, BLASTN (WU-BLAST-2).

Recombinant Host Cells

[0130] In an aspect the disclosure provides recombinant cells that comprise the polynucleotides and polypeptides described herein. In embodiments, the host cells can comprise any type of cell that is adaptable to genetic manipulation and/or expression of foreign genes and proteins. In some embodiments, host cells can include any species of filamentous fungus, including but not limited to any species of *Aspergillus*, which may be optionally genetically altered to accumulate and/or produce target, precursor, or intermediate indole alkaloid molecules. In embodiments, host cells may also be any species of bacteria, including but not limited to *Escherichia*, *Corynebacterium*, *Caulobacter*, *Pseudomonas*, *Streptomyces*, *Bacillus*, or *Lactobacillus*. In some embodiments, the host cell is a yeast cell capable of being genetically engineered and can be utilized in these embodiments. Non-limiting examples of such yeast cells include species of *Saccharomyces*, *Candida*, *Pichia*, *Schizosaccharomyces*, *Scheffersomyces*, *Blakeslea*, *Rhodotorula*, or *Yarrowia*.

[0131] These cells can achieve gene expression controlled by inducible promoter systems, natural or induced mutagenesis, recombination, and/or shuffling of genes, pathways, and whole cells performed sequentially or in cycles; overexpression and/or deletion of single or multiple genes and reducing or eliminating parasitic side pathways that reduce target compounds, intermediate or precursor concentrations.

[0132] The host cells of the recombinant organism may also be engineered to produce any or all precursor molecules

necessary for the biosynthesis of the target, precursor, or intermediate indole alkaloid compounds and can comprise any of the disclosed polynucleotide sequences, vectors or expression cassettes that are capable of expressing the recombinant enzyme encoded therein.

[0133] Construction of modified host cells such as *Saccharomyces cerevisiae* strains expressing the enzymes and regulatory proteins provided herein is carried out via expression of a gene which encodes for the enzyme. The gene encoding the enzyme can be cloned into vectors with the proper regulatory elements for gene expression (e.g. promoter, terminator) and the derived plasmid can be confirmed by DNA sequencing. As an alternative to expression from an episomal plasmid, the gene encoding the enzyme may be inserted into the recombinant host genome. Integration may be achieved by a single or double cross-over insertion event of a plasmid, or by nuclease-based genome editing methods, as are known in the art e.g. CRISPR, TALEN and ZFR. Strains with the integrated gene can be screened by rescue of auxotrophy and genome sequencing. See, e.g., Green and Sambrook (2012).

[0134] As described herein, the recombinant cell may be any species of yeast, including but not limited to any species of *Saccharomyces*, *Candida*, *Schizosaccharomyces*, *Yarrowia*, etc., which have been genetically altered to produce monoterpene indole alkaloid (MIA) molecules, including precursors and intermediates. Additionally, genetically engineered host cells may be any species of filamentous fungus, including but not limited to any species of *Aspergillus*, which have been genetically altered to produce precursor molecules such as GPP, geraniol, tryptamine, serotonin, 4-hydroxytryptamine, 5-methoxytryptamine, 4-methoxytryptamine, and strictosidine for use as precursors for the downstream MIA compound production in accordance with the present disclosure. Some of the species of yeast include but are not limited to: *Schizosaccharomyces cerevisiae*, *Schizosaccharomyces japonicus*, *Schizosaccharomyces pombe*, *Schizosaccharomyces cryophilus*, *Saccharomyces cerevisiae*, *Kluyveromyces lactis*, *Kluyveromyces dobzhanskii*, and *Yarrowia lipolytica*.

[0135] In various embodiments, the recombinant cell can comprise combinations of endogenously upregulated and/or codon optimized genes encoding for proteins which can improve: protein production, correct protein folding, protein secretion, proper protein intracellular localization, enzyme activity, correct post-translational modifications for protein features, mRNA stability, cell tolerance to stress, cellular metabolic activity, availability of cofactors for enzyme activity, glycolysis, fatty acid metabolism, feedstock conversion, amino acid biosynthesis, mevalonate pathway flux, coenzyme A (CoA) flux, acetyl-CoA production, tolerance to oxidative stress (e.g. H₂O₂) from increased protein production, tolerance to oxidative stress from monoterpene indole alkaloid (MIA) and related precursor enzymatic pathway steps, titer of alkaloids and alkaloid pathway precursors, intermediates, and compounds.

[0136] As discussed herein, the recombinant cells and modified strains express the genes for biosynthetic pathways that generate monoterpene indole alkaloid pathway products, such as demethylcorynantheidine, corynantheidine, 9-hydroxycorynantheidine, mitragynine, etc., and/or all intermediate monoterpene indole alkaloids compounds and precursors described herein. The monoterpene indole alkaloid pathway genes herein can be integrated into the genome

of the cell or maintained as an episomal plasmid. Recombinant host fermentation samples are: (i) prepared and extracted using a combination of fermentation, dissolution, and purification steps; and (ii) analyzed by HPLC for the presence of target molecules, precursor molecules, and/or intermediate molecules. In some embodiments, the product MIA pathway products are generated from an upstream pathway feed that comprises compounds such as geraniol, nepetalactol, loganic acid, loganin, secologanin, tryptamine, serotonin, 5-methoxytryptamine, 4-methoxytryptamine, 4-hydroxytryptamine, strictosidine, any of which can also be detected.

[0137] In some related embodiments, the genes and proteins that may be expressed by the recombinant cell (with reference to exemplary Genbank protein accession numbers) can include one or more of:

[0138] rRNA processing proteins for ribosome biogenesis, such as BFR2 (QHB07760.1)

[0139] 14-3-3 proteins which regulate vesicle and protein transport, such as BMH2 (CAA59275.1)

[0140] Transcription factors which upregulate the unfolded protein response, such as HAC1 (QHB08305.1).

[0141] Karyopherin overexpression to help the transport and translocation of hydrophobic and/or chimeric proteins, such as PSE1 (QHB11037.1)

[0142] Ribosomal proteins for maintaining translational productivity through cell stress, such as RPPO (QHB10483.1),

[0143] Oxidative stress protection, such as CCS1 (QHB10774.1), SOD1, (CAA89634.1), and/or SSA4, (KZV11871.1)

[0144] ER stress sensors which can upregulate protein folding chaperones and heat shock responses, such as IRE1 (QHB09072.1)

[0145] Proteins which can increase cellular glycosylation status, such as PSA1, (CAA98617.1)

[0146] Peroxide consuming proteins including catalase to protect from oxidative damage, such as CTT1, (CAA97090.1)

[0147] Protein folding chaperones and chaperonins, such as SSE1, (KZV07411.1), JEM1, (QHB09554.1), KAR2, (CAA89325.1), LHS1, (CAA81910.1), SCJ1, (CAA41529.1), SIL1, (QHB11576.1), GroEL (QEP09777.1), GeoES (QEP09776.1), and/or SSS1, (CAA98906.1)

[0148] Post-translational support, such as UBI4, (CAA97489.1)

[0149] Isomerases for proper protein folding, such as CPR5 (KZV12543.1) and/or CPR2 (KZV10787.1).

[0150] Heat shock response ER membrane proteins, including calnexin, such as CNE1, (QHB06590.1)

[0151] Endoplasmic reticulum redox balance for proper protein folding including disulfide bond formation, such as ERO1, (QHB10609.1), PDI1, (KZV12810.1), and/or EUG1, (KZV12759.1).

[0152] Protein translocation and protein folding enhancers, such as SBH1, (KZV11852.1), and/or SEC61 (CAA44215.1)

[0153] Enhancers of protein trafficking, vesicle formation, tethering, and fusion between organelles and membranes, SEC31 (CAA98772.1), SLY1, (CAA38221.1), BOS1, (CAA97636.1), BET1, (CAA86247.1), SEC22, (QHB10420.1), SED5,

- (CAA97549.1), COG7, (QHB08640.1), COY1, (QHB09825.1), IMH1, (QHB10458.1), SEC1, (QHB07623.1), SEC4, (QHB08330.1), SNC1, (KZV13437.1), SNC2, (CAA89974.1), SSO1, (CAA97949.1), and/or SSO2, (KZV09033.1)
- [0154] Secretory pathway processing enzymes, including glycosyltransferase, KEX2, (CAA96143.1), MNN1, (AAA53676.1), MNN2, (QHB06786.1), MNN9, (KZV07467.1), MNN10 (QHB07704.1), MNN11, (QHB09454.1), and/or OCH1, (CAA96740.1)
- [0155] Cell wall related proteins for stress tolerance and cell wall stability, such as CCW12 (KZV09357.1), CWP2, (CAA81937.1), YPS1, (KZV09366.1), MKC7, (KZV12382.1), and/or SED1 (BAI99734.1)
- [0156] Flavin cofactor synthesis and recycling proteins for FAD+/FADH use by enzymes expressed in a modified host, such as FAD1 (CAA98604.1) or FMN1 (NM_001180544.1).
- [0157] Nicotinamide cofactor synthesis and recycling proteins for NAD (P)+/NAD (P) H use by enzymes expressed in a modified host, such as UTRI (CAA89577.1), POS5 (CAA97900.1),
- [0158] Upregulation of the pentose phosphate pathway (PPP) such as increased expression of pentose phosphate genes encoding for enzymes in the PPP, such as GND1 (KZV10926.1) and/or ZWF1 (CAA96146.1)
- [0159] Enzymes to increase acetyl-CoA pool including alcohol and aldehyde dehydrogenases and acetyl-CoA synthetases such as ALD6 (QEP09776.1), ADH2 (QEP09776.1), and/or ACS (AAL23099.1)
- [0160] Enzymes to upregulate heme biosynthesis, such as delta-aminolevulinic acid dehydratase (Hem2) (CAA96742.1), porphobilinogen deaminase (Hem3) (CAA98783.1), uroporphyrinogen decarboxylase (Hem12) (QHB07515.1), oxygen-dependent coproporphyrinogen III oxidase (Hem13) (QHB07512.1), and/or ferrochelatase (Hem15) (CAA99385.1).
- [0161] Transporters to increase metal ion import, such as FET3 (QHB10789.1), FTR1 (QHB08222.1), and/or FET4 (QHB11048.1)
- [0162] Metal ion oxidoreductases to convert metals into usable forms, such as the ferric and copper reductases, FRE1 (QHB10365.1) and/or FRE3 (KZV08274.1)
- [0163] In various embodiments, the recombinant cell comprising endogenous genes encoding proteins for an improved effect(s) for the expression of a target compound (monoterpene indole alkaloid (MIA), intermediate, precursor, or metabolite thereof) may be adapted or manipulated (i.e., genetically modified) in a way that modifies endogenous gene expression. Non-limiting examples include swapping an endogenous promoter for the gene of interest with a stronger promoter (constitutive or inducible, such as a glycolytic TEFI promoter or a galactose inducible GALI promoter); codon optimized gene sequence encoding proteins (e.g., genetic integration, episomal plasmids, or artificial chromosomes of such genes in an expression cassette (e.g. promoter, coding region, terminator)) any of which can be accomplished using methods known in the art.
- [0164] The recombinant cell can contain combinations of modifications to host genes where genes or combinations thereof have downregulated or no (i.e. knocked out) expression, and/or silenced message, which can improve: protein production, correct protein folding, protein secretion, proper

protein intracellular localization, enzyme activity, correct post-translational modifications for protein features, mRNA stability, cell tolerance to stress, cellular metabolic activity, availability of cofactors for enzyme activity, glycolysis, fatty acid metabolism, feedstock conversion, amino acid biosynthesis, mevalonate pathway flux, coenzyme A (CoA) flux, acetyl-CoA production, tolerance to oxidative stress (e.g. H₂O₂) from increased protein production, tolerance to oxidative stress from monoterpene indole alkaloid (MIA) and related precursor enzymatic pathway steps, titer of alkaloids and alkaloid pathway precursors, intermediates, and compounds.

[0165] Such genes (with examples of Genbank nucleotide accession numbers) include combinations thereof:

- [0166] Transcription factors such as ROX1 (NM_001184162.1)
- [0167] Heme depletors, including oxygenases, such as HMX1 (NM_001182092)
- [0168] Proteases, such as PEP4 (NM_001183968.1), PRC1 (NM_001182806.1), PRB1 (NM_001178875.1)
- [0169] Ubiquitin ligases, such as BUL1 (NM_001182782.1) and/or BUL2 (NM_001182473.1)
- [0170] O-acetyltransferases, such as ATF1 (NM_001183797) and/or ATF2 (NM_001181306.1).
- [0171] Dehydrogenases and reductases, such as ARII (NM_001181022), ADH6 (NM_001182831.3), OYE2 (NM_001179310), and/or OYE3 (NM_001183985.1).
- [0172] Genes encoding for proteins involved in amino acid metabolism, such as BNA2 (NM_001181736.3), ARO10 (KU050081.1), PDC6 (NM_001181216.3)
- [0173] Genes that promote cell death in response to toxicity of high MIA precursor concentrations, such as YCA1 (AY692832.1).

Biosynthetic Methods

[0174] In some aspects, the disclosure provide a biosynthetic method for producing one or more target compounds (e.g., an iboga compound, an indole alkaloid, or an intermediate, precursor, or metabolite thereof), comprising: (i) generating a recombinant host cell; (ii) growing the recombinant host cell under conditions effective to produce the target compound(s), or precursor(s) thereof; and (iii) isolating the target compound(s) from the recombinant host cell. In some embodiments of the method, the method comprises growing the recombinant host cell under conditions effective to express the genes encoding the enzymes and/or proteins, and fermenting the recombinant host cell to produce the target compound(s). Endogenous pathways of the recombinant host can be modified by the systems and methods herein to produce high purity target compounds.

[0175] In various embodiments of the biosynthetic methods to produce the desired monoterpene indole alkaloids and related intermediates and precursors, the nucleic acid encoding the enzyme(s) and/or regulatory protein(s) are introduced into a host cell using standard cell (e.g., yeast) transformation techniques (Green and Sambrook, 2012). Cells are subjected to fermentation under conditions that activate the promoter controlling the synthesis of the enzyme and/or regulatory protein. The broth may be subsequently subjected to HPLC analysis to determine the presence or yield of the desired monoterpene indole alkaloid and/or related intermediate and/or precursor products.

[0176] In various embodiments, the host cells are provided with various feedstocks to drive production of the desired

monoterpene indole alkaloids and the related precursors or pathway intermediates, (e.g., feedstocks comprising glucose, fructose, sucrose, galactose, raffinose, maltose, ethanol, xylose, fatty acids, glycerol, acetate, molasses, malt syrup, corn steep liquor, dairy, flour, protein powder, olive mill waste, fish waste, etc., as is known in the art.

[0177] In some embodiments, the biosynthesis of MIAs, intermediates, and analogs, is carried out by cell-free lysate, crude or purified, or via purified enzymes from a recombinant host, or a combination of modified hosts, which has been modified to express MIA biosynthetic pathway enzymes. As described above, the modified strains express the genes for biosynthetic pathways that generate downstream biosynthetic pathways to generate, e.g., demethylcorynantheidine, corynantheidine, 9-hydroxycorynantheidine, or mitragynine, and/or other intermediate monoterpene indole alkaloids compounds and precursors.

[0178] Recombinant host cells expressing the MIA pathway to generate precursors, intermediates, and end products of the MIA pathway can be grown, fermented, and produce products on various carbon and nitrogen sources as described herein. In some embodiments, additions such as vitamin mixes and trace minerals are added to a fermentation media, which can include choline chloride, niacin, pyridoxine hydrochloride, riboflavin, calcium pantothenate, para-aminobenzoic acid (PABA), thiamine HCl, biotin, cyanocobalamin, and/or folic acid, and mineral mixes, which can include calcium chloride dihydrate, ferrous sulfate heptahydrate, manganese (II) sulfate monohydrate, copper sulfate pentahydrate, zinc sulfate heptahydrate, magnesium chloride, and solutes, such as glycerol.

[0179] In some applications, MIA precursors and products generated from a recombinant host are captured using in situ product removal (ISPR). In some embodiments, 1-30% volume to volume of fermentation broth to ISPR reagent is used. Such ISPR reagents can include, but are not limited to, isopropyl myristate, vegetable oils such as corn oil or olive oil, oleic acid, MTBE, 1-decanol, oleyl alcohol, bis(2-ethylhexyl) phthalate, alkanes such as octane and dodecane, styrene, silicone oil, non-polar macroporous resin such as C8 or C18 resin, ion-exchange resin, trioctylamine, activated carbon, ethyl oleate, lauryl acetate, farnesane, TWEEN-80, TX-100, squalene, glycerin, and/or propylene glycol. For example, ISPR and analysis of MIA products or precursors can be carried out by an ISPR process using dodecane. In this case, a 20% volume to volume dodecane layer is added to a recombinant host fermentation. After 72h of fermentation, the dodecane is removed, diluted 1:10 with ethanol, and filtered for volatile compound detection via HPLC. The broth extraction and subsequent HPLC detection can be carried out as described herein and/or as in WO/2021/248087 (PCT/US21/36031). The use of these reagents also contributes to increased MIA yield by reducing the amount of pathway intermediate lost to evaporation.

[0180] Isolation and detection of MIA products and precursors from recombinant cell, lysate, or co-culture fermentation broth can be performed by solvent extraction, filtration, and analytical methods including high performance liquid chromatography (HPLC), liquid chromatography mass spectrometry (LC-MS), high resolution mass spectrometry (HRMS), gas chromatography (GC), gas chromatography mass spectrometry (GC-MS), and/or nuclear magnetic resonance (NMR) methods, as known to those skilled in the art. For example, recombinant host strains expressing

the MIA biosynthesis pathway can be mixed with a solvent such as acetonitrile, ethanol, isopropanol, methanol, ethyl acetate, acetone, water, and mixtures thereof, which can include additives in amounts of 0.01-20% volume to volume formic acid (FA), ammonium carbonate, or trifluoroacetic acid (TFA). Such solvent mixtures can be applied to whole broth, cell pellets, or fermentation supernatant in combination with centrifugation, vortexing, bead-beating, homogenization, filtration, such as tangential flow filtration (TFF), and/or shaking and incubation, at varying temperatures from 4-100 degrees Celsius from 1-120 minutes. Such extractions can then be filtered for running on analytical equipment, such as dead-end filtration through a 0.1-0.22 micron filter of polytetrafluoroethylene (PTFE), polyvinylidene fluoride (PVDF), nylon, cellulose, or polyethersulfone (PES) to remove unwanted debris from sensitive analytical procedures.

Generating Downstream Mitragynine Alkaloids

[0181] As discussed throughout the disclosure, methods are provided for the biosynthetic production of molecules that are part of the downstream alkaloid biosynthetic pathway related to mitragynine, its precursors, and analogs (e.g., including isotopic analogs (e.g., deuterium analogs), substituted analogs (e.g., oxa- or thia-tryptamines; halogenated compounds), and stereoisomers).

[0182] In some embodiments, the methods for generating a downstream alkaloid (mitragynine) compound comprises one or a combination of polynucleotides, enzymes, proteins, or recombinant cells as described herein that are maintained under conditions that allow for the expression and/or activity of the polynucleotide(s) and/or enzymes or proteins and which are sufficient for the synthesis and production for the downstream alkaloid compound.

[0183] In some embodiments of the methods, the disclosure provides for an increase in the production yield and/or titer of one or more of the target compound(s) that are generated. In further embodiments, the increase in the production yield and/or titer of the target compound(s) is relative to the amount or titer of the target compound(s) that is present in a naturally-occurring source. In further embodiments, the increase in the production yield and/or titer of the target compound(s) is relative to the amount or titer of the target compound(s) that can be generated in alternative chemical synthetic methods. In further embodiments, the increase in the production yield and/or titer of the target compound(s) is relative to the amount or titer of the target compound(s) that can be produced from alternative recombinant host cells.

[0184] In some embodiments, the method comprises a DCS polynucleotide sequence including any one or more of the sequences DCS_In-26n as disclosed herein, or a sequence having at least 90% sequence identity to any such DCS polynucleotide sequence. In some embodiments, the method comprises a DCS amino acid sequence including any one or more of the sequences DCS_Ip-26p as disclosed herein, or a sequence having about 90% sequence identity to any such DCS amino acid sequence.

[0185] In some embodiments, the method comprises a M9OMT polynucleotide sequence including any one or more of the sequences M9OMT_In-64n as disclosed herein, or a sequence having at least 90% sequence identity to any such M9OMT polynucleotide sequence. In some embodiments, the method comprises a M9OMT amino acid

sequence including any one or more of the sequences M9OMT_Ip-64p as disclosed herein, or a sequence having about 90% sequence identity to any such M9OMT amino acid sequence.

[0186] In some embodiments, the method comprises an enolMT polynucleotide sequence including any one or more of the sequences enolMT_In-7n as disclosed herein, or a sequence having at least 90% sequence identity to any such enolMT polynucleotide sequence. In some embodiments, the method comprises an enolMT amino acid sequence including any one or more of the sequences enolMT_Ip-7p as disclosed herein, or a sequence having about 90% sequence identity to any such enolMT amino acid sequence.

[0187] In some embodiments, the method comprises a M9H polynucleotide sequence including any one or more of the sequences M9H_In-36n as disclosed herein, or a sequence having at least 90% sequence identity to any such M9H polynucleotide sequence. In some embodiments, the method comprises a M9H amino acid sequence including any one or more of the sequences M9H_Ip-36p as disclosed herein, or a sequence having about 90% sequence identity to any such M9H amino acid sequence.

[0188] In some embodiments, the method comprises a cytochrome P450 reductase (CPR) gene of the Sequence Listing provided herewith, including any one or more of sequences CPR_In-CPR_4n, or a cytochrome P450 reductase (CPR) amino acid sequence of the Sequence Listing provided herewith, including any one or more of sequences CPR_Ip-CPR_4p, or a sequence having at least 90% sequence identity to any of the CPR sequences.

[0189] In some embodiments, the method comprises a cytochrome b5 (CYB5) gene, which together with CPR can participate in electron transfer reactions. Coexpression of CPR and CYB5 proteins can enhance enzymatic oxidations, such as increasing the activity of the P450 enzymes described herein. In some embodiments, the method comprises a CYB5 gene of the Sequence Listing provided herewith, including CYB5_In, or CYB_2n, or a cytochrome b5 (CYB5) amino acid sequence of the Sequence Listing provided herewith, including CYB5_Ip, or CYB5_2p, or a sequence having at least 90% sequence identity to any of the CYB5 sequences.

[0190] In some embodiments, the method comprises a transporter (trxporter) gene of the Sequence Listing provided herewith, including any one or more of sequences trxporter_In-trxporter_6n, or a transporter (trxporter) amino acid sequence of the Sequence Listing provided herewith, including any one or more of sequences trxporter_Ip-trxporter_6p, or a sequence having at least 90% sequence identity to any of the trxporter sequences.

[0191] In some embodiments, the method comprises a tryptophan importer (TAT2) gene of the Sequence Listing provided herewith, including TAT2_In or a tryptophan importer (TAT2) amino acid sequence, or a sequence having at least 90% sequence identity to any of the TAT2 sequences.

[0192] In some embodiments, the method comprises a SAME importer (SAM3) gene of the Sequence Listing provided herewith, including SAM3_In or a SAME importer (SAM3) amino acid sequence, or a sequence having at least 90% sequence identity to any of the SAM3 sequences.

[0193] In some embodiments, the method comprises a methionine importer (MUP1) gene of the Sequence Listing provided herewith, including MUP1_In or a methionine

importer (MUP1) amino acid sequence, or a sequence having at least 90% sequence identity to any of the MUP1 sequences.

[0194] In some further alternative embodiments, the method comprises a tryptamine hydroxylase which can hydroxylate an indole, tryptamine, or tryptophan molecule, such as PsiH-type and other hydroxylase genes, such as a tryptamine-4-hydroxylase (T4H), disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM) and U.S. Pat. No. 11,441,164 (BIOSYNTHETIC PRODUCTION OF PSILOCYBIN AND RELATED INTERMEDIATES IN RECOMBINANT ORGANISMS).

[0195] In some further alternative embodiments, the method comprises an O-methyltransferase which can methylate hydroxyl groups on an indole or tryptamine molecule, such as the indole (or tryptamine)O-methyltransferases (IOMTs) disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM).

[0196] In some further alternative embodiments, the method comprises a kinase which can phosphorylate a hydroxylated indole, tryptamine, or tryptophan such as PsiK-type and other kinases genes disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM) and U.S. Pat. No. 11,441,164 (BIOSYNTHETIC PRODUCTION OF PSILOCYBIN AND RELATED INTERMEDIATES IN RECOMBINANT ORGANISMS).

[0197] Other useful sequences include cofolding tags CoFoldtag_1 (SEQ ID NO:358) and CoFoldtag_2 (SEQ ID NO:359); copper reductase oxidase gene CRO_In (SEQ ID NO:368); horseradish peroxidase gene HRP_In (SEQ ID NO:369); and histidine tag HHHHHH (SEQ ID NO: 232).

[0198] In some further alternative embodiments, the method comprises a tryptophan or tryptamine halogenase (TrpHalo) which can halogenate indole, tryptamine, and/or tryptophan molecules, such as the halogenases disclosed in U.S. patent application Ser. No. 18/000,931 (ENZYMES AND REGULATORY PROTEINS IN TRYPTAMINE METABOLISM).

[0199] In some embodiments, the method can include production of one or more labile intermediate compounds that can form spontaneously and which can feed into a separate portion of the pathway for production of indole alkaloids including mitragynine.

[0200] For purposes of illustration and clarity, FIG. 1 depicts the tryptamine and alkaloid skeletons. The portion of the molecule from the tryptophan indole is highlighted. The portion of the molecule from the monoterpenoid is highlighted. In embodiments, engineering of the native yeast tryptophan pathway to enable high flux toward tryptophan and tryptamine production is used in the systems herein, and is described in, e.g., WO 2021/248087 (PCT/US2021/036031).

[0201] The terpene moiety of strictosodine, secologanin, can be fermentation-derived by the enzyme-engineered host as described herein or fed into the media. Secologanin exogenously provided to a fermentation media can be crude extract material from plants which contain MIA precursors. For example, extracts from plants from the *Symphoricarpos* genus, such as snowberry (*Symphoricarpos albus*), and other plants such as Silvervine (*Actinidia polygama*), *Lonicera japonica* and Honeysuckle (*Lonicera caerulea*) can be a

source of exogenous MIA precursors such as secologanin. In other embodiments, material and extracts thereof from plants of the Gentianales order, which can include *Mitragyna speciosa* (kratom), *Uncaria tomentosa*, *Uncaria rhynchophylla*, *Voacanga africana*, *Voacanga thouarsii*, *Tabernanthe iboga*, *Tabernanthe mannii*, *Picralima nitida*, *Tabernaemontana elegans*, and/or *Catharanthus roseus*. In some embodiments the secologanin is provided by the plant and/or berry extract for the host cell to uptake as a starting substrate to feed into the biosynthetic pathway for MIA production. Likewise, other iridoid precursors for MIA production can be sourced and extracted, whether to high purity or remaining as a crude extract, from the plants listed herein. For example, snowberry (*Symphoricarpos albus*) extract and Honeysuckle (*Lonicera caerulea*) extract contain loganic acid and loganin. In some embodiments the loganic acid and/or loganin is provided by plant and/or berry extract for the host to uptake as starting substrate to feed into the biosynthetic pathway for indole alkaloid production.

[0202] In embodiments wherein plant material is incorporated as a fermentation media supplement for recombinant hosts to produce MIAs, the plant material comprising MIA precursors is processed by methods known by those skilled in the art. For example, in some embodiments, the plant material, such as Honeysuckle, is crushed up into powder using a mortar and pestle, a homogenizer, vortexing, or a blender, and can be extracted using a variety of organic solvents ranging in polarity, including but not limited to methanol, ethanol, butanol, ethyl acetate, acetonitrile, methyl tertiary-butyl ether (MTBE), and/or acetone. To lyse plant cells for efficient extraction of indole alkaloid pathway compounds, the powder is dissolved in water (e.g., ddH₂O) and subjected to either bead beating for 3-6 min. repeated one to five times, homogenization, or freeze-thaw cycles (e.g., alternating boiling water bath and frozen ethanol bath or liquid nitrogen) for three to six times. Organic solvent listed above is then added to the lysate and incubated with the lysate for 1-24 hrs at 20-50° C. The organic layer is then collected, dried and resuspended in acetonitrile for analytical detection and quantification. The leftover aqueous layer is also extracted with acetonitrile and subjected to analysis by HPLC, LC-MS, HRMS, and LC-MS/MS. The dried plant extract containing MIA precursors can be used as a fermentation media supplement for MIA production by recombinant host cells.

[0203] For selective detection and characterization of MIA compounds and precursors including ketones and aldehydes produced by recombinant hosts or present in plant extracts, derivatization techniques known to those skilled in the art can be used. Reagents (e.g., modifying or derivatizing agents) that can be used for detection and characterization of MIA compounds can include, but not be limited to, 2-amino benzamidoxine (ABAO), para-methoxy-2-amino benzamidoxime (PMA), 4-bromo-N-methylbenzylamine, 9-fluoromethylchloroformate (FMOC), 3-methyl-2-benzothiazolinone hydrazone (MBTH), 1-(5-fluoro-2,4-dinitrophenyl)-4-methylpiperazine (PPZ), 2,4-dinitrophenylhydrazine (DNPH) and 4-bromo-N-methylbenzylamine (4-BNMA). For example, in some embodiments, fermentation broth or plant material extracts are derivatized using 2-amino benzamidoxime (ABAO). In such embodiments, fermentation broth or plant material extracts (e.g., in acetonitrile) are subjected to reacting with 0.1-1 mM ABAO at pH 4-7. The reaction can be performed at 30-37° C. and incubated for at

least 15 min or allowed to incubate for hours (i.e., overnight). In embodiments comprising an ABAO derivatization of a MIA precursor and/or product, the derivatized product formation is monitored with fluorescence excitation at 360 nm and emission at 528 nm. Alternatively, both the ABAO reagent and derivatized adducts can be monitored with a UV absorbance at 405 nm. All derivatized products can be further analyzed with LC-UV-MS.

[0204] In some embodiments, sugars and nitrogen from plant and berry extracts also provide the nutrients necessary for the growth and fermentation of recombinant host cells in addition to providing iridoid precursors for MIA production, minimizing or eliminating any other additional media additives for the recombinant host.

[0205] In some embodiments, these precursors are converted to MIAs in a cell-free system, such as via crude or purified lysate, microsomes, or purified enzymes from a recombinant host or combinations of recombinant hosts which contain the biosynthetic enzymes for MIA production.

[0206] In some further downstream biosynthetic pathway embodiments, FIG. 3 shows example enzyme conversion of substituted indoles and tryptamines, such as from (A) a modified indole that leads to a substituted tryptophan, and (B) enzymatic modifications of tryptamine which can lead to downstream analogs of alkaloids including strictosidine-type analogs through mitragynine analogs. Such a pathway can be carried out in a recombinant host expressing combinations of sequences described here in, including genes encoding enzymes for enoMT, DCS, M9OMT, and/or M9H, in addition to other genes described herein (e.g., PsiH or T4H), IOMT, PsiK, TDC/AADC, and TrpHalo) and genes otherwise known in the art to generate MIA pathway precursors and intermediates.

[0207] FIG. 1C-1D depicts a stereoisomer (1C) and substituted analogs with a modified or substituted indole ring (1D). Sources of modified indole rings can stem from substituted tryptophan or tryptamine-like molecules can be found in WO/2021/248087. R substitutions on the indole ring positions include single or multiple independent substitutions of H, deuterium, OR2, NR2 R2, N+R2, SR2, S+R2, P+R2, P(O) R2, P(O) R2, S(O) R2, SO2R2, SO3R2, CO2R2, CN, oxo, a halide, a haloalkyl and substituted or unsubstituted alkyl, alkenyl, alkynyl, alkoxy, alkoxyalkyl, alkylamino, dialkylamino and alkylthio, wherein R2 independently comprises H, deuterium, alkyl, alkenyl, alkynyl, oxoalkyl, alkenyl, alkynyl, alkoxy, alkoxyalkyl, alkylamino, a halide, a haloalkyl, dialkylamino and alkylthio. Additional variants of the indole ring nitrogen (exemplified as "R" in the indole ring in FIG. 1D) can also be incorporated including, for example, the substitutions of the indole ring nitrogen with an oxygen atom (i.e. benzofurans), a sulfur atom, a carbon atom (optionally substituted), or a substituted ring nitrogen (such as with R2 as defined above (e.g., methylated, deuterated, or a halogenated)) as indicated. Such substitutions of the nitrogen of the indole ring can lead to substituted strictosidines, as well as derivatives to downstream indole alkaloids such as, for example corynantheidines and mitragynines. A nonlimiting example of these substituted MIAs where the indole nitrogen is replaced, for instance with an oxygen (e.g. benzofuran derivatives) can lead to 'oxa' versions of MIAs, including oxa-strictosidine and downstream alkaloid oxa-derivatives thereof (see FIG. 1D) and many other corynanthe-type MIAs, where the benzo-

furan moiety can originate from a benzofuran precursor added to a fermentation or cell free reaction. Another nonlimiting example can stem from a 1-(or N-) methylated substituted corynanthe molecule (see FIG. 1D) where the N-methylated indole moiety can originate from a 1-methylated tryptophan or tryptamine precursor added to a fermentation or cell free reaction.

[0208] FIG. 2 depicts the biosynthetic pathway to mitragynine from various starting strictosidine aglycone-type analogs. From strictosidine and strictosidine aglycone, as in FIG. 2A, the DCS, enolMT, M9H, and M9OMT yield mitragynine. From hydroxystrictosidine and hydroxystrictosidine aglycone, as in FIG. 2B, the DCS, enolMT, and M9OMT yield mitragynine. From methoxystrictosidine and methoxystrictosidine aglycone, as in FIG. 2C, the DCS, and enolMT yield mitragynine.

[0209] FIG. 3 depicts the routes leading to substituted tryptophan and modified tryptamines, which can go on to react with secologanin via the STR enzyme to create strictosidine analogs which can enter a pathway with SGD, DCS, enolMT, M9H, and/or M9OMT enzymes to yield analogs of mitragynine intermediates and analogs of mitragynine. In FIG. 3A, the modification originates from an indole, that is then condensed into a modified tryptophan, which can then enter the MIA pathway. In FIG. 3B, upstream enzymatic modifications to tryptamine yield modified tryptamines which can then enter the MIA pathway.

[0210] FIG. 4 depicts example embodiments comprising modified cell hosts expressing gene combinations for generating MIA pathway products, in accordance with the disclosure, and embodiments that utilize the compounds generated by the disclosed methods to feed into various downstream MIA biosynthetic pathways. FIG. 4A depicts a modified host cell which generates MIA intermediates, such as strictosidine, hydroxystrictosidine, methoxystrictosidine, among others, which can feed into or be utilized by downstream MIA pathway enzymes or by a recombinant host expressing the downstream MIA pathway leading to mitragynine and mitragynine analogs. FIG. 4B depicts a modified host cell which expresses both an upstream (as in FIG. 4A) and a downstream MIA biosynthetic pathway to generate mitragynine and mitragynine analogs.

[0211] To illustrate some additional embodiments, FIG. 5A depicts an example of bioconversion by a modified cell host of an MIA precursor into a MIA target compound. Exogenous pure precursor or crude extract of precursors, e.g., containing iridoids, loganic acid, loganin, secologanin, strictosidine, among others, can be fed to recombinant cells expressing the MIA biosynthesis pathway. FIG. 5B shows an example of bioconversion using multiple organisms via a modified cell host expressing a portion of the MIA pathway, for example producing up to strictosidine, while another modified host cell expressing a downstream MIA pathway uses the products from the first host cell to produce MIAs such as mitragynine. Precursors made by a modified cell host can be excreted or derived from crude or purified lysate and fed to cells expressing downstream pathway enzymes. FIG. 5C depicts an example of bioconversion by a modified cell host of an MIA precursor which includes modified tryptamines into a MIA target compound.

[0212] In embodiments, the de novo biosynthesis pathway of L-tryptophan and SAM/SAMe are utilized as precursor molecules and cofactors in the systems and methods herein. The precursor molecules lead to target molecules of the

tryptamine pathways, when incorporated on-pathway. In some embodiments of the systems (e.g., organisms) and methods herein, glycolysis leads to chorismate (e.g., comprising chorismate synthase) via the shikimate pathway; glutamate biosynthesis pathway leads to L-glutamine via L-glutamate; and the L-serine biosynthesis pathway leads to L-serine via 3-phospho-L-serine (i.e., dephosphorylation). Chorismate, glutamine, and L-serine can be combined to form L-tryptophan as a precursor molecule, and which can be incorporated into the biosynthetic pathway that can yield tryptamine. In some embodiments of the systems (e.g., organisms) and methods herein, L-methionine is a direct precursor that can produce SAMe, when combined with ATP in the presence of Sam2 and Adkl enzymes. Methods described or referenced herein to increase flux through the tryptophan and SAMe biosynthesis pathways, such as over-expressing genes involved in the shikimate pathway, chorismate pathway, and methionine pathway, can lead to increased MIA product yield.

[0213] Describing this further, in some embodiments, tryptophan accumulation in a recombinant host cell can be increased by one or more of: (a) overexpressing feedback-resistant versions of tryptophan-synthesizing enzymes; (b) knocking out and/or inhibiting expression of off-pathway tryptophan-consuming genes and enzymes; and/or (c) over-expressing a recombinant L-tryptophan transporter for exogenous tryptophan import (see also U.S. PGPub 2021/0147888 and PCT Patent Application Publication WO 2021/248087). In some embodiments the biosynthesis can comprise (over) expression of enzymes (e.g., TRP1, TRP2, TRP3, TRP4 and/or TRP5) which can increase intracellular tryptophan and maintain or direct tryptophan flux through the pathway. In some embodiments the biosynthesis can comprise (over) expression of one or more genes useful for tryptophan biosynthesis, such as ARO1, ARO4, AROL, SER1, SER3, SER33, and/or SER2, and including feedback resistance mutant versions thereof, which increase shikimate pathway flux and/or supply serine.

[0214] FIG. 11 illustrates the novelty of this newly discovered enolMT class in relation to other methyltransferases in the SABATH (SAMT, BAMT, and Theobromine synthase) family, which is a common family of small molecule methyltransferases in plants. FIG. 11 depicts a phylogenetic tree of SABATH family methyltransferases in comparison with enol methyltransferases (enolMTs). Abbreviations: IAMT for indole-3-acetic acid OMT, SAMT for salicylic acid OMT, JMT for jasmonic acid OMT, BAMT for benzoic acid OMT, XMT for xanthosine NMT, LAMT for loganic acid OMT, enolMT for enol OMT, FAMT for farnesic acid OMT, M9OMT for mitragynine indole ring OMT, C3OMT for caffeic acid 3-OMT, COMT for catechol OMT. The closest relatives, loganic acid methyltransferases (LAMTs) only show ~40% sequence similarity to enolMTs whereas farnesic acid methyltransferases (FAMT) show ~30% sequence similarity to enolMTs.

[0215] The following Examples illustrate several aspects and embodiments in accordance with the disclosure and in no way serve to limit the scope or spirit of the claims.

EXAMPLES

Example 1. Method of Cell Growth

[0216] Modified host cells that yield MIA precursors and compounds, such as the downstream alkaloid-producing

strains described herein, express engineered genes and enzymes. More specifically, the downstream alkaloid-producing strains herein are grown in a minimal, complete culture media containing yeast nitrogen base, amino acids, vitamins, ammonium sulfate, and a carbon source of glucose and galactose. The recombinant host cells are grown in 24-well plates or shake flasks in a volume range of 2 mL to 100 mL of media starting from an inoculation density of $OD_{600nm}=1$. Exogenous compounds (e.g., upstream precursor compounds, such as strictosidine, strictosidine aglycone, or coronaridine) up to 1% can be added to media to supplement the precursor pool for downstream alkaloid-production. In addition or alternatively, early (upstream) pathway precursors can be imported into host cells. Exogenous L-tryptophan can be taken up by strains expressing the TAT2 L-tryptophan importer protein and exogenous L-methionine can be taken up by strains expressing the MUP1 L-methionine permease protein. The strains herein can be harvested during a fermentation period ranging from 12 hours onward from the start of pathway enzyme induction.

[0217] Alternatively, strains such as *Saccharomyces cerevisiae* that are modified to produce downstream alkaloids can be grown and fermented in media containing in the juice of Snowberry (*Symphoricarpos albus*) or Honeysuckle (*Lonicera caerulea*) to provide the secologanin precursor. Platform strains are grown in berry juice supplemented with compounds essential for growth (20 mg/L adenine, 20 mg/L L-histidine, 30 mg/L lysine, 20 mg/L L-leucine, and 20 mg/L tryptophan, pH 5.8). Carbon source, such as sucrose, glucose, galactose, xylose, lactose, and/or glycerol at 10-60 g/L is added to the media. Alternatively, secologanin can be provided as a feed-in to the media at 2 mM final concentration.

[0218] Some embodiments can include, during the fermentation, addition of substituted substrate molecules (e.g., substituted tryptophan or tryptamine) that allow for the synthesis and production of substituted downstream MIAs. For example, oxa-tryptamine (i.e. a benzofuran) can be added to a fermentation at a concentration of 0.1-0.5 g/L, and maintained at such concentrations for 12-48 hrs by bolus addition of a concentrated oxa-tryptamine stock solution or fed-in at a rate to maintain the concentration of the substrate in the fermentation, to generate 'oxa' analogs of downstream MIAs. Similarly, 1-methyl-tryptamine (i.e., N-methylated at the indole ring N) or 4-methoxytryptamine can be added to a fermentation at a concentration of 0.1-0.5 g/L, and maintained at such concentrations for 12-48 hrs by bolus addition of a concentrated stock solutions or fed-in at a rate to maintain the concentration of the substrate in the fermentation to generate N-methyl or substitution analogs of downstream MIAs. In addition the non-limiting examples mentioned above, the methods can incorporate deuterated molecules (e.g., as feedstocks, solvents, precursors, etc. comprising deuterium), in order to generate and synthesize deuterated forms of the alkaloid compounds generated by the methods described herein. Similarly halogenated molecules can be incorporated in a similar fashion in order to generate and synthesize halogenated forms of the alkaloid compounds generated in accordance with the aspects and embodiments of the disclosure.

[0219] After fermentation, the broth can be extracted as described herein, for analysis of MIA pathway products.

Example 2—Detection of Isolated Product

[0220] To identify fermentation-derived monoterpene indole alkaloids (MIAs), their precursors, and all other products of a recombinant host expressing an engineered biosynthetic pathway for alkaloids, an Agilent 1100 series liquid chromatography (LC) system and Agilent 6125 coupled to an Agilent 1260 LC stack, equipped with a reverse phase C18 column (Agilent Eclipse Plus C18, Santa Clara, CA, USA) is used. For high accuracy small molecule analysis and high resolution mass spectrometry (HRMS), an Agilent 6230 LC system coupled with Time-of-Flight (TOF) is used with a reverse phase C18 column (Agilent Eclipse Plus C18, Santa Clara, CA, USA). A gradient is composed of mobile phase A (ultraviolet (UV) grade $H_2O+0.1\%$ formic acid) and mobile phase B (UV grade acetonitrile+0.1% formic acid). Column temperature is set at 30° C. Compound absorbance is measured at 210 nm, 235 nm, 270 nm, and 305 nm using a diode array detector (DAD) and spectral analysis from 200 nm to 400 nm wavelengths. A mass range of m/z 10-2000 is enabled for mass spectrometry analysis of small molecules. A 0.1 mg/mL analytical standard is made from certified reference material for each compound (Cayman Chemical Company, USA). Each sample is prepared by diluting fermentation biomass from a recombinant host expressing the engineered biosynthesis pathway 1:3 or 1:20 in 100% acetonitrile and filtered in 0.2 μ m nanofilter vials. The retention time and UV-visible absorption spectrum (i.e., spectral fingerprint) of the samples are compared to the analytical standard retention time and UV-visible spectra (i.e. spectral fingerprint) when identifying the monoterpene indole alkaloids, their precursors, and all other related compounds. In addition or alternatively, mass fragmentation of analytes can be performed for additional compound identification data (e.g., LC-MS and LC-MS/MS). LC-MS/MS as depicted in FIG. 9, and FIG. 10, was carried out using an Agilent Eclipse C18, 4.6 $\times$ 100 mm, 2.5 μ m particles column, at 40 degrees celsius, a flow rate of 1 ml/min, with mixtures of solvents A (H_2O with 0.1% Formic acid) and solvents B (Acetonitrile with 0.1% Formic acid), with UV-vis detection and with a Thermo LTQ Orbitrap. FIG. 9 and FIG. 10 depicts LC-MS/MS fragmentation pattern data identifying corynantheidine produced by recombinant *Saccharomyces cerevisiae* cells expressing the MIA pathway with an enolMT gene (enolMT_In) encoding an enolMT enzyme (cnolMT_1p).

Example 3—Construction of *Saccharomyces cerevisiae* Platform Strains with Accumulation of MIA Pathway Intermediates and Products

[0221] Construction of the *Saccharomyces cerevisiae* platform strain with accumulated intermediates in the mitragynine biosynthetic pathways is carried out via knocking out transporter genes that can be implicated in the export of aforementioned intermediates. MIA upstream biosynthetic pathway intermediates and products such as nepetalactol, loganic acid, loganin, secologanin, strictosidine, and strictosidine aglycone, as well as downstream mitragynine biosynthetic pathway compounds such as, corynantheidine, demethyl corynantheidine, 9-hydroxycorynantheidine, mitragynine, and/or other MIA products intermediates, can exit cells via transporters, permeases, and exporters. The loss of MIA pathway intermediates to the extracellular space decreases the availability of these pathway intermediates

inside the cells to act as substrates for the MIA biosynthetic enzymes and lowers the flux of the pathway towards mitragynine MIAs. To accumulate these key intermediates in the cells, the endogenous *Saccharomyces cerevisiae* genes, including but not limited to ERCl, TPO1, FLR1, SNQ2, STE6, YORI, AUSI, MDL1, MDL2, PTR2, NPF, PDR5, PDR10, PDR12 and PDR15, are deleted as single or combinatorial knockouts. Knockout of these transporter genes is performed by placement of an early stop codon in the coding region of the corresponding gene sequences using techniques known in the art. Such insertions can proceed via homologous recombination using the host machinery and make use of a synthesized DNA framing (oligonucleotide) with 40-800 nucleotides on either or both sides of a STOP codon that are complementary to the gene locus to be knocked out. Further verification of the modification in said strain can be carried out by genome sequencing and analyzed by the techniques disclosed in U.S. Pat. No. 10,671,632. Once the *Saccharomyces cerevisiae* platform strains with single and/or multiple transporter knockouts are completed, combinatorial episomal expression of MIA pathway enzymes, such as those encoded by, for example, M9H, enolMT, and M9OMT genes can consume the intracellular trapped MIA precursors to generate additional amounts of downstream MIA products, including mitragynine.

Example 4—Cell-Free Production of MIA Compounds Using a Recombinant MIA Pathway-Expressing *Saccharomyces cerevisiae* Strain

[0222] Cells expressing any membrane-associated/tethered enzymes and cytosolic biosynthetic enzymes encoded by the genes in accordance with certain embodiments of the disclosure are inoculated in 25-50 mL of appropriately selective media and cultured at 30° C. at 250 rpm for 40-56 hours, where the media contains 40 g/L glucose, 15 g/L ammonium sulfate, 2 g/L yeast nitrogen base (YNB), 1 g/L complete minimal dropout media (CSM) supplement, buffered with 60 mM KH₂PO₄ and 10 mM K₂HPO₄. An inducer such as 0.2-10% v/v galactose is added to the culture media and induction proceeds under the same conditions as growth for a further 18-48 hrs. Cell growth via optical density (OD) is measured and cells are harvested by centrifugation 11,000 rcf for 10 min. Cell pellets are normalized by OD. Cell pellets are washed once with 25 mL cold water. Pellets are resuspended in water and aliquoted at 30 OD. Cells are pelleted once again at 1 min at 8000 ref and then the supernatant removed. Remaining cells can be frozen and stored at -80° C. for MIA producing reactions as needed, or used immediately. Cells are then washed and lysed by adding 250 uL lysis buffer per pellet and bead beaten for 1-3 min over 1-3 intervals. Lysis buffer included 100 mM potassium phosphate buffer pH7, 200 mM sodium chloride, 5% glycerol, protease inhibitor cocktail tablet, and ImM DTT. If pellets were previously frozen, pellets are first thawed on ice. Resuspended pellets are added to a tube with glass beads and cells are disrupted in a bead beater 3x3 min at 2400 rpm with 1 min on ice in between. Once the lysate is prepared, an end-point activity assay is performed by incubating the lysate with 0.01-1 mg per ml of substrate and corresponding cofactors at 30° C. for 2 hr. Substrates can include 4-methoxytryptamine, corynantheidine, secologanin, strictosidine, methoxystictosidine, and/or 9-hydroxy-

corynantheidine. Samples are extracted, isolated, and analyzed via methods described herein (See Example 2).

Example 5—Expression of a NADPH-Dependent Reductase Dihydrocorynantheine Aldehyde Synthase (DCS) in a Modified Host Organism to Produce Demethylcorynantheidine and Related Pathway Intermediates

[0223] The removal of the glycosyl moiety from strictosidine triggers a cascade of chemically labile intermediates starting with strictosidine aglycone to spontaneously form unstable 4-ring intermediates such as dehydrogeissoschizine and more stable shunt products such as 5-ring cathenamine and related alkaloids. The DCS enzyme is believed to function as an NADPH-dependent reductase that iteratively reduces dehydrogeissoschizine from its iminium schiff base form to a tertiary amine to form demethylcorynantheidine, stereoisomers, and related side products. This reduction by DCS enzyme can stabilize the otherwise unstable and difficult-to-isolate dehydrogeissoschizine. Construction of *Saccharomyces cerevisiae* production of demethylcorynantheidine can be carried out via the expression of optimized STR, SGD and DCS genes that are designed and synthesized in techniques as described above. Alternatively, the set of genes can be integrated into the recombinant host genome by a single cross-over insertion event of the plasmid. The resultant strains can be verified by genome sequencing. Culture experiments where recombinant hosts were grown as described above and expressed DCS enzymes were performed by feeding secologanin and tryptamine as substrates (0.1-10 mM) during the fermentation and by monitoring the formation of demethylcorynantheidine and related intermediates that transiently form in the late-stage MIA pathway via LC-MS, HRMS, and/or LC-MS/MS. In FIG. 6, DCS activity is detected only in recombinant hosts expressing DCS and not in negative control strains. The DCS depicted are DCS_2n and DCS_3n genes encoding the DCS_2p (first from top in FIG. 6) and DCS_3p (second from top in FIG. 6) enzymes.

Example 6-Biosynthesis of Demethylcorynantheidine, Corynantheidine, 9-Hydroxycorynantheidine, and Mitragynine

[0224] Biosynthesis of demethylcorynantheidine. The removal of the glycosyl moiety from strictosidine triggers a cascade of chemically labile intermediates starting with strictosidine aglycone to spontaneously form unstable 4-ring intermediates such as dehydrogeissoschizine and more stable shunt products such as 5-ring cathenamine and related alkaloids. The DCS enzyme is incorporated into the recombinant cell to act as a NADPH-dependent reductase that iteratively reduces dehydrogeissoschizine from its iminium schiff base form to a tertiary amine to form demethylcorynantheidine and related side products. This reduction by DCS enzyme can stabilize the otherwise unstable and difficult-to-isolate dehydrogeissoschizine. Construction of a *Saccharomyces cerevisiae* strain for production of demethylcorynantheidine can be carried out via the expression of optimized STR, SGD and DCS genes that are designed and synthesized in techniques as described above, filling in the pathway depicted in FIG. 2. Alternatively, the set of genes can be integrated into the recombinant host genome by a single cross-over insertion event of the plasmid. The resul-

tant strains can be verified by genome sequencing. Culture experiments where recombinant hosts were grown as described above and expressed DCS enzymes were performed by feeding secologanin and tryptamine (or tryptamine derivatives) as substrates (0.05-0.2 mg/ml) during the fermentation and by monitoring the formation of demethylcorynantheidine and related intermediates that transiently form in the late stage mitragynine MIA pathway via LC-MS. An example pathway using hydroxytryptamine is shown in FIG. 12. Example product formation from such a hydroxytryptamine substrate can be observed in FIG. 13. An example pathway using methoxytryptamine is shown in FIG. 14. Example product formation from a methoxytryptamine substrate can be observed in FIGS. 15 and 16.

[0225] Biosynthesis of corynantheidine from demethylcorynantheidine. Generation of corynantheidine is achieved by incorporating an enol methyltransferase (enolMT) enzyme (e.g., either by integration into the host cell genome, or episomally incorporating it into the cell as described above) to methylate demethylcorynantheidine and form corynantheidine. Such biosynthetic reactions using a recombinant host are carried out as above. In FIG. 7, enolMT activity is detected only in recombinant hosts expressing enolMT and not in negative control strains. The enolMT depicted is enolMT_{In} genes encoding enolMT_{Ip} enzymes.

[0226] Biosynthesis of 9-hydroxycorynantheidine from corynantheidine. Generation of 9-hydroxycorynantheidine from corynantheidine is achieved by incorporating an indole ring hydroxylase (M9H) enzyme (e.g., either by integration into the host cell genome, or by otherwise incorporating it into the cell as described above) and optionally in the presence of an oxidase to hydroxylate corynantheidine and form 9-hydroxycorynantheidine. The generation of 9-hydroxycorynantheidine can also emerge from a previously hydroxylated product, which can result from feeding in or generating 4-hydroxytryptamine, whereby DCS and enolMT enzymatic activities yield 9-hydroxycorynantheidine and its stereoisomer from the hydroxylated precursors, as shown in FIG. 13.

[0227] Biosynthesis of mitragynine from 9-hydroxycorynantheidine. Generation of mitragynine from 9-hydroxycorynantheidine is achieved by incorporating an indole ring O-methyltransferase (M9OMT) enzyme (e.g., either by integration into the host cell genome, or by otherwise incorporating it into the cell as described above) to methylate the hydroxyl group at the 9-position of the indole ring of 9-hydroxycorynantheidine. The generation of mitragynine can also emerge from a previously methoxylated product, which can result from feeding in or generating 4-methoxytryptamine, whereby DCS and enolMT enzymatic activities generate mitragynine from the methoxylated precursors, as shown in FIG. 15 and FIG. 16.

[0228] In view of the above, it will be seen that several objectives of the invention are achieved and other advantages attained.

[0229] As various changes could be made in the above methods and compositions without departing from the scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

[0230] All references cited in this specification, including but not limited to patent publications and non-patent literature, and references cited therein, are hereby incorporated by reference. The discussion of the references herein is

intended merely to summarize the assertions made by the authors and no admission is made that any reference constitutes prior art. Applicants reserve the right to challenge the accuracy and pertinence of the cited references.

[0231] As used herein, in particular embodiments, the terms “about” or “approximately” when preceding a numerical value indicates the value plus or minus a range of 10%. Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the disclosure. That the upper and lower limits of these smaller ranges can independently be included in the smaller ranges is also encompassed within the disclosure, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the disclosure.

[0232] The indefinite articles “a” and “an.” as used herein in the specification and in the embodiments, unless clearly indicated to the contrary, should be understood to mean “at least one.”

[0233] The phrase “and/or.” as used herein in the specification and in the embodiments, should be understood to mean “either or both” of the elements so conjoined. i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion. i.e., “one or more” of the elements so conjoined. Other elements can optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B.” when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0234] As used herein in the specification and in the embodiments, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of.” or, when used in the embodiments, “consisting of.” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either.” “one of.” “only one of.” or “exactly one of.” “Consisting essentially of.” when used in the embodiments, shall have its ordinary meaning as used in the field of patent law.

[0235] As used herein in the specification and in the embodiments, the phrase “at least one.” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combina-

tions of elements in the list of elements. This definition also allows that elements can optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B.” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to

at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

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aactcatcca	tcatgtttga	gggtgatctc	tggaaacggga	cgcacaaggt	actggcgag	1200
aagcacataa	actggatcag	ggaactttac	gatgttatag	gtcagtatgt	tgacaacaat	1260
cccaggggag	catatgcgca	ctacagagac	ctagaccttg	gcgtgaataa	cattgatgag	1320
acctcagctg	aacaagcaag	gggtgtgggt	aaatcttact	tcaaaagata	ttttgataga	1380
ctggtaaatg	tgaaaaaccg	gggtggaccg	aataatttct	ttaagaatga	acagagcatt	1440
ccaccaagac	tagcccatag	cagctca				1467

SEQ ID NO: 5 moltype = DNA length = 1611
 FEATURE Location/Qualifiers
 source 1..1611
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 5

atggcagcca	tgcttagaac	aaggatgcta	cagttcgctc	tcgttttggt	attcagcttc	60
tgtggggtaa	gttcatcaag	tcagccagaa	accccgcttc	tgcaatgttt	attccaccac	120
agcgtagact	ccaccattgg	ttatatattt	tacacaccgg	aaaactccag	ttatagctca	180
atcctgaact	catgcatcca	gaactcccg	ttcatagttc	cagagacgcc	aaaaccgaaa	240
gtcattctaa	ctcccgtcag	gcagagccag	gtacaagctg	caattttact	ttccaagaag	300
catggaatgc	aaattagaat	ccgtagtggc	ggacacgatt	tcgaaggatc	atcatcacaca	360
tctaactgtc	cgtttcttgt	ccttgacatg	ttcaacttta	gatctatttc	cgttgactta	420
caaagcaaaa	ctgcatgggt	tggcgcagga	gccacgcttg	gagagacata	ttaccatata	480
tacaaaaaaa	actctacctt	aacgtactct	gcgggttact	ggccaactgt	agggataggt	540
ggacacgtct	ctggtgggtg	ttgtggacc	cttgtagctg	agtaggggt	ggcagcggac	600
aacgtcttgg	atgcgaggtt	agtcgatgtg	aagggcagga	tcttagatcg	taagagcatg	660
ggcgatgacc	tgttttgggc	tatacgttgt	ggtatgggag	cgtcctttgg	cgttataact	720
gcctacaaaa	taaagtgtgt	tgaagtacct	gaaaaagtcg	tggcttttaa	ggttcgtagg	780
acacttgaac	agaacgcaac	caagctagtt	cacaagtggc	aatttggggc	tccgaacta	840
ccagttcgtc	ttattatcag	cctgcagatt	gcattccata	actcagatca	aactggagga	900
agaaccatac	aggcctcctt	cgcttcctgc	tttcgtggag	gagtcgatga	gcttttccag	960
ataatggagg	aacactttcc	tgaactggga	ttaaagaaag	aggatgtaaa	ggaaatgagt	1020
tggatacaat	atttcagttt	ctatgccgat	gaacccatag	atgacgtagc	gcatttcccta	1080
actaatagag	tcagcaccac	caaacatac	tccaaagcca	aatccgattt	tgtccagaaa	1140
ccatttcctg	aggctggcat	agaaaaaata	tggaaagaaac	tatttgaaaa	agatcccttg	1200
gctgcacagc	tgcatgtggc	tcctttcgga	ggtaaaatgg	accagatctc	cgagtcagaa	1260
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ggcactgata	aggtcttagc	gcagaagcac	ataaactgga	ttagaaaaat	atatggtgta	1380
atcgggcagt	atgtggacaa	caacccacgg	ggtgcctacg	cggactacag	agatttggac	1440
ttgggtgtca	ataacattga	tgagacaagt	gtggaacagg	caagggtatg	gggtaagagt	1500

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tacttcaaaa	ataattttga	tcgtctagtt	aatgtgaaaa	ccagggtaga	ccccgataac	1560
ttcttcaaaa	acgagcaaa	cattccctct	agattggcgc	atagttcatt	t	1611

SEQ ID NO: 6 moltype = DNA length = 1611
 FEATURE Location/Qualifiers
 source 1..1611
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 6

atggcaacta	tgcttcgtac	caggatgcta	cagttcgcgt	ttgtcttatt	gattttcttc	60
tgctgcgtct	ccagtagtag	tcagccggag	atcccgttcc	tgcaagtgcct	tttccaccac	120
tcagacgaca	gcaccatcgg	gaatgtaatg	tataccccag	aaaatagttc	atacagtagc	180
atacttaaca	cttacattca	aaattcccgt	ttcataagtc	cagaagctcc	aaaacccaag	240
gtgattttta	caccgcgttag	ggaatcccaa	gtccaaacgg	ccatatactg	tagcaaaaag	300
cacgggatgc	agatttaggat	acgttcagga	ggccacgact	tcgaagggtc	atcctatatg	360
tcaaacgtac	cttttctagt	attggacatg	ttcaatttta	gaagcatctc	cgtcgatctg	420
aggagtaaaa	ctgcatgggt	cggtagcagga	gctaccttag	gcgaaacata	ttataacata	480
tatcagaaaa	actccaccct	aacgtactca	gccgggtatt	ggccgaccgt	aggaatcggc	540
ggatcatgtg	ctggaggtgg	gtgcggaccc	cttggtcgtg	agtagcgggt	agcagccgac	600
aatgtcttgg	atgcgagatt	aatcgacgta	aaggggaaaa	tcctagaccg	taaatctatg	660
ggagaggatt	tattctgggc	tatcgttggt	ggaatgggtg	ctaacttcgg	agtaattttg	720
gcgtataaga	ttaagctggg	ggaagtcca	gaaaaagtgg	tcgcttttaa	agtaaggagg	780
actcttgggc	agaatggcac	gaaattggta	cataaatggc	aatttggggc	tccgaaactt	840
cccgttaggc	tgatcatttc	actacaaa	gccagtatta	actctgatca	aacaggtgga	900
cgtaccatcc	aagcgtcatt	tcctctgtc	tttagaggtg	aggttgacga	attgctgcag	960
attatgggaag	agcatttccc	ggaattgggt	ctaaagaagg	aagacatcaa	agagatgtcc	1020
tggatacaat	acttttcatt	ctatggagat	gaaccaattg	atgatgtcgc	ccattttttg	1080
acaaacaggg	tgagtacaac	aaaaccgtat	ttcaaagcta	aatctgactt	tgttcaaaag	1140
ccaataccgg	aggctggaat	agagaaaa	tggttgaaaa	agacccactg		1200
gcggcacagt	tacagtggac	tccgtttggc	ggaaaaatgg	atcaaattag	cgacagtga	1260
actccctttc	ctcacagggc	cggtaattct	ttataaatgt	ttgaagtaat	cttctggaat	1320
ggaaccgaca	aagtgttggt	acaaaagcac	attaattgga	ttcgtgaact	atatgacggt	1380
attgggcaat	acgttgacaa	taatcctagg	ggagcatatg	cagactaccg	tgacctagac	1440
ttgggagtc	ataacattga	cgagacgtct	gctgagcaag	cgagagtcgt	gggcacaaag	1500
tacttcaaga	ataacttcga	caggttggtt	aatgtgaaga	ccagagtaga	tccgaataat	1560
ttctttaaga	atgaacagtc	aatacctcct	aggttggtct	actcttcaag	t	1611

SEQ ID NO: 7 moltype = DNA length = 1059
 FEATURE Location/Qualifiers
 source 1..1059
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 7

atggcagctg	ggaaatgcc	gcaagaggag	caaactgtga	aggcattcgg	atgggctgca	60
cgtgaagtat	caggagcact	gtcaccttac	gggttctccc	gtcgtgccac	gggtgaacgt	120
gacgtaaaaa	ttaatgtact	ttactctggc	atgtgttaatt	tcgatactga	gatgataaat	180
aataagtttg	gggttcacaaa	gtatccatac	gtgttggggc	atgagactgt	cggagtggtg	240
tccgaagtgt	gggtccaaggt	ccgttaagtc	aaagttggcg	ataaggttgg	cgttggcggg	300
ctggtgggct	cgtgccgtaa	gtgctacagc	tgactaacgc	accttgagaa	ttattgtact	360
aaggccgaat	tgaccgaggc	aggagccggg	ggatgtagta	acttcgtagt	aacggatgag	420
gatttcgtat	tccattggcc	tgaaaagctg	cccttagacc	ttggagcacc	tctactttgc	480
gcaggagtta	caacattctc	cccactaaag	actcacgggc	tggacaaaac	cggaatgcac	540
gtcggagtg	caggattagg	cgttctagtc	catgttgacg	ttaaattttt	aaaggcattc	600
ggagcaaaag	taaccgttat	tagtacctct	gacaacaaga	agaggggaag	gatagaaaag	660
tacggagccg	acgcctcttt	agttagcagc	aacccggaac	aatgcaaac	cgctgctggg	720
acactgcag	gaattataga	caccataacc	tctgcgcact	cccttgttcc	ccttttaaac	780
ctgctgatgc	cgcaagggaa	attgattgtc	ctaggggtcc	ctaccgaacc	gtttgtgctg	840
cctgtagtgc	caatgatcca	aggaggtaaa	gtcgtggctg	gatcttctct	tggcgggtta	900
aaactaatac	aggaaatgct	tgatttcgca	gcagaacata	atatagttgc	tgacgctgaa	960
gtaatacctg	tggattacat	caacacagct	attaaaagaa	tagagaccgg	agacatcaaa	1020
tatagattcg	tcacgatgtg	gggcaacacg	ctaaaaagt			1059

SEQ ID NO: 8 moltype = DNA length = 1080
 FEATURE Location/Qualifiers
 source 1..1080
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 8

atggcagccg	agaaatctcc	agaggaaaga	caccctgtaa	aggcgtttgg	ccttgcgtgc	60
aaagacagtt	ccgggatact	tagtccgttt	aacttctcca	ggagggctac	tggtgatcat	120
gatgtccagt	taagatgtct	ttactcggga	ttgtgttatt	acgatacgaa	gatgataaag	180
aataagaggg	gtgtcacgcg	ttatcctttc	gttttcggcc	acgagattgt	gggagaagtg	240
accgagatcg	gcagagaggt	acagaaatcc	aaagtgggcg	ataaggtagg	ggtaggttgt	300
atggtagcaa	gctgtaggtc	ctgcgaatca	tgtgcaata	actgtgagaa	ctactgcccc	360
aatgtgtctg	tgacggatgg	ggccttcttc	tttaaaacag	gggaggttct	atacggaggg	420
tgttctgaca	tcattggtagc	cgatgagaat	tttgtgatca	ggtggccaga	aaattttcca	480

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ctagacgcgcg gtgctccggt gttatgcgct ggtatcacaa catactctcc tctaaggaat 540
tttgggctgg acaagcctgg aatccacgta ggcattttacg gattgggtgg cttggggccac 600
gtagccgttc aatttgc aaa agcattcgggt gcgaagggtga cggtaataag cagtagtgac 660
aggaagagga tggaaagcgt tga aaaaatta ggccgagaca gtttctggt aaatagtaat 720
ctggaagaaa tgcaggcggtc tatgggcacc atgcacggca ttattgatac tgtgccagct 780
aatcattctc tagtccctct attggatctt ctgaaaccac aagggaatt gatcgctgta 840
ggtggaccgc agaagcgtt cgagcttctt gtgttccctc tattgcaggg tggcagattg 900
gtagcgggaa gcgcgacggg tggaaattaag caaacacaag aaatgatcga cttcgctgcc 960
gagcataaca ttcttccaca cgtagaagtc gtaagtgtcg attatgtgaa caccgcgatt 1020
gaaaggaccg agagaggcga tgttaagtat cgtttcgtga tagatattgg aaatacactt 1080

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SEQ ID NO: 9      moltype = DNA length = 951
FEATURE          Location/Qualifiers
source           1..951
                 mol_type = other DNA
                 organism = synthetic construct

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SEQUENCE: 9
atggcagccg aaaagtcccc agaggaggag cagacgggtca aggccatagg ctgggcagta 60
cttgaggctg cgggtgtcct gtctccattt gaggtaagca gaagagcaac tggagaacgt 120
gatgtacagg ttaaggctct tttactgcgg atctgcaact tcgatactga gatgattaac 180
aacaattctg gctttactcg ttatcccttt gtcttgggct gcaccaataa cctggagaac 240
tactgtccca aagcagctact tattgaagct aacaatggag gttgttctaa cttagtgggtg 300
gcagacgaag atttcgtatt tcattggcca gaaaagctgc ccttggatct aggcgctccc 360
ttactttgtg caggcattac gacatattcc ccaactaaaga attttggatt ggataagccc 420
ggactacacg tgggggtagc aggcctgggt ggcttgggtc atgtagccat aaagttcgca 480
aaggccttcg gagcgaaggt gacggtcctc tcatcatctg acaagaagaa gaaagagggt 540
atagaaaagt acgggggtga tgaagttcta gtgagtagta atcccagaca aatgcaagct 600
acggcaggaa cactaaatgg tataatcgac acggttccgt cccgcactc cttagtgcg 660
ttacttaate tctgtcttcc ccagggaag ctaatagttc tagtgctccc catcgaaact 720
tttgtgttac cgatctaccc ctgtgtacag ggtggtaggg tcgtggctgg ttccgccacg 780
ggaggtctaa agcaaatcca agaaatgctt catttcgagg cagagcattc catagctgct 840
gatgcagagg ttatacctat tgactacatc aatacagctg tcgagcgtat agaaaaggga 900
gacattaagt acagggtttgt tgttgatata ggcaatacac taaagtcagc a 951

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SEQ ID NO: 10     moltype = DNA length = 1080
FEATURE          Location/Qualifiers
source           1..1080
                 mol_type = other DNA
                 organism = synthetic construct

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SEQUENCE: 10
atggcagctg agcaatcccc acaggaccag caccagggtta agtttctagg ttgggctgct 60
agggatagtt ccggtatcct atccccgttc aaatttagca ggagagcaac cggcgagcat 120
gatgtcgacg taaaggttat ctattgcgga atttgcaact atgacgtcga gatgatcaag 180
aacaagtggg ggaatgacaaa ttaccacagta gtttttggac atgaagccgt ggggtgtgggt 240
acggaatctg gctccaaggt tcagaaatcc aaggtggggg atacagtcgg ttaggaact 300
caagtcggta cctgtgggga atgcgactgt tgcaccaacg acttaaatca gtattgtcct 360
gaaatgattg caacgggatgg ctccggttct aatgatgtga gttgcggcgc atgctgtgac 420
ataatggtag ttgatgaaaa ttctgttacc gtatggcctc agaaccttcc aatggatgct 480
ggagcacctc ttctgtgtgc ggggtattac acttacagtc cccttcgtaa ctttggctcg 540
gacaagccga gaatacatgt ggggatcggt ggacttgggg gtctagggtc tgtggcagtg 600
agatttgcca aagcgttcgg agccaaagtt accgttatat ctacatccat gaataagaaa 660
caagaggctt tgcataaata tggcgctgac agcctttagt tcagttccga acaagaacaa 720
atgcaagccg cgtgtctcaac aatgaacggg atcatcgaca cgtcacctgt gcctcatgcc 780
ctagtgcctc ttttgaagct attgaaaccg cagggttaagt tgatattgat tggagcacgg 840
gacaaaacct atgagttgcc tgtgttccct ctaatagaag gtggcgttat cgtggcaggt 900
agcgtctacc ggtcattcag acagacacaa gaaatgatag atttcgcagc taaacacaat 960
attgtagcag atgtcgaggt aatccctacc aattatctga atactgcgtt agaactgatt 1020
gagaagggtg atgttaataa cagatttgtc atcgacttgg gaaacggact taaactagca 1080

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SEQ ID NO: 11     moltype = DNA length = 1062
FEATURE          Location/Qualifiers
source           1..1062
                 mol_type = other DNA
                 organism = synthetic construct

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SEQUENCE: 11
atggcagctg ggaatgtgac acaggaagag cacacagtaa aagccttcgg ttgggcagca 60
agagaagcca cgggtgcact tagcccttat ggattttccc gtagagcgac aggagaacgt 120
gacgtaaagg ttaagattct gtattgcggg atctgcagga cagacgccga gatgattaac 180
aacaattttg gctttaccac ctaccctgtt gttcttgggc atgagatagt gggcgctcgta 240
tccgaggtcg gatcaaaagt gaagaaattt cagttcgggt ataaagtgg agtggggacg 300
atgattggag cttgcagaac ctgttacagc tgcactaata atttagaaaa ttattgtcca 360
aaggcagtc taatagaggg caacaatgga ggatgtagca atttggctgt cgtgacgaa 420
gatttcgtat ttagggtggc tgagaagctg ccattagatc ttggtgctcc gctgctatgt 480
gctggtatca caacctactc cccgttgaaa aacttcggat tggacaaacc tggactgcat 540
gttggtcgtg cgtgtttggg aggttttagt catatagcgg tcaagtttct taaagcgttt 600
ggtgcaagg tgactgttat ctcaagctct gataagaaga aaaagggaagc tatcgaaaaa 660

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tactcagccg	atgagttttt	ggtagttcc	aatcccgagc	agctgcagcc	ggctgctggg	720
actctaaatg	gtattataga	tacagtcccc	tcaccgcatt	ccttggtgcc	cttcttgaac	780
ttgctgcttc	ctcaggga	attaatagtt	ttgggtgccc	caattgaacc	attcggtttg	840
ccaatcttcc	cattgtctac	gggaggcaga	gttgtcgcgg	gcagcgcaac	ggggggccta	900
aaacaaatcc	aagaaatggt	attgtttgcg	gcggaacaca	acatagcagc	tgacgcagaa	960
gtgatcccca	tagattacat	taacactcgc	gtagaacgta	tagagaaggg	ggacatcaag	1020
tataggttcg	tggttgacat	acgtaataca	ttaaaaagtg	ca		1062

SEQ ID NO: 12 moltype = DNA length = 1086
 FEATURE Location/Qualifiers
 source 1..1086
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 12

atggcagctg	gcaagactcc	agagactgtt	cacccggtaa	aggccttcgg	ttgggctgct	60
cgtgatacta	gcggggaact	gagtcctttc	aagttctcaa	gaagagcaac	gggtgaacac	120
gacgtacaat	tcaaggctct	gtactgcgga	atatgccact	cagaccttca	tatgatcaag	180
aatgagtggtg	gaatgacagt	atatcccata	atccctggtc	acgaaatgtg	tggtgtgtgc	240
accgaagtgg	gtaacaaagt	ggaaaaggta	aaagtgggag	ataaggtcgg	cgttggtatg	300
ctggtaggtt	ctcggcgcaa	ctgtaatatg	tgcagtcagg	acctagaaaa	ctactgcccc	360
aaacagatcc	tgacttactc	cgcgatcgac	acagatggta	gcataacgta	cggaggctat	420
tccgacgtga	tggttgccaa	tgagcacttt	ataatcagat	ggccggaaaa	cttacctcta	480
gacgcgggtg	ctccactact	atgcgctggc	attacaacat	actcaccctc	acgttacttc	540
gggcttgata	aaccaggcat	gcacgtttgt	atcgtagggc	tggttggtgc	agggcacgtc	600
gctgttaagt	tcgcaaaagc	attcggttct	aaggttacag	tgatctcaac	aagtgagaat	660
aagaagaagg	aggcgataga	aaaacttggc	gctgacgctt	tcttaaacag	cagagatcca	720
caacaaatgc	aggcagcggt	cgttacatcc	gatgggatct	tggaacactg	ctctgctgtg	780
caccgcgatc	taccacttat	caacctagtt	aaatctcagg	ggaaagtaat	catggtcggg	840
gctcctgaaa	agccattaga	attggcgaca	tttcctgtac	tagtcggaag	aaagacgcta	900
agcggcagtg	caataggtgg	actgaaagaa	acacaagaga	tgatagattt	cgtgccaag	960
cacaacatct	tggccgatgt	ggaagtaata	cctatgaatt	acgtgaacac	cgcgatggag	1020
cgtttggtga	aggcggtatg	taagtaccgt	ttcgttattg	atataggtaa	cagtcttaaa	1080
tcagct						1086

SEQ ID NO: 13 moltype = DNA length = 1086
 FEATURE Location/Qualifiers
 source 1..1086
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 13

atggcagcag	gtaaatgcgc	acaagaagag	caccagtcac	agagcttttg	gtgggctgca	60
cgtgatacga	gcggcatcct	gagccctttt	aagttctcta	ggcgtgccac	tgatgagcac	120
gatgtccggt	tgaaagtctt	ttattgtgga	atatgtcact	cagacttaca	caccatcaaa	180
gacgagtggtg	gaattacgga	gtatccggtt	gtgccaggcc	atgagatagc	tgccgtagtc	240
actgaggtag	gtagcaaggt	tcagaagttc	aaggtcggcg	acaaagtcgg	agttgggtgt	300
tttgtaggct	cttgtcgtaa	ttgcaacatg	tgtgataaga	atttggaaaa	ttactgccca	360
gcgcagatat	ttacatatgg	tagcccgtag	acggatggca	ctccaacata	tgagggtcat	420
tctgatatga	tcgttagtaga	cgaacacttt	gttattagat	ggcctgaaaa	tttccgctg	480
gacgcaggag	cgccctttgt	attcgcaggc	attaccacgt	actccccact	tagatacttc	540
gagctggata	agcctggcat	ccatataggg	attgtgggac	ttggtggctt	aggtcatgta	600
gctgtgaaat	ttgctaaggc	tttaggggct	aaggttacgc	ttattagcac	atctgccagt	660
aaaagacaag	aggcatttga	aaagctcgga	gcagacagct	tttactgag	tagcgaccct	720
gagcagatag	aagcggctgt	tggaaacctg	gatggtataa	tagatactgt	gtcagcggtg	780
catccgatag	tgcccttgat	tgatctgttg	gctcctcatg	ggaaactaat	agtagtgggt	840
tctccagcca	aacccttgga	gctaccgatt	tataatctga	gcataggccg	taagatcgta	900
ggagcgtcct	ccgttgaggg	gcttaaaag	acccaagaga	tgatcgactt	tgctgcaaa	960
cataacatac	ttccggacgt	agaaataatt	cccctcgact	atgtcaatac	agctatggag	1020
agattagctc	gtggcgatgt	gaaatatagg	ttcgtgattg	atgtaggtaa	taccctaaaa	1080
tcaacc						1086

SEQ ID NO: 14 moltype = DNA length = 1086
 FEATURE Location/Qualifiers
 source 1..1086
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 14

atgtcatccg	cagagaaaag	ccccgaagaa	gagcaccggg	tcaaggcatt	tggtgctggc	60
gccaagattt	cctctggaat	attgagtcca	ttcaactttt	ctagaagggc	gaccggggac	120
catgacgtcc	aattgaggtg	tctttactgc	ggtctgtgct	actatgatac	aaaaatgatt	180
aagaacaaga	ggggagtaac	gcgttatcca	ttcgtattcg	gacatgaaat	tggtggagag	240
gtcacggaaa	tcgggctgta	agtcacagaa	tttaaggtag	gggacaaaag	gggtgtggga	300
tgcattgtag	cctcatggcg	ttcctgtgaa	tcttctgcga	acaactgcga	aaactattgt	360
cccaatgtta	gcgttacgga	tggtgcatcc	ttttttaaaa	cgggggaagt	tctgtacgga	420
ggttgcagcg	atattatggt	agcggatgaa	aacttcgtaa	tacgttggcc	ggagaacttt	480
cctcttgatg	ccggtgctcc	gctgctttgc	gccggcatta	ctacatactc	cccgtgctgc	540
aattttggtt	tagataaaac	tgggattcac	gtcggcattt	atggggttagg	cggattaggt	600

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catgtggcag tccaatttgc gaaagctttc ggcgctaagg taacagtcac aagttccagc 660
gaccgtgaaga gaattggaggc aatcgaaaag ttaggtgccc acagttttct tgtaaactcc 720
aatttggagg aaatgcaagc ggcattggga accatgcacg gtataataga cactgttccc 780
gctaatacatt ctctagtcct tctgctggac cttttgaagc cgcaggggcaa actaattgtt 840
gttggcggcc ccgaaaagcc ttttgaatta cctgtatttc ctctattgca gggcgggagg 900
cttgtggcgg gtatgcgccac tgggggcac c aaacaaactc aagaaatgat agacttcgcg 960
gctgaacaca acatattgcc ccacgtagaa gtggtaagtg tagattatgt gaacacggcc 1020
atagagcgta cagaaaagagg cgatgtgaag tacagattcg tgatagacat tggtaacacg 1080
ttgtat 1086

```

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SEQ ID NO: 15      moltype = DNA length = 1086
FEATURE           Location/Qualifiers
source            1..1086
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 15
atgtcatccg ccaagacgcc ggagatgggt catccagtc gagctttcgg atgggcagcc 60
agagataact ctggagaatt atcccccttc aagttctcaa gacgtgcgac tggggaacac 120
gacgttcaat ttaaggctct atactgcggc atttgtcatt ctgatctaca catgataaaa 180
aacgaatggg gcatgcacgt atatcctatt gtacctgggc acgaaatcgt gggggtagtgt 240
acggaggtgg ggtcaaaagg cgaaaaggtc aaagtaggag ataaggttgg tgcggatgt 300
cttgtaggat catgtgggaa ctgcgatatg tgcagtcagg acttggaaaa ttattgtcct 360
aaacaaattt tgacttacag cgccatcgac gaggatggat caattacata tggcgggtat 420
agcgcagctt tggtagccaa cgagcacttc atcatcagat ggccagaaaa ttgccttttg 480
gatcggggcg ctcccttgtt atgcgcaggg atcaccacat attcccatc aaggtacttt 540
ggactggaca aaccagggat gcactgcggt atcgttggac ttggtggttt aggtcacgtc 600
gctgtgaagt ttgcgaaagc attcggcgcg aaagtgcagg tgatctctac ttccgaaaac 660
aagaaaaaag aagcgatcga aaagctggga gcggatgcgt ttttaaattc cagggatcca 720
caacaaatgc aagcggcggt aggcactttt gatggcatcc tggacactgt atcagcagtt 780
caccgcagtg tcccgcgtgt taacctagtt aagtcacaag gtaaaatgat aatggtcggg 840
gcacccgaga agcctttaga actggcaact ttccattat tggtcggcag aaaaaacttg 900
gcgggggtcga cgtatggcgg acttaagaaa acccaggaga tgatcgattt tgcagctaa 960
cacaatatat tggcagaggt tgaagtgat cctatgaatt atgttaatac agcagtgagg 1020
cgtctactaa aagccgatgt caagataga ttcgtgattg acataggaaa ctctttaag 1080
agtgtc 1086

```

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SEQ ID NO: 16      moltype = DNA length = 1086
FEATURE           Location/Qualifiers
source            1..1086
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 16
atgtcatccg cgaaaacacc tgggactgag cattcagtta aagcattcgg gtgggcccgc 60
agggatacaa gtggtgagtt aagcccggtt aaattctcca ggcgtgaaac tggcaaacac 120
gatgtgcaat ttaaggttct gtattgcgga atatgtcact cagacttaca tatggtaaag 180
aatgagtggg atatgacccc ttaccctatt gtacctggcc atgagatagt gggagtgtgt 240
acggaaagtgg gcagtaaaagt tgaagaggta aaggtggggg acaaaagtag ggtcgggtgc 300
ctagtccgca gctgcagaaa ctgcgagatg tgttctcaag acctggagaa ctattgtgca 360
ggccagatcc ttacttactc ccgcatatac accgacggca ctataacata tgggtgggtat 420
tctgatctaa tggtcgcgga cgaacacttt gttatacgtt ggctgagaa ctcccttg 480
gacgcgggtg cgcccttttt atgcgcaggg ataacaacct actcaccttt gagatatttt 540
gggctagata aaccgggtac ccacgtgggc gtggtaggac ttgggggttt aggccacgta 600
gcagtgaagt ttgcgaaggg gttcggggcg aaggtcaccc tcatttctac gagcgaaaac 660
aaaaagaaag aggccatcga gaaattgggt gctgacgctt tcctggttag tcgtgatccg 720
gaacagatgc agccggcaat cggtaacaat gatggaataa tgcacccgt gagcgccgtg 780
caccgcgtag tacggttgat taacctagtt aaaagccagg gtaaaattgt gatggttgg 840
gctccggaga aacccttga atcgcctact tttccgttaa tagttggaag aaaaactcta 900
tcaggtagtg ctataggagg cctaaaggaa acgcaggaga tgatcgattt tgcagcgaaa 960
cataatatat tagctgatgt agaagttatt ccaatcgatt atgtgaacac tgcaatggat 1020
agactattga aggcagatgt gaagtatcgt tttgtaattg acataggtaa ctccctgaaa 1080
agcgca 1086

```

```

SEQ ID NO: 17      moltype = DNA length = 1089
FEATURE           Location/Qualifiers
source            1..1089
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 17
atgtcatccg ctaagagttt tgaagaagaa caccctgtta aggcctttgg ttgggtgccc 60
cgtgacagca gcggaatcct gagcccggtt aaattctcca ggagggcgac gggtaagaag 120
gatattagat tcaaatgtgt gtacactggg atctgtcaca gcgacctaca ccactgaag 180
aacgattggg gctcaagtaa gtaccctctt gtaccgggac atgaaatagc aggtatggta 240
actgaagtgg ggagttaggt gaagagcttt aaagtggag ataagatcgg ggttaggtgc 300
ttgattgggt catgtgggac ctgcgaggac tgtgccgatg gactagagaa ctattgccct 360
aagaagatcc taacatacga tgggtccaaa tactacgacg gcacaatgac ttccggcgcc 420
tattccgatc acatgatcgt ggacgagcat tttgcaatcc attggccaga aaacatgccc 480

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ctagacgggg gtgcaccctt gctttgcgcc ggaatcacca cttattctcc gatgaagtac 540
ttcggcttgg acaaacccagg gttacatata ggagtcgtgg gtttaggcgg cctggggccat 600
gtacgcgttg aattcgcaaa ggcttttgg gctaagggtta ctgtaatctc ctcaactcca 660
ggcaagaagg aggaagcgat taaaaacttt aaggctgatt catttctggt aagccgtgac 720
acagcageta tgcaagccgc agccggcaca ttacatggga tacttgacac agtcagcgcc 780
gttcatccca tcctaccatt actgagcctt ttaaagaatc atagtaataa cattatgctt 840
ggcgccaccg agaaccatt agagctacca gtctttccaa tgcttatggg caggaagatc 900
gtcgacggga gcaacatcgg gggcgtgaaa gaaactgagg aaatgcttaa cttcgccgca 960
gagcatggaa taacagctga tgttgaggta atoccaatcg actatgtcaa cactgcggtc 1020
gaacgtctgg gtaagaatga tgtgagatat aggttcgtaa ttgacgttgg taatacactg 1080
aaagcgact                                     1089

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SEQ ID NO: 18      moltype = DNA length = 1065
FEATURE           Location/Qualifiers
source            1..1065
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 18
atgtcatccg ccggaagtg cccacaagaa gaacagactg ttaaagcctt tggtcgggct 60
gcgagagaag ttagtgaggc attgtccctt tacgggtttt ctaggcgtgc gacgggggag 120
agagatgtca aaattaatgt actgtatagt gggatgtgca actttgatac cgaatgatc 180
aacaacaagt tcggttttac caaatatcct tacgtccttg ggcacgagac ggttggcgta 240
gtatccgagg tgggctcaaa agtccgtaaa tttaaggtgg gggataaagt aggcgtgggg 300
ggccttgcgc gttcttgcgc taagtgttac tctgcacgca atgactcgtg aaactattgc 360
actaaagctg aattaactga ggcggcgcca gggggttgtt ccaattttgt tgtaaccgat 420
gaagactttg ttttccattg gctcgagaaa ttgcccgttag acctgggagc accacttttg 480
tgccgtggtg tgacgacatt ttctcctttg aagacgcgat gtctagacaa gccgggaatg 540
catgttggtg tcgcaggcgt gggaggcctt ggccacgtgg ccgttaaatt tctaaaagca 600
ttcggcgcta aggttaactg gatttcaaca agtgataaca agaagcgtga agcgatagaa 660
aaatattggg ccgatgcgtt tctagtgtca tctaataccc acaaatgca gacagccgct 720
gggaccctga ccggtattat tgataccata cctagtgtct atagccttgt tcccttacta 780
aatattattg tgccacaagg gaagcttata gtgcttggtg tcccaaccga accattcgtt 840
ctgcccgttg tacgatgat tcaagtggtg aaagtcgttg ctgggtcctc cttcggtggc 900
ttaaggttaa tacaggagat gcttgacttt gccgctgaac ataacttgt agccgatgcg 960
gaagtcatac ctgttgatta tatcaatacc gctattaagc gtatcgaaac cggagacatt 1020
aagtacaggt ttgttaattga tgttggaaac acgctgaagt cacc 1065

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SEQ ID NO: 19      moltype = DNA length = 1065
FEATURE           Location/Qualifiers
source            1..1065
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 19
atgtcatccg ccgggaagtg tgctcaagaa gaacacaccc tcaaggcgtt tggtcgggct 60
gcgaggggag caagtgaggc tttaagcccc tatggcttct ccagaagggc cacaggtgag 120
agggatgtga gagtgaagat tctttactgt gggatatgta ggacggatgc cgagatgatc 180
tccgacaaat tctgcttcac taaatatcct catgtacccg ggcagagat agttggagtg 240
gtaagtgaag tcgggaacaa agtgacagaag ttcaaggctg gtgccaaggt gggagtaact 300
ggcatgattg gatcttgcgc tacttggtat agctgcacaa ataactctga aaactactgc 360
caaaaagccg tattaataga agccaataac gggggttgtt caaatcttgt cgttgcagac 420
gaagatttcg tattcagggt gccagagaag ctaccgcttg atctgggcgc acccttgcta 480
tgccgtggga tcactacgta ttctccgcta aaaaacttcg gtttagataa accaggcctg 540
catgtcgggg tagctgggtt aggaggacta ggtcacattg ccgtgaaatt tctgaaagct 600
tttgagcta aagtcaccgt cataagttcc tcagataaga aaagaaaaga agctatagaa 660
aaatacggcg cagatgagtt cctgttttct agcaatccc agcagatgca agccgcagcc 720
ggcacgctta acggtatcat cgacaccgta ccgtctccac atagtcttgt accctttctg 780
aaccttttac taccacagg caaacttata gttcttgggg ctcccataga gccctttgta 840
ctaccgattt accctctttt acaggggggg aaagtagtag cagggtctgc cactggagga 900
ttgaagcaaa ttcaagaaat gttgcacttc gcggtgaac ataatacgc cgcgcatgcg 960
gaggttatac cgattgacta tatttaatac gctgtagaac gtatagagaa gggggatatt 1020
aaatatcgtt tcgttgttga tattgttaac accctgaaga gtg 1065

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SEQ ID NO: 20      moltype = DNA length = 1056
FEATURE           Location/Qualifiers
source            1..1056
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 20
atgtcatccg ccggcaagag ccccgaaagaa gaacagactg ttaaagcgat aggttgggct 60
gcttccgagg ctgcccagggt tctatcccc tgcgagtttt ctaggcgtac gacggggagaa 120
catgacgtac aggtgaaggt cctatactgc gggatatgca acttcgatac cgaatagata 180
aacaataaat tgggcttcac cagataccca ttcgtactgg gacacgaaat cgttgggtgtg 240
gtatccgagg tggggtcaaa ggtcaaaaag ttccaggtgg gcgacaaggc ggggggtgggt 300
accatgatcg gcacgtgtag gacctgttac agttgcacta ataacttaga gaattactgt 360
ccaaaagcgg tactgattga ggtcaacaac ggtgggtgta gcaatcagat agttgctgac 420
gaggactttg tctttagggt gccggagaaa ttgccgttgg acttgggagc acctcttcta 480

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tgtgcaggga	ttacgacgta	ttcaccgcta	aagaacttcg	gtctagataa	gcttgatta	540
cacgtgggag	tcgcggtct	tggcggtg	ggcatgtcg	ccgtcaaatt	cgtaaggca	600
tttggtgcga	aagtgcagct	gatctctagc	tcagacaata	aaaagaagga	ggcaattgag	660
aaatacggag	ctgatgaatt	tttagtgta	tcaaacatcg	agcagatgca	agctgccgca	720
ggaaccctaa	acggcattat	tgatacggtg	ccttccccgc	attctctagt	accgtgtgtt	780
aatctattgc	tgccccagg	taaaactaat	gtccttggcg	cgctaccca	acccttgtt	840
ctgccaatat	atccgttct	tcagggggaag	tcaattatgc	catctttctg	ctttgcaatg	900
atttgtatag	gtcaaaaagg	gattaacaat	gtgcattatg	tgtttattgt	tgtttattac	960
agaaggaaaa	actgtagatg	gaagtgccac	tggcgttttg	aggcgaaact	cagaaacgca	1020
ccgttttgc	ccaggactca	gcattccagc	aggtgc			1056

SEQ ID NO: 21 moltype = DNA length = 1086
 FEATURE Location/Qualifiers
 source 1..1086
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 21

atgtcatccg	ctaagtcaact	ggaggagcaa	cacccgataa	aagccttcgg	atgggctgct	60
cgtgatgta	gtgggggtact	tagccccttt	agtttctcca	gacgtacaac	cggagagaaa	120
gacatccaga	ttaaaattct	ttattgcgga	atatgccact	cagatttaca	ttgcgtcaag	180
aacgagtggtg	gttttagcaa	ctatcccatt	gtcccaggtc	atgagattgt	cggtgtgtgc	240
acggaggtgg	gaagcaaat	tactaaattc	aaagtggggg	acaaggtagg	ggtgggtgc	300
atggtaggaa	gttgtttatc	ctgcgagaa	tgctcaaaac	acctggagaa	ttactgtaaa	360
aaggtcattc	caacttaca	ttcaacatac	tatgacaaga	cccacacata	tggtggatgc	420
tccgatatat	tggttagtaa	tgagcacttc	gtagtcagga	taccgaaaa	tatgccctta	480
gatgtcggag	cgccgttact	tggtgctggc	atcacgacat	acagtcacct	gcgttatatt	540
ggattagaca	agcctggcat	gcataatcga	gtgggtggcc	ttggaggact	gggccattgc	600
gcggttaagt	ttgcgaaggc	ttcggcggtt	aaagtaacgg	tcattagcac	atccctacat	660
aaaaagaaaa	aggccttgga	acacttggga	gccgatagtt	ttttagcttc	caccgataag	720
gaccaaatgc	aggcggcaat	ggaagtatg	gacggaattt	tagatacagt	gagcgccgta	780
cacccccctt	taccctcgt	caacctgctt	aaaagccatg	ggaatttgt	tatggtgggt	840
gtcctcaaac	gtccttggga	attaccgggt	tttcccatgt	tagctggtag	aaaaattttg	900
gcaggttcag	gtattgtgtg	tatgaaggaa	acacaagaga	tgcttgactt	cgctgctgaa	960
cacaacatcg	ttgctgacgt	ggaggttaatt	agtatggatt	acgtgaacac	cgcatggaa	1020
agactggccc	gtgccgatgt	aaagcacagg	ttcgttattg	acatagggaa	tacactgaaa	1080
actgct						1086

SEQ ID NO: 22 moltype = DNA length = 1089
 FEATURE Location/Qualifiers
 source 1..1089
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 22

atgtcatccg	ccaaatcatt	cgaagaggaa	catccagtta	ctgcatttgg	gtgggcagca	60
agggatacct	ccggtttcct	aagccccttc	aaattctcaa	gaagggttac	aggggagaca	120
gacgtgcgtt	ttaaggttct	attcactgga	atttgtcact	cagacttgca	ccactaaaa	180
aacgactggg	gtgcttccaa	gtatcccttg	gttcggggtc	atgagatcgt	aggtgtggtt	240
acagaggtgg	gggtcaagg	tcagaaatc	aaagcaggcg	acaaagttag	ggtaggatgt	300
ttgataggga	ctgttaggac	atgcgaggac	tggtcggaa	atcttgatc	ttactgccct	360
aaaaagataa	cactcgtaca	tggaacccag	tactatgacg	gcagcctgac	gttcggcggg	420
tactccgac	acatggctgt	cgatgagcat	ttcgttatac	actggccaga	gaagtggccc	480
ttggacgggg	gtgctccttt	gctatgtgct	ggtattacta	cgtacagccc	catgaagtat	540
tttggctctg	ttaaagccgg	cctgcacatc	gggtgggtgg	gtctgggctg	cttaggacac	600
gtcgcggtga	agtttgcgaa	agcctttggg	gccaaagtta	ccgtcatctc	ttcctcccc	660
agtaaaaaag	aggaggcaat	taaaaatttt	ggtgcagatt	cttttttagt	tagcagagac	720
acagaggcga	tgcaaggcgc	ggcaggcacg	cttcatggga	tcttggtatc	agtcagcgcg	780
gtacatcccc	tgcttctct	tctgtccttg	cttaaaaatc	actctaagta	tataatgttg	840
ggagcgctcg	aaaagccatt	agagctgcgc	gttttcccaa	tggtgatggg	aagaaaaata	900
gtcgcgggca	gtaatatagg	gggtatcaaa	gagacggagg	agatgcttaa	ctttgcagcc	960
gaacatggta	taactgcgga	tgtggaggta	ataccaatgg	aatacgtcaa	taccgcgatt	1020
gaaagactgg	gcaagaatga	tgttcgttat	cgtttcgtga	tcgacgttgc	gaatacccta	1080
aaagcgccc						1089

SEQ ID NO: 23 moltype = DNA length = 1068
 FEATURE Location/Qualifiers
 source 1..1068
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 23

atgtcatccg	ccggtaaagag	cccagagaag	gagcagacgg	taaaagccac	tggttgggct	60
gcgtctgagg	tttcaggcat	tttaagtcgc	atcgaaattct	ctagacgtgc	tacgggtgac	120
agcgatgtac	aaattaaagt	attgtactgt	ggaatttgca	gcttcgatac	cgagatgata	180
aacaacaagt	tcgggttcac	tcgtttatct	tatgttttgg	gacatgaaac	cgtaggggtg	240
gttagcggag	tcggctccaa	agtaagggaag	tttaaaagtag	gcgacaaagt	gggcgtcgga	300
ggactttag	gtagctgtg	tacatgctat	ggctgtacta	atggtttgga	gaattattgt	360
acaaaagagc	aaataactga	ggcaggagcg	ggcggttggt	ccaactttat	ggtcatggga	420

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gaagatttcg tttttcactg gctgaaaag ctaccgtag acctgggggc tccacttctg 480
tgtgccggtg ttaccaccta ttccccctg aagacccacg gggtggataa gcttgggtctt 540
cacgtcggga tagcagggtt tgggtggcatg ggtcatgtat ccgtggaagt cgcctaaagcg 600
ttcggcgcca aggttacgggt tataagtaca tcagacaata aaaaagaaga ggcatcaag 660
aagtacgggg cggacgcctt tctgaactcc tccaaccacg aacagatgcg tgetgctgcc 720
ggcacttttag ctgcaatagt ggataccatc cctctccac acagccttgt gcccttgcta 780
gacttgctat tggcgcattg caaggtcatt gtccttgccg ctccaagtga gcccttcgtt 840
ctgcccgctc tgcgcgtact gcaaggaggt agagttgttg ccgggtcctc cggtgcatct 900
cttaagcaaa tacaagaat gctggatttt gctgctgaac acaacattgt agcagatgca 960
gaggtcatat caatagatta catcaatag gctataaaaa ggatagagaa gggggacatc 1020
aagtacagat ttgttgttga tattgttaac accttgaaga gcgcctga 1068

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SEQ ID NO: 24      moltype = DNA length = 1065
FEATURE           Location/Qualifiers
source            1..1065
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 24
atgtcatccg caggaaagtc ccccgaaaaa gaacagacag tcaaagccac ggggtgggca 60
gcttcagagg ttacggggat cttatcccca atagagttct ctaggagggc gacaggagat 120
tctgatgttc agattaaggt cctgtactgt ggaatatgct cctttgacac cgagatgata 180
aacaataaat ttggattcac tagatacccg tacgtccttg gccatgagac agtcgggggtg 240
gtaagcgagg ttggctctaa ggttcgtaag ttcaaggtgg gggataaagt tggggttggc 300
gggttggttg ggtcctgttg gacttggtat ggctgtacga acgggcttga aaactactgt 360
accaaggaac aaatcacgga ggcgggtgct gggggatgtt ccaacttcac ggtaatggga 420
gaagacttgc tatttcattg gccagagaag ctaccattag acctgggtgc cccattgcta 480
tgccgtgggg ttacaactta ttccccgtta aagacgcgat gtttggaaca acccgggcta 540
cacgtcgggg tagccggcct gggaggactt gggcacgtag cagtgaattt cctaaaagct 600
tttggcgcta aggttaacgt tataagttct tctgacaata agaaaaaaga agcaatagag 660
aaatcgggtg ctaatgtttt tctggtttct tctaaccctc agcaaatgca agctgcggcg 720
ggcacccttt caggcatcat agatacgatc cctcaccgc atagcttact accgttacta 780
gatttggtac ttccgagagg aaaattaatt gtattaggag cccctataga accctttgtc 840
ctaccaattt atcctctgct acaagggggt cgtgtcgtgg caggatccgc aacgggcggt 900
cttaaacaga tccaagaaat gttggaattt gcggcggaac acaacgtcgc ggccggatggt 960
gaggtgatca caattgatca tgttaatacg gctgtcaaaa ggatcgaaac agggaaatatc 1020
cgtcataggt tcgttattga catcggaat acactaaagt ccgct 1065

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SEQ ID NO: 25      moltype = DNA length = 1065
FEATURE           Location/Qualifiers
source            1..1065
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 25
atgtcatccg cgggaaaaac ccccgaggaa gagcaaacgg tcaaagcgat aggggtgggc 60
gcgtccgagg ttcccggtgt gctttcaccc tttgaagtc gtaggcgtgc caccggagaa 120
agggacgtcc aagttaaaggc cctatatgct ggtatctgtt cctttgacac agaaatgatt 180
aacaacaaat ttgggttcac gaaataccct tttgtgcttg gacatgaaat tgcggagta 240
gcgtcagaag tcggcagtaa ggtgcaaaaag tttcaagtc gtagcaaggt tgggtgcgtt 300
gctttgatcg ggtcctgtag aacttgctac agttgcacga acgatctaga gaactattgt 360
ccgaaggcgg ttttgaccga agccaactca ggagggtgtt caaacttcgt agttgcggat 420
gaagactttg tattcaggtg gcccgagaaa ttgocacttg atttaggtgc cctctgctg 480
tgcccggaat ttaccaccta ctcccctctg aagaattttg ggctggataa gccgggacta 540
cacgtagggg tggcaggttt aggcggggtt ggtcatgtcg cggtaaaagt tttgaaagct 600
ttcggtgcaa aggttacggt tatctccagc agtgacaata agaagaaaga ggcaatcgaa 660
aaatatggcg cgaacgcggt ttgggtctca tctaaccgag aacagatgca agctgcggcc 720
gggacattat ctggcataat cgacactata ccgagtcctc acagtcttct acctttgtta 780
gatttgctat taccgcaagg gaagctgatt gtgctggggg ctcccattga gccctttgtc 840
ctgctctatt accctttatt gcaaggaggg agagttgttg caggtagcgc gacgggtgga 900
cttaaacaga tccaggagat gttggaattc gcggctgaac acaatgttgc ggccggacggc 960
gaagtaatac caatcgacca cgtcaacacg gcagtgaagc gtatcgagac cggtaatata 1020
agacacagat tcgttattga tattgttaac acgttgaat cagct 1065

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SEQ ID NO: 26      moltype = DNA length = 1086
FEATURE           Location/Qualifiers
source            1..1086
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 26
atgtcatccg cggagcagtc cccagaggaa caacaccccg tgaaggtcct gggttgggca 60
gcacgtgact catcaggaaat cttatcaccc tccaagtttt ctaggcgtgc caccggagaa 120
catgatgttc agattaaggt gatataattg ggtatatgta cgtatgatgt ggagatgata 180
aaaaataagt ggggcatgac aaactacccg gtagttttcg gccatgaagc cgttggtatg 240
gtcacagaaa tagggtcaaa ggttcagaag tccaaggtcg gggatacagt cggagtggtt 300
acgcaagtgg gttcttggtg caagtgcgag tgetgcacca acgatctaga ccaatactgt 360
cccgaatga tagctacgga cgggttttgt agtaatgatg tgagctatgg cgcttggtgc 420
gacataatgg ttgttgacga aaacttcgta atagtttggc ccagaaacct accaatggat 480

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gcaggcgcac ccttactatg cgcggggatc acgacctata gcccgcctgag gaactttggt 540
cttgacagac cgggaatoca tgtcggagtt gtaggactgg ggggcttggg tcatgtggcc 600
gtgaggttcg cgaaggcggt cggggcgaag gttacgggtga tcagtacatc aatgaataaa 660
aaacaagaag ccgttcataa gtatggagcg gactccttcc ttgtctctac tgatcaagaa 720
cagatgcagg taaatacggc ctgggtcgct gccatagtgg ataccatacc atcaccgcat 780
tcacttgtac cgctgctaga ttgtctacta ccacacggaa aggtaaatgt gcttggagct 840
ccgtccgagc catttgttct acccgttatg ccactgttac aggggtggag agtagtggcc 900
gggtcctccg gggcgagttt aaaacagatt caggagatgc tggatttcgc ggcagagcac 960
aatatagtgg cggatgcaga ggtoatcccc attgattaca tcaatacagc tattaaaaga 1020
atcgagaagg gggatataaa atacagggtt gttgtggaca ttggaaatac ccttaaaagt 1080
gcttag                                     1086

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SEQ ID NO: 27      moltype = DNA length = 1032
FEATURE            Location/Qualifiers
source             1..1032
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 27
atgtcatccg aaagagtga ggacctgacg gagctattag cagcccagca ccatgtcgga 60
aatcaaatgc tgaattttcg taagtacgac tccctaaaat gcgccattga attagggata 120
ccagacgcca ttaaccaaca tggaaagccg attactttat gcgagctagt ttcagctctt 180
ccaattaatc catccaaagc taaccacata tacagggtta tgagggttct atcaaatgcc 240
ggattcttca tactacaaga tcagggttat gctcttactg ctgcccggag gctactttta 300
aaagaagagc cttttaacct aagggctttt attttctata tgcagacccc agcggtagtc 360
aaaccttggc actccctgac aagaatggtt agtaacgacg accctttccc gttcgatacg 420
gcccacggca aaaattttct ggccctacgct gggaggagc ctaattttgc gaatctcttt 480
aatgaagcta tggcaaatga ttccactcta atcgtccagg taatgatgac ccaatgcaaa 540
tttgtattcg acggattgac ttctcttgca gatgtgggtg gtgtaccgg agcggtagct 600
cgtgctatag ctcaaaactt tccaaaccta aaatgtgtag tctgcgacct accgcatgct 660
atcgtctgac aggaggggac agaaaaatct gacttcgtgg caggagatat gctagagaag 720
gttccggcag ctgacgccat tcttttaaag tggattctac atgattggtc agacgaggat 780
tgtgtaaaaga tccttaagaa ctgtaaggaa gcgattccag gaagagataa ggggtgggaag 840
gtgataatta tagacatgat tatggaaaat caaatgaagg atgacgaaag cgtcgaaact 900
caagttgggg tgaacatgca gatgttgatg gggtacggag cgaaagagag gtcagaaaaa 960
gaatggggca aattgtttca agacgcgggg ttttcgattt acaagggtgct tcctcttctt 1020
tgcgtgtctc at                                     1032

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SEQ ID NO: 28      moltype = DNA length = 1080
FEATURE            Location/Qualifiers
source             1..1080
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 28
atgtcatccg atgccatgaa gtcagcggaa ctattcaagg cacaagcaca tatcttttaa 60
caagtctttt gctttactaa cggagcctca cttaaatgcg cggtgcaact aggaatccct 120
gatgccatcg acaaccacgg gaaggcaatg accctaagtg agcttacgga cgcgcttccc 180
attaaccctt ccaaaagcac acacatacat agattgatga gaatccttgt aacggctggg 240
ttcttcgtag aagagaggtt agggaatggg aaggaggaga aagcaaacgg gtatgcttta 300
acacctttct caaggctact attgaagaat aaacccttat ccttacgtgc atcagcgttg 360
actatggttag accctgtcac agtaaaaaca tggaacgccc tgagttagtg gtttcagaat 420
gaagatcaaa ctgcctttga aactcgcgac ggaagaata tgtgggattt ctttgcgtga 480
gatccgggtc tatccaagaa gttcaacgaa agtatggcct ccgacagcca gttggttact 540
gaagttcttg taactaaatg taagttcgtc ttccagggtc ttacatctat ggtggacgtg 600
gggtggaggaa cagggactgt ggcaggtgca attgcgaaaa ctttcccgtc tctaagggtg 660
accgtctttg atctaccgca ttgtggttgc aacttggaac ctacggagaa cttagacttc 720
gttgacggcg atatgtttgg caaaattcct ccagcaaatg caattttctt aaagtgggtg 780
ttgcacgact ggaacgatga agactgtgtg aagatcctga agaattgtaa gcgtgctatt 840
ccagggaagg agaagggagc caaagtattt atagttgata tcatcatgga gactgagaag 900
cacgatatcg atgagtttga ctacgcaaaag atgtgcatgg acatggaaat gctgggtattg 960
tgcaattcca agaacgtac cgaaaaggag cttgccatgt tagtctctga ggcgggtttc 1020
agcggctaca aaattttccc tgtattgggc attaggagtc ttattgaggt atatccgtaa 1080

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SEQ ID NO: 29      moltype = DNA length = 1080
FEATURE            Location/Qualifiers
source             1..1080
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 29
atgtcatccg ggagcgtgta ggaagaagca agcctgttcg ccatgcagct agtgacatcc 60
gcatccctac caatgggtgt atacaccgca gtcaaagtaa atctgtttga aatcatagct 120
aaagtctgtc ctggtgcaaa gttatcacc cagaaaattg cttgtaggct gcctacaacg 180
aagaacctcg acgtgcac c aatgctggac aggatgctaa gattgctaag ttctcactca 240
ctattgacat atgatgtggt cgaagtgtga ggtggagccg aaaccgacat aggatatgag 300
cgtgtgtacg gctgagttcc ggtgggtgag taccttatac cagatgaaga aggaaattct 360
gtggctcctt tagtcgagtt gctgcttgac aaagtctga ttaatacatg gtacgaacta 420
gggaacgctg tcctagaagg aggcattcca ttaactcgtg ttcacgggac ccatgcattt 480

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gactacccat caaggaatcc cagatacaac gagctgttta acaatggaat ggtgggaccg 540
actgccatca ccatgaagga acttctacaa cagtataaag gggtcgagct acttcagact 600
cttgttgacg ttggtgtgtg tttgggtata acgcttcata agattattgc taaatacccc 660
agcatccgtg gaattaattt cgatctgcca cacgttattg aggatgctcc gccgtatcct 720
ggcatggagc atataggagg tgacatgttc gagtctgttc caggtggcga tgcaattttc 780
atgaagatga ttctacacga ttggtctgat gatcattgct taaaactatt aaagaactgc 840
tatagggcac tgccagacca tggtaaagta atagtagtgg acctgatctt gcctgtaaag 900
ccggatacga atatcttcgt caagggtata ttccaggcag atgtcattat gatgacgcag 960
aatcctgggt gcaaggagag atcagaagca gatgtgcgtg cgctggcgat cggagctgga 1020
tttaaaggta tcaaaactga atgctgtgct ggaaacgtag ggatattaga attgtacaaa 1080

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SEQ ID NO: 30      moltype = DNA length = 1044
FEATURE           Location/Qualifiers
source            1..1044
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 30
atgtcatccg acgttcaatc agctgagcta ctgagggctc aagcccacgt gtggtctcaa 60
acttttaact tcaagaactc agctagtttg aaatgcgctg tccaactagg aatcgctgat 120
gcgatccaaa agcatggaaa acctatcacg ttaagtgaat tggctgcagc gctaccgatc 180
aatccttcca aagctaatta tatttacagg ctaatgcgta tcttagtcaa cgctggcttc 240
tttgctcaag agaaggaggg ctactccctt accgcagctg gaagactatt attgagtgat 300
gaaccgggtc atgcaagggc ctttacccta gttgttctgg acctgacac tatgaagcct 360
tggaatttct taagttagctg gttccagaac gatgacctat caccctttga caccgcacat 420
gggaagaatt tctgggactt cactgcggaa gaaccocaaat tcagtaagct gttcaacgag 480
ggcatggtaa gcgattcaca cttataaacy atggctcttg tcaccgaatg taagtacgtt 540
ttcgagggac ttacctcatt agtggacgtt ggagggtgga cgggaacagt cgctagggtc 600
atcgctgagg cttttccaaa ctgaaagtgt actgtatttg acctccaca cgttgtcgct 660
aaccagaag gaacattaaa tttagacttc gttgctggtt acatgctgga aaaggtaacct 720
cccgcgaatg caatcttact taaatggata ttgcatgact ggtcagacga agattgcgta 780
aagatattga agaattgtta gaaggcaatc ccgggtaggg acataggcgg taaggtaatt 840
ataattgata tgggtgatgga aagtcacatt gaggggtgatg agagcacaga agcacaaatc 900
tggttggaca tgcagatgct ggttttgtag cgtgcaaaag agagatcaga aaaggagtgg 960
gcgaatttat tctctgatgc tctcttccca tcatataaag tattccctgc actgggaacg 1020
cgtgctataa tagaggctca tccc 1044

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```

SEQ ID NO: 31      moltype = DNA length = 1044
FEATURE           Location/Qualifiers
source            1..1044
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 31
atgtcatccg atgtccaatc agctgaatta ctgagagcgc aagcgcacgt gtggtctcag 60
acgtttaatt tcaaaaatc tgccctatta aagtgtgcgg tacagttggg aattgccgat 120
gttatccaga aacatgggaa acccattacc ctatctaagt tggcagctgc gctaccaata 180
aatcccaagta agcgaacta tatttaccgt ttaatgagaa tccctgttaa tgcgtgattc 240
ttcgcccaag agaaggaggg atactcatta actagcgtg gtcgtctatt gttaaagcag 300
gagccagtg aagctcgtgc ctttatatta gttgtgttgg acctgacac aattaaacct 360
tggaatttct tatccgactg gtttcaaaat gatgacccta gtcggttcga caccggtcat 420
ggaaagaact tctgggattt caccgcagag gaaccaaaat ttagtaaat gttcaacgag 480
ggaatggttt ctgattcaca ctttaactat atgggtgtag tgactgagtg caaatatgtt 540
ttcgagggat tgacaagcct agtggatgta ggaggtggga cgggaactgt cgccaggagc 600
atcgccgagg tactccctaa cttaaaatgc acagtttttg acctgcctca tgtggttagc 660
aaccaggaag gaacattgaa cctagatttt gtcgctggaa atatgttgga aaagggtccc 720
cctgctaagc ctattctact taaatggatt cttcatgact ggtctgatga agcatgcgtg 780
aagatcttga agaactgtta gaaagcaatc ccaggggcgtg acatcggagg taaggttatc 840
ataatagaca tggatcatgga atcacaattg gaggcagcag agtctacaga agcccagatc 900
tgcttagaca tgcaaatgct tgtgttatac agagccaaag agagatcaga aaaggagtgg 960
gcgaatttat tttctgatgc aggtttccca tctataaag tctttccagc ccttggtacc 1020
agggccatca tagaggttta cccc 1044

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SEQ ID NO: 32      moltype = DNA length = 1074
FEATURE           Location/Qualifiers
source            1..1074
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 32
atgtcatccg acgtacaggg aggcgaactg ttacaagcac aggcacacat ttggaaccac 60
atattcaatt tcatcaactc tatgtccctt aaatgcgccg tgcagttaga tatacctgac 120
atcacacata agcaggggga accaatgact ctgacacagc taatacatgc attgccaatc 180
agtagcgcta aggtctcctt cgtccaccgt ctaatgagaa tactgattca ctcaggattc 240
ttcgtgaagg caaagatttc aggtaatgag ctggaagagg gctacgagct tacatctgca 300
agcaagttgc tgcttaagtc taatcccta tccgtgaccc cattcctgtt agccatgcta 360
gatccgatct taactgatcc ctggcaccac tttttctcaat ggttccaatg cgacgacgaa 420
acgcggttgg atacttgcca cggaaaggcg ttctggaaac tagcaggtga gcagccaagg 480
ctgaatcaat tctttaatga gggcatggcc tctgacgcga gactattttc gagcttgggt 540

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ataaagcact gcaaacatgt cttcgaggga cttaaatagtt tagtagatgt cggtaggagg 600
actgggacat tggcgatggc tatagcggat gcgttcccg acctgaaatg cacagtcta 660
gacttgccgc atgtggtggc gggccttgat ggcagtagca aaaatcttgc atacgttgcc 720
ggagacatgt tcgaggcgat tccgcctaca gacgctgtct tactgaagtg gatcttgcc 780
gactggtctg acgaagaatg cgtccagatt ctgaagagat gcaaaagaag cattccttct 840
aagggtaaag gaggttaagt tataattatt gatatacttg tcaaggatca gcacagaggt 900
gatgacaatg atgaagctat cgagactcag ctattctttg atatgctgat gatggtcctg 960
gttaaaggta ggggagaggaa tcagaaagat tgggccaac tttcttcga ggtgggttc 1020
aacgattata agatcacacc agtcttgggt ctgcttccct taatcgaagt atac 1074

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```

SEQ ID NO: 33      moltype = DNA length = 1071
FEATURE           Location/Qualifiers
source            1..1071
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 33
atgtcatccg atgtgcacgc gggtaggttg ctacaagcac aggcgcacat ctggaatcat 60
atatccaact tcattaattc aatgtctcta aaatgtgcc aatcagcttg gataccggac 120
ataatccaaa agcacgggtca gcccatgacc ttggcacac taatccatgc cttgcctata 180
tctaaccgca aagcgcgcat cgtttataga ctgatgagaa tactgatcca ctacaggtttc 240
ttcataaaag ccaagatccc aggtaatgaa ttggaggaag gttacgagct gactcctgcc 300
tcaaaactac ttcttaagtc aaaccctat agcgtcactc cttcttatt agctatgttg 360
gaccgcgacc tgacgggacc atggcacaa ttctcacagt ggttccaatc taacgacgag 420
acccctttct atacatgcca cgggagagaa ctatgggaat tagctggcga gaagcctagg 480
ctaaatcagt tctttaacga gggcatggcc tctgacgctc gtctaacttc ctccctagtc 540
ataaaacact gcaagcatgt gtttaggggt cttaacagcc tagtcgatat aggaggtggg 600
actgggacgc ttgctaaccg catagcggac gcttttccag atttgaagtg tacagtgttg 660
gacttgccac cgtgtgtgac ggggttagat ggatcatcca agaacttagc ctatgtcggc 720
ggcgatatgt tcgagggcat acctcctacc gatgctgttc tgtaaaatg gattcttcac 780
gattggtcag acgaggagtg tgtacaaa taaaaaaagt gtaagggaag aattccgtca 840
aaggagaaag gtggttaagt cataattatt gacatgttaa tgaagaacca acagaaaggt 900
gatgatgaag acaaaagctat agagacacaa cttttctttg acatgcttat gatggtgcta 960
gttaagggtg aagagcgtaa ccaaaaggac tgggtcaaac tttcttcga agcaggttc 1020
aacgactata aaataactcc tgtgttggga ctacgttcac ttattgaat c 1071

```

```

SEQ ID NO: 34      moltype = DNA length = 1080
FEATURE           Location/Qualifiers
source            1..1080
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 34
atgtcatccg acgttcaggc tgataaattg ctggaggctc aggcctcacat ctggaatcat 60
acatttaact ttataaattc tatgtgcctt aaatgtgcga ttcaattagg gataccggac 120
attatcaaga agcatggcgc tccgatgact cttgaggagc taaccgatgc actaccaatc 180
aacaatgcaa agagcccggt catctatcgt ttgatgagga tcttgatata ttctgggttc 240
tttattaaag cgaaagtaac agacacgcag ctggaaaaag gttatatgct aacgagcgcc 300
tctgagctac ttctggcaaa tgacctttc tccatgaaga ctttctact ggcaatgtta 360
gatccaattc taacggctcc ctggcagcat ttctctcaat ggttccaatc caacgatgag 420
aacctttct atacctgtca cggaagagcc ttctgggagc tagctgggca ggaaccgaga 480
cttaaccaat tctttaatga agcaatggct aacgatgcac gtcttgtgag cagcatcgta 540
attaaacact gcaaacacgt cttcagcggc ctgaacagtt tgcttgacgt tggaggtggg 600
actggaacgt tcggcaaggc aatcgagct gctttccag atctgaaatg cacagtacta 660
gatcttccc acgttgttga cggattggac gtgggtggca agaaccttgc ctatattggc 720
gggtgatatg tcgaggacat acctccagca gatgccgtac ttttgaatg gattttcat 780
gattggacgg atgaagaatg tgtcaggatt ctggaaaaat gcaaacaggc tatccgtca 840
aaagaaaatg gaggcaaggt tatcatcata gacatgggtc gtaacgacca gcaaaaagaa 900
gcggatgatg acgcacgtaa gagtatagaa actcagttat tctttgatat ttaaatgatg 960
gtcgatacca aaggtaggga aaggaatgaa aaagagtggg ccaaaactgt ctatgaagcc 1020
ggctttaaag attataagat tacgcaggta gtaggactga gatctctaag agaggtatat 1080

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```

SEQ ID NO: 35      moltype = DNA length = 1074
FEATURE           Location/Qualifiers
source            1..1074
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 35
atgtcatccg acgtacaggc tggcgaacta ctacaggcgc aggcctcagat atggaaccac 60
atatttaact tcataaatag tatgtccttg aaatgtgctg tacaattaga tattcccgac 120
ataattcaca agcatggtca gcccatgaca ctagcccaat taatacacgc cttccaata 180
tctagcgcta aagctccatt cgtctatagg ttaatgagga ttttgattca tagtgggttc 240
ttcttgaaa agcaaatgag agggcaatgag ctagaggaag gttacagatt gacgagcgcc 300
tctaagctac tattgaaaag caatccattt tcagtgcgc ctttcttgcgt gtctatgtta 360
gaccaaatc taacggaccc atggcaccag ttttccaggt ggtttcagtg cgatgacgaa 420
acgccatttt atacatgcca ccagaaggca ttctggaac tagctggtga acaccacagg 480
gtgaatcaat tcttcaacga aggcattggc tccgacgcaa gactaatcag tagcttggtc 540
atgaagcact gtaagcatgt tttcgaagcc ttaaacagtc tagtcgatgt aggtggaggc 600

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acaggaaccc tagcgatgac cattgtctgat gcttttccgg atctgaaatg cacagtatta 660
gaccttccctc atgtgggtggc tgggcttgac ggtagcagta agaacttagc ttatgttggg 720
ggagacatgt ttgaggcgat cccacctact gacgctgttt tgttgaagtg gattcttcat 780
gattgggtccg atgaggaatg cgtccaaata ttaaagaaat gcaaagaggc cattccgagc 840
aaaggtaaag gcggaaaaggt gataataatt gatatgttag tgaaggatca gcaaagggga 900
gatgataaag acgaggcgat cgaacccag ttgttctttg acatgttgat gatggattg 960
gtcaaagggc gtgaaaggaa tcaaaaggat tgggcaaagc tattcttcga ggcggggttc 1020
aatgattaca agattacccc tgtattggga ctgcgttcct taatcgagggt atat 1074

```

```

SEQ ID NO: 36      moltype = DNA length = 1080
FEATURE           Location/Qualifiers
source            1..1080
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 36
atgtcatccg atgtgcaagc cgacaaactt cttgaagccc agggctcatat ttggaaccac 60
acattcaact ttataaaactc tatgtgccta aaatgcgcca ttacgttggg cataccagac 120
attattaaga aacatgggag acctatgaca ttggtcgaac ttacggatgc actaccaatc 180
aattccgcga aatcaccatt catataagg ttaatgagaa tactgattca ctcaggattc 240
ttcatcaaag tcaaaagtac cgataccgat ttggagaaaag gatacatgtt aacgtctcgc 300
tcagagctac tactggctaa tgacccttc agtatgaaaa ccttcttact agcaatgtta 360
gaccccatcc taactgatcc ttggcagcat ttacgccaat gggttcaatc aaacgatgaa 420
aacccgtttt acacgtgcca tgggagggct atttgggaac ttgcaggcca ggagccacgt 480
ctgaaccagt tcttcaacga ggcgatggcc aatgacgcta ggcttgtgtc tagcatcggt 540
attaacacat gcaaacatgt gttcagcggc ttaaatagcc tattagacgt tggtaggagt 600
acgggtacct tcgctgaagc gatagctgcc gcatttccag acctaaagtg tactgtgcta 660
gacttaccgc acgtcgtaga cggattagac gttagttcca agaactctgc ctacatagga 720
ggtgatatgt ttgaagatat acctccggca gacgcggtcc ttttgaagtg gattttacat 780
gattgggtctg atgaggaatg cgttagaata ctggagaaat gtaaacaggc gataccctca 840
aaagagaacg gaggtaaagt aatcatcata gatattgtca gaaatgacca acagaaggag 900
gccgatgacg acgcgcacaa aagcattgaa acgcagctat tctttgacat cctaattgatg 960
gtggacacaa aaggcaggga aagaaatgag aaggagtggg cgaagtgtgt ctacgaggca 1020
ggattcaaag actacaagat caccacaagta gtgggcctta ggagccttat cgaggcttat 1080

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```

SEQ ID NO: 37      moltype = DNA length = 1077
FEATURE           Location/Qualifiers
source            1..1077
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 37
atgtcatccg atgtacaggc ggataagtta ttagaggcac aggcacatat atggaatcac 60
atatccaact tcataaaatc aatgtccttg aagtgtgcca tccaaatggg tattcccgat 120
ataataaaga atcatggaag gccaatgaca ttgaaggagt taacggatgc cttacctata 180
aacaacgcga agagtccggt cgtctatagg ttgatgagaa tcttaattca ctcaggattc 240
tttgctaaag ctaaggttac cgataacgat gttgaggaag gatacgtact aacctccgca 300
tccgaacttt tattggcgga tgaccattc tcattaacgc ctctgctact agcgtatgcta 360
gatcccataa tgacagatcc ctggcaaacac ttctctcagt gggtccaaaa taacgatgaa 420
actccgtttt atacttgcca tggcaggact ttctgggagt tggctagtca ggagccgaga 480
ttaaaccaat tcttcaatga ggcgatggcc agagacgcgt ggctggtagg ctctgtgcta 540
attaaccatt gtaaacatgt ttccagcggg ttaaatcttc tgggtgtagt cgggtggagg 600
actggtacgc ttgcgaaggc gatagctgac gctttccagg atttgaaatg tacagctctt 660
gatctgccat acgtcgtgga cgggttagac gggttctagca agaactcggg ttacattgga 720
ggtgacatgt ttgaggccat tcctccggct gacgctgtcc tacttaaatg gatactacat 780
gactggactg atgaagaatg cgttcgtatt ttagtaaaat gtaagcaggc aatcccagat 840
aaggagaaacg gtcggaaagt aatcattatt gacatggtta ttatcattga tatggtgaag 900
gggtcgaaacg accatagaag cattgagact cagttattct tcgacttgtt gatgctagtt 960
gacgtcaagg gtaaaagag gaatgaaaag gaaagagcga aacttttcta tgaagctggc 1020
tttaatgatt ataaaaaac tcaagtccga gggcttagga cttgataga agtttac 1077

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SEQ ID NO: 38      moltype = DNA length = 801
FEATURE           Location/Qualifiers
source            1..801
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 38
atgtcatccg ccgagctata taatcgtcaa gccaaagcagt atagttagat tagaccgacg 60
taccagaag agttatttca attcatagcc tccaagactc cttgtcatga cctagcgtgg 120
gatgtcggaa ctggaagcgg ccaggctgcg aggtccttag caggcatcta taagaatgtg 180
gtcgaaccgc atataagccc taagcagcta gaatttgcag tgaagttacc gaatataagg 240
tatcattgta cccacacaaa aatgccgata gctgaggttg aaaaaaacat cgctgccgag 300
tcctctgtgg atctagtcac aatagccact gcgatccatt gggttgactt accaaacttt 360
tatcagcagg tcaaatgggt tttgaagaaa cctcacgggg ttatgcgagc ctggagctac 420
atcaatgttg agatcaatac atgtagcgtc aacaccttgt tgcaaaaact atatctggat 480
tgtgtccgtt attgcgaacc acaagctaag tcagtcgctg atgacaaata tagaaccatt 540
gatttcccat ttcagcgggt agacgggctt gacaatacgg gtccgttcga gttcaagatg 600
gaaaaattga tggatttggg tgggttcttc acctttatca agagctggag cctataccag 660

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accgccaag ataaaggtgt agaattattg aatgacgatg tggttcagga ctttactag 720
gcatggaatg aagatggcaa tcctcagaaa atagecgacgt ttccaatata ccttcgtatt 780
ggcaaatgtag gcaactggta t 801

```

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SEQ ID NO: 39      moltype = DNA length = 1065
FEATURE           Location/Qualifiers
source            1..1065
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 39
atgtcatcgc attttgctag aaatgggtgac catactgggtg aattgtttca agctcaagca 60
catattttgga atcatttggt taatttttatt aattctatgt cattgaaatg tgctattcaa 120
ttgggtattc cagatattat tcataaacat ggtcaaccaa tggcattggc tcaattaatt 180
gatgctttgc caattaataa tgctaaagca cttttgtttt acagattgat gagaattttg 240
attcattcag gtttctttat taaagctaaa atgccagata atgaaggcca tgaagggttac 300
gctttgactt cagttctctaa attgttggtt gcaaatgac cttttctgt tacaccattt 360
ttattggcta tgttagatcc aattttgact gatccgtggc atcatttttc tcaatgggtt 420
caagataatg aagaaacacc atttcattca tgtcatggta cttctatgtg ggaattggct 480
ggtagacaac caagattgaa tcaatttttc aatgaaggta tggcatcaga tgctagattg 540
gtttgtacaa tgggtattta aaattgtaaa gatgttttta tgggtttaaa ttctttgatt 600
gatgttggtg gtggtacagg tactgttggt aaagcaattg ctgatgcatt tccacatttg 660
aaatgttctg ttttagattt gccacatggt gttgatgggt ctgaatcacc taaaaatttg 720
gcttacgttg gtggtgacat gtttgaagct attccaccat ctgatgcagc tatgaatcaa 780
tggattttgc atgattgggc agatgaaaaa tgtgttcaag ttttgaaaaa gtgtaaaaga 840
gcaattccat ctaaaagaaa aggtgggtaaa gttattatta ttgatattg ggttaaaaat 900
caacaaaaag gtgacgataa tgatgaacca attgaaacac aattgttttt cgatatgttg 960
atgatgattt tgtacaaagg tagagaaaaga aatcaaaaag attgggaaaa attatttttg 1020
caagctgggt ttaattacta taaaattatt ccagttttag gttaa 1065

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SEQ ID NO: 40      moltype = DNA length = 840
FEATURE           Location/Qualifiers
source            1..840
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 40
atgtcatcca gatttttgggt taatgggtggt ttctttatcc aacaagatca atgttacgct 60
ttgacaaactg caggtagattt gttgttaaaa gatgaacata ttaatgttac tggtttggtt 120
gctttatctc ctgacccaat ttgatgatg ccgtgggattt tcttgactga atgggtttaaa 180
aatgatgac cactccatt tgatactaca tacggtaaat ctttttggga ttacactgct 240
caagaatcta gatttggtta agtttttgat gaagcaatgg ctaatgcacc taatttgcca 300
attcaagttc caattactaa atgttagattt gtttttgaat cattgacttc ttgggctaatt 360
gttggtgggt gtactggtaa agttgcaaaa gctatttgcta aaaattttcc aaatttgaaa 420
tgtgttggtt gtgatttgcc aagagttggt gctaatacag aaggtactaa aaatttggat 480
tttggttcag gtgacatggt ggaaaaagtt ccaccagcaa atgctgcatt gttaaaaatgg 540
gttttgcatg attggtctga tgaacatggt ttgaaaaatt tgaaaaattg taaaaatgct 600
attccaacta gagaaaaagg ttgtaaaagt attattattg atattattat tgaatctcat 660
aaagaagata atgaattgggt tgaacctcaa attggtatgg atatgcaaat gttgggtttta 720
tacgttgcaa aagaaagaac tgaagaagaa tgggctactt tgtttttgga tgctgggttt 780
tcttcataca aagtttttcc agttttgggt gtttcttggt tgattgaagt ttatccataa 840

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SEQ ID NO: 41      moltype = DNA length = 1098
FEATURE           Location/Qualifiers
source            1..1098
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 41
atgtcatcgc atttggctag aaatgggtgac caagctgggtg aattattgca agctcaagca 60
caaatttgga atcatatttt taatttttatt aattcaatgt cattgaaatg tgctgttcaa 120
ttggatattc cagatattat tcataaacat ggtcaaccaa tgacattggc acaattaatt 180
catgctttac caatttcttc agcaaaaagct ccatttggtt acagattgat gagaattttg 240
attcattctg gtttcttttt gaaagctaaa attccaggta atgaattgga agaagggttac 300
gaattgacat cagcttctaa attattgttg aaatctaacc cttttctgt tacaccattt 360
ttattgtcta tgttagatcc aattttgact gatccgtggc atcaattttc acaatgggtt 420
caatgtgatg atgaacacc attttataca tgtcatcaaa aagcattttg gaaattagct 480
ggtagaacatc caagagttta tcaatttttc aatgaaggta tggcttcaga tgcaagattg 540
atttcattct tggttatgaa acattgtaaa catgtttttg aagggtttaaa ttctttagtt 600
gatgttggtg gtggtacagg tactttgggt atgacaattg cagatgcttt tccagatttg 660
aaatgtacag ttttagattt gccacatggt gttgctgggt tggatgggtt atctaaaaat 720
ttggcttacg ttggtggtga catgtttgaa gctattccac caacagatgc tgttttggtta 780
aaatggattt tgcatgattg gtcctgatga gaattgttcc aaattttgaa aaagtgtaaa 840
gaagcaattc catcaaaagg taaagtggtt aaagttatta ttattgatg gttgggttaa 900
gatcaacaaa gaggtgacga taaagatgaa gctattgaaa ctcaattggt ttctgatag 960
ttgatgatgg ttttggttaa aggtagagaa agaaatcaaa aagattgggc taaattgttt 1020
ttcgaagctg gttttaatga ttataaaatt actccagttt taggtttgag atctttgatt 1080
gaagtttact attactaa 1098

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SEQ ID NO: 42 moltype = DNA length = 519
 FEATURE Location/Qualifiers
 source 1..519
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 42

atgtcatccg	aaggtgctgg	tggtactttg	caaacatggt	gttggctctcc	aggtccaaga	60
gctgttttcta	aatgtaatgg	tcatgaatgt	tttccaagat	ggtttgcagc	tgttttggca	120
aaataccccag	tttgggttat	gaatgttgtt	ccagctcatt	ctgatccaga	tactttgtct	180
attatttaacg	aaagaggttt	tattggtact	taccatgatt	gggtggaagc	attttctact	240
taccacaagaa	cttacgattt	gattcatgct	gatggtgttt	tctctattta	ccaagataga	300
tgtgatatta	cttacatttt	attggaatg	gatagaattt	tgagaccaga	aggtactgtt	360
atttttagag	atggtgttga	agttttggtt	aaaattaaag	ctattactga	tggtatgaga	420
tgggaatcaa	gaatttttga	tcatgaatct	gggtccattca	atccacaaaa	gattttgggt	480
gcttttaaat	cttactggac	tgcaaaagct	aatgattaa			519

SEQ ID NO: 43 moltype = DNA length = 1089
 FEATURE Location/Qualifiers
 source 1..1089
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 43

atgtcatccg	atactaattgt	tgaagatca	aaattgttac	atggccaagc	tgaaatttgg	60
aaacatactt	ttagatttgc	tgattctatg	gctttgaaat	gtgctgttga	attgggtatt	120
gcagatatta	ttcattctca	tggtcatcca	ttaacttttg	cagaaattgc	tactaatatt	180
acaaaattcat	cttctccaca	tattccatat	ttgtctagaa	ttatgagatt	tttgggtaga	240
aataagattt	ttacttctac	agatattcca	tctgatcatg	gtggtggttc	tccatcaact	300
acattgtatg	gttttaacagc	agcttcttgg	tggttattga	atgataatga	tggtttgtct	360
ttagctccat	ttttgttaat	ggaaaatcat	ccgtgggtgg	tttctccgtg	gcatacaatta	420
tcatcttgtg	tttagaagag	tggtatttga	tttcaaaaag	ctcatggtaa	agaaatttgg	480
gaattggctt	ctgaaaatcc	agaatttaat	acaattttta	atgatgggat	gggtttgtact	540
ggtaaaaatta	caatgaaagc	aattttgtct	ggttttaaag	atgcttggaa	ctctattgaa	600
tcatgtggtg	atgttgggtg	tggtacaggt	caaactattg	ctgaaattgt	taaagtttac	660
ccacatatta	gaggtattaa	ttttgattg	ccacatgttg	ttgctgcagc	accaaaatat	720
gatggtgttt	gtcatgttgg	tggtgacatg	tttgattcta	ttccaactgc	tcaagcagtt	780
tttatgaaat	ggattatgca	tgattgggtg	gacgaagatt	gtgttaaaat	tttgaatttt	840
tgtagaaaag	ctattccaga	aaagacaggt	aaaattttta	ttgttgatgt	tggttttaaaa	900
ccagatggtg	acggtttgtg	tgattcaatt	gggtatgatt	ttgatttgg	aatgattgct	960
cattcatctg	gtggtaaaga	aagaaactga	ccagaatgga	aaaagttatt	ggaaaaaggt	1020
ggttttccaa	gatacaatat	tactgaaatt	ccagcttggt	tttctattat	tgaagcatat	1080
ccagttttaa						1089

SEQ ID NO: 44 moltype = DNA length = 840
 FEATURE Location/Qualifiers
 source 1..840
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 44

atgtcatcca	gatttttgg	taatgggtg	ttctttatc	aacaagatca	atgttacgct	60
ttgacaactg	caggtagatt	gttgttaaaa	gatgaacata	ttaatgttac	tggtttgggt	120
gctttatcta	ctgatccaat	tatgatgatg	ccgtggattt	tcttgactga	atgggttaaa	180
aatgatgatc	catctccatt	tgatactaca	tacggtaaat	ctttttggga	ttacactgct	240
caagaatcta	gatttggtaa	agtttttgat	gaagcaatgg	ctaattgcac	taatttgcca	300
attcaagttc	caattactaa	atgtagattt	gtttttgaat	cattgacttc	tttggctaatt	360
gttgggtggt	gtactggtaa	agttgcacaa	gctattgcta	aaaattttcc	aaatttgaaa	420
tggtttgttt	gtgatttgcc	aagagttggt	gctaatacaag	aaggtactaa	aaatttggat	480
tttgttgcat	gtgacatgtt	ggaaaaagtt	ccaccagcaa	atgctgcatt	gttaaaatgg	540
gttttgcatt	attggtctga	tgaacatggt	ttgaaaattt	tgaaaaattg	taaaatgtct	600
attccaacta	gagaaaaagg	tggtaaagtt	attattattg	atattattat	tgaatctcat	660
aaagaagata	atgaattggt	tgaaactcaa	attggtatgg	atatgcaaat	gttgggtttta	720
tacgttgcaa	aagaaagaac	tgaagaagaa	tgggctactt	tggttttggg	tgctgggttt	780
tcttcataca	aagtttttcc	agttttgggt	gtttcttgg	tgattgaagt	ttatccataa	840

SEQ ID NO: 45 moltype = DNA length = 1053
 FEATURE Location/Qualifiers
 source 1..1053
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 45

atgtcatccg	ctacttttga	aaattctgct	gaattgttaa	gagcacaagc	tcatgtttgg	60
tcacaaactt	ttaattttta	aaattcagca	tctttgaaat	gtgctgttca	attgggtatt	120
gctgatgtta	ttcaaaaaa	tggttaacca	attactttgt	ctaaattggc	agctgcatta	180
ccaattaatc	catctaaagc	taattacatt	tatagattga	tgagaatttt	agttaatgct	240
ggtttctttg	ctcaagaaaa	agaaggttat	tcattgacat	ctgcaggtag	attgttattg	300
tctgatgaac	cagttaaatg	aagagctttt	attttgggtg	ttttagatcc	agatactatt	360
aaaccgtgga	atttcttctg	tgattggttt	caaaatgatg	atccatctcc	atttgatact	420

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gctcatggta aaaatttttg ggattttact gctgaagaac caaaattttc taaattgttt 480
aatgaaggta tggtttcaga ttctcatttg attactatgg ttttggttac agaattgtaaa 540
tacgtttttg aagggttgac ttcatatgtt gatgttggtg gtggtactgg tacagttgct 600
agatcaattg ctgaagcatt tccaaatttg aaatgtacag tttttgattt gccacatgtt 660
gttgcaatc aagaaggtac tttaaatttg gattttgttg ctggtaatat gttggaaaaa 720
gttcaccacg caaatgctat tttattgaaa tggattttgc atgattggct tgaatgaagca 780
tgtgttaaaa tttgaaaaa ttgtaaaaag gctattccag gtagagatat tgggtgtaaa 840
gttattatta ttgatattgt tatggaatct caattggaag atgatgaatc tactgaagct 900
caaatttgtt tggatatgca aatgttggtt ttatatagag ctaagaaaag atctgaaaaa 960
gaatgggcta aattgttttc agatgctggt tttccatctt ataaagtttt tccagctttg 1020
ggtagtagag caattattga agtttaccct taa 1053

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SEQ ID NO: 46      moltype = DNA length = 1101
FEATURE            Location/Qualifiers
source             1..1101
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 46
atgtcatcgg atttggctag aaatgggtgac agagctgggt aattgttaca agctcaagca 60
catatttggg atcatatttt taattatatt aattcaatgt ctttgaaatg tgctgttcaa 120
ttgggtattc cagatattat tgaaaaacat ggtcaaccaa tgactttggc tcaattgtgt 180
catgctttgt ctatttcaaa tactaaagca tcatttttgt acagattaat gagagttttg 240
attcattctg gtttctttat taaagctaaa attccaggtg atgaattgga agaaggttat 300
gaattgactt ctgcatcaaa attgttattg aaatctaact cattttcagt tacaccattt 360
ttgttagcaa tgttggtccc aatttcaaca gatttgggtg atcatttttc taaatgggtt 420
caatctgatg atgaaactcc atttcataca tgctatggta cttctggttg ggaaatggct 480
tctcaagaac cgtggattaa tcaatctttt aatgaagcaa tggcttcaga tgctagattg 540
ttatcatcta ttttgattga acattgtggt catgttttct ctggtttaa tttttgtgtt 600
gatgttggtg gtgtacagg tactttggct aaagcaattg ctgatgcatt tccagatttg 660
aaatgactct ttttgattt gccatattgt gttgatgggt tggatgggtg tagaagaagt 720
ttggctttta ttggtgggtg catgtttgaa gctgttcac cagctgatgc agttttgtta 780
aaatggacat tacatgattg gtcagatgaa gattgtgtta gaattttgga aaaatgtaaa 840
caagctattc catctaaga aaatgtgtgt aaagtattta ttattgatat ggttattaat 900
gatcaacaaa aaggtgtgtg tgatgggtgaa cataaatcta tgaaactca attgttttct 960
gatattgtga tgatgggtga tacaaaaggt agagaaagaa atgaaaaaga atgggtgtga 1020
attttctcag ctgcaggttt taaagattac aaaattactc atgcattggg ttttaagtct 1080
ttgattgaaa tttattatta a 1101

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SEQ ID NO: 47      moltype = DNA length = 1350
FEATURE            Location/Qualifiers
source             1..1350
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 47
atgtcatcgg gaagcgcaga ggaagattat ccttctcgca agaggcccaa ccagcatacc 60
cagacaaaagc acaagttcaa gatccttctg gtcgttatac taaccaactt attaacatt 120
tatatattcg caggctccctc actgcccata acaagaaatt taagcctacc gacatgggat 180
tacactacaa aactgttaca cgaactgaac gtcacaaaaa gagaattagc gaatagtaga 240
gcacagattt cagacttgca gcaacacttg aaggcgtcaa atatattgac taaatcacta 300
cttactcagc ttactcgttt gaatggagcc atagatcagc acagcactat cgaaagttagc 360
cttttcaact acgtgtattt gcttacgat ttaagttctg aagcaaaact gggcattggg 420
ccacataaac tgcctttagg atactccctc agatctggtt ccgaggaaat ttacagtagt 480
gtaggtggcg gctgtctgag gtacaaaagag gaattagccc aatacatgac ttacaaaatc 540
gggtggagaat gtccggtcga tgatgtcttt gtgcagcgct ttatgctgaa gggctgcgag 600
ccgcttccac gtagaagatg tcatccgaaa agccctgcag gttacaaaga accaacccca 660
cttccgaaat ctctttggac cactcctccg gacacatcta tcatctggga cccatatacg 720
tgcaagaact actcctgtct agtagagcgt aagaaattgc ccgggttcta tgattgtaag 780
gactgttttg acctacaagg aagggaaatt agccgttggc tgtttgacaa tggaggctta 840
gacttcggta ttgatcaagt cctagggagc aaaccacacg gcacaatcag gatcggattg 900
gacatcggag gagggacggg cactttcgct gcccgatatg aagaaaggaa tgttaccatt 960
gtgactacct ccattgaact ggatggacct tttaactctt tcatagcgtc cagaggattg 1020
atctcaatgc acgtttcagt aagccagcgt ctacctttct ttgaaaacac tcttgacatt 1080
gttcacagta tgcacattct tagcaactgg attccggaca ccatgcttga gttcacactg 1140
tatgacatat accgtgttct aagacctgga gggatcttct ggttggtatc tttctctgct 1200
ctaggctcac agcttaacgc aacatacgtc ccatggttag accgtgttgg ttttaagaaa 1260
cttcgttggg atgctggaat gaagctagat agaggcattg ataaaaacga gtggtatttt 1320
agttctttac tggagaagcc tacgtcatct 1350

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SEQ ID NO: 48      moltype = DNA length = 1095
FEATURE            Location/Qualifiers
source             1..1095
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 48
atgtcatcgg ggagtgcggc accggaacaa tcaaatagta tggctgggtg tgaggaagcc 60
tgcctatttg ctatgcaatt ggccagtgct tccgtactgc ctatggtatt gaaaagtgtc 120

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atcgagcttg atctactaga actgatcaag aaagcgggtc aagacgctta cgtttctcct 180
agtgaattgg cagcgaggtt acctactgaa aaccgggatg caccgctcat gctggacaga 240
atcttaagac tactggccag ttactcagtt cttaactgta ctttacgtaa tctgcccgcac 300
ggtggggtag aaaggttata cgccctggcc cctgtttgca agttcctaac taagaatgaa 360
gacgggggtt ctatgggtgc gttgttgctt atgaaccaag acaaaattct tatggagtcc 420
tggtatcacc taaaggacgc agtcctagag ggaggaatcc ctttcaataa ggccacggc 480
atgaccgcct ttgagtacca cggcaccgac ccgaggttca acaaagtgtt taaccaaggt 540
atgtccaacc attctacgat tattatgaag aaaattcttg aaatctacaa tggcttcgaa 600
gggcttaaga cgggttgtga tgttggtgga ggtactggag cgactttgag catgatcggt 660
tccaagtatc cggcaatcaa gggaatcaac ttgacctgc cacatgtgac gggcgatgcg 720
cctacttacc ctggcattga gcacgttggt ggcgatatgt tcttctcagt cccgaaggga 780
gatgctatct tcatgaagtg gatttgccat gattggtcag atgagcattg cctgaaattt 840
ttgaagaatt gttaccaagc gctaccggag aatggtaaa tgatactagc tgaatgtttg 900
ttgccgaag cctctgctgc gacctgtgc acccaaaaca cagttcatgt ggacgtaatc 960
atgcttgccc ataacctggt tggaaaagaa agaacagaga aagaatttga ggctcttgca 1020
aaggaagcag gttttaaagg attccgtaaa atatgttgcg cttataattc ttggataatg 1080
gaactattga agtga                                     1095

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SEQ ID NO: 49      moltype = DNA length = 1098
FEATURE           Location/Qualifiers
source            1..1098
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 49
atgtcatcca actctccagt ccaagaccag atctcaata caccagcgga agaagaagat 60
ttctttaagg tgatgcaatt atctagcggc ttaattcttc ctatgatctt aaaagctgcc 120
attgatctcg aattgtttga gcttatagca aaagctggtc ctggagctaa actttctgcg 180
gtcgaaattg cctcacaaat ctcgaccaa aatccaaatg cagctgtcac gcttgatcgg 240
atgatgaagt ttcttgccag ttctctttt cttaattgca gtttgggtac agatgcagaa 300
ggacatccta ggagattata cagttttgca ccgacatgca aaaactttct tccggatgaa 360
gacggagttt cttttgctcc cgtgttgatg ttgctgtgtg acaaagtaca cttcgagtct 420
tggtatcact taaaagatgc agttcttgag ggaggtattc cgttcacaa ggctaaccgg 480
gacgatgcac ttgcacatcc agttaaagac accagattct atgaggtctt caacaaggca 540
ttgcacaatc atcgatgcac agtcacgaaa aggtctctta aagctacagc gggattcgag 600
ggagtgaaac agctgggtgga tgttggtggc ggacatggtg gcacactagg ctgcacgtt 660
tccaagtatc ctcatattca tgcatacaac tttgatgttc ctccggtaat cgaaggtgcc 720
ccaccacgcc caggtgtgga gttcggttgg ggagacatgt ttgaaagagt tcccaaggga 780
gaggttcttc tcatgaagtg ggtacttcac gattggagtg atgaacattg cctgaagata 840
ctgaagaact gttatgagc tttaccggaa tccggcaaac taatagtgat cgaattgggt 900
ctgccggaaa ctcaccaaac gataccgttg tcgaagattg ggtatcaatt tgacatgaat 960
atgatgagcg taaaccttgg aggaaaagag agaacagaga aagaatttga agatcttgca 1020
aaacaatctg gattcgcttc tattaacctc atctgtcgcg caggttgatg ttgggttatt 1080
gagttctata aggattaa                                     1098

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SEQ ID NO: 50      moltype = DNA length = 1845
FEATURE           Location/Qualifiers
source            1..1845
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 50
atgtcatccg cgactaaggg aaatccgggt gataaacagat ccagatctag catgcccatt 60
ttcattgcgc ctggcctttg ttgcgtattt tatttgcctg gtgcttggca gcgttctgga 120
tctggttaagg gagataacat agccctacgag ataacaaaga aagcagaaga ctgttcaata 180
ctagccaatc tagagtttga aactcatcac ggagataggg cagcgaacga tagtaagctg 240
gaagtataag agtacgagcc ttgtgacgac aggtacatag actacacacc ctgtcatgat 300
caagcaagag ccatgacatt ccctagggag aacatgaact atagagagag gcactgccct 360
ccggagcaag agaaattcca ctgcctaata cctgctccca agggatacgt cacacctttc 420
agatggccga agagttagaga ctatgtacca ttgcctaacc ctccacacaa gtcattaacc 480
gtagagaagg cgggtgcaaaa ctgggtgcaa tatgaaggtg atgtattccg ttcccagggt 540
ggaggcactc aattcccgcg cggagccgac gcttatattg agcagttggc ttctgtgatc 600
cccattaatg accgcacggt gcgtacagct ctggatacac gttgcggtgt ggcatcttgg 660
gggtgcttatt tattcaaaaa gaattgtata acgatgagct ttgcccgcag ggacagccac 720
gaggcgcagg tccagttcgc cttggaacgt ggagtaccag cggtaatagg cgtgctaggt 780
acgatcaagc tgccttatcc ttctagggcg ttcgatatgg ctcatgttcc acgttgtttg 840
atcccttggg gtgcctaatga cggtaagtat atgatggaag tagatagagt tctgagaccc 900
ggaggctact ggatttttaag cggacctcct attaaactgga gaaattctta tgaagcgtgg 960
cagagaccaa aagagggaat ggaagaagag cagaggaaga tcgaagaaat tgctgaactg 1020
ttatgctggg aaaagaaaca tgagaagggt gaaacggcta tctggagaaa aaggatcaac 1080
aatgagaact gcaatgacgc tgactctagg gtgacgatgt gtaagttggc ggacgcggat 1140
gatgtatggt ataagaagat ggaggtttgt gtaaccctct accccgaaac caactctcag 1200
gaagaagtgc ccggtggaga attgaaaccc ttcccgcaga ggcttaacgc gatccctcct 1260
aggggtggcg gagcgtctgt cttaggcgct accatagact cattccagga agataaacgg 1320
ctgtggaaga agcacgtgag ccgctaccag aggatcaaca agttgctaga ttccgggaga 1380
taccgtaacg tgatggacat gaacgcaggt ctaggtgggt ttgccgctgc aatcgaaagc 1440
tctaatttat ggttgatgaa cgttattccc accatcgagc agaaagacac cagtgaatc 1500
atctttgaac gtggactgat cggtatattac catgattggt gtgaagcctt cagcacttat 1560

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ccaagaacat acgacctgat tcatgtctgag ggcgtatctt cactgtacaa ggacaaatgc 1620
aacgcccagg atatattatt agaaatggac agaatccttc gtccctgagg tgccgttata 1680
ttcagggatc atgcagacgt gttgtctgaa gtaaagagga ttgtggcagg tatgaagtgg 1740
aatacgaaga tgtacgacca cgaggatggt cccttagtcc cggaaaaaat acttgcctgc 1800
gtaaaacaaat attgggtcgc aggagataca aacagcactt ttagc 1845

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SEQ ID NO: 51      moltype = DNA length = 1140
FEATURE           Location/Qualifiers
source            1..1140
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 51
atgtcatccg caaccagcga gaatactcag cttaagtcgg tctctccttc aagtatggaa 60
aatggaaaag cgggtgagga acaggatcac tttagtgcag tcatgcaggt agtcacotct 120
gcagccctgc caatgggtctt ttacactgtt gtgaagtgtga acctatttga aataattgct 180
aaggcaggac ctggcgctaa gttatctcct tctgagattg caagcagact accagcaacg 240
aagaatcctg atgctgcctc catgcttgac aggatgttga ggctattgtc ctcacactca 300
ttactaacct gcgaaatcgg cgagggtggc ggtggagggt aaacagaagt gggttacgag 360
cgtgtgtacg gtctgtctcc caccggtgag tacttaatac cagacgaaga gggcaacagt 420
gttgccctc tagttgaact tctgttggat aaagtcttaa cgcacacctg gtacgaactg 480
ggtaatgcgg ttttagaagg tggatcccg ttcaataggg ttcacgggac ccacgcgttc 540
gactaccctt ccaggaaacc tgagtacaac gaattgttta acaaggggat ggttggtccg 600
acagccatta caatgaagaa gctactgcag caatacaagg gctttgaaca cttacagacc 660
ctagttgatg tgggaggagg cttaggaatc actctacaca aaattgttag caaatacccc 720
aacataagtg ggatcaactt cgacctacct catgtaatag aagatgcacc atcataccct 780
ggtgtcgcag atataggcgg tgatatgttt gaaagtgtcc caggtgggtga tgccatcttc 840
atgaaaatgg ttcttcatga ctggagcgac gaccactgtt taaaattact aaagaattgt 900
tttaaagcct tgcgggatca tggaaaagt atcgtagcag acctaatct accgatgaaa 960
cccaacacct ccgcatctgt aaaaggcacc ttccaggctg atgtaataat gatgactcaa 1020
aatcctggag gcaaggaaag atcagagaca gacgtacgtg cacttgcaat cggtgctggg 1080
ttcaaagtgg taaagtggga gtgccaagct ggcaaccttg gaatcttaga gctgtacaag 1140

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SEQ ID NO: 52      moltype = DNA length = 1140
FEATURE           Location/Qualifiers
source            1..1140
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 52
atgtcatccg aaacgtccga aaatgctcaa ctgaagtcgt cctctccaag ttcaattgag 60
aatggtaagg agggagaaaa gcaggaccac tttagccatg taatgcagct agtaacgagc 120
gccagtttac caatgggtgtt gtacaccgct gttaaaactga acctatttga gataatcgcc 180
aaagctggac cagaagctaa gctatctcca agtgaattg catatagact tcttacgacc 240
aaaaatccag acgtcgcgtc tatgttggat agaatgctta gattactgtc ctcacacagc 300
ctgctaacct gtgacatcgt ggaagtcgct ggtggtgccc aaatagatat aggatatgaa 360
agagtcctat gcttgagtcg cgttggtgaa tacctaatcc cggacgagga gggtaacagc 420
gttgctccat tggtggaact attactggat aaagtctctg tagacacctg gtatgaaact 480
ggaaacgcag ttttagaagg tggcataccg tttaaccgtg tccatggaac acatgccttc 540
gattatccaa gtcgtaaccc cagatataat gagcttttca acaatggaat ggttggtccg 600
acggctatca cgatgaagga actattacaa caatacaaaag gctttgaact tcttcaaacc 660
ctagtggacg taggaggtgg actgggtatc acattacaca agatcattgc aaaaataccca 720
tcaattagag gcatacaact tgacctacca cacgtcatcg aagatgctcc tccataccct 780
ggaatggaa atattggtgg tgatatgttt gagtcogtgc caggtggaga cgctatcttt 840
atgaagatga tcttgcatga ttgggtccgc gaccattgtc taaagctgct aaagaactgc 900
taccgtgctt taccagatca tgggaaagt attgtcgttg atttaattat gcctgttaag 960
ccagacacga acattttcgt gaaaggtatt ttccaagcag atgtaattat gatgacccag 1020
aacctcgttg gcaaggagag atctgaagca gacgtgaggg ctctggcgat tgggtgctggg 1080
tttaagggca tcaagtggga gtgctgcgct gggaacctgg gtattctgga gttatataag 1140

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SEQ ID NO: 53      moltype = AA length = 366
FEATURE           Location/Qualifiers
source            1..366
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 53
MAAGKSPEEE HPVKAYGWAV KDRRTGILSP FKFSRRATGD NDIRIKILYC GICHTDLTSV 60
KNEYEFLLSY LVPQMEIVGI ATEVGSKVTK IKVGEKVAVA AYLGTGCKCY NCVNDLENYC 120
PEVIIGYGTP YHGTITINYGG LSNETVVNER FVLRPFPEKLS PAGGAPLLSA GITAYSAMRN 180
HGLDKPGQL GVVGLGGLGH LAVKFAKAFG VRVTVISTTP SKKDEAINNL GADAFLLSRD 240
DKQMRRAIGT FDAIDTLAV VHPPIAPLLDL LRSHGKLVLV GAPSKPLELP TIPLLSGGKS 300
LIGSAAGNVK QTQEMLDFAA EHDITANIEV IPIDYINTAM ERLDKGDIRF RFVVDIENTL 360
TPPPEP

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SEQ ID NO: 54      moltype = AA length = 354
FEATURE           Location/Qualifiers
source            1..354
                  mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 54
MAAGKSQEDG QTVKALGWAA REVSGAISPF DFSRRAPGER DVQVKILYCG ICSFDTEMIN 60
NKFGFTRYPF VLGHIEIVGV SEVGRKVQKF KIGDKVGVGT MIGSCRTCYS CTHNLENYCP 120
KVTLTTEATSG GCSNLVIAD EDFVFWHPVNL PLDLGAPLLC AGITVYSPLK NFELDKPGLR 180
IGVVLGLGIG HIAVKFAKAF GAKVTVISSS ESKKVEAIEK YGADSLVSS DPGQMLAAAG 240
TLDGVIDTVP APHSILPFLD LLLPRGKLI L GAPMEPFVL PIYPLLQGG VVAGSATGGL 300
KQIQEMLHFA AEHNIVADGE VIPIDDINTA IKRIEKGDKV YRFVVDIGNT LKSA 354

SEQ ID NO: 55                moltype = AA length = 354
FEATURE                      Location/Qualifiers
source                       1..354
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 55
MAAGKCAQEE HTVKAFGWAA REASGALSPY GFSRRATGER DVRVKILYCG ICRTDAEMIS 60
DKFCFTKYPH VPGHEIVGV SEVGNKVQKF KVGAKVGVGT IIGCCRTCYS CTNGLESYCP 120
NVALTEAGEG GCSNFIVLDE DFVFRWPEKL PLDLGAPLLC AGAASYSPLK NFGLDKPGLH 180
IGIAGLGGMG HVAVKFAKAF GAKVTVISTS DNKKEEAIEK YGADAFLNSS NPEQMRAAAG 240
TLAAIVDTIP SPHSLVPLLD LLLPHGKVIV LGAPSEPFVL PVMPLQGG VVAGSSGASL 300
KQIQEMLDFA AEHNIVADAE VIPIDYINTA IKRIEKGDIK YRFVVDIGNT LKSA 354

SEQ ID NO: 56                moltype = AA length = 489
FEATURE                      Location/Qualifiers
source                       1..489
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 56
MAYTPENSSY SSILNTYIQN SRFISPEAPK PKVILTPVHE SQVQTAIYCS KRHGMQIRIR 60
SGGHDPEGSS YMSNVPLVF DMFNFRSISV DLRSKTAWVG AGATLGETYY NIYQKNSTLT 120
YSAGYWPVTG IGGHVSGGGC GPLVREYGLA ADNVLDARLI DVKGRILDRK SMGEDLFWAI 180
RGGMGANFGV ILAYKIKLVE VPEKVVAFKV RRTLQVQATK LVHKKQFVAP KLPVRLIISL 240
QIASINSDQT GGRTIQASFA SVFRGEVDEL LQIMEEHFPE LGLKKEDIKE MSWIQYFSFY 300
GDEPIDDDVAH FLTNRVSTTK PYFKAQSDVF QKPIEAGIE KIWKKLFEKD PLAAQLQWTP 360
FGGKMDQISD SETPPPHRAG NSFIMFEVIF WNGTDKVLQA KHINWIRELY DVIGQYVDNN 420
PRGAYADYRD LDLGVNNIDE TSAEQARVWG KSYFKDNFDR LVNVKTRVDP NNFQKNEQSI 480
PPRLAHSSS 489

SEQ ID NO: 57                moltype = AA length = 537
FEATURE                      Location/Qualifiers
source                       1..537
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 57
MAAMLRTTML QFAFVLLFSF CGVSSSSQPE TPFLQCLFHH SDDSTIGYII YTPENSSYSS 60
ILNSCIQNSR FIVPETPKPK VILTPVSESQ VQAAIYCSKK HGMQIRIRSG GHDFEGSSYT 120
SNVPFLVLDL FNFRSISVDL QSKTAWVGAG ATLGETYYHI YQKNSTLTYS AGYWPTVGIG 180
GHVSGGGCGP LVREYGLAAD NVLDARLVDV KGRILDRKSM GDDLFWAIRG GMGASFGVIL 240
AYKIKLVEVP EKVVAFKVRR TLEQNATKLV HKWQFVAPKL PVRLIISLQI ASINSDQTGG 300
RTIQASFASV FRGVDDELQ IMEEHFPELG LKKEDVKEMS WIQYFSFYAD EPIDVAHFL 360
TNRVSTTKPY FKAQSDVFQK PIPEAGIEKI WKKLFEKDPL AAQLQWTPFG GKMDQISESE 420
TPFPHRAGNS FIMFEVIFWN GTDKVLAQKH INWIRKLYGV IGQYVDNNPR GAYADYRDLD 480
LGVNNIDETS VEQARVWGKS YFKNNFDRLV NVKTRVDPDN PFKNEQSIPS RLAHSSS 537

SEQ ID NO: 58                moltype = AA length = 537
FEATURE                      Location/Qualifiers
source                       1..537
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 58
MATMLRTRML QFAFVLLISF CCVSSSSQPE IPFLQCLFHH SDDSTIGNVM YTPENSSYSS 60
ILNTYIQNSR FISPEAPKPK VILTPVRESQ VQTAIYCSKK HGMQIRIRSG GHDFEGSSYM 120
SNVPFLVLDL FNFRSISVDL RSKTAWVGAG ATLGETYYNI YQKNSTLTYS AGYWPTVGIG 180
GHVSGGGCGP LVREYGLAAD NVLDARLIDV KGKILDRKSM GEDLFWAIRG GMGANFGVIL 240
AYKIKLVEVP EKVVAFKVRR TLVQNATKLV HKWQFVAPKL PVRLIISLQI ASINSDQTGG 300
RTIQASFASV FRGEVDELLQ IMEEHFPELG LKKEDIKEMS WIQYFSFYGD EPIDVAHFL 360
TNRVSTTKPY FKAQSDVFQK PIPEAGIEKI WKKLFEKDPL AAQLQWTPFG GKMDQISDSE 420
TPFPHRAGNS FIMFEVIFWN GTDKVLAQKH INWIRELYDV IGQYVDNNPR GAYADYRDLD 480
LGVNNIDETS AEQARVWGKS YFKNNFDRLV NVKTRVDPNN PFKNEQSIPP RLAHSSS 537

SEQ ID NO: 59                moltype = AA length = 353
FEATURE                      Location/Qualifiers
source                       1..353
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 59

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MAAGKCPQEE	QTVKAFGWAA	REVSGALSPY	GFSRRATGER	DVKINVLYSG	MCNFDTEMIN	60
NKFGFTKYPY	VLGHETVGVV	SEVGSKVRKF	KVGDKVGVGG	LVGSCRKCYC	CTNDLENYCT	120
KAELTEAGAG	GCSNFVVTDE	DFVFHWPEKL	PLDLGAPLLC	AGVTTFSPLK	THGLDKPGMH	180
VGVAGLGGGL	HVAVKFLKAF	GAKVTVISTS	DNKKREAIEK	YGADAFVSS	NPEQMOTAAG	240
TLTGIIDTIP	SAHSLVPLLN	LLMPQGKLIV	LGVPTEPPVL	PVVPMIQGGK	VVAGSSFGLL	300
KLIQEMLDFA	AEHNIVADAE	VIPVDYINTA	IKRIETGDIK	YRFVIDVGNT	LKS	353

SEQ ID NO: 60 moltype = AA length = 360
 FEATURE Location/Qualifiers
 source 1..360
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 60

MAAEKSPREE	HPVKAFGLAA	KDSSGILSPF	NFSRRATGDH	DVQLRVLYCG	LCYYDTKMIK	60
NKRGVTRYPF	VFGHEIVGEV	TEIGREVQKF	KVGDKVGVGC	MVASCRSCES	CANNCENYCP	120
NVSVTDGAFF	FKTGEVLYGG	CSDIMVADEN	FVIRWPENFP	LDAGAPLLCA	GITYSPLRN	180
FGLDKPGIHV	GIYGLGGLGH	VAQVFAKAFG	AKVTVISSSD	RKRMEAIEKL	GADSFVNNSN	240
LEEMQAAMGT	MHGIIDTVPA	NHSLVPLLDL	LKPQGKLIVV	GGPEKPFELP	VFPLQGGRL	300
VAGSATGGIK	QTQEMIDFAA	EHNILPHVEV	VSDVYVNTAI	ERTERGDKVY	RFVIDIGNTL	360

SEQ ID NO: 61 moltype = AA length = 317
 FEATURE Location/Qualifiers
 source 1..317
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 61

MAAEKSPREE	QTVKAIGWAV	LEAAGVLSPP	EFSRRATGER	DVQVKVLYCG	ICNFDTEMIN	60
NKFGFTRYPF	VLGCTNNLEN	YCPKAVLIEA	NNGGCSNLVV	ADEDFVFHWP	EKLPLDLGAP	120
LLCAGITTYC	PLKNFGLDKP	GLHVGAVAGL	GLGHVAIKFA	KAFGAKVTVI	SSSDKKKKEA	180
IEKYGADEFL	VSSNPEQMQA	TAGTLNGIID	TVSPHSLVP	LLNLLLPGQG	LIVLGAPIEP	240
FVLPIYPLLQ	GGRVVAGSAT	GGLKQIQEML	HFAAEHSIAA	DAEVIPIDYI	NTAVERIEKG	300
DIKYRFVVDI	GNTLKSA					317

SEQ ID NO: 62 moltype = AA length = 360
 FEATURE Location/Qualifiers
 source 1..360
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 62

MAAEQSPQDQ	HQVKVLGWAA	RDSSGILSPF	KFSRRATGEH	DVQLKVIYCG	ICTYDVEMIK	60
NKWMGTNPYP	VFGHEAVGVV	TEIGSKVQKF	KVGDTVGVGT	QVSGCGECDC	CTNDLNQYCP	120
EMIAIDGFGS	NDVSCGACCD	IMVVDENFVI	VWPQNLPMDA	GAPLLCAGIT	TYSPLRNFGI	180
DKPRIHVGIV	GLGGLGHVAV	RFAKAFGAKV	TVISTSMNKK	QEAHLKYGAD	SFVVSSEQEQ	240
MQAAVSTMNG	IIDTSPVPHA	LVPLKLLKLP	QGKLILIGAP	DKPYELPVFP	LIEGGRIVAG	300
SATGSFRQTQ	EMIDFAAKHN	IVADVEVIPT	NYLNTAVERI	EKGDKYRFV	IDLGNGLKLA	360

SEQ ID NO: 63 moltype = AA length = 354
 FEATURE Location/Qualifiers
 source 1..354
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 63

MAAGKCAQEE	HTVKAFGWAA	REASGALSPY	GFSRRATGER	DVRVKILYCG	ICRTDAEMIN	60
NKFGFTTYPF	VLGHEIVGVV	SEVGSKVKKF	QFGDKVGVGT	MIGACRTCYS	CTNNLENYCP	120
KAVLIEANNG	GCSNLVVADE	DFVFRWPEKL	PLDLGAPLLC	AGITYSPLK	NFGLDKPGLH	180
VGVAGLGGGL	HIAVKFLKAF	GAKVTVISSS	DKKKKEAIEK	YSADEFLVSS	NPEQLQAAAG	240
TLNGIIDTVP	SPHSLVPFLN	LLLPQGKLIV	LGAPIEPFVL	PIFPLQGGGR	VVAGSATGGL	300
KQIQEMLLFA	AEHNIAADAE	VIPIDYINTA	VERIEKGDIK	YRFVVDIRNT	LKSA	354

SEQ ID NO: 64 moltype = AA length = 362
 FEATURE Location/Qualifiers
 source 1..362
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 64

MAAGKTPETV	HPVKAFGWAA	RDTSGELSPF	KFSRRATGEH	DVQFKVLYCG	ICHSDLHMIK	60
NEWGMTVYPI	IPGHEIVGVV	TEVGNKVEKV	KVGDKVGVGC	LVGSCGCNCM	CSQDLENYCP	120
KQILTYSAID	TDGSITYGGY	SDVMVANEHF	IIRWPENLPL	DAGAPLLCAG	ITTYSPRYF	180
GLDKPGMHVG	IVGLGGLGHV	AVKFAKAFGS	KVTVISTSEN	KKKEAIEKLG	ADAFNLSRDP	240
QQMQAAVGTG	DGILDTVSAY	HPIVPLINLV	KSQGVIMVVG	APEKPLELAT	FPLLVGKRTL	300
SGSAIGGLKE	TQEMIDFAAK	HNILADVEVI	PMNYVNTAME	RLKADVKYR	FVIDIGNSLK	360
SA						362

SEQ ID NO: 65 moltype = AA length = 362
 FEATURE Location/Qualifiers
 source 1..362

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mol_type = protein
organism = synthetic construct

SEQUENCE: 65
MAAGKCAQEE HPVKSFGWAA RDTSGILSPF KFSRRATDEH DVRLKVLYCG ICHSDLHTIK 60
DEWGITEYPV VPGHEIAGVV TEVGSKVQKF KVGDKVGVGC FVGSCRNCNM CDKNLENYCP 120
AQIFTYGSPPY TDGTPTYGGY SDMMVVDEHF VIRWPENFPL DAGAPLLCAG ITTYSPLRYF 180
ELDKPGIHHG IVGLGGLGHV AVKFAKALGA KVTVISTSAS KRQEAPEKLG ADSFLLSSDP 240
EQIQAAVGTM DGIIDTVSAV HPVPLIDL L APHGKLIVVG SPAKPLELPI YNLSIGRKIV 300
GASSVGLLKE TQEMIDFAAK HNILPDVEII PIDYVNTAME RLARGDVKYR FVIDVGNTLK 360
ST 362

SEQ ID NO: 66      moltype = AA length = 362
FEATURE           Location/Qualifiers
source            1..362
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 66
MSSAEKSPPEE EHPVKAFLA AKDSSGILSP FNSRRATGD HDVQLRVLYC GLCYDTKMI 60
KNKRGVTRYYP FVFGHEIVGE VTEIGREVQK FKVGDKVGVGC CMVASCRSCE SCANNENYCP 120
PNVSVTDGAF FFKTGEVLYG GCSDIMVADE NFVIRWPENF PLDAGAPLLC AGITTYSPLR 180
NFGLDKPGIHH VGIYGLGGLG HVAVQFAKAF GAKVTVISSS DRKRMEAIK LGADSFVNS 240
NLEEMQAAMG TMHGIIIDTV ANHSLVPLLD LLKPQGLIV VGGPEKPFEL PVFPLQGG 300
LVAGSATGGI KQTQEMIDFA AEHNILPHVE VVSDYVNTA IERTERGDVK YRFVIDIGNT 360
LY 362

SEQ ID NO: 67      moltype = AA length = 362
FEATURE           Location/Qualifiers
source            1..362
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 67
MSSAKTPPEMV HPVRAFGWAA RDTSGELSPF KFSRRATGEH DVQFKVLYCG ICHSDLHMIK 60
NEWGMTVYPI VPGHEIVGVV TEVGSKVEKV KVGDKVGVGC LVGSCGNCNM CSQDLENYCP 120
KQILTYSAID EDGSITYGGY SDVMVANEHF IIRWPENLPL DAGAPLLCAG ITTYSPLRYF 180
GLDKPGMHVG IVGLGGLGHV AVKFAKAFGA KVTVISTSEN KKKEAIEKLG ADAFLNSRDP 240
QQMQAAVGTG DGIIDTVSAV HPVPLINLV KSQGVIMVG APEKPLELAT FPLLVGRKTL 300
AGSAIGGLKE TQEMIDFAAK HNILAEVEVI PMNYVNTAME RLLKADV KYR FVIDIGNSLK 360
SA 362

SEQ ID NO: 68      moltype = AA length = 362
FEATURE           Location/Qualifiers
source            1..362
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 68
MSSAKTPGTE HSPVKAFLGAA RDTSGELSPF KFSRRATGKH DVQFKVLYCG ICHSDLHMK 60
NEWDMTRYPI VPGHEIVGVV TEVGSKVEKV KVGDKVGVGC LVGSCRNCNM CSQDLENYCA 120
GQILTYSAIY TDGITYGGY SDLMVADEHF VIRWPENFPL DAGAPLLCAG ITTYSPLRYF 180
GLDKPGTHVG VVGLGGLGHV AVKFAKAFGA KVTVISTSEN KKKEAIEKLG ADAFLVSRDP 240
EQMQAAIGTM DGIIDTVSAV HPVPLINLV KSQGVIMVG APEKPLELAT FPLLVGRKTL 300
SGSAIGGLKE TQEMIDFAAK HNILADVEVI PIDYVNTAMD RLLKADV KYR FVIDIGNSLK 360
SA 362

SEQ ID NO: 69      moltype = AA length = 363
FEATURE           Location/Qualifiers
source            1..363
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 69
MSSAKSPFEE HPVKAFLGAA RDSSGILSPF KFSRRATGKK DIRFKVLYTG ICHSDLHHLK 60
NDWGS SKYPL VPGHEIAGMV TEVGSEVKSF KVGDKIGVGC LIGSCGTCE CADGLENYCP 120
KKILTYDGAK YYDGTMTFGG YSDHMIVDEH FAIHWPENMP LDGGAPLLCA GITTYSMPKY 180
FGLDKPGLHI GVVGLGGLGH VAVKFAKAFG AKVTVISSTP GKKEAIAKNF KADSFVSRD 240
TAAMQAAAGT LHGILDVSA VHPILPLLSL LKNH SKYIML GAPEKPLELP VFPMLMGRKI 300
VAGSNIGGVK ETEEMLNFAA EHGITADVEV IPIDYVNTAV ERLGKNDVRY RFVIDVGNTL 360
KAT 363

SEQ ID NO: 70      moltype = AA length = 355
FEATURE           Location/Qualifiers
source            1..355
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 70
MSSAGKCPQE EQTVKAFLGAA AREVSGALSP YGFSRRATGE RDVKINVLYS GMCNFDTEMI 60
NNKFGFTKYP YVLGHETVGV VSEVSGKVRK FKVGDKVGVGC GLVSGCRKCY SCTNDLENYC 120
TKAELTEAGA GGCSNFVVT DEDVFWHPEK LPLDLGAPLL CAGVTTFSPL KTHGLDKPGM 180

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SEQ ID NO: 71      moltype = AA  length = 355
FEATURE            Location/Qualifiers
source             1..355
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 72      moltype = AA  length = 352
FEATURE           Location/Qualifiers
source            1..352
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 73      moltype = AA length = 362
FEATURE            Location/Qualifiers
source             1..362
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 74      moltype = AA length = 363
FEATURE           Location/Qualifiers
source            1..363
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 75      moltype = AA length = 355
FEATURE            Location/Qualifiers
source             1..355
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 76          moltype = AA  length = 355
FEATURE                Location/Qualifiers
source                 1..355
                        mol_type = protein
                        organism = synthetic construct
```

-continued

SEQUENCE: 76
MSSAGKSPEK EQTVKATGWA ASEVSGILSP IEFSSRRATGD SDVQIKVLYC GICSFDEMI 60
NNKFGFTTRY YVLGHETVGV VSEVSGSKVRK FKVGDKVGVG GLVSGCGTCY GCTNGLENYC 120
TKEQITEAGA GGCSNFMVMG EDFVFHWPEK LPLDLGAPLL CAGVTYSPL KTHGLDKPGL 180
HVGAVAGLGG L GHVAVKFLKA FGAKVTVISS SDNKKKEAIE KYGANAPLVS SNPEQMQAAA 240
GTLSGI IDTI PPSHSLPLLL DLLLPOGKLI VLGAPIEPFV LPIYPLLQGG RVVAGSATGG 300
LKQIQEMLDF AAEHNVAADG EVIPIDHVNT AVKRIETGNI RHRFVIDIGN TLKSA 355

SEQ ID NO: 77 moltype = AA length = 355
FEATURE Location/Qualifiers
source 1..355
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 77
MSSAGKSPEE EQTVKAIGWA ASEVSGVLSP FEVSRRATGE RDVQVKVLYC GICSFDEMI 60
NNKFGFTKYP FVLGHEIVGV ASEVSGSKVQK FQVGDKVGVG ALIGSCRTCY SCTNDLENYC 120
PKAVLTEANS GGCSNFVVD EDFVFRWPEK LPLDLGAPLL CAGITTYSPL KNFGLDKPGL 180
HVGAVAGLGG L GHVAVKFLKA FGAKVTVISS SDNKKKEAIE KYGANAPLVS SNPEQMQAAA 240
GTLSGI IDTI PPSHSLPLLL DLLLPOGKLI VLGAPIEPFV LPIYPLLQGG RVVAGSATGG 300
LKQIQEMLDF AAEHNVAADG EVIPIDHVNT AVKRIETGNI RHRFVIDIGN TLKSA 355

SEQ ID NO: 78 moltype = AA length = 361
FEATURE Location/Qualifiers
source 1..361
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 78
MSSAEQSPPE QHPVKVLGWA ARDSSGILSP FKFSRRATGE HDVQIKVIYC GICTYDVEMI 60
KNKWGMTNYP VVFGHEAVGM VTEIGSKVQK FKVGDTVGVG TQVGSCGKCE CCTNDLDQYC 120
PEMIATDGF S NDVSYGACC DIMVVDENFV IWPQNLPMD AGAPLLCAGI TTYSPLRNFG 180
LDRPGIHVGV VGLGGLGHVA VRFKAFGAK VTVISTSMNK KQEAHVHYGA DSFLVSTDQE 240
QMQVNTAWVA AIVDTIPSPH SLVPLDLLL PHGKIVLGA PSEPFVLPVM PLLQGGRVVA 300
GSSGASLKQI QEMLDFAAEH NIVADAEVIP IDYINTAIKR IEKGDIKYRF VVDIGNTLKS 360
A 361

SEQ ID NO: 79 moltype = AA length = 344
FEATURE Location/Qualifiers
source 1..344
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 79
MSSSERVEDLT ELLAAQHVG NQMLNFRKSA SLKCAIELGI PDAINQHGKP ITLCELVSAL 60
PINPSKANHI YRLMRFLSNA GFFILQDQGY ALTAAGRLLL KEEPFNLRAF IFYMSDPAVV 120
KPWNSLTWEF S NDPPFPFDT AHGKNFWAYA AEEPNFANLF NEAMANDSTL IVQVMMTQCK 180
FVFDGLTSLA DVGGGTGAVA RAIAQNFPNL KCVVCDLPHV IADQEGTENL DFVAGDMLEK 240
VPAADAILLK WILHDWDED CVKILKNCKE AIPGRDKGGK VIIIDMIMES QMKDDESSET 300
QVGVNMQMLM GYGAKERSEK EWAKLFQDAG FSDYKVLPLL CVSH 344

SEQ ID NO: 80 moltype = AA length = 359
FEATURE Location/Qualifiers
source 1..359
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 80
MSSDAMKSAE LFKAQAHIFK QVFCFTNGAS LKCAVQLGIP DAIDNHGKAM TLSELTDALP 60
INPSKAPHIH RLMRLIVTAG FFVEERLNG KEEKANGYAL TPSSRLLLN KPLSLRASAL 120
TMLDPVTYKT WNALSEWFQ EDQTAFETAH GKNMWDFFAE DPGLSKKFNE SMASDSQLVT 180
EVLVTCKEFV FEGLTSMVDV GGGTGTVAGA IAKTFPSLRC TVFDLPHVVA NLEPTENLDF 240
VAGDMFGKIP PANAIPLKWV LHDWNEDCV KILKNCKRAI PGKEKGGKVI IVDIIMETEK 300
HDIDEFDYAK MCDMEMLV L CNSKERTKE LAMLVSEAGF SGYKIFPVLG IRSLEIVYP 359

SEQ ID NO: 81 moltype = AA length = 360
FEATURE Location/Qualifiers
source 1..360
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 81
MSSGSAEEEA SLFAMQLVTS ASLPMVLYTA VKLNLFEIIA KAGPGAKLSP SEIACRLPTT 60
KNPDAASMLD RMLRLSSHS LLTYDVVEVA GGAETDIGYE RVYGLSPVGE YLIPDEEGNS 120
VAPLVELLLD KVLINTWYEL GNAVLEGGIP FNRVHGTHAF DYPSPRNPRYN ELFNNGMVG 180
TAITMKELLQ QYKGFELLQT LVDVGGGLGI TLHKIIAKYP SIRGINFDLP HVIDEAPPYP 240
GMEHIGGDMF ESVPGGDAIF MKMILHDWSD DHCLKLLKNC YRALPDHGKV IVDLILPVK 300
PDTNIFVKGI FQADVIMMTQ NPGGKERSEA DVRLAIGAG FKGIKLECCA GNVGILELYK 360

SEQ ID NO: 82 moltype = AA length = 348
FEATURE Location/Qualifiers

-continued

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source                1..348
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 82
MSSDVQSAEL LRAQAHVWSQ TPNFKNSASL KCAVQLGIAD AIQKHGKPIT LSELAAALPI 60
NPSKANYIYR LMRILVNAGF FAQEKEGYSL TAAGRLLLSL EPVNARAFIL VVLDPDTMKP 120
WNFLSDWFQN DDPSPFDTAH GKNFWDFTAE EPKFSKLFNE GMVSDSHLIT MVLVTECKYV 180
FEGLTSLVDV GGGTGTVAR S IAEAFPNLKC TVFDLPHVVA NQEGTLNLDF VAGNMLEKVP 240
PANAILLKWI LHDWSDEDCV KILKNCKKAI PGRDIGGKVI IIDMVMSQL EGDESTAQI 300
CLDMQMLVLY RAKERSEKEW AKLFSDAGFP SYKVFPALGT RAIIEVYP 348

SEQ ID NO: 83         moltype = AA length = 348
FEATURE              Location/Qualifiers
source               1..348
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 83
MSSDVQSAEL LRAQAHVWSQ TPNFKNSASL KCAVQLGIAD VIQKHGKPIT LSKLAAALPI 60
NPSKANYIYR LMRILVNAGF FAQEKEGYSL TSAGRLLLSL EPVNARAFIL VVLDPDTIKP 120
WNFLSDWFQN DDPSPFDTAH GKNFWDFTAE EPKFSKLFNE GMVSDSHLIT MVLVTECKYV 180
FEGLTSLVDV GGGTGTVAR S IAEAFPNLKC TVFDLPHVVA NQEGTLNLDF VAGNMLEKVP 240
PANAILLKWI LHDWSDEACV KILKNCKKAI PGRDIGGKVI IIDMVMSQL EDESTAQI 300
CLDMQMLVLY RAKERSEKEW AKLFSDAGFP SYKVFPALGT RAIIEVYP 348

SEQ ID NO: 84         moltype = AA length = 358
FEATURE              Location/Qualifiers
source               1..358
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 84
MSSDVQAGEL LQAQAHWNH IFNFINSMSL KCAVQLDIPD IIHKHGQPM T LAQLIHALPI 60
SSAKAPFVHR LMRILIHSGF FVKAKIPGNE LEEGYELTS A SKLLKSNPL SVTPFL LAML 120
DPILTDWVHH FSQWFQCDDE TFPDTCGKA FWKLAGEQPR LNQFFNEGMA SDARLISSLV 180
IKHKHVFEG LNSLVDVGGG TGT LAMAIA D AFPDLKCTVL DLPHVVAGLD GSSKNLAYVG 240
GDMFEAIPPT DAVLLKWILH DWSDEECVQI LKRCKEAIPS KGKGGKVIII DILVKDQHRG 300
DDNDEAIE TQ LFFDMLMMVL VKGRERNQKD WAKLFFEAGF NDYKITPVLG LRSLIEVY 358

SEQ ID NO: 85         moltype = AA length = 357
FEATURE              Location/Qualifiers
source               1..357
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 85
MSSDVHAGEL LQAQAHWNH IFNFINSMSL KCAIQLGIPD IIQKHGQPM T LAQLIHALPI 60
SNAKAPFYR LMRILIHSGF FIKAKIPGNE LEEGYELTPA SKLLKSNPY SVTPFL LAML 120
DPILTDWVHH FSQWFQCDDE TFPYTCHGRE LWELAGEKPR LNQFFNEGMA SDARLISSLV 180
IKHKHVFEG LNSLVDIGGG TGT LANAIA D AFPDLKCTVL DLPHVVAGLD GSSKNLAYVG 240
GDMFEAIPPT DAVLLKWILH DWSDEECVQI LKKCKEAIPS KEKGGKVIII DMLMKNQQKG 300
DDEDKAIE TQ LFFDMLMMVL VKGKERNQKD WVKLFFEAGF NDYKITPVLG LRSLIEI 357

SEQ ID NO: 86         moltype = AA length = 360
FEATURE              Location/Qualifiers
source               1..360
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 86
MSSDVQADKL LEAQAHWNH TPNFINSMCL KCAIQLGIPD IIKKHGRPM T LEELTDALPI 60
NNAKSPFIYR LMRILIHSGF FLKAKIPGNE LEEGYELTSA SELLLANDPF SMKTFL LAML 120
DPILTAPWQH FSQWFQSNDE NPFYTCHGRA FWELAGQEPR LNQFFNEAMA NDARLVSSIV 180
IKHKHVFEG LNSLVDVGGG TGTFGKAI AAFPDLKCTVL DLPHVV DGLD VGGKNLAYIG 240
GDMFEDIPPA DAVLLKWILH DWTDEECVRI LEKCKQAIPS KENGKGVIII DMVRNDQQKE 300
ADDDARKSIE TQLFFDILMM VDTKGRERNE KEWAKLFYEA GFKDYKITQV VGLRSLIEVY 360

SEQ ID NO: 87         moltype = AA length = 358
FEATURE              Location/Qualifiers
source               1..358
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 87
MSSDVQAGEL LQAQAHWNH IFNFINSMSL KCAVQLDIPD IIHKHGQPM T LAQLIHALPI 60
SSAKAPFYR LMRILIHSGF FLKAKIPGNE LEEGYELTSA SKLLKSNPF SVTPFL LAML 120
DPILTDWVHH FSQWFQCDDE TFPYTCHQKA FWKLAGEHPR VNQFFNEGMA SDARLISSLV 180
MKHKHVFEG LNSLVDVGGG TGT LAMTIA D AFPDLKCTVL DLPHVVAGLD GSSKNLAYVG 240
GDMFEAIPPT DAVLLKWILH DWSDEECVQI LKKCKEAIPS KGKGGKVIII DMLVKDQQRG 300
DDKDEAIE TQ LFFDMLMMVL VKGRERNQKD WAKLFFEAGF NDYKITPVLG LRSLIEVY 358

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SEQ ID NO: 88 moltype = AA length = 360
 FEATURE Location/Qualifiers
 source 1..360
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 88
 MSSDVQADKL LEAQAHWNH TFFNFINSML KCAIQLGIPD IIKKHGRPMT LVELTDALPI 60
 NSAKSPFIYR LMRILIHSGF FIKVKVTDTD VEKGYMLTSA SELLLANDPF SMKTFLLAML 120
 DPILTDPWQH FSQWFQSNDE NPFYTCHGRA IWELAGQEPR LNQFFNEAMA NDARLVSSIV 180
 IKHCKHVFSG LNSLLDVGGG TGTFAEAIAA AFPDLKCTVL DLPHVVDGLD VGSKNLAYIG 240
 GDMFEDIPPA DAVLLKWILH DWSDEECVRI LEKCKQAIPS KENGGKVIII DMVRNDQQKE 300
 ADDDAHKSIE TQLFFDILMM VDTKGRERNE KEWAKLFYEA GFKDYKITQV VGLRSLIEVY 360

SEQ ID NO: 89 moltype = AA length = 359
 FEATURE Location/Qualifiers
 source 1..359
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 89
 MSSDVQADKL LEAQAHWNH IFNFINSMSL KCAIQMGIPD IIKNHGRPMT LKELTDALPI 60
 NNAKSPFVYR LMRILIHSGF FAKAKVTDND VEEGYVLTSA SELLLADDPF SLTPLLLAML 120
 DPIMTDPWQH FSQWFQSNDE TPFYTCHGRT FWELASQEPR LNQFFNEAMA RDAWLVGSVV 180
 INHCKHVFSG LNSLVDVGGG TGTAKAIAD APPDLKCTAL DLPYVVDGLD GSSKNLVYIG 240
 GDMFEAIPPA DAVLLKWILH DWTDEECVRI LVKCKQAIPS KENGGKVIII DMVIIIDMVK 300
 GANDHRSIET QLFFDILMLV DVGKGERNEK ERAKLFYEAG FNDYKITQVG GLRTLIEVY 359

SEQ ID NO: 90 moltype = AA length = 267
 FEATURE Location/Qualifiers
 source 1..267
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 90
 MSSAELYNRQ AKQYSDSRPT YPEELFQFIA SKTPCHDLAW DVGTSQGQAA RSLAGIYKNV 60
 VATDISPKQL EFAVKLPNIR YHCTPPKMPI AEVEQNIAAE SSVDLVTIAT AIHWFDLPTF 120
 YQQVKWVLKK PHGVIAAWSY INVEINTCSV NTLLOKLYLD CAPYCEPQAK SVADDKYRTI 180
 DFPFQPVVGL DNTGPFEFKM EKLMDLDGFF TFIKSWSLYQ TAKDKGVELL NDDVVQDFTF 240
 AWNEDGNPQK IATFPPIYLR I KGVGNWY 267

SEQ ID NO: 91 moltype = AA length = 101
 FEATURE Location/Qualifiers
 source 1..101
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 91
 MSSAMNQWIL HDWSDEKCVQ VLKKCKEAIK SKEKGGKVII IDMLVKVQQK GDDNDEPIET 60
 QLFFDMLMMI LYKGRERNQK DWEKFLQAG FNYKIIIPVL G 101

SEQ ID NO: 92 moltype = AA length = 354
 FEATURE Location/Qualifiers
 source 1..354
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 92
 MSSDFARNGD HTGELFQAQA HIWNHLENFI NSMSLKCAIQ LGIPDIIHKH GQPMALAQLI 60
 DALPINNAKA HFVYRLMRIL IHSGFPIKAK MPDNEGHEGY ALTSASKLLL ANDPFSVTPF 120
 LLAMLDPILT DPWHHFSQWF QDNEETPFHS CHGTSMWELA GRQPRLNQFF NEGMASDARL 180
 VCTMVIKNCK DVFMGLNSLI DVGGGTGTVA KAIADAPPHL KCSVLDLPHV VDGSESSKNL 240
 AYVGGDMFEA IPPSDAAMNQ WILHDWSDEK CVQVLKKCKE AIPSKKGGK VIIIDMLVKV 300
 QQKGDDNDPE IETQLFFDML MMILYKGRER NQKDWEKLFQ QAGFNYYKII PVLG 354

SEQ ID NO: 93 moltype = AA length = 365
 FEATURE Location/Qualifiers
 source 1..365
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 93
 MSSDLARNGD QAGELLQAQA QIWNHIFNFI NSMSLKCAVQ LDIPDIIHKH GQPMTLAQLI 60
 HALPISSAKA PFVYRLMRIL IHSGFPIKAK IPGNELEEGY ELTSASKLLL KSNPFSVTPF 120
 LLSMLDPILT DPWHHFSQWF QCDDETPFYT CHQKAFWKLA GEHPRVNQFF NEGMASDARL 180
 ISSVMKHCK HVFEGNLNSL DVGGGTGTVA MTIADAPFDL KCTVLDLPHV VAGLDGSSKN 240
 LAYVGGDMFE AIPPTDAVLL KWILHDWSDE ECVQILKKCK EAIPSKKGGK KVIIDMLVK 300
 DQQRGDDKDE AIETQLFFDM LMMVLVKGRE RNQKDWAKLF FEAGFN DYKI TPVLGLRSLI 360
 EYYYY 365

SEQ ID NO: 94 moltype = AA length = 172
 FEATURE Location/Qualifiers

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source 1..172
mol_type = protein
organism = synthetic construct

SEQUENCE: 94
MSSEGAGGTL QTCWSPGPR AVSKCNGHEC FPRWFAAVLA KYPVWVMNVV PAHSDPDTLS 60
IIYERGFIGT YHDWCEAFST YPRTYDLIHA DGVFSIYQDR CDITYILLEM DRILRPEGTV 120
IFRDVVEVLV KIKAITDGMR WESRILDHES GPFNPQKILV AFKSYWTAKA ND 172

SEQ ID NO: 95 moltype = AA length = 362
FEATURE Location/Qualifiers
source 1..362
mol_type = protein
organism = synthetic construct

SEQUENCE: 95
MSSDTNVERS KLLHGQAEIW KHTFRFADSM ALKCARELGI ADIIHSHGHP LTLAEIATNI 60
TNSSSPHIPY LSRIMRFLVR NKIFTSTDIP SDHGGGSPST TLYGLTAASW WLLNDNDGLS 120
LAPFLLMENH PVLVSPWHQL SSCVREGGIA FQKAHGKEIW ELASENPEFN TIFNDGMGCT 180
GKITMKAILS GFKDAWNSIE SLVDVGGGTG QTIAEIVKVY PHIRGINFDL PHVVAAPKY 240
DGVCHVGGDM FDSIPTAQAV FMKWIMHDWG DEDCVKILKI CRKAIPEKTG KIRIVDVVLK 300
PDGDGLFDSI GMIFDLLMIA HSSGGKERTE PEWKKLEKKG GFPRYNITEI PACFSIIEAY 360
PV 362

SEQ ID NO: 96 moltype = AA length = 279
FEATURE Location/Qualifiers
source 1..279
mol_type = protein
organism = synthetic construct

SEQUENCE: 96
MSSRFLVNGG FFIQQDQCYA LTTAGRLLLK DEHINVTGLV ALSTDPIMMM PWIFLTEWFK 60
NDDPSPPFTT YGKSFWDYTA QESRFGKVPD EAMANASNLP IQVPITKCRF VFESLTSANL 120
VGGGTGKVAK AIAKNFNNLK CVVCDLPRVV ANQEGTKNLD FVAGDMLEKV PPANAALLKW 180
VLHDWSDEHG LKILKNCKMS IPTREKGGKV IIDIIEESH KEDNELVETQ IGMDMQMLVL 240
YVAKERTEEE WATLFLDAGF SSKVFPVVLG VSCLIEVYP 279

SEQ ID NO: 97 moltype = AA length = 350
FEATURE Location/Qualifiers
source 1..350
mol_type = protein
organism = synthetic construct

SEQUENCE: 97
MSSATLENSA ELLRAQAHVW SQTFFNFKNSA SLKCAVQLGI ADVIQKHGKP ITLSKLAAL 60
PINPSKANYI YRLMRILVNA GFFAQEKEGY SLTSAGRLLL SDEPVNARAF ILVLDPDTI 120
KFWNFLSDWF QNDPPSPFDT AHGKNFWDFE AEEPKEFKLF NEGMVSDSHL ITMVLVTECK 180
YVFEGLTSLV DVGGGTGTVA RSIAEAFPNL KCTVFDLPHV VANQEGTLNL DFGVAGNMLEK 240
VPPANAILLK WILHDWDEA CVKILKNCKK AIPGRDIGGK VIIIDMVMS QLEDDESTE 300
QICLDMQMLV LYRAKERSEK EWAKLFSDAG FPSYKVPFAL GTRAIIEVYP 350

SEQ ID NO: 98 moltype = AA length = 366
FEATURE Location/Qualifiers
source 1..366
mol_type = protein
organism = synthetic construct

SEQUENCE: 98
MSSDLARNGD RAGELLQAQA HIWNHIFNYI NSMSLKCAVQ LGIPDIEKH GQPMTLAQLC 60
HALSISNTKA SFLYRLMRVL IHSGFFIKAK IPGNELEEGY ELTSASKLLL KSNPFVTPF 120
LLAMLAPIST DLWHHFSKWF QSDDETPFHT CHGTSGWEMA SQEPWINQSF NEAMASDARL 180
LSSILIEHCG HVFSGLNSLV DVGGGTGTLA KAIAADFPDL KCTVLDLPYV VDGLDGGKRN 240
LAFIGGDMFE AVPPADAVLL KWTLHDWDE DCVRILEKCK QAIPSKENGK KVIIIDMVIN 300
DQQKGADDGE HKSJETQLFF DMLMMVDTKG RERNEKEWAE IFSAAGFKDY KITHALGLRS 360
LIEIYY 366

SEQ ID NO: 99 moltype = AA length = 450
FEATURE Location/Qualifiers
source 1..450
mol_type = protein
organism = synthetic construct

SEQUENCE: 99
MSSGSAEEDY PSCKRPNQHT QTKHKFKILL VVILTNNLLTI YIFAGPSLPI TRNLSLPTWD 60
YTTKLLHELN VTKRELANSR AQISDLQQHL KASNILTKSL LTQLTRLNGA IDQHSTIESS 120
LFNYADLLTD LSSEAKLGIG PHKLPLGYSP RSGSEEIYSS VGGGCLRYKE ELAQYMTYKI 180
GGCEPVDVDF VQRLMLKGCE PLPRRRCHPK SPAGYKEPTP LPKSLWTTTP DTSIIWDPYT 240
CKNYSCLVER KKLPGFYDCK DCFDLQGREI SRWLPDNGGL DFGIDQVLGT KPHGTIRIGL 300
DIGGTGTFA ARMKERNVTI VTTSMNLDGP FNSFIASRGL ISMHVSVSQR LPFFENTLDI 360
VHSMHILSNW IPDTMLEFTL YDIYRVLRPG GIFWLDHFFC LGSQNLNATYV PMLDRVGFKK 420
LRWNAGMKLD RGIDKNEWYF SSLLEKPTSF 450

-continued

SEQ ID NO: 100 moltype = AA length = 364
 FEATURE Location/Qualifiers
 source 1..364
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 100

MSSGSAAPEQ	SNSMAGDEEA	CLFAMQLASA	SVLPMVLKSA	IELDLLELIK	KAGQDAYVSP	60
SELAQALPTE	NPDAVVMLDR	ILRLLASYSV	LNCTLRNLDP	GGVERLYGLA	PVCKFLTKE	120
DGVSMALLL	MNQDKILMES	WYHLKDAVLE	GGIPFNKAYG	MTAFEYHGTD	PRFNKVFNQG	180
MSNHSTIIMK	KILEIYNGFE	GLKTVVDVGG	GTGATLSMIV	SKYPAIKGIN	FDLPHVTGDA	240
PTYPGIEHVG	GDMFFSVPKG	DAIFMKWICH	DWSDEHCLKF	LKNCYQALPE	NGKVILAEC	300
LPEAPDATLS	TQNTVHVDVI	MLAHNPGGKE	RTEKEFEALA	KEAGFKGFRK	ICCAYN swim	360
ELLK						364

SEQ ID NO: 101 moltype = AA length = 365
 FEATURE Location/Qualifiers
 source 1..365
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 101

MSSNSPVQDQ	ISNTPSEED	FFKVMQLSSG	LILPMILKAA	IDLELFELIA	KAGPGAKLSA	60
VEIASQLSTK	NPNAAVTLDR	MMKFLASFSP	LNC SLVTD AE	GHPRLLYSFA	PTCKNFLPDE	120
DGVSFAPVLM	LPVDKVFHES	WYHLKDAVLE	GGIPFIKANG	DDAFAPHPVD	TRFYEVFNKA	180
LHNHACIVTK	RLLKVYRGFE	GVKQLVDVGG	GHGGTLGCIV	SKYPHIHAIN	FDVPPVIEGA	240
PRPGVEFVVG	GDMFERVPKG	EVLMLKWLH	DWSDEHCLKI	LKNCYEALPE	SGKLIVIELV	300
LPETPNSDPL	SKIGYQFDMN	MMSVNLGGKE	RTEKEFEDLA	KQSGFASIKL	ICRAGCDWVI	360
EFYKD						365

SEQ ID NO: 102 moltype = AA length = 615
 FEATURE Location/Qualifiers
 source 1..615
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 102

MSSATKGNPG	DNRSRSSMPM	FIAAGLCCVF	YLLGAWQSRG	SGKGDNIAYE	ITKKAEDCSI	60
LANLEFETHH	GDRAANDSKL	EVKEYEPCDD	RYIDYTPCHD	QARAMTFPRE	NMNYRERHCP	120
PEQEKFHCLI	PAPKGYVTPF	RWPKSRDYVP	FANAPHKSLT	VEKAVQNWVQ	YEGNVFRFPG	180
GGTQPPHGAD	AYIEQLASVI	PINDGTVRTA	LDTGCGVASW	GAYLFKKNVI	TMSFAPRDSH	240
EAQVQFALER	GVPAVIGVLG	TIKLPYPSRA	FDMAHCSRCL	IPWGANDGKY	MMEVDRVLRP	300
GGYWLISGPP	INWRNSYEAW	QRPKEELEEE	QRKIEEIAEL	LCWEKKHEKG	ETAIWRKRIN	360
NENCNDRDSR	VTMCKLADAD	DVWYKKMEVC	VTPYPETNSQ	EEVAGGELKP	FPERLNAIPP	420
RVAGASVLGV	TIDSFQEDNR	LWKKHVSAYQ	RINKLLDSGR	YRNVMMDNAG	LGGFAAAIES	480
SNLWWMNVIP	TIAEKDTSGI	IFERGLIGIY	HDWCEAFSTY	PRTYDLIHAE	GVFSLYKDKC	540
NAEDILLEMD	RILRPEGAVI	FRDHADVLSE	VKRIVAGMKW	NTKMYDHEDG	PLVPEKILAA	600
VKQYVWAGDT	NSTFS					615

SEQ ID NO: 103 moltype = AA length = 380
 FEATURE Location/Qualifiers
 source 1..380
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 103

MSSATSENTQ	LKSVSPSSME	NGKAGEEQDH	FSDVMQLVTS	AALPMVLYTV	VKLNLFELIA	60
KAGPGAKLSP	SEIASRLPAT	KNPDAASMLD	RMLRLLSSH	LLTCEIGEVA	GGGETEVGYE	120
RVYGLSPTGE	YLIPDEEGNS	VAPLVELLLD	KVLTDTWYEL	GNVAVLEGGIP	FNRVHGTHAF	180
DYPSRNPEYN	ELFNKGMVGP	TAITMKKLLQ	QYKGFELHQT	LVDVGGGLGI	TLHKIVSKYP	240
NISGINFDLP	HVIEDAPSYP	GVEHIGGDMF	ESVPGGDAIF	MKMVLHDWSD	DHCLKLLKNC	300
FKALPDHGKV	IVADLILPMK	PNTSAFVKGI	FQADVIMMTQ	NPGGKERSET	DVRALAIGAG	360
FKVVKLECA	GNLGILELYK					380

SEQ ID NO: 104 moltype = AA length = 380
 FEATURE Location/Qualifiers
 source 1..380
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 104

MSSETSENAQ	LKASPSSEIE	NGKEGEKQDH	FSHVMQLVTS	ASLPMVLYTA	VKLNLFELIA	60
KAGPEAKLSP	SEIAYRLT	KNPDAASMLD	RMLRLLSSH	LLTCDIVEVA	GGAEIDIGYE	120
RVYGLSPVGE	YLIPDEEGNS	VAPLVELLLD	KVLIDTWYEL	GNVAVLEGGIP	FNRVHGTHAF	180
DYPSRNPRYN	ELFNNGMVGP	TAITMKELLQ	QYKGFELHQT	LVDVGGGLGI	TLHKIIAKYP	240
SIRGINFDLP	HVIEDAPPYP	GMEHIGGDMF	ESVPGGDAIF	MKMILHDWSD	DHCLKLLKNC	300
YRALPDHGKV	IVVDLILPVK	PDTNIFVKGI	FQADVIMMTQ	NPGGKERSEA	DVRALAIGAG	360
FKGIKLECCA	GNVGILELYK					380

SEQ ID NO: 105 moltype = AA length = 380
 FEATURE Location/Qualifiers

-continued

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source                1..380
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 105
MSSEISENAQ LKSASPSSIE NGQEGEKQDH FSHVMQLVTS ASLPMVLYTA VKLNLFEIIA 60
KAGPGAKLSP SEIACRLPTT KNPDAASMLD RMLRLLSSHS LLTYDVVEVA GGAETDIGYE 120
RVYGLSPVGE YLIPDEEGNS VAPLVELLLD KVLINTWYEL GNAVLEGGIP FNRVHGTHAF 180
DYPSPRNPRYN ELFNNGMVG P TAITMKELLQ QYKGFELLQT LVDVGGGLGI TLHKIIAKYP 240
SIRGINFDLP HVIDEAPPYP GMEHIGGDMF ESVPGGDAIF MKMILHDWSD DHCLKLLKNC 300
YRALPDHGKV IVVDLILPVK PDTNIFVKGI FQADVIMMTQ NPGGKERSEA DVRALAIGAG 360
FKGIKLECCA GNVGILELYK                                     380

SEQ ID NO: 106        moltype = AA length = 371
FEATURE              Location/Qualifiers
source                1..371
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 106
MSSATSSENTQ LKSVSPSSME NGKVEEEEQL FSYAMQLVTS VSLPRVLLAA FRLRVFEIIE 60
EAGPGAQLSA SEIAAQVSGK NKDAAAMLDL MLRLLASHSV LTCGARSQGR VYGLAPVAKY 120
FVQSKADGGG TLGSLALLIQ DKVFIDSWFQ LEDAVREGGV PFDRVHGTHA FEYPGRDPRF 180
NEVFNKAMLD HTTIAINRIL ERYKGFENLK TLVDVGGGLG VTLNVITTKY PNLKGINFDL 240
PHVIQRAPGY PGVEHIGGDM FESVPEGEAI FMKWILHDWD DGHCLKLLKN CYKALSDNGK 300
VIVVDALVPV IPDTSPCVKT TCQSDLIMMT QNPGGKERSE PEFLALATAA GFKGIRLECY 360
VCNLWVMEFY K                                             371

SEQ ID NO: 107        moltype = AA length = 371
FEATURE              Location/Qualifiers
source                1..371
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 107
MSSATSSENTQ LKSVSPSSME NGKVEEEEQL FSYAMQLVTS VSLPRVLLAA FRLRVFEIIE 60
EAGPGAQLSA SEIAAQVSGK NKDAAAMLDL MLRLLASHSV LSCGARSQGR VYGLAPVAKY 120
FVQSKADGGG TLGSLALLIQ DKVFIDSWYQ LEDAVREGGV PFDRVYGTHA FEYPGRDPRF 180
NEVFNKAMLD HTTIAINRIL ERYKGFENLK TLVDVGGGLG VTLNVITTKY PNLKGINFDL 240
PHVIQRAPGY PGVEHIGGDM FESVPEGEAI FMKWILHDWD DGHCLKLLKN CYKALSDNGK 300
VIVVDALVPV IPDTSPCVKT TCQSDLIMMT QNPGGKERSE PEFLALATAA GFKGIRLECY 360
VCNLWVMEFY K                                             371

SEQ ID NO: 108        moltype = AA length = 377
FEATURE              Location/Qualifiers
source                1..377
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 108
MSSATSSENTQ LKSVSPSSME NGKAGEEEEH CSYALQLATS AALPVVLNAI IKLNVLEIIA 60
KAGPGAQLSP SEIAAHLPT E NSGAAVMLDR MLRLLASYSV LTCSVVGGGD KDNQHQRAYG 120
LAPVAKYFLR DEDGGSLSL LDDLQDKVLV DIWYELEPAV LKGGTAFNRT HGTHIFQYTG 180
SDQKFNETFN KAMIDHTFIV GRKVLQFYKG FEHLKTLVDV GGGLGVNLSM ITSKYPSIKG 240
INFDLPHVVQ NAPAYPGEVH VGGDMFESIP KGDAIFMKWI LHDWDDHCL TLLKNKYKAL 300
PDNGKVIADV AILPMVVDNS ASCKSNQVD LFTLALYFPG GKERTQEFLA ALAKEAGFKG 360
IRLECFACNM WVIEFYK                                       377

SEQ ID NO: 109        moltype = AA length = 374
FEATURE              Location/Qualifiers
source                1..374
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 109
MSSATSSENTQ LKSVSPSSME NGKGSQAQEEA SLFAMQLASA SVLPMVLKSA IELDLELIA 60
KAGPGAYVSP SKLAAQLPTH NPQAPVMLDR ILRLLATYSV LNCKLRDLPD GGVERLYGLA 120
PVCKFLTKNA DGVSMAPLLL MNQDKVLMS WYHLKDAVID GGIPFNKAYG MTAFEYHGTD 180
PRFNKVFNQG MSNHSTITMK KILEVYKGE GVKTVDVVG GTGATLNMII DKYPTIKGIN 240
FELPHVVEDA PSYPGVEHVG GDMFVSVPEG DAIFMKWICH DWSDDHCRKL LKNKYHALPE 300
GGKVILAECI LPDAPDTSLA TQNVVHIDVI MLAHNPGGKE RTEKEFEALA KGAGFKEFRK 360
ICSAVNTWIL ELCK                                           374

SEQ ID NO: 110        moltype = AA length = 377
FEATURE              Location/Qualifiers
source                1..377
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 110
MSSATSSENTQ LKSVSPSSME NGKAGEEEEH CSYALQLATS AALPVVLNSI IKLNVLDIIA 60
KAGPGAQLSP SEIAAHLSP E NSGAAVMLDR MLRLLASYSV LTCSVVGGGD KDNQHQRAYG 120

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LAPVAKYFLR DEDGSSLGSL LDLLQDKVLV DIWYELEPAV LKGGTAFNRT HGTHIFQYTG 180
SDQKFNETFN KAMIDHTFIV VRKVLQFYKG FGHKLTLVDV GGGLGVNLGL ITSKYPSIKG 240
INFDLPHVVQ NAPAYPGVEH VGGDMFESIP KGDAIFMKWI LHDWDDHCL KLLKNCYKAL 300
PDNGKVIADV AILPMVPDNS ASCKSNQVD LFTLALYFPF GKERTQEFL ALAKEAGFKG 360
IRLECFACNM WVIEFYK 377

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SEQ ID NO: 111      moltype = AA length = 374
FEATURE            Location/Qualifiers
source             1..374
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 111
MSSATSENTQ LKSVSPSSME NGKGSAREEA SLFAMQLASA SVLPMVLKSA IELDLELIA 60
KAGPGAYVSP SKLAAQLPTH NPQAPVMDLR ILRLATYSV LNCKLKNLPD GGVERLYGLA 120
PVCKFLTKNA DGVSMAPLLL MNQDKVLMES WYHLKDAVID GGIPFNKAYG MTAFEHYHGT 180
PRFNKVFNQ MSNHSITTMK KILEVYKGF GVKTVVDVGG GTGATLNMII DKYPTIKGIN 240
FELPHVVEDA PFYPGVEHVG GDMFVSVPEG DAIFMKWICH DWSDDHCRKL LKNKYHALPE 300
GGKVILAECI LPDAPDTSLS TQNVVHIDVI MLAHNPGGKE RTEKEFEALA KGAGFKEFRK 360
ICSAVNTWIM ELCK 374

```

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SEQ ID NO: 112      moltype = AA length = 375
FEATURE            Location/Qualifiers
source             1..375
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 112
MSSATSENTQ LKSVSPSSME NGKAGERQEE LQKLLQLTGT LTVHVALTAA IELNLFLEIMA 60
KSGPNAQLSS AQIVSLRLTK NPDAPAILDR ILQLLGSYSI LKCTIVSNDQ NGHTTKLFGL 120
TPMAEYLVDP SNGISFAPAI LMAREKAPAK CGYFLKDAIL EGGTPPHKVN GMHLFEYIGS 180
SSRLNEVFN R IMACHTSVFM GAVLETYKGF EDLKLVDVVG GGHGTTLMKI LSKYPRIRGI 240
NFDLFPFVKN APACPGVEHV EGDMPQNVPS VEAILLKNVL HDWNDDRCLK LLKNCWDALP 300
ESGKVIAIDA IPELPSTDS ASQFAYISDV EILIAAPGGR ERTFKEFNAL ATAAGFTACK 360
LVARTSQYGV LEFYK 375

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SEQ ID NO: 113      moltype = AA length = 337
FEATURE            Location/Qualifiers
source             1..337
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 113
MSSATSENTQ LKSVSPSSME NGKAGEEGEQ CSYAMQLATS AALPNVLNAI IRLNVLEIIA 60
RAGPGAQLST SEIAHLHTQ NSNAAVMLDR MLRLLASYSV LTCVSVGGGG GGGDDKDDQH 120
QRVYGLAPVA KYFLQNEDEG SLRPLDLFQ GKVLTDIWC LEPAVLEGGT AFNRKHGIHI 180
FEYTGDRDQKF NESFNKAMID HTFIVVKNVL QFYRGFEHLK TLVDIGGGLG VTLGMITSKY 240
PNIKGINFDL PHVQNAPTY PDNGKVIADV AILPTIPDNS ASSKSNIQMD LFTLASVFPF 300
GKERTQEFL ALANEAGFKG IRLECLVCNM WMEFYK 337

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SEQ ID NO: 114      moltype = AA length = 615
FEATURE            Location/Qualifiers
source             1..615
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 114
MSSVTGNPG DSRNKSSVSM FIVAGLCCVF PLLGAWQSRG SGKGDNIAYE ITKKAKACSI 60
LGNLEFETHH GNKAVNDSML EVKEYKPCDD QYIDYTPCQD QARAMTFPRE DMNYRERHCP 120
SEQEKLHCLI PAPKGYVTFP RWPKSRDYVP YANAPHKSLT VEKAVQNVVQ YEGNVFKFPG 180
GGTQPPHGAD AYIEQLASVI PINDGTVRTA LDTGCGVASW GAYLFKKNVI TMSFAPRDSH 240
EAQVQFALER GVPVIGVLG TIKLPYPSRA FDMAHCSRCL IPWGANDGKY MMEVDRVLRP 300
GGYWILSGPP INWRNSYQAW QRPKEELQEE QRKIEEIAEL LCWEKKHEKG ETAIWRKRIN 360
EDYCSHGDSR VTKCRSDAD DIWYKKMENC VTPYPETRSQ EEVAGGELKP FPERLHAIPP 420
RVDGSGIPGV SIDSFQEDNR LWKKHMSAYR RINKLLDSGR YRNVMDMNAG LGSFAAFES 480
PKLWVMNVMP TIADRDTLGV IFERGLIGIY HDWCEAFSTY PRTYDLIAE GVFSLYKDKC 540
NMEDILLEMD RILRPEGAVI LRDHADVLTE VKRIIAGMRW NMKMYDHEDG PLVPEKILVA 600
VKKYWVAGDK NSTFS 615

```

```

SEQ ID NO: 115      moltype = AA length = 451
FEATURE            Location/Qualifiers
source             1..451
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 115
MSSSGDDDDH PSSKRPIQHT QTKYKLKILL LVILTNNLLTI YVFTGPSLPI AQNLSLPTWD 60
HTKLLNELNI TKRELANSRA QNSDLQOHLK ASNILTKSLT TELSRNLAAI DQSSSSKEGS 120
SIFSFDLLS DLSTEAKIGM GPQKLPLGYS PRSGTDELYP SVGGGCLRYK EELAQYMTYK 180
IGGECPVDDV FAQRLMLKGC EPLPRRRCHP KSPAGYKEPN PLPKSLWTPD PDSIVWDYPY 240
TCKSYGCLVE RKKLPVFYDC KDCFDLEGRE KSRWLFNDGG LDFGINEVLG TKPHGTIRIG 300

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LDIGGGTGTF	AARMKENNVT	IVTTSMNLDG	PFNSFIASRG	LISMHMSVSQ	RLSFFENTLD	360
IVHSMHILSN	WIPDTMLEFT	LYDIYRVLRP	GGIFWLDHFF	CLGSQNLNSTY	VPMLDRVGFK	420
KLRWNAGMKL	DRGIDKNEWY	FSALLEKPVK	S			451

SEQ ID NO: 116 moltype = AA length = 1411
FEATURE Location/Qualifiers
source 1..1411
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 116

MSSESPGQWL	EKALLDLCKK	IETGLDLDGD	IISGLVSYCE	LASPLDAKEY	LDNIIGQEAG	60
KNVTEEYLKR	RGHSDNYNST	SDAPTSKLHA	YVKPPSNEVL	VSGTRKPSRA	PKELAASSNQ	120
EKRAPTETVE	LRNMNKGNGQ	SSRKKKAGKI	VSLAEAAGKS	IVFQQGKPCS	CQARRHRLVS	180
NCLSCGKIVC	EQEGEGPCTF	CGALVLKEGS	SYAGLEEGAV	PLSDAEATAE	AYTKRLVEYD	240
RNSAARTTVI	DDQSDYYEIE	GNSWLSMEEK	ELLRKKKEEI	EEAERTKRSK	LIMTFDLVGR	300
KVLLNEDEV	ELEMENRILR	PPDEREANRI	KPNPTLKVQP	IFVDPGTSRK	QPVKGNHLNK	360
GLSNGLCLEI	SGRVQHGTH	LKHFMDDDKL	ETASNGGLWL	GPSVNDLSQI	VISLLRSARR	420
IMATKGNSGD	NKTRSSVSIF	IVAGLCCFFY	ILGAWQSRGF	GKGDSIAQEI	TKKADCSILS	480
NLNFETHHGD	EAGNDDSKSE	VEIFKPCNDR	YVDYTPCQDQ	MRAMTFPREN	MNYRERHCPP	540
EEEEKLHCLIP	APKGYVTPFP	WPKSRDYVPY	ANAPYKSLTV	EKAVQNWIQY	EGNVFRFPFG	600
GTQFPFHGDA	YIDQLASVIP	IDNGMVRTAL	DTGCGVASWG	AYLFKRNVAI	MSFAPRDSHE	660
AQVQFALERG	VPAVIGVLGT	IKLPYPSRAF	DMAHCSRCLI	PWGANDGIYM	TEVDRVLRPG	720
GYVWLSGPPI	NWRTHYEAW	RPEELEEEBQ	RKIEEIAKLL	CWEKKHEKGD	IAIWRKRINK	780
QYCNERDSHV	TMESSNAND	VWYKKMEACV	TPYPESTSE	EVAGGELQPF	PERLNAVPPR	840
ISRGSIPIGIS	VESQEDNNL	WKKHLKAYKS	INKIIDTGRY	RNVMDMNAGL	GSFAAAFESP	900
KLWVMNVMP	IAERDTLGVI	YERGLIGIYH	DWCEAFSTYP	RTYDLIHANG	VFSLYKDKCN	960
AEDILLEMDR	ILRPEGAVIF	RDQVNVLIKV	KRIVAGMRWN	AKMEDHEDGP	LVPEKILVAV	1020
KQYVWVAEET	PCPITIGRIK	TKESEISVFG	ADKYFNMKIN	CENPGLADNA	GRKHGHQKED	1080
CRADLQGLKP	NFRSGTPSTI	SESSWNSQTA	LPPSLLRKSS	QNKQKRAIGK	RFPLGFGCSV	1140
PCLDKKSYYI	DGSVSHGVVK	GTKQSEPGLK	SGEHVDPFIL	NFGMENLAVK	KQLEVEKLEE	1200
EQKRSLEVFG	SHMMRKGDIE	AKNLERKLSM	LTWDAIPAAQ	SLPRTLASSR	LCDDIESDAS	1260
SDLFELENIS	STGQYSLLTR	QASDSMSGCM	TPTTQYAPSE	ASIEWSVVTA	SAADFSVASD	1320
YDAEHMISSN	TSKMAKTNI	NIGGKAPKGL	PSGLLSCKSQ	KAVNVAETAA	YRANEKTKSM	1380
SLGKPDADQT	RVKDFEFSRV	QHPFAARVIA	P			1411

SEQ ID NO: 117 moltype = AA length = 449
FEATURE Location/Qualifiers
source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 117

MSSGSPDDDN	LSRKQTHKSI	NDQTKYKLKS	LVLIIILTNLF	TIYIFTRPSL	NLHPISKYDS	60
FSDSSFLLEH	LNATMKQLET	SRSQIIVLQQ	QLKSTNQLVN	ELLAQLNILN	QSINRTPDST	120
LLNPNLDDLD	LSNEAKLALG	PHKLPLGYSP	RSQTDEVYAS	VGGGCLMYKE	DLAQYMTYRV	180
GKECPVDDVF	AQRMLKGCE	PLPRRRCHPK	APVGYVEPTP	LPKSLWSTPP	DTSIIWDPYT	240
CKNYTCLIER	RKFPGFYDCK	CFDQLQFREK	TRWQFDNGGL	DFGMDQVMAT	RRQGTIRIGL	300
DIGGGTGTF	ARMKERNVTI	VTTSMNLDGP	FNSFIASRGL	ISMHVGVVSQR	LPPFFENTLDI	360
VHSMHILSNW	IPDAMLEFTL	YDIYRVLRPG	GLFWLDHFFC	LGSQNLNVTV	PMLERVGFKK	420
LRWNAGMKLD	RGIDKNEWYF	SALLEKPIS				449

SEQ ID NO: 118 moltype = AA length = 493
FEATURE Location/Qualifiers
source 1..493
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 118

MSSAAAGQE	REMOKSYWIE	HTTDLSEAM	MLDSKAADLD	KEERPEVLSL	LPPYEGKSIL	60
ELGAGIGRFT	GELAKKAGQI	IALDFIEGAI	KKNERINGHC	KSASFMCADV	TSPELDFAE	120
SMDLVFSNWL	LMYLSDEEVE	ALAKRMVKWL	KVGGHIFPRE	SCFHQSGDHK	RKNPNTHYRE	180
PRFYTKVPKE	CHFRDESGNS	FELTLVGCKC	IGAYVKNKKN	QNQICWTWKK	VRSEDDRAFQ	240
RFLDNVQYKC	TGILRYERVF	GQGYVSTGGI	DTTREFVAKL	DLQPGQKILD	VGCGIGGGDL	300
YMAEKFDVHV	VGIDLSINMV	SFALERAIGL	QCSVEFEVAD	CTKKTYPDDT	FDVIYSRDTI	360
LHIQDKPALF	KFFYKWLKPG	GRVLISDYCK	RAGSPSEDFA	RYIKQRGYDL	HDVEAYGQML	420
KDAGFDEVIA	EDRTSQFTNV	LQTELDLVEK	DREAFIRDFS	BEDYDDIVGG	WKAKLIRSSS	480
GEQRWGLFIA	KKK					493

SEQ ID NO: 119 moltype = AA length = 301
FEATURE Location/Qualifiers
source 1..301
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 119

MSSGKSGRDW	TQIYAIYGMD	DWQTPFFLLI	HAVFFSALSI	LFLLYFEPIC	FFTQHFFPFI	60
GAGLARFVAG	FTGVSVALSA	VCLFFAAANF	FYSSVSLHWD	MAQRIVNAV	DWSSVKQALD	120
LCGGRGILLN	AVAMQLKKS	SSGRVVGLDP	TRAHTNGPTT	LSTLRTAGLE	GVQEYVTCRA	180
GDPRKLPSD	NYFDVVVSSV	FVHTVGKEFG	QKSAAGAER	MRLVGEVVRV	LKSGGVGVVW	240

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DLVHVPEYVQ	RLQLKLMEDI	RVSERVTAFM	LSSHIVSFKK	PNQHFVGSNE	VRVDWRFNNL	300
C						301

SEQ ID NO: 120	moltype = AA	length = 921
FEATURE	Location/Qualifiers	
source	1..921	
	mol_type = protein	
	organism = synthetic	construct

SEQUENCE: 120						
MSSAIGKARS	NKRSSSSSR	STLTTTILVA	LCIFGLWLLA	PNTLVYPRTT	TRTSATTSSQ	60
SSSMPSHEI	HPTEPIPKS	QPVFQDSQGD	LPEDAIQTDH	GATHNDHTDE	SPELPNQSST	120
GSKAQAEADN	VSEQDSGPEN	NGDSOSSSDN	NNEAEGDGGG	GDGGNEEVDQ	KSDVVEGNPQ	180
ELHGMESVEE	LEKQKLETQ	NSEESSITQN	QEAETREDQ	TKASANKDFE	GKENQHTADD	240
KKIENHDSSE	NKEHEISDAD	QQRLEQHQQ	QEDDQIQHGK	SQRTEGTTSI	SNQDVQSTVG	300
GSTDPENHES	QHNILHNKNA	TESISTEED	KKEKPSKWDT	SDSFPSGDSS	EIPKESRESK	360
KSWSTQADES	ENQKERRKVG	TDDQDGSSIN	GYKWQLCNVT	AGPDYIPCLD	NEEAIAKLQS	420
RRHYEHRERH	CPKEAPTCLV	PLPKGYKRPI	EWQSRDKIW	YYNVPHTKLA	DVKGHQNWVK	480
VTGEFLTFFG	GGTQFIHGAM	HYIDFLEEAV	PDIANGKHSR	VLLDVGCQVA	SFGGYLFEKD	540
VLAMSFAPKD	EHEAQVQFAL	ERGIPAISAV	MGTRQLPFPS	RVFDVVHCAR	CRVPWHADGG	600
ALLLELNRVL	RPGGYFVWSA	TPVYQTLLED	VKIWKEMSNL	TVSMCWELVT	IKKDINSIG	660
AAVYHKPDNS	ECYHKRKHNN	PPMCKSDDDP	DAAWYIPLQA	CMHKVPAEGN	ERGSDWPEEW	720
PRRLQTPPYW	LNRSQMGIYG	KPAPEDTAD	HEHWKQVSVK	SYLSGLGISW	SDVRNIMDMR	780
AVYGGFAAAL	KDLKVWVLNV	VNIDSPDTLP	IYERGLFGI	YHDWCESFST	YPRTYDLLHA	840
DHLFRLKKR	CKLIPVIAEV	DRIVRPGGKL	IMRDDSSSTIG	EVENLLKSLH	WEVHLTFSKN	900
QEGILSAQKS	EWRPDSYVAS	S				921

SEQ ID NO: 121	moltype = AA	length = 343
FEATURE	Location/Qualifiers	
source	1..343	
	mol_type = protein	
	organism = synthetic	construct

SEQUENCE: 121						
MSSASSLVSG	AEKQLIGIA	PAGLGFISD	LHGKHFPKLS	LVSCSGISGD	RSKILVPRCS	60
LSERPTSQP	RFIQHKKKAF	WFYRFLSIY	DHVINPGHWT	EDMRDEALEP	ADLSSRDMIV	120
VDVGGGTGFT	TLGIVKHVDA	KNVTILDQSP	HQLAKAKQKE	PLKECKIIEG	DAEDLPFRTD	180
YADRYVSAGS	IEYWPDPQRG	IKAYRVLKL	GGKACLIGPV	HPTFWLSRFF	ADMWMLFPKE	240
EYIEWFQKA	GFKDVQLKRI	GPKWYRGVRR	HGLIMCCSVT	GVKPSAGDSP	LQLGPKAEDV	300
SKVPNPITFL	LRFVLGAMAA	TYFVLVPIYM	WLKDQIVPKG	QPI		343

SEQ ID NO: 122	moltype = AA	length = 630
FEATURE	Location/Qualifiers	
source	1..630	
	mol_type = protein	
	organism = synthetic	construct

SEQUENCE: 122						
MSSATLQPPV	SQLPQKSPAP	KLLNKPLFNI	LLLTIFFCSLS	YFIGSYTNSI	YSISPSPSLI	60
RRDTCGFEV	KNNLNTTVEN	VQEPLNFEPL	HTLSLPMEDT	PESHQPPPYC	PKNYTNYCPC	120
QDPNKERLFT	IDRRFHRERH	CPEKGEALRC	LVQKPAGYRR	PPWPCKSRDY	AWFKNVPFKR	180
LTEYKKSQNW	VRQEGDKLFF	PGGGTSFPKG	VKGYIDLKLR	VVPLNSGAIR	TVLDVGCQVA	240
SFGASLMDYN	IVTMSIAPRD	IHEAQVQFAL	ERGVPALLGV	LSTYRLPFPS	RCFDMIHCSR	300
CLVQWTDYGG	LYLMEIDRL	RPGGYVWLSG	PPISWRISYK	GWERNASDLE	IEQNNLEDLA	360
RKLCWKKVRE	SDPIAVWQKP	TNHIHCAQKL	KTWKAPKFC	HDFPDHGMYK	KMEPCITPLP	420
KVEHVRSTSG	GYLEKWPKRL	KTAPPRIRSA	TVEGVTMRTF	DEDNKIWKRR	VLYYGSVLKF	480
LFKGYRNMIM	DMNSGLGGFA	AVLSEYPWV	MNVVPYDAKN	NTLGIVYERG	LIGTYMDWCE	540
PFSTYPRTYD	LIHSDGLFSL	YMEKCDILD	LFEMYRILRP	BGAIIRDHV	DVIVKIKEIT	600
DRMRWNSKIS	HNEISPFHSE	KILFIDNSPE				630

SEQ ID NO: 123	moltype = AA	length = 386
FEATURE	Location/Qualifiers	
source	1..386	
	mol_type = protein	
	organism = synthetic	construct

SEQUENCE: 123						
MSSDYNIVTM	SIAPRDIHEA	QVQFALERGV	PALLGVLSY	RLPFPSCRCD	MIHCSRCLVQ	60
WTDYGGLYLM	EIDRLLRPGG	YWVLSGPPIS	WRISYKGWER	NASDLEIEQN	NLEDLARKLC	120
WKKVRESDEI	AVWQKPTNHI	HCAQKLKTWK	APKFCSHDFP	DHGMYKKMEP	CITPLPKVEH	180
VRSTSGGYLE	KWPRLKTAP	PRIRSATVEG	VTMRTFDEDN	KIWKRRVLYY	GSVLKFLFKG	240
KYRNIMDMNS	GLGGAFAVLS	EYPAWVMNVV	PYDAKNNTLG	IYVERGLIGT	YMDWCEFFST	300
YPRTYDLIHS	DGLFSLYMEK	CDILDILFEM	YRILRPEGAI	IIRDHVDVIV	KIKEITDRMR	360
WNSKISHNEI	SPFHSEKILF	IDNSPE				386

SEQ ID NO: 124	moltype = AA	length = 348
FEATURE	Location/Qualifiers	
source	1..348	
	mol_type = protein	
	organism = synthetic	construct

-continued

SEQUENCE: 124
MSSSKPSALD LATGLGGKIA KDDVISAVDK YEKYHGYGG EEEERKANYT DMVNKYDDL 60
TSFYEYWGGE SFHFAARMKG ESLKESIKRH EHFLALQLGL KPGQKVLVDG CGVGGPLREI 120
AQFSLTSVTG LNNNEYQISR GKALNRVAGV DRTCDFVKAD FMKMPFPDQ FDAVYAVEAT 180
CHAPDAVGCY KEVYRVLKPG QCFAYEWCW TDSFDPNNKE HQKIKAEIEL GNGLPDIRLT 240
GQCLEAVKKA GFEVIWAKDL AADSPVPWYL PLDTSHFSSS SFRLTAIGRL ITKNLVKGLE 300
YIGLAPSGSQ RVQDFLEKAA EGLVAGAKD IFTPMYFFLA RKPLSGSE 348

SEQ ID NO: 125 moltype = AA length = 337
FEATURE Location/Qualifiers
source 1..337
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 125
MSSASSLLSG AENLKHITGS HFPKGEHQLK YFPKKGVPSS AKSSKFSTLA PRCSLSSSRP 60
VSQPRFIQHK QEAFFWYRFL SIVVDHIINP GHWTEDMRDE ALEPADLYSR HLTVDVVG 120
TGFTTLGIVQ HVDAKNVTIL DQSPHQLAKA KQKEPLKECK IIEGDAEDLP FRDADRYI 180
SAGSIEYWPD PQRIKEAYR VLKIGGKACV IGPVYPTFWL SRFFADVWML FPKKEEYIEW 240
FOKAGFKDVK LKRIGPKWYR GVRRHGLIMG CSVTVGKPMG GDSPLQLGPK QEDVKKPVNR 300
FVFLRLFILG AMAAAYFVLV PIYMWLKDQI IPKGMPI 337

SEQ ID NO: 126 moltype = AA length = 385
FEATURE Location/Qualifiers
source 1..385
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 126
MSSAFCSQSK LNNLNFWIKT SIGSSTNNVG HDSMRKRTSR NHANLVVVKM AASTSGSGGQ 60
GTVAPEEGQL ERPQWTGETL LSLRLRALIS FKPLFSIMKF GARQVLSTA EKTNPWREM 120
TRQILESVDY KEMESIEDTS LVYPDYLLNP FHAYDEGNLS WLAAAEAEAA TMSMVRAIP 180
DASSLDEANQ IVRGNWLKAI EEHNKYSQN VIIRDILDIG CSVGVSTRYL ADKFPAAKLT 240
GLDLSPYFLA VAQPKKKSN PRKNTICWIIH ANGESTGLPS KSYDIVSIAY VFHECPAGAI 300
KNIVKEAFRL LRPGGTFALT DNSPKSKILQ ELSPVLFTLM KSTEPFLDEY YLTDIEKVM 360
EAGFINVRTV LTDPRHRTVT ATPVY 385

SEQ ID NO: 127 moltype = AA length = 824
FEATURE Location/Qualifiers
source 1..824
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 127
MSSALGKYSR VDRGKPSYNY CSTVTIVVFV ALCLVGLWMM TSSSVVPVQN SDVASQQESK 60
NDVNENTRVS KSNANENDSD NSSTVNKNND TSESSMNDNE GKSKQFEDNP GDLPEDATKG 120
DNTVSSDQEE KNPNPQEKPT EENQEEKSGE ENREEGGSKS EDGSKIEKEN EENGELKSDE 180
KVSEAGEANR SNASGGESRD GNKPDEFNDE TKDANKEDGQ KVKNKMDTQS DQDSGVKKES 240
DQTTEVFPSG AQSELLNETT TQNGAFSTQA TESKNEKEAQ QSSELENQSG TSWKLCNVTA 300
KSDYIPCLDN LEAIRSLRST KHYEHRERHC PDNPPTCLVA FPEGYQRSIE WPTSREKIY 360
HNVPHTKLAE VKGHQNWVKV TGEYLTFFPG GTQPKHGALH YIDFIQQSVP DIAWGKRSRV 420
VLDVGCQVAS FGGFLFDRDV LTMSLAPKDE HEAQVQFALE RGIPASAVM GTKRLPYPA 480
AFDIVHCARC RVPWHIEGGK LLELNLRLR PGGYFVWSAT PIYQKIPEDV EIWEAMKLT 540
KSMCWVVSII TKDRVNGVGI AVYRKPIITNE CYEQRSQNEP PLCQESDDPN AAAYVPLQVC 600
MHKVPAAESE RGSQWPELWP ARLDKPPYWL SSSQVGVYK PAPERFATDF QHWRVVTKS 660
YVNGLGINWS AVRNVMDMRA VYGGFAAALK DMNVWVMNIV SIDAPDTLPI IYERGLFGIY 720
HDWCESFSSY PRSYDLLHAD HLFSEIKKKC NFNAFVAEVD RLLRPEGKII VRDNVETINE 780
LESIFKSLHW DIRMTYSKNK EGLLCAQKTM WRPKEFETVT YAI 824

SEQ ID NO: 128 moltype = AA length = 237
FEATURE Location/Qualifiers
source 1..237
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 128
MSSEACLTPV PEVSDINEIA GGPLAKWPER LTAVPPRIVS GSLDGITAAT FEEDTNLWKK 60
RVAHYKVLDS QLAEPGYRNR LLDMMAYLGG FAAALTNDFV WVMNVVPEA KLNTLGAIYE 120
RGLIGTYQTW CEAMSTYPT YDFIHADAVF SLYQDRCDME DILLEMDRIL RPQGSVRI 180
DVDILNVNKS VVDRLQWDSR MVDHEGGPHV REKLLIAVKQ YWTAPAPSED QEQSRTA 237

SEQ ID NO: 129 moltype = AA length = 603
FEATURE Location/Qualifiers
source 1..603
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 129
MSSKHKDGKP STQGTGSSRT VPMAIMFVVL CGLSFYLGGI FCSEKDKYAS AKVEDTVESP 60
KSTASGSLQV KDVSFPECGL EYQDTPCTD PRRWKYGLH RLTFLEHCP PAFERKECLV 120
PLPDGYKIPI RWPKSRDQCV YRNVPYDWIN KQKSNQHWLK KEGDKFLFPG GGTMPFNGV 180

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RYVDLMQDLI	PGMKDGTIRT	AIDTGCQVAS	WGGDLLDRNI	LTVSLAPRDN	HEAQVQFALE	240
RGIPAILGVI	STQRLPFPNS	SFDMAHCSRC	LIPWTEFGGI	YLLEIHRILR	PGGFWVLSGP	300
PVNYENRWRG	WNTTIEEQKS	DYDKLQDLLT	SLCFKFYKKK	DDISVWQKSL	DNNCYKKFDS	360
SDNYPTKCDD	GTEPDSAWYT	PLRPCVVVPN	HKYKNLGLKS	LPKWPQRLHV	APERVADVRG	420
GSEGAFNHDD	SKWKTRVKHY	KKLLPAVGTD	KIRNVMDMNT	LYGGFAAALI	DYPLWVMNVV	480
SSYTSNTLAV	VFDRGLIGTY	HDWCEAFSTY	PRTYDLLHLD	GLFTAESHRC	EIKYVLLEMD	540
RILRPNGYAI	IRESSYLLDA	VATVAKRMRW	GCRKVETEDK	LDKEKILICQ	KKLWYSNTKK	600
SSR						603

SEQ ID NO: 130 moltype = AA length = 441
 FEATURE Location/Qualifiers
 source 1..441
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 130						
MSSGRKYKEY	ITISSLLHS	FPSFSPLSFP	FLGLVPMATW	ANLPYYLTLK	IKKTNLYYMA	60
MITVLCMSY	LIGIWQHGG	KLAANTATST	LPQLPCNPLK	KKFTTIGFPL	DFTAHNADD	120
LVPLSTSARV	QQFPCKPEL	SEYTPCEDAK	RSLKFDRDKN	IYRERHCPEK	EELLKCRIPA	180
PYGYKTPFRW	PASRDMVWYA	NVPHKHLTVE	KARQNWVQFK	GDRFKFPGGG	TMFPRGADAY	240
IDDIGKLINL	RDGSIRTAID	TGCGVASWGA	YLLSRNLSV	SFAPRDTHVA	QVQFALERGV	300
PALIGVMASI	RLPYPSRAFD	IAHCSRCLIP	WGKFDGLFLI	EVDRILRPGG	YWILSGPPIN	360
WENHWKGWDR	KPEDLKAEQD	GIENVAKMLC	WKKLIQKNDI	AIWQKPTNHK	HCKINRNVFK	420
NPPFVRHKIQ	TKHGTPEWKP	A				441

SEQ ID NO: 131 moltype = AA length = 493
 FEATURE Location/Qualifiers
 source 1..493
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 131						
MSSDLNKEQE	REMQKSYWVE	HTADLTVEAM	MLDSKAADLD	KEERPEVLSL	LPPYDGKSIL	60
ELGAGIGRFT	GELAKKAGHL	VALDFIESAI	EKNESSNGHY	KNAKFICADV	TSPELSFPVE	120
SMDLIFSBNL	LMYLSDEEVE	ALAERMVKWL	KVGGHIFPRE	SCPHQSGDHK	RKNNPTHYRV	180
PRFYTKVPKE	CHLRDGSNGS	FELTLVGCKC	IGAYVRNKKK	QNQICWVWQK	VRQDDRGFQ	240
RPLDNVQYKC	NGILRYERVF	GHGFVSTGGI	ETTREFVSKL	DLQPGQKVL	VGCGIGGGDF	300
YMAEYDVHV	VGIDLSNMI	SFALERAIGL	KCSVEFEVAD	CTKKTYPDGS	FDVIYSRDTI	360
LHIQDKPALF	KSFYKWLKPG	GRVLISDYCK	RSVSPSEDF	KYIKQRGYDL	HDVEAYRQML	420
RDAGFDEVIA	EDRTDQFIKV	LQRELDSEVK	DREAFIKDFS	EEDYNDIVGG	WKAKLIRSSS	480
GEQRWGLFIA	KKN					493

SEQ ID NO: 132 moltype = AA length = 354
 FEATURE Location/Qualifiers
 source 1..354
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 132						
MSSGSABEEA	SLFAMQLASA	SVLPMVLKSA	IELDLLELIA	KAGPGAYVSP	SKLAAQLPTH	60
NPQAPVMLDR	ILRLLATYSV	LNCKLKNLDP	GGVERLYGLA	PVCKFLTUNA	DGVSMAPLLL	120
MNQDKVLMES	WYHLKDAVID	GGIPFNKAYG	MTAFEYHGTD	PRFNKVFNQG	MSNHSTITMK	180
KILEVYKGEF	GVKTVVDVGG	GTGATLNMII	DKYPTIKGIN	FELPHVVEDA	PFYPGVEHVG	240
GDMFVSVPGE	DAIFMKWICH	DWSDDHCRKL	LKNCYHALPE	GGKVILAEIC	LPDAPDTSLA	300
TQNVVHIDVI	MLAHNPGGKE	RTEKEFEALA	KGAGFKEFRK	ICSAVNTWIM	ELCK	354

SEQ ID NO: 133 moltype = AA length = 545
 FEATURE Location/Qualifiers
 source 1..545
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 133						
MSSIDDSESN	VKEIKPCHPR	YTDYTPCQDQ	NRAMKFPREN	MNYRERHCPP	QEEKLHCLIP	60
APKGYVTPFR	WPKSRDYVPY	ANAPYKSLTV	EKAIONWIOY	EGNVFRFPGG	GTQFPQGADK	120
YIDQLASVIP	IDNGIVRTAL	DTGCGVASWG	AYLWKRKVIA	MSFAPRDSHE	AQVQFALERG	180
VPAVIGVLGT	IKMPYPSPRA	DMAHCSRCLI	PWGANDGMYM	KEVDRVLRPG	GYWVLSGPPI	240
NWKTNKAWQ	RSKEDLEBEQ	RKIEDIAKQL	CWEKKSENAE	IAIWQKRIDT	DACRAEQENS	300
GAAPCQAADP	DDVWYKMEV	CITPSDRKGD	EAYGERIKPF	PERLYAIPPR	IASGSVPGFS	360
AESYEEDNKK	WKKHVSAYKR	INKILDSGRY	RNIMDMNAGL	GGFAAALESP	KLWVMNVVPT	420
IAEKNTLQVI	YERGLIGIYH	DWCEGFSTYP	RTYDLIHANG	LFSLYKDKCD	FEDILLEMDR	480
ILRPEGAVII	RDEVDVLRNV	QKIVRGIRWD	FKLTDHEDGP	LVPEKILIAV	KKYWVAGENN	540
STSTQ						545

SEQ ID NO: 134 moltype = AA length = 263
 FEATURE Location/Qualifiers
 source 1..263
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 134

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MSSSGYQAKE	YSTGRPTYPE	EIFQFVASKT	PCHDLAWDVG	TGSGQAARSL	ARIYKNVVAT	60
DVSPQLECA	VKLPNIKYYC	TPPKMSMTEL	EQNVATESSV	DLVSIATAMH	WFDLPTFFYQQ	120
VKWLKPKPDG	VIAAWSYTLF	QINKNVDTLL	RKLFFADAVQ	FFGPQAKLII	EENYRTIDFP	180
FEFVDGLEHT	GPFEKMEKL	INLDGYFTLI	KWSAYKTAK	EKGVELLSED	LVKEFTLAWN	240
EDGKQEKIAT	WPLYMRIGKV	GKQ				263

SEQ ID NO: 135 moltype = AA length = 238
 FEATURE Location/Qualifiers
 source 1..238
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 135						
MSSDVEAK	MDAEKLKRG	AEFYDESSGI	WEDIWGDHMH	HGFYDPSSTV	SLLDHRSAQI	60
RMIEEALQFA	SVSDALNQPF	PDGKFDLVWS	LESGEHMPNK	AKFVNELVVRV	AAPGATIIIV	120
TWCHRNLS	EQSLHPDEKK	LDDKLCDAAY	LPPWCSTDDY	VKLLESFSLQ	DIKAADWTEY	180
VAPFWPAVVS	SALSWKGITS	LILSGWKTIK	GALAMLLMIK	GYKKGLIKFA	VITCRKPE	238

SEQ ID NO: 136 moltype = AA length = 512
 FEATURE Location/Qualifiers
 source 1..512
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 136						
MSSAKDNGS	KPHHLETQR	KRLTWILAVS	GLCVLFYVLG	AWQSTNPAPS	SQSDVWERVG	60
CQDGSPDQKV	NNPDLSTSTL	SLSSWSSTP	LDFFESHQLE	VNSSEEPKRF	PPCDTSYTEY	120
TPCQDPPRGR	KFDRNMLKYR	ERHCPEKEEL	LHCLIPAPPN	YKTPFKWQOR	RDYAWYANIP	180
HKHLSVEKAN	KNWIQFEGDR	FKFPGGGTME	PRGADAYIDD	INELIPLSNG	TIRTAVDTCG	240
GVASWGAYLF	QRNIIAMSFA	PRDTHEAQVW	FALERGVPM	IGVMGSQRLP	YPARAFDMAH	300
CSRCLIPWHK	YGGHLHIEVD	RILRPGGYWI	LSGPPIRWKN	YWRGWERTQE	DLKQEQDAIE	360
DVAKNICWKK	VIEKDDLAUV	QKPINHIECI	KSRAIHHATP	MCKSDNADVA	WYKDMETCIS	420
PLPEVSNKDE	VAGGSLEKWP	ERVFAVPPRI	SSGSVAGITM	ENFRDDNDVW	KERVAHYKRV	480
VGPLDQGGYR	NVMDMNAFLG	GLQQFWQNIQ	CG			512

SEQ ID NO: 137 moltype = AA length = 625
 FEATURE Location/Qualifiers
 source 1..625
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 137						
MSSTRGRSDG	AQKKRLTSTV	GIIAIFLLFL	YVYFGSKSNG	ESALEYGSRS	LRKLGSSYLG	60
GDEDGDLGSK	QDDASSKFAL	DDGEDGLVPK	SFLVCDDRHS	ELIPCLDRHL	IYQMRKLKDL	120
SLMEHYERHC	PLPERRFNCL	IPPPSGYKVP	IKWPKSRDEV	WKANIPHTHL	ASEKSDQNMW	180
VVKGEKIIFP	GGGTHPHYGA	DKYIASIANM	LNFSNNLNN	EGRLRTVLDV	GCGVASFGAY	240
LLSSDIIAMS	LAPNDVHQNQ	IQFALERGIP	AYLGLVGTMR	LPYPSRSFEL	AHCSRCRIDW	300
LQRDGILLLE	LDRLLRPGGY	FAYSSPEAYA	QDEEDLRIWR	EMSALVERMC	WKIAAKRNQT	360
VIWVKPLTND	CYMEREPGTQ	PPLCQSDNDP	DAVWGVKMEG	CITRYSDHDH	RTRGSGLAPW	420
PARLTSPPPR	LADFGYSSEM	WEKDMQVWRR	RVENYWNLLS	PKISSDSLNR	VMDMKANLGS	480
FGAALKEKDV	WVMNVVPEDG	PNTLKLVDYR	GLIGTIHNWC	EAFSTYPTRY	DLLHAWSVLS	540
DIEKKGCSGE	DLLLEMDRIL	RPTGFVIIRD	KQPVDFVKK	FLGALHWEAV	GTADSSSDPD	600
QEGDELVFVI	QKKLWLTS	SGS	FRDTE			625

SEQ ID NO: 138 moltype = AA length = 254
 FEATURE Location/Qualifiers
 source 1..254
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 138						
MSSATLQPPV	SQLPQKSPAP	KLLNKPLFNI	LLLTIFFCLS	YFIGSYTNSI	YSISPSPSLI	60
RRDNTCGFEV	KNNLNTTVEN	VQEPLNFEPL	HTLSLPMEDT	PESHQPPPYC	PKNYTNYCPC	120
QDPNKRERLT	IDRRFHRERH	CPEKGEALRC	LVQKPAGYRR	PPWPKSRDY	AWFKNVPPFKR	180
LTEYKKSQNW	VRQEGDKLFF	PGGGTSFPKG	VKGYIDLKLR	VVPLNSGAIR	TVLDVGC	240
INGCSKIFIV	FVRF					254

SEQ ID NO: 139 moltype = AA length = 523
 FEATURE Location/Qualifiers
 source 1..523
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 139						
MSSEPSTGKP	SFFRNILVRL	FLFCVFILGV	RFAYVVTIRG	ESCDLGDFCF	FSLPDNLNTI	60
VSGVGQLGTK	SFSSSAIIAV	GKSAPAKPHK	LPDLWATQGF	QKAAQFYSSV	FQDLISEGFL	120
TPNSKALCVD	TPTGADVVAL	KEIGVSDSIG	IFKKSSKPLV	ISGQAIKQPF	GDDKDFDFVS	180
GAGMIEKSVK	AADFAAEVCR	TLKPEGFFV	HTGSKDTYSY	NSFLDLFNCC	KLKFRDIDG	240
FDSKLPFIRE	IVMQKKFNEN	VDDEIQMPGG	NSDKCFVPG	YKQELIRNAE	PLIVEEPPKP	300
WITLKNITQN	IKYLPMSADI	SPQRVYVVD	VGARSYGSSI	VSWFKKQYPK	QNRTFEIIAI	360
EADKHPHEEY	KYKKGVKLLP	YAAWVRNETL	FFEINEDPID	KDVVKGRGMG	RIQPVQSSSG	420

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VSDGNVDEIQ	GFDFAEWLKS	TVSEKDFVVM	KMDVEGTEFD	LIPRLIETGT	ICLIDEVFLE	480
CHYNRWQKCC	PGERTSKYQK	TYDQCLELFT	SLRESGVLVH	QWW		523

SEQ ID NO: 140 moltype = AA length = 704
 FEATURE Location/Qualifiers
 source 1..704
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 140
 MSSSRPLHRG VSGGGRFSPN SHDLWDDDD S QMKDKAEKD GDRKSTNADH TYLSIRQPPR 60
 FLLADNSSSK HGVTKDGFVS DPFSPSPGVS RNRHKFILL LRFGLVVIVM LALAGSFVWT 120
 ISITTSSRGQ IYRGYRRLQQ KLVSDLVDVG KLSLGSTRLK DLEYCSPESE HYVPCFNVSE 180
 NLALGISGGN EYDRHCGPGS RQSCFVLPVP NYKTPLRWFP GRDLIIVANI NITAEVVRSS 240
 GSLTKRMMY DEDQISFRSA SPGYDDMEDY AHQIAEMIGL RNESNFVQSG VRTILDIGCG 300
 YGNLGAHLFS NELLITICIAN YEASGSQVQL TLERGLPAMI SSFTSKQLPY PSLSFDMIHC 360
 AGCGIDWDQK DGILLVEVDR VLRPGGYFVW TSQLTSSQQF LRKNENQKEW NFVHGFAKSL 420
 CWEMLSQQVD TVVWKKTSKK NCYSARKSGS GPSLCSRGH D VESPHYRPLQ GCIGGTQSQR 480
 WPIEERTKW PARARLNAKE LAIHGLNSDD FAEDTLNWS AVRNYWSLLS PVIFSDHPKR 540
 PGDEDPSPPF NMVRNVLDNM AHFGGFNAAL LEAKKSVVWM NVIPTTGMNH LPLILDRGFV 600
 GVLHDWCEAF PTYPTYDLV HGEGLSLEF GEEQRCEMLD VFTEIDRILR PEGWFILRDT 660
 TPLIESARTL TARLKWDMARM IETENNSDER LLICQKPPAK RQAM 704

SEQ ID NO: 141 moltype = AA length = 102
 FEATURE Location/Qualifiers
 source 1..102
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 141
 MSSKWILHDW DDGHCLKLLK NCYKALSDNG KVIVVDALVP VIPDTSPCVK TTCQSDLIMM 60
 TQNPGGKERS EPEFLALATA AGFKGIRLEC YVCNLWVMEF YK 102

SEQ ID NO: 142 moltype = AA length = 140
 FEATURE Location/Qualifiers
 source 1..140
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 142
 MSSKIHSRSL CPSRSYGQGE VEVFQIARTM MQSIILYSII CSLHCFHAFP SSNSSTPAYN 60
 ALWNKSLTLP NVTNNRYFLF IFTPFKHTK VKHATTIVK LNNYISIVND VRNLSVIDFI 120
 FVIYIHSSRC HYTYRISSNK 140

SEQ ID NO: 143 moltype = AA length = 385
 FEATURE Location/Qualifiers
 source 1..385
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 143
 MSSQPQRGRK RERERDREEM ESVSQNSSSS DQFAMKGGDD DFSYTKNSTH QRDALQATKF 60
 FIQESIAEKL DVNKFPCGKAF CVDALGCSVG PNTLIAMQNI VEAVELKFKN RKGFSPTIP 120
 EFQVFFNDHT VNDENTLERS LPTGHDKRY Y GVGVPGSFYG RLFPCDSIHI MHTSFSTPFL 180
 SQVPKEVMDK NSAAWNGRI HHNYAKADVL KAYEAQHAED IDCPLTARAK ELVHGGLMD 240
 VTSFRPDGVP HTHVLTNIGM EVLGYCLMDL AKKGLIDEEN VDSYNVPVYL QSPPELKQAV 300
 QRNKYFSIEK MESVPMIDS DVSAKAQQYS LGMRAVMGDV IREQFGAEIV DKLFDFLKKK 360
 LEEHPNFAKG VVLDMFVLLK RNAED 385

SEQ ID NO: 144 moltype = AA length = 366
 FEATURE Location/Qualifiers
 source 1..366
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 144
 MSSSESSQNN RFCDQFATKG GDDDFSYTKN SIHQDAIEA TKLLIQEAIA EKLDINKFCG 60
 KTFCTAHLGC STGPTTLTAM QNIIEAVELK FQNKQGFNFP NFPEFQVFFN DHTVNDNFAL 120
 FRSPLTGHDK RFLYGVGVPGS LYGRLPFCNS IHITHTSFST PFLSLVPKEV MDKNSAAWNE 180
 GKIHYNHAK E DVLKAYBGKH AEDMDCFFMA RAKELVHGGL MLIVTSLRPD GIPHVLTNIG 240
 MEVLGSC LMD LAKKGLIDEE NVDSFNVPVY LESPEELKQA IQRNKYFSME KVESVPMIMD 300
 SDISCAQQY SLAIRAVMED VFREQFEAEI LDQLFDLPFK KLEGHPNFAS AVVLDLFLVLL 360
 KRNAED 366

SEQ ID NO: 145 moltype = AA length = 368
 FEATURE Location/Qualifiers
 source 1..368
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 145
 MSSESLPGNN RSYDQFAMKG GDDDFSYTKN SIHQDAIEA TKLLIEEAVA EKLDISNFCG 60

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KAFCIADLGC	SVGPNTLIAM	QNIVEEVKLG	FRNRKGLNFP	NIPEFQVFFN	DHTVNDFNAL	120
FRSLPTGHDK	SYYGVGVP	FYGRLFPCDT	IHMHTSFST	PFLSQVPKEV	MEKNSAANK	180
GRIHYNHAK	DVLKAYEGKH	AEDMDCFFMA	RAKELVHGGL	MLIVTSLRPD	GVPHTRVLTN	240
IGMEVLGSC	MELAKKGLID	EEKVDSFNAP	VYLESPEELK	QAIQKNKYFS	IEKMETVPMI	300
MDSDISCRAQ	QYSLAIRAVM	EDVFREQFEA	EILDQLFDLF	KKKLEGHFNF	ASAVVLDLFLV	360
LLKRNAED						368

SEQ ID NO: 146 moltype = AA length = 383
 FEATURE Location/Qualifiers
 source 1..383
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 146

MSSGAPKKKR	KVGSSGGMES	VQSNSSSSDQ	FAMKGGDDDF	SYTKNSTHQR	DAIQATKFFI	60
QESIAEKLDV	NKFCGKAFCV	ADLGCSVGP	TLIAMQNIVE	AVELKFKNRK	GFHSPTIPEF	120
QVFFNDHTVN	DFNTLFRSLP	TGHDKRYYG	GVPGSFYGR	FPCDSIHIMH	TSFSTPFLSQ	180
VPKEVMDKNS	AAWNKGRHH	NYAKADVLKA	YEAQHAEDID	CFLTARAKEL	VHGGLMDVT	240
SFRPDGVPH	HVLTNIGMEV	LGCLMDLAK	KGLIDEENVD	SYNVPVYLQS	PEELKQAVQR	300
NKYFSIEKME	SVPNMIDSDV	SAKAQQYSLG	MRAVMGDVIR	EQFGAEIVDK	LFDLFKKKLE	360
EHPNFAKGVV	LDMFVLLKRN	AED				383

SEQ ID NO: 147 moltype = AA length = 385
 FEATURE Location/Qualifiers
 source 1..385
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 147

MSSQPQRGRK	RERERDREEM	ESLQNNNRYC	DQFAMIGGDD	DFSYTKNSIH	QRDAIEATKF	60
LIQGSIAEKL	DINNFCGKAF	CVADLGCSVG	PNTLIAMQNI	VDAVELKFKN	RKGFHSPTIP	120
EFQVFFNDHT	VNDFTLFRS	LPTGHDKRY	GVGVPGFFLR	SIISLDSIHI	MHTSFSTPFL	180
SQVPKEVMDK	NSAAWNKGR	HNNYAKADVL	KAYEAQHAED	IDCFLTARAK	ELVHGGLMD	240
VTSFRPDGV	HTHVLTNIGM	EVLGYCLMDL	AKKGLIDEEN	VDSYNVPVYL	QSPEELKQAV	300
QRNKYFSIEK	MESVPMIDSD	DVSAKAQQYS	LGMRAVMGDV	IREQFGAEIV	DKLFDLFKKK	360
LEEHPNFAKG	VVLDMFVLLK	RNAED				385

SEQ ID NO: 148 moltype = AA length = 382
 FEATURE Location/Qualifiers
 source 1..382
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 148

MSSQPQRGRK	RERERDREEM	YKAPFVLC	AVITDHMMNL	SGAGNYGRDA	IEATKFLIQE	60
STAEKLDINK	FCGKAFPCVAD	LGCSAGPNTL	IAMQNIVEAV	ELKFKNRKGF	HSPNIPEFQV	120
FFNDHTVND	NTLFRSLPTG	HDKRYYG	GVGV	PVLFTVDYFL	VTLFTHIMHT	180
PKEVMDKNSA	AWNKGRIHNN	YAKADVLKAY	EAQHAEDIDC	FLTARAKELV	HGGLMDVTS	240
FRPDGVPHTH	VLTNIGMEVL	GYCLMDLAKK	GLIDEENVDS	YNVPVYLQSP	EELKQAVQRN	300
KYFSIEKMES	VPMIDSDV	AKAQQYSLGM	RAVMGDVIRE	QFGAEIVDKL	FDLFKKKLEE	360
HPNFAKGVVL	DMFVLLKRNA	ED				382

SEQ ID NO: 149 moltype = AA length = 291
 FEATURE Location/Qualifiers
 source 1..291
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 149

MSSQNIVEAV	ELKFKNRKGF	HSPTIPEFQV	FFNDHTVND	NTLFRSLPTG	HDKRYYG	60
PGSFYGRLF	CDSIHMHTS	FSTPFLSQVP	KEVMDKNSAA	WNKGRIHHNY	AKADVLKAYE	120
AQHAEDIDCF	LTARAKELVH	GGLMDVTSF	RPDGVPHTHV	LTNIGMEVLG	YCLMDLAKKG	180
LIDEENVDSY	NVPVYLQSP	ELKQAVQRNK	YFSIEKMESV	PMMIDSDVSA	KAQQYSLGMR	240
AVMGDVIREQ	FGAEIVDKLF	DLFKKKLEEH	PNFAKGVVLD	MFVLLKRNAE	D	291

SEQ ID NO: 150 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 150

MAALPLLI	SLFLALLLKL	LLKHKNRNS	KLPPCPPKLP	IIGNLHLLGN	MPQESLHQLA	60
KKYGPVIFLQ	LGEVPTIVIS	SARLAKEAMK	TFDLALSSRP	QIFSAKHLFY	NCTDVVFSPY	120
GAYWRNIRKI	CILELLSAKR	VQSFSYVREE	EV AHLVQVRA	EAYPGTTNLT	KMLGLYANNL	180
VCRVAFGKDF	SQAGDNHRHN	FQKLLLEYQV	LLGGFSIGDF	FPSMEFVHKF	TGMESRLKDT	240
FRFPDQFFDE	ILQEHRNPES	RKEHKDLVDV	LLEIQKNGDP	ETPLTTDNIK	SIILDMFAAG	300
TDTNFITLDW	AMTELMNPG	VLKRAQAEVR	NIVGERRSVS	ENDLPQLPYV	KAVVKETFRL	360
HPPAPVLVPR	ESMEDVTIDG	YDIPAKTRFF	INAWAIGRDP	KSWENPDVFE	PERFMNSHID	420
FKGQDYELIP	FGAGRRSCPA	ITPSAVTVEL	ALAQLLHSFD	WELPPGVAK	DLDMTVEVFI	480
TMHKIYPLMV	VAKPYFSQGL	N				501

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SEQ ID NO: 151 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 151

MAALPLLIFS	SLFLALLLKL	LLKHTKLVKL	PPGPRPLPII	GNLHQLNGSL	PHRILTDLAK	60
KYGPLMHLQL	GEVSTVIYSS	PAVAKEVMKD	HDIIFCYRPL	LLSPAILNYG	GTDIAFAPYG	120
EYWRQLRKIC	TIELLSKRV	QTFRSIREQE	VSNLFRSIS	EEGSVVNL	MIYSLTYSIT	180
AQAAGKESK	YQDKFISVMQ	DAIKLMGGFN	IADVPSIKI	LEKITGIRHK	LEETHNQIDE	240
IVQNILIEHK	AKRAESKLG	RGAKEDLDV	LLNIQSSGEF	EPPLTDDNLK	AVIYDIFSAG	300
GETSSTTVIW	AMSEMIRNPK	VMKKAQDEVR	EVFAGNGNVD	ESRLHELKYL	QAVIKETMRL	360
NPAVPLLLPR	ECSEHCQING	FDPVPAKTRVI	VNAWAIGRDP	KQWTEPEKFK	PDRFLDSEID	420
FRGKTFSFIP	FGAGRRICPG	ISFALPLVEL	PLAQLLYHFD	WKLPGQSKPD	ELDMTEIFGL	480
TVGRKHDNL	VPPIYRCSV	L				501

SEQ ID NO: 152 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 152

MAELLFTLCA	LLVFFSFAVK	IVRKTKEPKL	PPGPRPLPII	GNLHQLNGSL	PHRILRNLA	60
KYGPLMHLQL	GEVSTVIYSS	PALAKEIMNE	HDIVFCYRPF	LLSPSILNYG	CADIAFAPYG	120
EYWRQLRKIC	TMELLTSRRV	QTFRSIREVE	VSNLIRSIS	EEGSVVNL	KIYSLTYSIT	180
AQAAGKESK	YHDKFISIMQ	DVVKLMGGFK	IADMYPSLKI	LEKITGIRHK	LEMVHNQVDE	240
VENILSEHK	VKRADSNLGN	REAKEDLDV	LLNIQSSGEF	GVPLTDHNIK	AVIYDVFSAG	300
GETSSTTVIW	TMSEMIRNPK	VMKKAQDEVR	EVFAGNGNVD	ESRLHELKYL	QAVIKESMRL	360
HPAAPLLPR	ECSEHCQING	YDIPAKTRVI	VNAWAIGRDS	KAWTEPEKFN	PNRFLDSEID	420
FRGKTFSFIP	FGAGRRICPG	ISFALPLVEL	PLAQLLYHFD	WKLPGQSKPE	ELDMTEIFGL	480
AVGRKHDLKL	IPIYRCSR	E				501

SEQ ID NO: 153 moltype = AA length = 223
 FEATURE Location/Qualifiers
 source 1..223
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 153

MAIAKDVLT	HHPSFADRP	NLATEIIRYD	QEDMGFSPYG	DYWKQMRKIC	MMQLSMTNV	60
ASFSSIRQDE	ISKLVFAIR	TPGVPVNLTE	KIFACTSSII	CRAAFGQVCQ	DKENMIKNLK	120
IAPVLLAGFN	LADVPPSLKF	IPITGFKQK	LKVVHKKMDE	ILGDVIKEHI	LKHENGKKCN	180
AESGDEDLID	VLLQQPKNGN	LHIPITARNI	KALILVSTIS	LCF		223

SEQ ID NO: 154 moltype = AA length = 446
 FEATURE Location/Qualifiers
 source 1..446
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 154

MALQLGSKVT	LUVSSAEAAH	EIMKTHDLNF	LNRPEKSVNR	KLLYDFKDL	VAPYGEYWRQ	60
MKSICVLHLL	SNKRVQSIRY	IREETSFLI	TKIREGSLAS	PSSGAVNLSE	MFVTLTNDVV	120
SRAAFGRKYS	EGESGRKFRK	LMSEFVSVLG	GPDFGTVLPC	LGWIDRLNGL	TARVERIAKE	180
MDKFLEGVLE	EHLSDSRKNN	ESRVVMTRKI	HGENGEDFAD	VLLGIYKKA	AGISIDRDSI	240
KGLLLDIFAG	GTDTTYTVLE	WAMTELLRHP	KALKELQDEV	RGLVGGGREI	TEQDLEKMKY	300
LKAVIKETLR	LHPPPIPLVP	RKAINDAKIM	DYDISAGTMV	ITNAFAIGRD	PSLWEEPDEF	360
RPERFLNSCI	DFRGHDFQLI	PFSGSRRGCP	GISFAMATNE	HVLANLVLFK	DWDLPDGAKG	420
TDLDMTECTG	LTIHRRKFPIF	AVATPV				446

SEQ ID NO: 155 moltype = AA length = 340
 FEATURE Location/Qualifiers
 source 1..340
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 155

MAFMTLTNDV	MSRAAFGWIC	SEGESGRKFR	KLMSESARVF	GVFDFGTYLP	CLGQIDSVNG	60
LMVRVERIAK	EMDEFLEVL	EEHLSDSRKN	NESRVVMTSK	IHGENGEDFA	DVLFGIYKKN	120
AAGSSIDRDS	IKALLDMFA	GGTDTNYTFL	EWAITELLRH	PKALKELQDE	VRGLVDGGRE	180
ITEQDSEKMK	YLKAVIKETL	RLHPPPLPSV	RKAINDVKIM	DYDISAGTTV	ITNAFAIGRD	240
PSLWEEPDEF	RPERFLNCCV	DFRGHDFQLI	PFSGSRRGCT	GISFAMATNE	HVLANLVLFK	300
DWDLPDGAKG	TDLNMTKSTG	LTIHRRKNSS	CCNSSSLTQ			340

SEQ ID NO: 156 moltype = AA length = 374
 FEATURE Location/Qualifiers
 source 1..374
 mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 156
MAELLSPKRV QTFRSIREGE VLNMIKSIAL HMGSSVDLSR KIFSLPYGIT AQAAF GKKGK 60
YQEQYIMLID EMVKLSGGFS LVDVYPSIKL LERISGTRTK LEKHLKQIDG ILQIIVNEHK 120
KKLEKEELGS GEKKVDLVDV LLNIGKRGEF GSHLADNNIK AIIYDIFSAG SETSSTTVEW 180
AISEMIKSRIT IMKRAQDEVR NIFKERGNVD ESNLHELKYL QAVIKETMRL HPGAPLLLPK 240
ESIEKCEYKVG YEIPAKTRVI INAWALGRDP MYWSEAEKFN PERPLDSELD FRGNHFEYIP 300
FGAGRRICPG ISFALPNTEL PLAQLLFHFD WKLPGLDLKQE QLDMTEVFGV TVRRKNDLRL 360
IPVPYSHSSL NKDL 374

SEQ ID NO: 157      moltype = AA length = 257
FEATURE            Location/Qualifiers
source             1..257
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 157
MALAKIVMKS KKNKSGKKLP PGPRQVPIIG NMHQLYGLPL HHKLRDLAKK YGPLMHLQLG 60
EISTIVVSSP EFAKEVYRTH DIIFAERAEL LSISIVSYNF LGIVFAPYAD YWRQLRKICT 120
MELLSPKRVQ TFRSIWEGEV LNMIKSIALQ KGSSVNLRSK IFSLTYGITA QAAF GKKGKY 180
QEYQYVMLVDE MVKLSGGFSL VDLYPSIKLL ERISGTRTKL EILHKQIDGI LQTIVNEHKK 240
KLEKEELGSG EKKSLSG 257

SEQ ID NO: 158      moltype = AA length = 521
FEATURE            Location/Qualifiers
source             1..521
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 158
MAAIFSKIGI VGAGISGIAA AKQLRHYKPV VFEATDSLGG VWRHCSYRST KLQTPRCDYE 60
FSDFPWTERD NSSFPSYMEI LDYLNYSASN FGVLELIKFN SKVVEIKFTG NRDQSNYSDS 120
GNSKEYGNL LGGQPVWEVA VQTSDDTIE WYAFEFVLIC TGKYGDIPII PKFPNNKGPE 180
VFRGQVMHTL DYSKLDEEAS VNLLKDKKVV VIGYKKSaid LAVECAKANE GPEGKPCTVV 240
VRTLHWTVPH YSIWGLPFYL FYSTRASQFL HERPNQGILR NILCKLLSPV RKAMSKIIES 300
YLVWKLPLKK YGLKPDHPFE EDYGCQMAI LPEEFFAEAD KGMIEFKRAS QWWFCEGGVQ 360
FDDNSKLEAD VVLLATGFDG KRKLRSILPE PPRSLETIDG VMPLYRGTH PLIPNMAFVG 420
YIESVSNLHT AEIRCIWLSR LVENYFKLPS LENMLEQTQK EMEIMRRTTR FFKRSCISTF 480
SINHSDICE EMGWKSWRKN SWLAFAFSPY NSQDYQEKKV N 521

SEQ ID NO: 159      moltype = AA length = 329
FEATURE            Location/Qualifiers
source             1..329
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 159
MAAMSKVLSL FFLQFLCVLC ISDLSNGQGL DVNFYKKTCP SLEDIVKDTT ARFISRAPSL 60
AAPLLRMHFF DCFVRGCDGS VLLNSTSSNQ SEKAAIPNQS LRGFQVIDAV KSELEKKCPG 120
VVSCADIVAL VARDAITLIK GPSWQVELGR RDGRVSIASQ VLNKLPSPFM NIAQLKASFQ 180
SLGLSVKDLA VLSGGHTIGI SHCASVNNRL YNFTGKGDT PTLDPKYVAT LKRKCQPGDV 240
TTILEMDPGS AKTFDITYFT TVSKRRGLFQ SDAALLDDTE TNTVQQHLS SSGSGSFFDD 300
FGVPMVKMGR IGVLTGNQGE IRKQCALVN 329

SEQ ID NO: 160      moltype = AA length = 329
FEATURE            Location/Qualifiers
source             1..329
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 160
MAAFPRHLAV FMLLQIILFP TAFQHCNAQG LKLGIFYKKTG PGAEATVKQI MTGVISRAPT 60
LAAPLLRMHF HDCFVRGCDG SVLLNSTKNN KSEKEAIPNQ SLRGFQIIDN IKSLEKKCP 120
GVVSCADILA LVARDAVSLL NGPFWIVPLG RRDGKVSIMN EALTNLPPPF ANITQLKAFP 180
SAKGLNTKDL AVLSGGHTIG TAHCLGFTNR LYNFTGKGDT DPSLDPNYIP LLKRCKCPGD 240
VTTLVEMTPG TFKTFDEYY TLITKRRGLF QSDAALLDDS QTRAYVKLQA ATHGSTFFKD 300
FAVSMVKMGK VGVLTGKAGE IRRHCALVN 329

SEQ ID NO: 161      moltype = AA length = 316
FEATURE            Location/Qualifiers
source             1..316
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 161
MAASSIVPTI DLDPFLREGD EEAKKNAIE ICQACSEYGF FQVVNHGVPL DLMTRATDMF 60
TEFFNYPIEE KFKSISKPGV SPPPGYFNSP DHSKEGNEHF LILTPSLGLN VYPENPPEFR 120
KVLIEIFPNL AKLALLIERI VSVSLGLAPN FLKEYNNDRS CDFLMGLHYF PAEETENNGK 180
SAHEDGNCIT FVFQDEVGGL EVLKNQWIP ITPAQGLTVV NIGDVVQVLS NNKFKSATHR 240
VVRPKGSRY SYVFHNLHG DKWVEPLPQF TKDIGESAKY RGFVYKEYMQ HRIRNRTHPP 300
ARPEDLIHIT HYSIST 316

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SEQ ID NO: 162 moltype = AA length = 317
 FEATURE Location/Qualifiers
 source 1..317
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 162

MAPLLQMPVT	DLSSFFTGD	VGQKNVLSKI	SQACSDYGFF	QIVNHGLCLD	LMNRALEISR	60
VFFESPTTEK	LKCSPPRGAS	PPAGYNRKPN	PEYEFSEYFI	MFPFGSDFNR	FPPNLPQLRE	120
VMEELFSQFL	KTASVISSIL	IESLGLPQSI	DEGFNFADRS	SDLLIALFYL	SAKEKENMGV	180
NPHKDVSCFT	MVLQDEVGGL	GVEKDGEWLP	LGPTTEGALIV	NIADVLQVLT	NDKFKSPNHR	240
VLRPKARSRS	SFAFFHDLPG	KTWLEPLPEF	TREIGELPKY	RKFLYKDYKQ	MRVRNKSNDP	300
SRLEDEITIS	HYAIASK					317

SEQ ID NO: 163 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 163

MAALPLLIFA	SVSLIVLVKF	VSKPKNNRNS	KLPPCPPKLP	IIGNLHLLGN	MPQESLHQLA	60
KKYGPVIFLQ	LGEVPTIVIS	SARLAKEAMK	TFDLALSSRP	QIFSAKHLFY	NCTDVVFSFY	120
GAYWRNIRKI	CILELLSAKR	VQSFSYVREE	EVAHLVQRVA	EAYPGTTNLT	KMLGLYANNL	180
VCRVAFGKDF	SQGGDNHRHN	FQKLLEEQV	LLGGFSIGDF	FPSMEFVHKF	TGMESRLKDT	240
FRRFQDFPDE	ILQEHNPES	RKEHKDLVDV	LLEIQKNGDP	ETPLTTDNK	SIILDMFAAG	300
TDTNFITLDW	AMTELMTNGP	VLKRAQAEVR	NIVGERRSVS	ENDLPQLPYV	KAVVKETFRL	360
HPPAPVLVPR	ESMEDVTIDG	YDIPAKTRFF	INAWAIGRDP	KSWENPDVFE	PERFMNSHID	420
FKGQDYELIP	FGAGRRSCPA	ITFSAVTVEL	ALAQLLHSFD	WELPPGVEAK	DLDMEVFGI	480
TMHKIYPLMV	VAKPYFSQGL	N				501

SEQ ID NO: 164 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 164

MAALPLFIFA	SISLVPVLVI	LLMKKDKRSV	KLPPSPPKLP	IIGNLHQIGN	MPHESFHNLA	60
KKFGPIMFLW	LGQVPTVVIS	SARLAKEALK	TSDDLALSSRP	ELFSAKHLHY	NCTDLAFSPY	120
GTYYWRNIRKI	TTLQLLSAKR	VQSFSFVRKD	EVAHLVKRIT	ESYPSTINLS	HMLEVYANNN	180
ICRVAFGKDF	SQGGDNVRHN	FQKVLLEEQM	LLGGFSIGDY	FPSMEFLHKF	TGMESRLKDT	240
FRRFDHICDK	ILEEHRKTEN	RKDQKDLVDV	LLEIQHNGDL	ETPLTTDNK	AIIDMFVGG	300
TDTNFITLDW	TMTIELINPK	VMQRAQAEVR	SIVGERRFVS	ENDLSHLGYM	KAVVKETFRL	360
HPPAPLLAPR	ESIQDVTIDG	YNIPAKTRFF	INAWAIGRDS	ELWENPEMFE	PERFMNSHID	420
FKGQDFELIP	FGAGRRICPA	ISFTVATVEL	ALAQLLHGFD	WELPPGVEAK	DLDMTEAFGI	480
TMHRTSPLMV	VAKPYFSDHL	N				501

SEQ ID NO: 165 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 165

MAALPLFIFA	SISLVPVLVI	LSMKKDKRSV	KLPPSPPKLP	IIGNLHQIGN	MPHESFHNLA	60
KKFGPIMFLW	LGQVPTVVIS	SARLAKEALK	TFDLALSNRP	ELFSAKHLHY	NCTDLAFSPY	120
GTYYWRNIRKI	TTLQLLSAKR	VQSFSFVRKD	EVVHLVKRIT	ESYPSTINLS	HVLEVYANNN	180
ICRVAFGKDF	SQGGDNVRHN	FQKVLLEEQM	LLGGFSIGDY	FPSMEFLHKF	TGMESRLKDT	240
LRRFDHICDK	ILEEHRKTEN	RKDQKDLVDV	LLEIQHNGDL	ETPLTTDNK	AIIDMFVGG	300
TDTNSITLDW	TMTIELINPK	VMQRAQAEVR	SIVGERRFVS	ENDLSHLGYM	KAVVKETFRL	360
HPPAPLLAPR	ESMQDVTIDG	YNIPAKTRFF	INAWAIGRDS	ELWENPEMFE	PERFMNSHID	420
FKGQDFELIP	FGAGRRICPA	ISFTVATVEL	ALAQLLHSFD	WELPPGVEAK	DLDMTEAFGI	480
TMHRTSPLMV	VAKPYFSDHM	N				501

SEQ ID NO: 166 moltype = AA length = 497
 FEATURE Location/Qualifiers
 source 1..497
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 166

MAALPLFIFA	SISLVPVLVI	LLMKKDKRSV	KLPPSPPKLP	IIGNLHQIGN	MPHESFHNLA	60
KKFGPIMFLW	LGQVPTVVIS	SARLAKEALK	TSDDLALSSRP	ELFSAKHLHY	NCTDLAFSPY	120
GTYYWRNIRKI	TTLQLLSAKR	VQSFSFVRKD	EVAHLVKRIT	ESYPSTINLS	HMLEVYANNN	180
ICRVAFGKDF	SQGGDNVRHN	FQKVLLEEQM	LLGGFSIGDY	FPSMEFLHKF	TGMESRLKDT	240
FRRFDHICDK	ILEEHRKTEN	RKDQKDLVDV	LLEIQHNGDL	ETPLTTDNK	DMFVGGTDTN	300
FITLDWTMT	LIINPKVMQR	AQAEVRSIVG	ERRFVSENDL	SHLGYMKAVV	KETFRLHPPA	360
PLLAPRESIQ	DVTIDGYNIP	AKTRFFINAW	AIGRDSELWE	NPEMFEPERF	MNSHIDFKGQ	420
DFELIPFGAG	RRICPAISFT	VATVELALAQ	LLHGPDWELP	PGVEAKDLDM	TEAFGITMHR	480

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TSPLMVVAKP YFSDHLN 497

SEQ ID NO: 167 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 167

MAALPLLIFS	SLFLALLLKL	LLKHKNNRNS	KLPPCPPKLP	IIGNLHQIGN	MPHESFNNLA	60
KKFGPIMFLW	LGQVPTVVIS	SARLAKEALK	TFDLALSSRP	ELFSAKHLHY	NCTDLAFSPY	120
GTYWRNIRKI	TTLQLLSAKR	VQSFSFVRKD	EVAVHLVKRIT	ESYPSTINLS	HMLEVYANNN	180
ICRVAFGKDF	SQGGDNVRHN	FQKVLEEQM	LLGGFSIGDY	FPSMEFLHKF	TGMESRLKDT	240
FRSFDHICDK	ILEEHRKTEN	RKDQKDLVDV	LLEIQHNGDL	ETPLTTDNIK	AIIDMFVGG	300
TDTNLITLDW	TMTIELINPK	VMQRAQAEVR	SIVGERRFVS	ENDLSHLGYM	KAVVKETFRL	360
HPPAPLLVPR	ESIQDVTIDG	YYIPAKTRFF	INAWAIGRDP	ELWENPEMFD	PERFMNSHID	420
FKGQDFELIP	FGAGRRICPA	ITFSIANVEL	ALAQLLHSFD	WELPPGVEAK	DLDMTEAFGI	480
TMHRISPLMV	VAKPYLS DYL	N				501

SEQ ID NO: 168 moltype = AA length = 497
 FEATURE Location/Qualifiers
 source 1..497
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 168

MAALPLFIFA	SISLVVLVI	LSMKDKRSV	KLPPSPKLP	IIGNLHQIGN	MPHESFNNLA	60
KKFGPIMFLW	LGQVPTVVIS	SARLAKEALK	TFDLALSSRP	ELFSAKHLHY	NCTDLAFSPY	120
GTYWRNIRKI	TTLQLLSAKR	VQSFSFVRKD	EVVHLVKRIT	ESYPSTINLS	HVLEVYANNN	180
ICPVAFGKDF	SQGGDNVRHN	FQKVLEEQM	LLGGFSIGDY	FPSMEFLHKF	TGMESRLKDT	240
LRRFDHICDK	ILEEHRKTEN	RKDQKDLVDV	LLEIQHNGDL	ETPLTTDNIK	DMFVGGTDTN	300
SITLDWTMTE	LIMNPKVMQR	AQAEVRSIVG	ERRFVSENDL	SHLGYMKAVV	KETFRHLPPA	360
PLLAPRESMQ	DVTIDGYNIP	AKTRFFINAW	AIGRDSSELWE	NPEMFEPERF	MNSHIDFKGQ	420
DFELIPPGAG	RRICPAISFT	VATVELALAQ	LLHSFDWELP	PGVEAKDLDM	TEAFGITMHR	480
TSPLMVVAKP	YFSDHMM					497

SEQ ID NO: 169 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 169

MAALPLLIFA	SLSLIVLVKF	LLKPKNNRNS	KLPPCPPKLP	IIGNLHLLGN	MPQESLHQLA	60
KKYGPPIIFLQ	LGEVPTVVIS	SARLAKEAMK	TFDLALSSRP	QIFS AKHLFY	NCTDVVFSPY	120
GAYWRNIRKI	CILELLSAKR	VQSFSYVREE	EVAVHLVQVA	EAYPGTTNLT	KMLGLYANNL	180
VCRVAFGKDF	SQGGDNHRHN	FQKLLEEQV	LLGGFSIGDF	FPSMEFVHKF	TGMESRLKDT	240
FRRFQDFPDE	ILQEHRNPES	RKEHKDLVDV	LLEIQKNGDP	ETPLTTDNIK	AIILD MF AAG	300
TDTNFITLDW	AMTELMTNPG	VLKRAQAEVR	NIVGERRSVS	ENDLPQLPYV	KAVVKETFRL	360
HPPAPVLVPR	ESMEDVTIDG	YDIPAKTRFF	INAWAIGRDP	KSWENPDVFE	PERFMNSHID	420
FKGQDYELIP	FGAGRRSCPA	ITFSAVTVEL	ALAQLLHSFD	WELPPGVEAK	DLDMTEVFGI	480
TMHKIYPLMV	VAKPYFSQGL	N				501

SEQ ID NO: 170 moltype = AA length = 517
 FEATURE Location/Qualifiers
 source 1..517
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 170

MAALIEDHST	QASVFILMEL	FLTLCALLAF	FSIALKIVRK	TKLVKLPPGP	RPLPIIGNLH	60
QLNGSLPHRI	LTDLAKKYGP	LMHLQLGEVS	TVIVSSPAVA	KEVMKDHDII	FCYRPLLLSP	120
AILNYGGTDI	AFAPYGEYWR	QLRKICTIEL	LSSKRVQTFR	SIREQEVSNL	FRSIKSAEGS	180
VVNL SKMIYS	LTYSITAAQA	FGKESKYQDK	FISVMQDAIK	LMGGFNIA DV	YPSIKILEKI	240
TGIRHKLEET	HNQIDEIVQN	ILIEHKAKRA	ESKLGNRGAK	EDLVDVLLNI	QSSGEFEPPL	300
TDDNLKAVIY	DIFSAGGETS	STTVIWMASE	MIRNPKVMKK	AQDEVREVFA	GNGNVDESRL	360
HELKYLQAVI	KETMRLNPAV	PLLLPPRECSE	HCQINGYDVP	AKTRVIVNAW	AIGRDPKQWT	420
EPEKFKPDRF	LDSEIDFRGK	FSFIFPGAG	RRICPGISFA	LPLVELPLAQ	LLYHFDWKLP	480
GQSKPDELDM	TEIFGLTVGR	KHDLNLVPI	YRPCSVL			517

SEQ ID NO: 171 moltype = AA length = 508
 FEATURE Location/Qualifiers
 source 1..508
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 171

MAALFEWEKE	GFPYPYLLIF	ASLSLIVVLK	FLLKPKNNRN	SKLPPCPPKL	PIIGNLHLLG	60
NMPQESLHQL	AKKYGPPIIFL	QLGEVPTVVI	SSARLAKEAM	KTFDLALSSR	PQIFS AKHLF	120
YNCTDVVFSP	YGAYWRNIRK	ICILELLSAK	RVQSFSYVRE	EEVAVHLVQR	AEAYPGTTNL	180
TKMLGLYANN	LVCRAVFGKD	FSQGGDNHRH	NFQKLEEQYQ	VLLGGFSIGD	FFPSMEFVHK	240

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FTGMESRLKD	TFRFRDQFFD	EILQEHRNPE	SRKEHKDLVD	VLEIQKNGD	PETPLTTDNI	300
KAIILDMFAA	GTDTNFITLD	WAMTELMTNP	GVLKRAQAEV	RNIVGERRSV	SENDLPQLPY	360
VKAVVKETFR	LHPPAPVLVP	RESMEDVTID	GVDIPAKTRF	FINAWAIGRD	PKSWENPDVF	420
EPERFMNSHI	DFKGQDYELI	PFAGARRSCP	AITFSAVTVE	LALAQLLHSF	DWELPPGVEA	480
KDLDMTEVFG	ITMHKIYPLM	VVAKPYFS				508

SEQ ID NO: 172 moltype = AA length = 321
 FEATURE Location/Qualifiers
 source 1..321
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 172

MAALIEDHST	QASVFILMEL	FLTLCALLAF	FSIALKIVRK	TKLVKLPPGP	RPLPIIGNLH	60
QLNGSLPHRI	LTDLAKKYGP	LMHLQLGEVS	TVIVSSPAVA	KEVMKDHDII	FCYRPLLLSP	120
AILNYGGTDI	AFAPYGEYWR	QLRKICTIEL	LSSKRVQTFR	SIREQEVSNL	FRSIKSAEGS	180
VVNLSKMIYS	LTYSITAQAA	FGKESKYQDK	FISVMQDAIK	LMGGFNADV	YPSIKILEKI	240
TGIRHKLEET	HNQIDEIVQN	ILIEHKAKRA	ESKLGNRGAK	EDLVDVLLNI	QSSGEFEPPL	300
TDDNLKAVIY	VSLFYSPSHI	K				321

SEQ ID NO: 173 moltype = AA length = 465
 FEATURE Location/Qualifiers
 source 1..465
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 173

MAALDPPWQK	ESFLHPFIPA	SISLVPVLVI	LLMKKDKRSV	KLPPSPPKLP	IIGNLHQIGN	60
MPHESFHILA	KKGPIIMFLW	LGGVPTVVIS	SARLAKEALK	TSIDLALSSRP	ELFSAKHLHY	120
NCTDLAFSPY	GTYWRNIRKI	TTLQLLSAKR	VQSFSFVRKD	EVAHLVKRIT	ESYPSTINLS	180
HMLEVYANNN	ICRVAFGKDF	SQGGDNVRHN	FQKVLVEEYQM	LLGGFSIGDY	FPSMEFLHKF	240
TGMESRLKDT	FRFRDHICDK	ILEEHRKTEN	RKDQKDLVDV	LLEIQHNGDL	ETPLTTDNIK	300
AIIDIMFVGG	TDTNFITLDW	TMTELIINPK	VMQRAQAEVR	SIVGERRFVS	ENDLSHLGYM	360
KAVVKETFRL	HPPAPLLAPR	ESIQDVTIDG	YNIPFGAGR	ICPAISFTVA	TVELALAQLL	420
HSFDWELPPG	VEAKDLDMTE	AFGITMHRTS	PLMVVAKPYF	SDHLN		465

SEQ ID NO: 174 moltype = AA length = 536
 FEATURE Location/Qualifiers
 source 1..536
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 174

MAANVKANKH	NIVCILCLFL	LTMAKQIFVL	DLPPFSSLFPL	ALLLLTLVIW	FYASSRPQKR	60
LPPSPPKLPI	VGHLEFRLGIY	PHRTLQRLSK	KYGPLMMIHL	GSKPILVASS	ADVACEIMKN	120
HDLVFANRPK	RSINSKLLYD	CKDIATAQYG	EYWRQIRSI	VLQLLSNKR	QSFHVRREEE	180
TSMMIQKIRE	RCSSSSSSLI	NLSDALITLT	NDVICRVALG	RKYSNDENGR	KSMKIMKEFG	240
ELLGTFSVGD	YLPWLAWNR	FNLGLDAKLEK	VAKHIDQFLE	GVIIEHKFRK	KGEDYVDSTE	300
KKALDLVDIL	LDIQKDNSVG	LALGLDSVKA	IILDMFAGGT	DTSYTAMEWA	MSELLRDPKT	360
LEKLQVEVRK	VAQKKEITE	DDLDMHYLK	AVIKETLRH	TPIPLLVPRE	STQHVKILGY	420
DIPAGTEVMV	NAWAIARDPV	QWKKPEEFLP	ERFLNSTIDF	RGFNFEFIPF	GAGRRGCPGA	480
TFAIAVNELA	LAKLLHEFDF	ALPDGAKPED	LDMSDATGFT	VHRKFPLLAV	AIPHSC	536

SEQ ID NO: 175 moltype = AA length = 539
 FEATURE Location/Qualifiers
 source 1..539
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 175

MAAVRILKKK	KFQQTILNTK	KEMSIFQFEE	ITAGLSKHFP	SLLLLLLISL	LLLRWKNSTS	60
DTTQNLPPSP	PQLPIIGNLH	QLGSIPQRL	KSLAQKYGSL	MLLRLGSKPV	LISSADAAE	120
EVLKTHDLIF	ANRPKPDFVG	RLLYNFKDIA	FSPYGGYWRH	IRSICVLQLL	TNKRVSQFRN	180
IREEEVELML	EKIRESCASS	SIILIGEILA	TLTNNIVSRV	ATGKRYSEEE	SGSKFKDLFE	240
EFSMLLGVEN	VGDYIPWLAW	INHINGLEAK	VKRVGKDFDE	YMDKIVQEGI	IRQEKRGSG	300
GDHEKEQQNL	VDVLELQRT	NSAGFAIGRD	SIKAIITNMF	VGGTDTTFSL	LEWAISELLR	360
NTNAMHKLQK	EVQRFLGRKS	RTEDDLLENL	HYLKAVIKEA	LRLHPPAPLL	IPRESSKAVK	420
LMGYNIAAGT	QVIINAWAIG	RDPKLWDDAG	EFRPERFLNS	SVDINGQHFE	LIPFGSGRRS	480
CPGTAFAMIT	AELALANLIC	NPDFELAGGA	RPEDLDMTEA	PGIITPRKVP	LRLVASLPN	539

SEQ ID NO: 176 moltype = AA length = 515
 FEATURE Location/Qualifiers
 source 1..515
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 176

MAVSFPAFQD	EAKRWLSNHS	VAVSLLLPSF	MIFILFLSKW	WFLTTSKKKN	LPPSPPLRLPI	60
IGNLHQIGSV	AHSFKSLAE	KHGPLMLLHV	GRQKLLVSS	AETAREVLKT	HDVAFCGRPD	120
SEVTRIRIFYN	LKNISLSPYG	DYWKIVRSIA	VNQILNQKRV	QSFRSVREEE	ISLLVERIKE	180
SCASSVICM	NTLLTTLVND	IVSRITIGKR	SSGKGSRFA	ELLFASTELF	GSPLAGDFFP	240

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GLGWLGVIYTG	FEAKTKKIAK	DLDQCLNLI	EEQINRNKRG	DNQKDNQNF	LQVLEIQR	300
DSSGHALDRE	SIKAIIMDMV	AGGFDYSL	EWATSLLVKH	PDVMKKLQNE	VKEVAGSKSF	360
ISEDLSKLQ	YLKAVIKETF	RLCPGAFIVR	VSTKDVKIMG	YDIAAGTQVL	VNALAIGRDP	420
TLWEKPEEFR	PERFLNSSID	LKGHHFELL	FGSGRRSCPG	STFALVIDEL	VLANLVCNFN	480
YALPGGAKAE	DLDMTEAPGA	MPPRRTPLLL	VPSLC			515

SEQ ID NO: 177 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 177

MAELFLTLC	LLAFFSIALK	IVRKTCLVKL	PPGPRPLPII	GNLHQLNGSL	PHRILTDLAK	60
KYGPLMHLQ	GEVSTVISS	PAVAKEVMKD	HDIIFCYRPL	LLSPAILNYG	GTDIAFAPYG	120
EYWRQLRRIC	TIELSSSKRV	QTFRSIREQE	VSNLFRSIS	AEGSVVNLK	MIYSLTYSIT	180
AQAAGKESK	YQDKFISVMQ	DAIKLMGGFN	IADVPSIKI	LEKITGIRHK	LEETHNQIDE	240
IVQNILIEHK	AKRAESKLG	RGAKEDLVDV	LLNIQSSGEF	EPPLTDDNLK	AVIYDIFSAG	300
GETSSTTVIW	AMSEMIRNPK	VMKKAQDEVR	EVFAGNGNVD	ESRLHELKYL	QAVIKETMRL	360
NPAVPLLLPR	ECSEHCQING	YDVPAKTRVI	VNAWAIGRDP	KQWTEPEKFK	PDRFLDSEID	420
FRGKTFSFIP	FGAGRRICPG	ISFALPLVEL	PLAQLLYHFD	WKLPGQSKPD	ELDMTEIFGL	480
TVGRKHDNL	VPPIYRPSV	L				501

SEQ ID NO: 178 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 178

MAELFLTLC	LLAFFSIAIK	IVRKTCPVKL	PPGPRPLPII	GNLHQLNGSL	PHRILTDLAK	60
KYGPLMHLQ	GEVSTVISS	PAVAKEVMKD	HDIIFCYRPL	LLSPAILNYG	GTDIAFAPYG	120
EYWRQLRRIC	TIELSSSKRV	QTFRSIRERE	VSNLFRSIS	EESVNVLSK	RIYSLTYSIT	180
AQAAGKESK	YQDKFISVMQ	DAIKLMGGFN	IADVPSIKI	LEKITGIRHK	LEEVHNQVDE	240
IVQNILIEHK	AKRAESKLG	RGAKEDLVDV	LLNIQSSGEF	EAPLTDDNLK	AVIYDIFSAG	300
GETSSTTVIW	AMSEMIRNPK	VMKKAQDEVR	EVFAGNGNVD	ESRLHELKYL	QAVIKETMRL	360
NPAVPLLLPR	ECSEHCQING	YDVPAKTRVI	VNAWAIGRDP	KQWTEPEKFK	PDRFLDSEID	420
FRGKTFSFIP	FGAGRRICPG	ISFALPLVEL	PLAQLLYHFD	WKLPGQLKPD	ELDMTEIFGL	480
TVGRKHDNL	VPPIYRPSV	L				501

SEQ ID NO: 179 moltype = AA length = 535
 FEATURE Location/Qualifiers
 source 1..535
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 179

MAISSSIKSI	LNKLITIAQLA	LFLLQSSMDF	IIFFGALLLF	YLMVAKIHRK	STGENSKLNL	60
PPGPKPLPII	GNMHQLHGSP	IHHLLRDLAK	KYGPLMHLKL	GEISAIIVTS	PEMAKEIYKT	120
NDLLFASRPS	THLAFKIISY	NFTGIIYSAY	GNHWRQLRKI	CTAELLSPKR	VQTFKSIRE	180
EVFNLIQLIY	SQKGSIIINLS	RCIFLLSYSI	TSRIAFGKRN	KYTERFIQLV	EEINKLVSFG	240
SLADMYPSVK	LLQVLSIMRY	KLIAHRQVD	QILENIINDH	KQKITEAKQE	GGERKEDFVD	300
VLLIQEGGD	SEPQLTDAAI	KAVIFDIFSA	GSETSSSTAIE	WALSELIRNP	EVIKRAQDEV	360
RNVYDERGNV	DESSLHELKY	LHAIFKETLR	LHPSAPLLLP	RECGEQCEIN	GYVIPAKARI	420
LVNAWAIGRD	PMYWEAEKF	IPERFLDSEI	DFKGNNFEYI	PFGAGRRICP	GISFSQPHVE	480
LVLQQLLPHF	DWKLPGLKQ	EELDMTEDFG	VTTKRSNDLH	LIPYINRSC	LRMDA	535

SEQ ID NO: 180 moltype = AA length = 503
 FEATURE Location/Qualifiers
 source 1..503
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 180

MAEFIIFLGA	LLLFYFVISL	GKSSKLKLPP	GPKQLPIIGN	MHQLHGSQHL	HLLRDLAKIY	60
GPLMHLKLGE	ISTIIVTSPE	MAREIYKTND	LLFASRPSHY	LAFKIFSYNF	ADIAFSPYGN	120
YWRQMRKICT	LELLSTKRQV	TFQSIREDEV	FNLMKSISTQ	KGSIIINLSR	IFTVIYTITS	180
RAAFGKRNDK	KERYIRLVEE	VTKLASWFSL	ADVYPSVKLF	QLMSTVRHKA	AIVHKQVDEI	240
IENILNDHKE	KIEBAKQASG	EAKEDLVDVL	LSIQNSGDFE	PQLTDNIKA	IIFDMFTAGG	300
ETSSTVVEWA	ISEMIKNPET	IKRAQNEVRN	AFDEKGNVDE	SKLHELKYL	AIKETLRLH	360
PSAPLLVPRE	CTEQCEINGY	VIPAKARLFV	NAWAIGRDPK	YWIDAENFNP	ERFLDSKVYD	420
RGNDFEYIPF	GAGRRICPGI	TFSRPGIELA	LAQLLFHFDW	KVPGLKQEE	LDMTENFGMT	480
VRKNDNLNLI	PVLYNRSCLS	KDN				503

SEQ ID NO: 181 moltype = AA length = 512
 FEATURE Location/Qualifiers
 source 1..512
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 181

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MAALFEWEKE	GFPSPYLLIF	ASVSLIVVLK	FVSKPKNNRN	SKLPCCPKL	PIIGNLHLLG	60
NMPQESLHQL	AKKYGPVIFL	QLGEVPTIVI	SSARLAKEAM	KTFDLALSSR	PQIFSAKHLF	120
YNCTDVVFSP	YGAYWRNIRK	ICILELLSAK	RVQSFSYVRE	EEVAHLVQRV	AEAYPGTTNL	180
TKMLGLYANN	LVCRAVAFGKD	FSQGGDNHRH	NFQKLEELYQ	VLLGGFSIGD	FFPSMEFVHK	240
FTGMESRLKD	TFRFRDQFFD	EILQEHRNPE	SRKEHKDLVD	VLLLEIQKNGD	PETPLTTDNI	300
KSIIIDMFAA	GTDTNFITLD	WAMTELMTNP	GVLKRAQAEV	RNIVGERRSV	SENDLPQLPY	360
VKAVVKETFR	LHPPAPVLVP	RESMEDVTID	GVDIPAKTRF	FINAWAIGRD	PKSWENPDVF	420
EPERFMNSHI	DFKGQDYELI	PFAGARRSCP	AITFSAVTVE	LALAQLLHSF	DWELPPGVEA	480
KDLDMTEVFG	ITMHKIYPLM	VVAKPYFSQG	LN			512

SEQ ID NO: 182 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 182

MAELFLTICA	LLAFFSIALK	IVRKTKLVKL	PPGPRPLPII	GNLHQLNGSL	PHRILTDLAK	60
KYGPLMLHLQ	GEVSTVIVSS	PAVAKEVMKD	HDIIFCYRPL	LLSPAILNYG	GTDIAFAPYG	120
EYWRQLRKIC	TIELLSSKRV	QTFRSIREQE	VSNLFRSIS	AEGSVVNLK	MIYSLTYSIT	180
AQAAPGKESK	YQDKFISVMQ	DAIKLMGGFN	IADVYPSIKI	LEKITGIRHK	LEETHNQIDE	240
IVQNILIEHK	AKRAESKLG	RGAKEDLVDV	LLNIQSSGEF	EPPLTDDNLK	AVIYDIFSAG	300
GETSSTTVIW	AMSEMIRNPK	VMKKAQDEVR	EVFAGNGNVD	ESRLHELKYL	QAVIKETMRL	360
NPAVPLLLPR	ECSEHCQING	YDVPKATRVI	VNAWAIGRDP	KQWTEPEKFK	PDRFLDSEID	420
FRGKTFSFIP	FGARRRICPG	ISFALPLVEL	PLAQLLYHFD	WKLPGQSKPD	ELDMTEIFGL	480
TVGRKHDNL	VPPIYRCSV	L				501

SEQ ID NO: 183 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 183

MAELFLTICA	LLAFFSIAIK	MVNSKPDNL	PPGPMQLPII	GNIHQLFGSL	AYRILTGLAK	60
KHGPLMLHLQ	GEISTIVVSL	AEVAKEVMNE	HDIIFASREN	LLGTRIHSYN	ATKIAFSPYG	120
DYWRQLCKIC	TMELLSPKRV	ETFRSIREEA	VLDMIKSIAL	QRGSDVNLK	EIPSLTYRIT	180
AQAAPGKRNK	HQEEFISVMV	DIVELVAGFN	IADMYPSVKI	LERISGMRQK	LERTHKQVDE	240
ILENITLIEHK	GKRDGLQPGK	GDSKEDLVDV	LLKVQKSGEF	GAPLSNNNLK	AVILDIFSAG	300
DESSSTTAQW	TMLEMMKNPG	TIKRSQEEVR	EVFGDKGNVD	ESRLHELKYL	QAVIKDALRL	360
RPPVPLLLPR	ECAEPCEIHG	YEIPAKARM	VNAWAIGRDS	RRWTELEKFI	SERFLDSYD	420
FRGQNFYIP	FGARRRICPG	IAPALPNVEL	LLAQLLYHFD	WKLPGEIKEQ	QLDMTETFG	480
AVRPKHVDNL	VPISFSPSDE	N				501

SEQ ID NO: 184 moltype = AA length = 509
 FEATURE Location/Qualifiers
 source 1..509
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 184

MADLIIFLSA	FLLFYLVVVK	LLRKPTGKNS	KLNIIPGPKP	LPIIGNMHQL	HGSPIHLMR	60
DLAKKYGPLM	HLKGEISAI	IVTSPMAKE	IYKTNLLFA	SRPSSHLAFK	IISYNFADII	120
FSAYGNHWRQ	LRKICTVELL	SPKRVQTFKS	VREYEVFKLI	QSISSKRGS	INLSRSIFPL	180
SYSITSRIAF	GKRNKYTERF	IQIVEEINKL	VSGFSLADMY	PSVKLLRVMS	IMRYKLVIAH	240
KQVDQILENI	LNDHKQNVTE	AKQEGGERKE	DLVDVLLNIQ	EGGDFEPELT	GTAIKAVLFD	300
IFSAGSETSS	TAIEWALSEM	IRNPEVIRRA	QDEVNRVYDE	RGNVDESSLH	ELKYLRAIHK	360
ETLRLHPSAP	LLLPRECEQ	CEINGYVIPA	KARIIVNAWA	IGRDPYWAE	AEKFNPERFL	420
DSEIDFKGNN	LEYIPFGAGR	RICPGISFSQ	PHMELVLAQL	LFHFDWKLPG	ELKQEEELMT	480
EDFGVTIRRT	NDLHLIPIY	NRSCLRMDG				509

SEQ ID NO: 185 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 185

MAELFLTICA	LLAFFSIAIK	MVAITLKNK	PRDLPPGPMQ	LPIIGNIHQL	FGSLPHRILT	60
DQAKKYGPLM	HLQGEISTF	VVSSAEVAKE	VMNEHDITFA	SRPNLLGTRI	ISYNATDIAF	120
STYGDYWRQL	RKICTMELLS	PKRVQTFRSI	REEAVLDMVK	SLALQSGSIV	NLSKEIFSLT	180
YRITAQAAGF	KQKHQBEFI	SVMVDIVELV	AGFNIADMY	SVKILERISG	MSQKLERTHK	240
QVDEILENII	AEHKGKRDGL	HPVKGEVKED	LVDVLLKVQN	SGDFGAPLSD	SNPKRSSYAG	300
GETSSTTAQW	SMLEMMKNPG	IIKRAQEEVR	EVFGDKGNVD	ESRLHELKYL	PAVIKETLRL	360
HPPVPLLLPR	EGCEPCEIHG	YEIPAKARVL	VNAWAIGRDP	RHWTEPKFI	PERFLDSYD	420
FRGQNFYIP	FGARRRICPG	IAPGLPNVEL	PLAQLLYHFD	WNLPGEIGE	QVDMTETFG	480
AVRPKHGVHL	VPFSFSNSSE	N				501

SEQ ID NO: 186 moltype = AA length = 688
 FEATURE Location/Qualifiers

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source                1..688
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 186
MTTLLNPYFG EFGGMYPQI LMPALRQLEE AFVSAQKDPE FQAQFNDLLK NYAGRPTALT 60
KCQNITAGTN TTLYLKREDL LHGGAHKTNQ VLGQALLAKR MGKTEIIAET GAGQHGVASA 120
LASALLGLKC RIYMGAKDVE RQSPNVFRMR LMGAEVIPVH SGSATLKDAC NEALRDWSGS 180
YETAHYMLGT AAGPHPYPTI VREFQRMIGE ETKAQILERE GRLPDAVIAC VGGGSNAIGM 240
PADFINETNV GLIGVEPGGH GIETGEHGAP LKHGRVGIYF GMKAPMMQTE DGQIEESYSI 300
SAGLDFPSVG PQHAYLNSTG RADYVSITDD EALEAFKTLK LHEGIIPALE SSHALAHALK 360
MMRENPDKEQ LLVNVNLSGRG DKDIFTVHDI LKARGEIGGG SGEGRGSLLT CGDVEENPGP 420
MERYESLFAQ LKERKEGAFV PFVTLGDPGI EQSLKIIDTL IEAGADALEL GIPFSDPLAD 480
GPTIQNATLR AFAAGVTPAQ CFEMLALIRQ KHPTIPIGLL MYANLVFNKG IDEFYAQCEK 540
VGVDSVLVAD VPVEESAPFR QAALRHNVPV IFICPPNADD DLLRQIASYG RGYTYLLSRA 600
GVTGAENRAA LPLNHLVAKL KEYNAAPPLQ GFGISAPDQV KAAIDAGAAG AISGSAIVKI 660
IEQHINEPEK MLAALKVFVQ PMKAATRS 688

SEQ ID NO: 187        moltype = AA length = 131
FEATURE              Location/Qualifiers
source                1..131
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 187
MAAEQTTFTF SEIEKNNDKD KVLIIILHDKV FNVHRFLNEH PGGEEVLDDH KGKDGSEDFD 60
DVGHSKDAME LMKTYQGVVL VESERSNKPP KKGWVAGYNT KQSEKYVQPG VPFYLLIGGI 120
ALVVALFFYA Y 131

SEQ ID NO: 188        moltype = AA length = 135
FEATURE              Location/Qualifiers
source                1..135
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 188
MAASDQKLHK FDEVSKHNKT KDCWLIINGK VYDVTPFMDD HPGGDEVLLS ATGKDATNDF 60
EDVGHSDSAR EEMDKYYIGE MDMATVPLKR TYIPPQQAQY NPDKTPEFVI KILQFLVPLL 120
ILGLAFVRH YTKEK 135

SEQ ID NO: 189        moltype = AA length = 679
FEATURE              Location/Qualifiers
source                1..679
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 189
MAEKTEAILE TEHISEASDS LFSSLDIFLV ALIVGGISWY FINKYKKTIT VTSESYSIS 60
PSPITLENPS DSSFIKKLQS SGRSLVVFGY SQTGTAEFFA GRLAKEGARY RLKGMVADLE 120
DYEMEELVNL KNIPNSLAVF CLASYGEGEP TDNAREFCDW LQNSEPDLKG LTYTVFGLGN 180
KSYEYYNRVA INVDKRLKEL GATRIYDLGL GDDDANIEDD FITWKDQFWP TVCEHFGIEP 240
TEEEINMHQY RLEEFEEAPD KVFVGEAKL DSLKNQKPPH DAKNPFLAKV KVNRLDNNSD 300
RSLCHIEFDI KGSKMRYDSG DHTAVYPTND SELVEKIGKL CAKNLDITFT LINNDEESNK 360
KHPPFPCPSY RTALTHYLDI TQNPRTVLK ELAEYCTDEA NKEKLRLMSS TPEGKALYQ 420
SWIIEDNRSV VHVIEDMPCC RPSLDYLCEL LPRLQPRFYS ISSAKLYPD TVHISAVVVE 480
YDTPTRGRHNK GVAITWLKEK VPDPEKEAPV VPVFIRKSQF RLPTKPTPI IMIGPGTGLA 540
PFRGFIQERN ICKSEGKPVG ETVLYFGCRK RSEDFIYGDE LLEYEKNGLL TLHLAFSRDQ 600
PHKVYVTHLL QQHAEDEVVRV IGENNGHLYI CGDAKSMAVD VRVILTKILO EKQMQTEQEA 660
VAFLLKMETQ RRSADVWS 679

SEQ ID NO: 190        moltype = AA length = 682
FEATURE              Location/Qualifiers
source                1..682
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 190
MAENSEGVKI ESTVVTEEPL ISALDIALFV VIITVAYFWY FKRDQSSSS EKKPYTIQPS 60
SLSSIEQTSN SSFIKKLKST GRSLVVFYGS QTGTAEFFAG RIAKEGARYK MKGMVADPEE 120
CDMADLVEMK EIQKSMAIFC VATYGEQDPT DNAMDFYEWL QNGDADLEGL NYAVFGLGNK 180
TYEHYNEIAI YIDQRLEELG ATRVHEIGLG DDDANIEDDF VSWKEKLWES VCSHYGIEET 240
GEESNIRQYK LVDCSEILPE RIFSGEISRL KSYENQRPFP DVKNPYLSKI AVNRELHKSG 300
DRSCMHIEFD IDGSKMRYDT GDHVAVYPKN SSELVEKIGE LLNADLDTVF SLLNTDEESS 360
KKHPPPCPCT YRTALTYLTD ITSNPRTIM KELIEYASDP KDQEKLLKMA SSTPEGKKEF 420
HDWILDRNRN IVHILEDLPS VKPDLHLCE LLPRLQCRYI SISSPKVYP KSIHITAVLV 480
NYTPTNRVN KGVATNLLAQ LKPTNDELPO PTIPIYIRS QFRLPPKSQT PIIMIGPGTG 540
LAPFRGFIQE RDYARKEGRE IGMVLYFGC RKKDEDFIYE NELQEYVANG TLTKHLAFS 600
RDQPEKIYVT HLLQONADEL WNIIGEKNKH LYVCGDARSM AKDVHNIIVK VVMEKQGMTN 660
SQALNYVKKM EQQKRYSDV WS 682

SEQ ID NO: 191        moltype = AA length = 715

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FEATURE	Location/Qualifiers
source	1..715 mol_type = protein organism = synthetic construct
SEQUENCE: 191	
MADSSSEKLS PFELMSAILK GAKLDGSNSS DSGVAVSPAV MAMLLENKEL VMILTTSVAV	60
LIGCVVVLIIW RRSSGSGKKV VEPKLIIVPK SVVEPEEIDE GKKEFTIFFG TQTGTAEFGA	120
KALAEAAKAR YEKAVIKVID IDDYAADDEE YEEKPRKETL AFFILATYGD GEPTDNAARF	180
YKWFVEGNDR GDWLKNLQYG VFGLGNRQYE HFNKIAKVVD EKVAEQGGKR IVPLVLGDDD	240
QCIEDDFAAW RENVWPELDN LLRDEDDTV STTYTAAIPE YRVVFPDKSD SLISEANGHA	300
NGYANGNTVY DAQHPCRSNV AVRKELHTPA SDRSCTHLDF DIAGTGLSYG TGDHVGVCYD	360
NLSETVEEAE RLLNLPPETY FSLHADKEDG TPLAGSSLPP PFPPTLRTA LTRYADLLNT	420
PKKSALLALA AYASDPNEAD RLKYLASPAG KDEYAQSLVA NQSLLEVMA EFPSAKPPLG	480
VFFAAIAPRL QPRFYSISS PRMAPSRIHV TCALVYEKTP GGRIHKGVCS TWMKNAIPLE	540
ESRDSCSWAPI FVRQSNFKLP ADPKVPVIMI GPGTGLAPFR GFLQERLALK EEGAELGTAV	600
FFFGCRNRKM DYIYEDELNH FLEIGALSEL LVAFSREGPT KQYVQHKMAE KASDIWRMIS	660
DGAYVYVCGD AKGMARDVHR TLHTIAQEQG SMDSTQAEFG VKNLQMTGRY LRDVW	715
SEQ ID NO: 192	moltype = AA length = 697
FEATURE	Location/Qualifiers
source	1..697 mol_type = protein organism = synthetic construct
SEQUENCE: 192	
MAHNMDLPTV DASQPKAIR AGSPMLENES GPEDEQPFLL ITDIMLILGI LLAMMFVYVR	60
SRMRREAAPA PKSYSIQPTS FAVAPSSENS FIKKLKTSGR SLVVFGYSQT GTAEFPAGRL	120
AKEGIRYMKM GMVADPEECN MEELVHLKTI PNSLAVFCLA TYGEGDPTDN AMEFIDWLKN	180
GDADLTLENY AVFGLGNKTY EHYNEVGIVV DHRLEQLGAT RVFELGLGDD DANIEDDFIT	240
WKDKFWPAVC EFFGIEGAGE DVSIRQYKLT EHVDMPAERI YTGEFARLHS FKNQRPYPDA	300
KNPFLAPVIV NRELHGPTSE RSCMHIEFYI EDSKIRYDAG DHLAVYPVNN AKLVNKIGEQ	360
CGANLDIVFT LTNTDEESTK KHFPFPCPTY RTALTYYLDI TSNPRTHILK ELAEYATDPA	420
DKEKLKLMAS TTAEGKAAVQ QWIIQENRNI VHILEDIPSL KPPLDHLCEI LPRLQCRYYS	480
ISSSPKLHPT TIHITAVVVE YKTPTGRINR GVTTSWLKEK HPSDPPCLVP IFLRKSQFRL	540
PTRPNIPIM VPGTGLAPF RGFIQERDFA RKEGKEVGDT LLYFGCRKSQ EDFLYRDELE	600
EYVKNGLTTL HTAFSREQSH KVVVTHLEK NKEEIIWHVIG EQNGHIYVCG DARNMARDVH	660
NILLKVMER GNMSELDAA N YIKKMSQKR YSSDVWS	697
SEQ ID NO: 193	moltype = AA length = 588
FEATURE	Location/Qualifiers
source	1..588 mol_type = protein organism = synthetic construct
SEQUENCE: 193	
MAEVEKTEKN IAEDDESKII YRGWKVMPFI IGNETFEKLG IVGSSSNLVI YLTTVFNMKS	60
ITAAKVNIIY GGTSNFGTIV AAFLCDSYFG RYKTLSFAMI ACFLGVSAMD LTAVIHPLHP	120
AQCAKEIGSV CNGPSIGQIM FLAGAMVLLV IGAGGIRPCN LPFGADQFDP KTEGKRGIE	180
SFFNWYFFTF TFAQMVSLTL IVYVQSNVSW SIGLAIPAIL MLLGCIIFFA GSKLYVKVKA	240
SGSPIHSITR VIVVAIKRR LKPVGPNELY NYIASDFKNS KLGTEQFRF LDKSAIQTDQ	300
DKLNKDGSPV DAWKLCMQQ VEEVKCIVIR LPVWLSAALF YLAYIQTTY TIFQSLQSDR	360
RLGPGSFQIP AGSYTVFLML GMTIFIFIYD RVLVPFLRKY TGRDGGITQL QRVGAGLFLC	420
ITSMVSAIV EQYRRKVALT KPTLGLAPRK GAISMSGMW LIPQLVLMGI ADALAGVGQM	480
EFYKQFPEN MRSFAGSLYY CGIGLASLYS TFLLSAVHDT TEGFSGGSWL PEDLNKGRLE	540
YFYFLVAGMM TLNLAYFLLV SHWYRYKDVV AKDKMDKTS AEFDKVSV	588
SEQ ID NO: 194	moltype = AA length = 596
FEATURE	Location/Qualifiers
source	1..596 mol_type = protein organism = synthetic construct
SEQUENCE: 194	
MAAEKSSSID ATENGFKV K EPNHRGVKAM PFIVGNETFE KLGTLGTSAN LLIYLTTFVH	60
MKSITATNLI NIFNGTCNFG TLLGAFFSDT YFGRYKTLGF ASVSSTLGML ILTLTAAISK	120
LHPPSCEKNS KACSGPTSGQ FAFLLTAFLF LIIGASGIRP CNLAFGADQF NPNTESGRKG	180
TNSFFNWYFF TQTFAMVSL TVIVYVQSSI SWALGLAIPT FLMLLSCVFF FVGSRIYKV	240
LPEGSPLTSM MQLVIAAFRK RKLNLDPHQF WDSLFNYISP SSINSKLSYT DQFRWLNKAA	300
IKTSKDHLYA EGSANPWRL CSMQQVEEVK CVIRSIPIWT SGVLYYVVLN QMSTYLVFQA	360
LQSDRRIGKT NFKIPAASYT VFSMLSLSIW IPIYDRIIV FLRKITKTEQ GITILQKMGF	420
GLVIAVITMV VSAIIEKRR VIALTRPIVG IEERKGGISS LSGFWLVPQL ALAGISEAFT	480
VIAQVEFFYK QFPENMRSPG GSPLFCGFAI SSYLSSFLIS IVHKVTRNSS GGNWLDLEDN	540
KARLDYFFYL VAGLEVNLFG YFLVCAHWYR YKETQSGNEI AMEQILDEPK KNLDV	596
SEQ ID NO: 195	moltype = AA length = 589
FEATURE	Location/Qualifiers
source	1..589 mol_type = protein organism = synthetic construct

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SEQUENCE: 195
MAEENQEHKA IKNHEPNYRG IKAMPFVIGN ETFEKLGTIG TSTNLLIYLT SVFHMKTITA 60
TNLVNIFNGT CNFGTLVGAF LCDTYFGRYN TLGFASVSSF LGMLVLTTLTA AIKNLHPEAC 120
GNEEGSKCIS PSGGQMAFL LSGFAFLVGA SGIRPCNLAF GADQFNPTD SGRRGINSFF 180
NYYYFTYTFM MMVSLTIIVY VQSNISWSIG LAIPTFLMFL SCATFFVGTTR IYVRVLPESG 240
PLISVARVIV SAFKKRKLLE PDEPKLSLFP HSASSTSSIN SKISYTDQFR FLNKAIIITS 300
EDEFNSDGSV KNPWRICITQ QVEEVKCLLK LIPVWISCII YYISVVQQQN YVVFQALQSD 360
RRLFGTSKFK IPAASYIVIA MLALTIWIPI FDRIIIVPCLQ KITGKEGGIT ILQRMGIGMV 420
LSVLTMITSA LVESKRRKFA LNRPTLGFEP RKGAISSMSG VWLIPQLALS GLSEAFITIG 480
ENELFYKECP ENMRSIAMAL VVGFAGSSY LSSLLSSIVQ KTTGGNESWL AEDLNRGRDL 540
YFYLLIGALE TLNLIYFLVC SKWYKYKQVE KIPQVAMEKL NNVEEKSIV 589

SEQ ID NO: 196 moltype = AA length = 594
FEATURE Location/Qualifiers
source 1..594
mol_type = protein
organism = synthetic construct

SEQUENCE: 196
MAEKNKLVLP NGTKEDGKVA AKNEPNYRGI KAMPFVLGNE TCEKLGITG SSNLSVYLST 60
VFHMTMSAN NVINIFNGTC YLGSVLGAF CDTYFGRYKT LGLGSVASFL GMLMLTLTAA 120
ISTLHPTNCD NSGDCGKPSA GQMAFLVGGF VLMVIGASGI RPCNLAFGAD QFNPTESGR 180
RGINSFFNWW YFTYTFVAVM SATIIIVYVQS DISWSIGFAI PLSLMLFLSCA LFFAGTLYV 240
KVLPGQSPMT SLVQVIVVSV KKRRLKLPGE PSLLFNHIPS NAINSKLPRT NNPRFLDKAA 300
ILTPEDKVP DGSAAKWK LCSIQQVEEVK CVLRVLPWI AGVIFQISVT QQSNYVFPQA 360
FQSDRHLIKG NKQIPAGTY NVFSMLALTI WLPYIDRII PLLRRWTKKE DGITLLQRMG 420
VGIVLSILSM LVSALVESRR RNALTRPTL GIQEGQSGS SMSALWFVPQ LALMGLAEGF 480
TFIGENELFY KEPPENMRSI AASLVLCGVA VASYLTSPIA EIVHKTRSG TRDWLAQDLN 540
KARLDYFFYM VAVLEALNLV YFLVCAKWYR YKGTGDLGKL EVAMEKLETE KHLV 594

SEQ ID NO: 197 moltype = AA length = 683
FEATURE Location/Qualifiers
source 1..683
mol_type = protein
organism = synthetic construct

SEQUENCE: 197
MLNHPSQGS DAQDEKQGD PVIEEEKTQA VTLKDSYVSD DVANSTERYN LSPSPEDEDF 60
EAPTEEMQT LRHVGGKNEP NYRGIKAMPF VLGNETCEKL GTIGTSSNLS VYLSTVFHNM 120
TMSANNVINI FNGTCYLGSL VGAFCLDTYF GRYKTLGLGS VASFLGMLML TLTAISTLH 180
PTNCNDSGDC KGPSAGQMAF LVGGFVLMVI GASGIRPCNL AFGADQFNPN TESGRRGINS 240
FENWYFFTYT FAVMVSATII VYVQSDISWS IGFAIPTSLM FLSCALFFAG TWLYVKVLPQ 300
GSPMTSLVQV IVVSVKKRRL KLPGEPSLLF NHIPSNAINS KLPRTNNFRF LDKAAITLPE 360
DKVNDGSA NKWKLCSIQQ VEEVKCVLRV LPIWIAGVIF QISVTQQSNY VPFQAFQSDR 420
HLIKGNKFQI PAGTYNVFSM LALTIWLPY DRIIPLLR WTKKEDGITL LQRMGVGIVL 480
SILSMLVSAL VESRRRLNAL TRPTLGIQEG QGSISSMSAL WFPVQLALMG LAEGFTFIGE 540
NELFYKEPPE NMRSIAASLV LCGVAVASYL TSFIAEIVHK TTRSGTRDWL AQDLNKARLD 600
YFFYMAVLE ALNLVYFLVC AKWRYKGTG DLGKLEVAME KLETEKHLVE EDEFDLNPI 660
SAPKANDIEI LEPMSLRST AKY 683

SEQ ID NO: 198 moltype = AA length = 649
FEATURE Location/Qualifiers
source 1..649
mol_type = protein
organism = synthetic construct

SEQUENCE: 198
MLNHPSQGS DAQDEKQGD PVIEEEKTQA VTLKDSYVSD DVANSTERYN LSPSPEDEDF 60
EAPTEEMQT LRHVGGKNEP NYRGIKAMPF VLGNETCEKL GTIGTSSNLS VYLSTVFHNM 120
TMSANNVINI FNGTCYLGSL VGAFCLDTYF GRYKTLGLGS VASFLGMLML TLTAISTLH 180
PTNCNDSGDC KGPSAGQMAF LVGGFVLMVI GASGIRPCNL AFGADQFNPN TESGRRGINS 240
FENWYFFTYT FAVMVSATII VYVQSDISWS IGFAIPTSLM FLSCALFFAG TWLYVKVLPQ 300
GSPMTSLVQV IVVSVKKRRL KLPGEPSLLF NHIPSNAINS KLPRTNNFRF LDKAAITLPE 360
DKVNDGSA NKWKLCSIQQ VEEVKCVLRV LPIWIAGVIF QISVTQQSNY VPFQAFQSDR 420
HLIKGNKFQI PAGTYNVFSM LALTIWLPY DRIIPLLR WTKKEDGITL LQRMGVGIVL 480
SILSMLVSAL VESRRRLNAL TRPTLGIQEG QGSISSMSAL WFPVQLALMG LAEGFTFIGE 540
NELFYKEPPE NMRSIAASLV LCGVAVASYL TSFIAEIVHK TTRSGTRDWL AQDLNKARLD 600
YFFYMAVLE ALNLVYFLVC AKWRYKGTG DLGKLEVAME KLETEKHLV 649

SEQ ID NO: 199 moltype = AA length = 424
FEATURE Location/Qualifiers
source 1..424
mol_type = protein
organism = synthetic construct

SEQUENCE: 199
MGGASASGVD NKFNKERVIA IGEIMRLPNL NSLQVAFIN SLRDDPSQSA NLLAEAKKLN 60
DAQAPKGGSS ASSAGGSSVD NKFNKEAQT GVEIMELPNL NTRQLLAFIQ SLRDDPSQSA 120
NLLAEAKKLN DAQAPKTS GGSSASSAGSS VDNKFNKERV IAGEIMRLP NLNSLQVAF 180
INSLRDDPSQ SANLLAEAK LNDAPKGG SSASSAGSS VDNKFNKEAQ TAGVEIMELP 240

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NLNTRQLLAF	IQSLRDDPSQ	SANLLAEAKK	LNDAQAPKTS	GSASGSGAG	GSSVDNKFNK	300
ERVIAIGEIM	RLPNLNSLQV	VAFINSLRDD	PSQSANLLAE	AKKLNDAPAP	KSGASGSGA	360
GGSSVDNKFN	KEAQTAGEI	MELPNLNTRQ	LLAFIQSLRD	DPSQSANLLA	EAKKLNDQA	420
PKTS						424
SEQ ID NO: 200	moltype = AA length = 70					
FEATURE	Location/Qualifiers					
source	1..70					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 200						
MVDNKFNKEL	GWATWEIFNL	PNLNGVQVKA	FIDSLRDDPS	QSANLLAEAK	KLNDAPAPKG	60
SSSSGSSSSG						70
SEQ ID NO: 201	moltype = AA length = 70					
FEATURE	Location/Qualifiers					
source	1..70					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 201						
MVDNKFNKET	IQASQEIRLL	PNLNGRQKLA	FIHSLDDPS	QSANLLAEAK	KLNDAPAPKG	60
SSSSGSSSSG						70
SEQ ID NO: 202	moltype = AA length = 56					
FEATURE	Location/Qualifiers					
source	1..56					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 202						
MQLLRCSIF	SVIASVLAQE	LTTICEQIPS	PTLESTPYSL	STTTILANGK	LDKRG	56
SEQ ID NO: 203	moltype = AA length = 72					
FEATURE	Location/Qualifiers					
source	1..72					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 203						
MKVLDTLLTVL	SASSLLSTFA	AAESTATADS	TTAASSTASC	NPLKTTGCTP	DTALATSFSE	60
DFSSSLDKR	GS					72
SEQ ID NO: 204	moltype = AA length = 51					
FEATURE	Location/Qualifiers					
source	1..51					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 204						
MLSLKTLTCT	LLTVSSVLAT	PVPARDPSSI	QFVHEENKKR	YYDYDHGSLG	E	51
SEQ ID NO: 205	moltype = AA length = 33					
FEATURE	Location/Qualifiers					
source	1..33					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 205						
MKFAYSLLLP	LAGVSASVIN	YKRDGDSL	DK	RGS		33
SEQ ID NO: 206	moltype = AA length = 66					
FEATURE	Location/Qualifiers					
source	1..66					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 206						
MKLQSLVSA	AVLTSLTENV	NAWSPNNSYV	PANVTCDDDI	NLVREASGLS	DNETEWLKKR	60
DAYTKE						66
SEQ ID NO: 207	moltype = AA length = 20					
FEATURE	Location/Qualifiers					
source	1..20					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 207						
MLLQAFLL	AGFAAKISAS					20
SEQ ID NO: 208	moltype = AA length = 99					
FEATURE	Location/Qualifiers					
source	1..99					

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mol_type = protein
organism = synthetic construct

SEQUENCE: 208
MKLSATTLTA ASLIGYSTIV SALPYAADID TGCTTTAHGS HQHKRAVAVT YVYETVTVDK 60
NGQTVTPSTST EASSTVASTT TLISESSVTK SSSLDKRGs 99

SEQ ID NO: 209 moltype = AA length = 92
FEATURE Location/Qualifiers
source 1..92
mol_type = protein
organism = synthetic construct

SEQUENCE: 209
MRFPsIFTDV LFAASSALAT PVNTTTEDET AQIPAEAVIG YSDLEGDFDV AVLPPFSNSTN 60
NGLLFINTTI ASIAAKEEGV SLEKREATNF GS 92

SEQ ID NO: 210 moltype = AA length = 100
FEATURE Location/Qualifiers
source 1..100
mol_type = protein
organism = synthetic construct

SEQUENCE: 210
MEEQREILEQ LKKTQLMLTV EPSKNNQIAN EEKEKKENEN SWCILEHNYE DIAQEFIDFI 60
YKNPTTYHVHV SFFAELLDKH NPKYLSEKSN WQDSIGEDGG 100

SEQ ID NO: 211 moltype = AA length = 75
FEATURE Location/Qualifiers
source 1..75
mol_type = protein
organism = synthetic construct

SEQUENCE: 211
MFRIQLRTMS SKTCKSDYPK EFVSFLNSSH SPYHTVHNIK KHLVSNGFKE LSERDSWAGH 60
VAQKGKYFVT RNGSS 75

SEQ ID NO: 212 moltype = AA length = 297
FEATURE Location/Qualifiers
source 1..297
mol_type = protein
organism = synthetic construct

SEQUENCE: 212
MSDSSPTINF INFNQTGTCT SLGTSKGFKI FNCEPFGKFY SEDSGGYAIV EMLFSTSLLA 60
LVGIGDQPAL SPRRLRIINT KKHsiICEVT FPTSILSVKM NKSRLVVLLQ EQIYIYDINT 120
MRLHHTIETN PNPRGLMAMS PSVANSYLVY PSPPKVINSE IKAHATTNNI TlSVGGNTET 180
SFKRDQQDAG HSDISDLQY SSFTKRDDAD PTSSNGGNS IIKNGDVIVF NLETLOPTMV 240
IEAHKGEIAA MAISFDGTLM ATASDKGTII RVFDIETGDK IYQPRRGTYA TRIYSIS 297

SEQ ID NO: 213 moltype = AA length = 57
FEATURE Location/Qualifiers
source 1..57
mol_type = protein
organism = synthetic construct

SEQUENCE: 213
MIALPVEKAP RKSLWQRHRA FISGIVALII IGTFFLTSGL HPAPPHEAKR PHHGKGP 57

SEQ ID NO: 214 moltype = AA length = 35
FEATURE Location/Qualifiers
source 1..35
mol_type = protein
organism = synthetic construct

SEQUENCE: 214
MFSLKALLPL ALLLVsANQV AAKVHKAKIY KHELs 35

SEQ ID NO: 215 moltype = AA length = 99
FEATURE Location/Qualifiers
source 1..99
mol_type = protein
organism = synthetic construct

SEQUENCE: 215
MKLENTLFTL GALGSISAAL VIPNLENAAD HHELINKEDH HERPRKVEFT KDDDEEPSDS 60
EDKEHGKPHK KGRKGQDKES PEPNGKRASG SHGSAHEGG 99

SEQ ID NO: 216 moltype = AA length = 164
FEATURE Location/Qualifiers
source 1..164
mol_type = protein
organism = synthetic construct

SEQUENCE: 216

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MKAFTSLLCG LGLSTTLAKA ISLQRPLGLD KDVLLQAAEK FGLDLDLHL LKELDSNVLD	60
AWAQIEHLYP NQVMSLETST KPKFPEAIKT KKDWDVVKV DAIENYQLRV NKIKDPKILG	120
IDPNVTQYTG YLDVEDEDKH FFFWTFESRN DPAKDPVILW LNNG	164
SEQ ID NO: 217 moltype = AA length = 75	
FEATURE Location/Qualifiers	
source 1..75	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 217	
MTKNFIVTLK KNTPDVEAKK FLDSVHHAGG SIVHKFDIIK GYTIKVPDVL HLNKLKEKHN	60
DVIENVEEDK EVHTN	75
SEQ ID NO: 218 moltype = AA length = 78	
FEATURE Location/Qualifiers	
source 1..78	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 218	
MKSFITRNTK AILATVAATG TAIGAYYYN QLQQQQQRGK KNTINKDEKK DTKDSQKETE	60
GAKKSTAPSN PPIYPVSS	78
SEQ ID NO: 219 moltype = AA length = 117	
FEATURE Location/Qualifiers	
source 1..117	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 219	
MAMFCTPFKE HHRKWDILLE KSTGVMEAMK VTSEEKEQLS TAIDRMNEGL DAFIQLYNES	60
EIDEPLIQLD DDTAELMKQA RMYGQEKLN EKLNTIIKQI LSISVSEEGE KEGSGSG	117
SEQ ID NO: 220 moltype = AA length = 171	
FEATURE Location/Qualifiers	
source 1..171	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 220	
MADRRDRSGIY GGTYGQHQQG GGRPMGEQVK GMIQDKGPTA SQALT VATLF PLGGLLLVLS	60
GLALAASVVG LALATPVFLF FSPVLVPAAL LIGTAVTGFL TSGALGLGGL SSLTCLANTA	120
RQAFQRTPDY VEEARRRMAE AAAHAGQKTA QAGHAIQSRA QEASGGAGA G	171
SEQ ID NO: 221 moltype = AA length = 93	
FEATURE Location/Qualifiers	
source 1..93	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 221	
MSTLQRRRVN RADSGDTSSI HSSANNTKGD KIANIAVDGD DDNGTNKKIA YDPEESKL RD	60
NINIPTLTLM EEVLLMGLRD REGYLSFWND SIS	93
SEQ ID NO: 222 moltype = AA length = 88	
FEATURE Location/Qualifiers	
source 1..88	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 222	
MSLPLRTIVH LVKPFAC TAR FSARYPIHVI VVAVLLSAAA YLSVTQSYLN EWKLD SNQYS	60
TYLNIKPDEL FEKCTHYRS GGNAAIRS	88
SEQ ID NO: 223 moltype = AA length = 30	
FEATURE Location/Qualifiers	
source 1..30	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 223	
MDSKGSSQKG SRLLLLLVVS NLLLCQGVVS	30
SEQ ID NO: 224 moltype = AA length = 23	
FEATURE Location/Qualifiers	
source 1..23	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 224	
MKAFTSLLCG LGLSTTLAKA ISL	23
SEQ ID NO: 225 moltype = AA length = 73	

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FEATURE	Location/Qualifiers	
source	1..73	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 225		
MSATKSIIVGE ALEYVNIGLS HFLALPLAQR ISLIIIIIPFI YNIWVQLLYS LRKDRPPLVF	60	
YWIPWVGSAV VYG	73	
SEQ ID NO: 226	moltype = AA length = 46	
FEATURE	Location/Qualifiers	
source	1..46	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 226		
MPFGIDNTDF TVLAGLVLA V LLYVKRNSIK ELLMSDDGDI TAVSSG	46	
SEQ ID NO: 227	moltype = AA length = 33	
FEATURE	Location/Qualifiers	
source	1..33	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 227		
MELTMASTMS LALLVLSAAY VLVALRRSRS SSS	33	
SEQ ID NO: 228	moltype = AA length = 70	
FEATURE	Location/Qualifiers	
source	1..70	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 228		
MSSVAENIIQ HATHNSTLHQ LAKDQPSVGV TTAFSILDTL KSMSYLKIPA TLICILLVWD	60	
QVAYQIKKGS	70	
SEQ ID NO: 229	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 229		
MEELLKQALQ QAQQLLQQAQ ELAKK	25	
SEQ ID NO: 230	moltype = AA length = 251	
FEATURE	Location/Qualifiers	
source	1..251	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 230		
MVSKGEELFT GVPVILVELD GDVNGHKFSV SGEGECDATY GKLTCLKICT TGKLPVPWPPT	60	
LVTTLGYGLQ CFARYPDHMK QHDFFKSAMP EGYVQERTIF FKDDGNYKTR AEVKFEGDTL	120	
VNRIELKGID FKEDGNILGH KLEYNYNSHN VYITADKQKN GIKANFKIRH NIEDGGVQLA	180	
DHYQQNTPIG DGPVLLPDNH YLSYQSALS K DPNEKRDHNV LLEFVTAAGI TLGMDELYKG	240	
GSGSAGSGGS S	251	
SEQ ID NO: 231	moltype = AA length = 387	
FEATURE	Location/Qualifiers	
source	1..387	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 231		
MKIEEGKLVI WINGDKGYNG LAEVGKKFEK DTGIKVTVEH PDKLEEKFPQ VAATGDGPD	60	
IFWAHDRFGG YAQSGLLAEI TPDKAFQDKL YPFTWDAVRY NGKLIAYPIA VEALSLIYNK	120	
DLLPNPPKTW EEIPALDKEL KAKGKSALMF NLQEPYFTWP LIAADGGYAF KYENGKYDIK	180	
DVGVDNAGAK AGLTFLVDLI KNKHMNADTD YSIAEAAFNK GETAMTINGP WAWSNIDTSK	240	
VNYGVTVLPT FKQGPSKPFV GVL SAGINAA SPNKELAKEF LENYLLTDEG LEAVNKDKPL	300	
GAVALKSYEE ELAKDPRIAA TMENAQKGEI MPNIPQMSAF WYAVRTAVIN AASGRQTVDE	360	
ALKDAQTNSS SNNNNNNNNN NLGIEGR	387	
SEQ ID NO: 232	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 232		
HHHHHH	6	
SEQ ID NO: 233	moltype = DNA length = 1140	

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FEATURE
source

Location/Qualifiers
1..1140
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 233

atgtcatccg	aaattttcaga	gaatgctcag	cttaaatcag	cttcaccctc	cagcattgag	60
aatggacagg	agggtgagaa	gcaggatcac	ttagtcacg	ttagtcaggt	ggttacatcc	120
gctagtcttc	cgatgggtct	ttatacggca	gtcaagtga	atttggttga	gatcatagcc	180
aaagcaggtc	cagggtgctaa	gttaagtccg	agtgaatatg	cctgcccgtc	accgacgaca	240
aagaaccctg	atgctgcctc	cgtatggtg	cgatgctac	gtttactatc	ctctcatagt	300
cttctaacct	atgatgtggt	tgaggtagca	ggtaggagcg	aaacggacat	cgggtatgaa	360
aggggtctatg	gcctgtcccc	tgtgggagaa	tatctaatac	cagacgaaga	gggaaattct	420
gtcgctccac	ttgtcgaatt	gttactggac	aagggtactaa	ttaatacttg	gtatgaactt	480
ggtaaatgctg	tactggaagg	agggataacct	ttcaatcggtg	tccacggaac	gcacgcattc	540
gattacccta	gtagggaaccc	aagggtacaat	gaacttttca	ataatggtat	ggttggtcca	600
acggcaatca	ccatgaagga	gttacttcag	caatataagg	gattgaact	actacaaaca	660
ttagtggacg	ttggtggtgg	tctaggtata	acactgcaca	aaatcatcgc	gaaatatccg	720
agcatcaggg	gaattaaact	tgacctgccc	cacgtcatcg	aagacgctcc	tccctatcct	780
ggaatggaa	atataggtgg	cgatgtggtt	gaatcagtc	ctggaggcga	tgctattttt	840
atgaaaaatga	tacttcatga	ttgggtccgat	gatcattgcc	ttaaaactact	taagaattgc	900
tatagagctc	tacctgatca	tggaaaagggt	atagtcgtag	atctaattct	tcccgtaaaa	960
cctgacacca	acatctttgt	caaagggtatt	ttccaggccg	atgtgatcat	gatgactcaa	1020
aacctgggtg	gaaagggaacg	ttccgaagca	gacgtgctg	cggtggcaat	aggtgcagga	1080
tttaaggga	taaaacttga	gtgttggtgca	gggaacgtag	gtatactgga	gttgataag	1140

SEQ ID NO: 234

FEATURE
source

moltype = DNA length = 1113
Location/Qualifiers
1..1113
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 234

atgtcatccg	caacaagcga	gaatacgcag	ttaaagagtg	tttcccatc	aagcatggag	60
aacggaaaaa	tagaggagga	agagcagctt	ttctcttacg	caatgcagct	ggtaacatca	120
gtaagtctac	cgagagttct	ttagctgccc	ttcagggtaa	gagtattcga	aataatcgaa	180
gaagctggac	cagggtgctca	gttatcagcc	agtgcagtag	ccgcacaggt	ttccggaaa	240
aataaagacg	cagctgcaat	ctagaccgt	atgttacgct	tgcttgctc	acatagcgtt	300
ttaacatgcg	gagcgagaag	ccagggtcgt	gtatacgggt	tggcacctgt	tgctaataat	360
ttcgttcagt	ccaaggcaga	cggaggtggc	acccttggct	cattactggc	tttaatacac	420
gataaggtat	ttattgatcc	atgggttcaa	ctagaagatg	ccgtaagaga	gggaggtgta	480
cccttttgata	gagttcaagc	gacgcacgca	tttagtagcc	caggacgtga	ccctagattc	540
aatgaggtgt	ttataaaggc	catgcttgat	cataccacca	tcgcaattaa	caggatctta	600
gaaaggtata	aaggtttoga	aaaccttaag	accctggctg	atgtaggtgg	aggcttggga	660
gttactttga	acgttataac	aactaagtag	ccaaatctta	aggaatcaa	cttcgatcta	720
ccacacgtta	ttcagagagc	acccggttac	ccagggtgct	aacatattgg	tggagatag	780
tttgaaatctg	ttcctgaagg	agaggcgatc	ttcatgaaat	ggattttaca	cgattgggac	840
gatgggcatt	gcttgacagc	gtttaaagaat	tgctataaag	ccttatctga	caatggcaca	900
gttatcgtgg	ttgatgcgct	ggtgccagta	ataccagata	caagtccttg	tgtcaaaaca	960
acctgccaga	gtgatttaat	aatgatgaca	cagaacctgt	gtgggaaaga	acgtagttaa	1020
cctgagtttc	tggccctagc	tactcttgca	gggtttaaag	gaatcagact	ggaatgctac	1080
gtatgtaacc	ttggggtgat	ggaattttat	aaa			1113

SEQ ID NO: 235

FEATURE
source

moltype = DNA length = 1113
Location/Qualifiers
1..1113
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 235

atgtcatccg	ccactagcga	aaacacacag	cttaaaaagtg	tctctccatc	ttcaatggag	60
aacgggaagg	tcgaagagga	ggagcagtta	ttttcatacg	caatgcaatt	ggtagacctca	120
gtctccttgc	cgaggggttt	gcttgcagcg	ttagactac	gtgtatttga	gataaattgct	180
gaagctggac	cagggtgcaca	gttatcagcc	ctgaaatcg	ctgcacaagt	gagtggttaag	240
aacaaggatg	ctgctgcaat	gcttgacaga	atgcttcggt	tattggccag	tcatagtgtt	300
ctaagttgtg	gtgcgagatc	ccagggaaga	gtttatggat	tagctcctgt	cgcgaagtag	360
ttcgttcagt	caaaggcaga	tggaggtggg	acgctgggta	gcctattagc	acttattcaa	420
gacaagggtt	tcacgcattc	ttggtatcag	ctagaggatg	cggtcaggga	aggtggagtc	480
ccctttgatc	gtgtctatgg	tacacacgct	tttagtagtc	ccggtcgtga	cccaagattc	540
aatgaggttt	tcaataaggc	catgctagat	catactacaa	tcgcatcaa	caggattctg	600
gaacgttaca	aaggattcga	aaaccttaaa	acactgggtg	acgtcggagg	cggtttagga	660
gtaacgctga	acgtgatcac	taccaaata	cctaatttga	aaggcataaa	ttttgacctca	720
ccccacgtaa	ttcagagggc	accggctac	ccgggagttg	agcatattgg	aggtgatatg	780
ttcagagtg	ttccagaagg	tgaagctatc	tttatgaagt	ggatacttca	tgactgggat	840
gacggacatt	gtctaaaaat	attaaagaat	tgctataagg	cgcttagcga	taatgggaag	900
gtaatcgtag	tagacgctct	tgttccggtc	atccagata	cgtctccttg	cgtaaagact	960
acgtgtcaaa	gtgatttaat	catgatgact	cagaatccag	gtgggaagga	gaggagtgag	1020
ccagaatttc	ttgcactagc	gacggctgca	ggtttcaaag	ggatcagact	ggagtgttat	1080
gtttgtaacc	tgtgggtaat	ggagttctac	aag			1113

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SEQ ID NO: 236      moltype = DNA length = 1131
FEATURE            Location/Qualifiers
source             1..1131
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 236
atgtcatccg cgacgagcga gaatacacag cttaaaagcg taccaccag ttcaatggaa 60
aacggcaagg ctggtgaaga ggaagagcac tgttcatacg cctgcgaact ggccactagc 120
gctgcgcttc ccgtgtgttc aaacgcgatt attaatgtaa atgtgcttga aataatcgcc 180
aaagccggtc caggcgctca attaatgcca agtgaaattg cggctcacct gccaacggag 240
aactcagggtg ctgccgtaat gcttgacaga atgttacgct tcttgccag ctactcagtg 300
cttacttgca gtgtagtgtg aggtggtgat aaggataacc agcaccagag agcgtatggg 360
ttggtcccg tagccaaata ttcttgagg gatgaggacg gaggtagttt aggcctccct 420
ctggacctac ttcaggataa agtggtgtgt gatatttggt atgagttgga gcctgcggtc 480
ttgaaagggtg gcaactgcgtt caaccgtacg catgggacac acatatttca atacacgggt 540
agcgatcaaaa agtttaacga aacatttaac aaagctatga ttgatcacac gttcatagtg 600
ggaaggaaag ttctacaatt ctataaaggg ttcgagcctc taaagacgct tgttgatgtc 660
ggaggtgggtc taggcgtcaa ttatcaatg attacttcaa aatattcctt tattaaaggg 720
attaattttg atttacctca cgtgtgtgag aatgcaccgc cttatccagg cgtagagcat 780
gtcggagggtg atattgttga atcaattcca aaggagagac ccattttcat gaagtggatt 840
cttcacgatt gggatgacga ccattgcctg accctactta agaactgcta taaagcggtt 900
ccagataaacg ggaaggttat tgcgtgtggc gctattttac ctatgggtcc ggataaacgc 960
gcatcttgta aatccaattg ccaagtggat ttattcacgc tggctttata ttcccccggt 1020
ggaaaggaaa ggacggaaca agaattctta gctttggcaa aggaagcggg ttccaaaggc 1080
atcagattgg agtggtttgc ttgtaatatg tgggttatag agttctacaa g 1131

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SEQ ID NO: 237      moltype = DNA length = 1122
FEATURE            Location/Qualifiers
source             1..1122
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 237
atgtcatccg caacctcaga gaatacacag ttaaagagcg tttctccatc aagtatggag 60
aacggtaagg gttcagcaca agaggaagcc tccctgtttg cgatgcaact tgcgtccgca 120
tctgtattgc cgatggttct gaagagtgcg atagaactag acttgtaga acttattgct 180
aaagccgggtc cgggtgcata cgtttcacgc tccaaattag ctgcccaatt acccaccat 240
aacctcaag caccagttat gctagacaga atcctgcgtc tattagctac ctatagtgtg 300
ttaaactgta agctacgtga cttacctgat ggtggggctg agaggtcata cggccttgct 360
ccagcttgta agttccttac caagaacgca gatggtgtga gtatggctcc attgctgttg 420
atgaatcaag acaaggttct tatggagagc tgggtatcac ttaaggacgc agtgattgac 480
ggaggtatac cgttcaataa ggcctatggg atgaccgcct tcgagtacca tggaaactgac 540
ccacgtttca acaaagtott caatcagggt atgtcaaatc atagcaactat aacaatgaag 600
aaaattctgg aagtctacaa aggggttcgag ggagtttaaga ctgtggtaga tgtaggtgga 660
gggaaccgggt cagcgttaaa tatgatcata gataagatc ccaccatcaa ggggatcaat 720
ttcagattac cacatgtgtg agaagatgct ccattctacc caggtgtaga acacgttggt 780
ggtgacatgt ttgtctctgt gccagaggga gacgctatct ttatgaagtg gatatgccat 840
gactggtcag acgatcattg tcgtaagtgt ctaaaagact gctatcacgc actgcccga 900
ggtggcaga tcatctctgc tgaatgtatt ttaccagatg ctcctgacac ctccctggct 960
acgcaaatg tcgtccacat agatgttacc atgttagccc ataactcggg tgggaaagag 1020
agaacggaga aggagtttga ggccttagcc aaaggtgcgg gctttaagga gttcagaaaa 1080
atatgttcag cagtaaacac atggatctta gagctgtgta aa 1122

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SEQ ID NO: 238      moltype = DNA length = 1131
FEATURE            Location/Qualifiers
source             1..1131
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 238
atgtcatccg ctacaagcga gaatacgcaa ttaaaatcag tgtctccctc atcaatggag 60
aacggtaaaag caggtgagga ggaagagcac tgttcctatg cattacaatt agccacttct 120
gctgcacttc ctgtcgtgtt gaatagcctc ataaaaacta acgtattgga cataattgca 180
aaagctggcc caggggctca gctgtcccg tctgagattg ctgccacctc aagtccggag 240
aactctgggtg cggcagtgat gctagacaga atgttgaggc tcttgccctc ctactccgta 300
ttgacatgca gcgtgtgggt tgggtgagac aaggataatc aacaccaaag ggcgtacggg 360
ttggctccag tcgccaataa cttcttgaga gatgaggatg gaggtcatt aggtagccta 420
cttgacttat tgcaggataa ggtacttgta gatatatggt atgagctgga accagctgtg 480
ttgaaagggtg gaacagcctt caatagaacc cagggaacgc atattttcca gtacaccggc 540
tccgatcaga aattcaacga gacgttcaac aaggcgtatg ttgatcatac ttctatagtg 600
gttaggaaag ttcttcagtt ctacaaaggc ttcggacacc ttaaaacctc agtcgatgta 660
ggaggtggat taggcgttaa tcttgggctg atcacttcaa agtatccgtc tatcaaggga 720
attaatttcg atttgcocca cgtcgtgcag aacgctccag cgtatccggg tgtggagcac 780
gtcggtgag atattgttga aagcattcct aagggggacg ctatctttat gaaatggatt 840
ttgcacgact gggatgatga acactgccta aagttattga agaactgcta taaggccttg 900
ccgacaatg gaaaagtatt ccgcgttgac gccatcctc ctatgggtcc agacaattca 960
gcatcctgta agagtaattg ccagggtgac ttgttcacct tggccttgta ctcccccggt 1020

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ggcaaggaac gtacggaaca ggagttctta gccctagcca aagaggctgg cttcaagggt 1080
atacgttttag aatgcttcgc ttgcaacatg tgggtaattg agttctacaa a 1131

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SEQ ID NO: 239      moltype = DNA length = 1122
FEATURE            Location/Qualifiers
source              1..1122
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 239
atgtcatccg cgacaagtga gaatacgcga ctgaaaagtg tgtctccatc cagtattgga 60
aatggaaaagg gtagtgcgga agaggaagct agtttggtcg ccatgcaatt agccagtgc 120
tccgtacttc cgatgggtct taaatccgct atcgagctgg atttactgga acttattgca 180
aaagctggtc ccggagcgta tgtttcccg agcaagctgg ctgctgagct tccgaccac 240
aatccgcaag cactgtttat gcttgataga atcctaagat tgctggcaac ttattccgtc 300
ctaaactgca agttaaagaa cctgcccgac ggaggtgttg agaggtata cggattagca 360
ccggtgtgca aatttctaac caagaatgca gacggcgtct caatggcacc actgctgtta 420
atgaatcaag acaaatgtgt aatggagtc tggtaccact tgaaggacgc ggtgatagac 480
ggaggtatcc catttaacaa ggcgtatggg atgacggcat ttgaatacca cggaacagat 540
ccacgtttca acaaggtggt taatcagggg atgtctaact actctacaat tacgatgaag 600
aagatattag aggtttacaa aggcctttgag ggagtaaaga cagtggtcga tgttgagggt 660
ggcacagggt cgacattaaa catgattatc gacaagtacc ctactataaa gggataaaac 720
ttcgaattac ctacgctggt taagatgcc cctttctatc ctggggtgga acatgtaggg 780
ggtgacatgt ttgtctctgt gcccgaaagga gacgctatat ttatgaaatg gatattgcat 840
gattggtcgg atgactactg tagaaagtta ttgaagaatt gctaccacgc ttgcccagaa 900
ggtggcaagg taatcttagc agagtgcata ctgccggatg ctcccgatac atctcttgcg 960
acccagaacg tggtccacat agacgtaatc atgctggcac acaatcctgg tgggaaagaa 1020
agaactgaga aagaatttga ggccttagca aaaggtgcag gattcaagga gttcagaaga 1080
atatgttcag ctgtcaatac atggattatg gagctatgca ag 1122

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SEQ ID NO: 240      moltype = DNA length = 1125
FEATURE            Location/Qualifiers
source              1..1125
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 240
atgtcatccg ccacatcaga aaacacccag ttaaaatctg tgagccctag ttctatggag 60
aacgggaagg ccggcgagag gcaggaagaa ttacaaaagc tcttacaatt aactgggacc 120
ttgacgggtcc acgtagcggt aactgccgag atagagttaa atctattcga gattatggcc 180
aagagtggac caaacgcgca gttatccagt gcacagattg tcagtctctc aagaactaag 240
aatcctgatg ctccggctat cttagatagg atattacaat tacttggttc ctatagcata 300
ttgaagtgca cgatagtgtc caacgatcag aatggccaca caactaagct cttcggattg 360
acgcccatgg ctgaatatct agtgcccgat tcaaatggta tatccttcgc ccagctatc 420
ttaatggcga gggaaaaggg attcgctaaa tgcggctatt tccttaaaga cgccatatta 480
gagggaggta cgccttttca taaagtaaac gggatgcact tattcgagta catcgggtca 540
agctcccgtc taaacgaggt ctttaacaga atcatggcct gccatacctc tgtgttcacg 600
ggagcgggttc tagaaactta taagggtttc gaggacctta aagaattggt ggacgttgga 660
gggtgtcacg gcacgacgct gaagatgatt ctgtctaagt acccactgat aagaggtatc 720
aattttgatc tgcccttcgt aattaagaat gcgcctgcct gtccgggagt cgagcacgta 780
gaaggagaca gttttcaaaa ttaccgtccc gtagaggcaa tattacttaa gaacgtgtcg 840
catgactgga acgatgatag gtgcttgaag ctactgaaga actgttgga cgcactgcca 900
gagagcggaa aagttaattgc gattgacgct atcatccagc agttaccgct aacggacagt 960
gcgtcacagt tcgctgaact tagtgatgtc gaaatactta tcgctgcacc aggtggaaga 1020
gaaagaactt tcaaggaggt taacgcacta gccacggcag ctggtttcac tgctgttag 1080
ttggtcgcta gaacttccca atacggagtg ctggaatttt acaaa 1125

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SEQ ID NO: 241      moltype = DNA length = 1011
FEATURE            Location/Qualifiers
source              1..1011
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 241
atgtcatccg caacgtctga gaatacacag ttaaagagtg tttcaccaag ttctatggag 60
aacggtgaagg caggtgagga ggggtgaacaa tgttcatacg ctatgcaatt agccacctcc 120
gctgctcttc ccaatgtttt gaacgcaatc attaggctaa acgttttgga aataatagcg 180
aggcgaggtc caggcgctca gctatctact agcgagatag ctgctgaccc gcatactcag 240
aacagcaacg ccgcagtaat gttagatcgt atgctgagac ttctggcatc ttacagcggt 300
cttacctgct cagttgttgg tggaggcgga ggtggagggt acgataagga cgatcagcac 360
caaagagctc atggactggc tccggttgca aagtatttct tgcaaaatga agacggcggt 420
agcctaaggc cctgcttaga cctattccaa ggaaaggtgc tgacggatat ttggtgcgaa 480
ttagaaccag cagtgttgga ggggtggcag gcatttaaca ggaaacacgg catccatc 540
ttgaataaca cgggttaga tcaaaaattc aatgagagtt ttaacaaggc aatgatcgac 600
cacaccttca tcgttggtta gaattgtatt cagttttatc gtgggtttga acatttaaa 660
acccttgggg acatcggcgg tggacttggc gtcactctgg gcatgataac aagcaagtac 720
cctaacaata aaggcataaa ttctgacctt ccacatgttg ttcaaaacgc tcccacttat 780
ccgataacg gtaaggtagt agcggtagat gccatactac ctacaatccc ggacaattca 840
gcaagtagta aatccaatat tcagatggat cttttcaccc tagcatcata ttccctggt 900

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ggtaaggaaa ggactgagca agagttctta gcgttggcta atgaggtctgg cttcaaggga 960
atccgttttag aatgtctgggt atgtaatatg tgggtcactgg agttttacaa g 1011

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SEQ ID NO: 242      moltype = DNA length = 1848
FEATURE            Location/Qualifiers
source              1..1848
                   mol_type = other DNA
                   organism = synthetic construct

```

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SEQUENCE: 242
atgtcatccg ttacaaaagg taatccaggt gactctagaa ataagtcacg tgtttcaatg 60
tttattgttg ctgggtttgt ttgtgttttc tttttattgg gtgcttggca aagatcaggt 120
tctggtaaa gtagacaatat tgcttacgaa attactaaaa aggctaaagc atgttctatt 180
ttgggttaatt tggaaattga aactcatcat ggtaataaagg ctgttaatga ttctatgttg 240
gaagttaaa gaaataaaacc atgtgatgat caatacattg attacacacc atgtcaagat 300
caagctagag caatgacttt tccaagagaa gatatgaatt atagagaaag acattgtcca 360
tcagaacaag aaaaattgca ttgtttaatt ccagcaccaa aaggttacgt tactccattc 420
agatggccaa aatcaagaga ttatgttcca tacgcaaatg ctccacataa atcattgact 480
gttgaataaa cagttcaaaa ttgggttcaa tatgaaggta atgtttttaa atttccaggt 540
gggtgtaact aatttccaca tgggtgcagat gcttacattg aacaattggc ttcagttatt 600
ccaattaatg atggtacagt tagaacagca ttggatactg gttgtggtgt tgcattcttg 660
gggtgcttatt tgtttaaaaa gaattgttatt actatgtctt ttgctccaag agattctcat 720
gaagctcaag tccaatttgc tttagaaaga ggtgttccag ctgttattgg tgttttaggt 780
actattaaat taccataatc atctagagca ttgatattgg ctcatgttgc tagatgtttg 840
attccgtggg gtgctaatag ttgtaaatat atgatggagg ttgatagagt ttgagacca 900
gggtgttact ggattttatc aggtccacca attaattgga gaaattctta ccaagcatgg 960
caaagaccac aagaagaatt gcaagaagaa caaagaaaaa ttgaagaaat tgcgtgaattg 1020
ttatgttggg aaaaagaaca tgaaaaagggt gaaactgcaa ttggagaaaa aagaattaat 1080
gaagattatt gttctcatgg tgactcaaga gttactaatg tagatctgtg tgatgctgat 1140
gatatttggg tataaaaagg ggaattttgt gttactccat atccagaaaa tagatctcaa 1200
gaagaagtgt ctggtgtgta attgaaacca ttccagaaaa gattgcatgc tattccacca 1260
agagttgatt ccggtttcat tccaggtgtt tctattgatt catttcaaga agataataga 1320
ttgtggaaaa agcatatgtc tgcttacaga agaattaata agttgttaga ttcaggtaga 1380
tacagaaatg ttatggatat gaatgctggt ttgggttctt ttgctgtctg atttgaatct 1440
ccaaaattgt ggggttatga ttgtatgcca actattgctg atagagatac tttgggtgtt 1500
atttttgaaa gaggtttgat ttgtatttat catgattggg gtgaagcatt ttctacttat 1560
ccaagaacat acgatttgat tcatgctgaa ggtgttttct ctttgtaaaa agataaatgt 1620
aatatggaa g atattttatt ggaaatggat agaattttga gaccagaagg tgctgttatt 1680
ttgagagatc atgctgatgt ttgacagaaa gttaaaaaaa ttattgctgg tatgagatgg 1740
aatatgaaaa tgtatgatca tgaagatggt ccattagttc cagaaaaagat tttagttgca 1800
gttaaaaagt actgggttgc ttgtgacaaa aattcaactt tttcttaa 1848

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SEQ ID NO: 243      moltype = DNA length = 1356
FEATURE            Location/Qualifiers
source              1..1356
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 243
atgtcatccg gttctggtga cgatgatcat ccatcttcaa aaagaccaat tcaacatact 60
caacacaaaat acaaaattgaa aattttgtta ttggttattt taactaattt gttaacaatt 120
tacgttttta caggtccatc tttgccaatt gctcaaaatt tgtctttacc aacatgggat 180
catactaaat tattgaatga attaaatatt acaaaaagag aattggctaa ttctagagca 240
caaaaattctg atttgcaaca acatttgaaa gcatcaataa ttttgactaa atctttattg 300
actgaattat caagattgaa tgcgtcaatt gatcaacatt catcttctaa agaaggttca 360
tctattttct catttgatga ttgttatca gatttgtcta cagaagctaa aattggtatg 420
gggtccacaaa aattaccatt gggttactca ccaagatctg gtactgatga attgtatcca 480
tctgttgggt gtggttgttt gagatacaaa gaagaattag ctcaatatat gacttacaaa 540
attggtgggt aatgtccagt tgatgatgtt ttgtcacaaa gattaatgtt gaaaggttgt 600
gaaccattgc caagaagaag atgtcatcca aaatctccag caggttataa agaaccacaa 660
ccattgcccac aatctttgtg gactatgcca ccagatacat ctattgtttg ggaccatat 720
acttgtaaat cttacggttg ttgggttgaa agaaaagaaat tgccagtttt ctatgatgt 780
aaagattggt ttgattttgga aggtagagaa aaatctagat ggtgtgttga taatgggtgt 840
ttggattttg gtattaatga agttttgggt acaaaaaccac atggtactat tagaattgg 900
ttagatatgt gtggtgttac aggtactttt gctgcaagaa tgaaagaaaa taatgttact 960
attgttacta catctatgaa ttgtgatgtt ccattcaatt cttttattgc ttcaagaggt 1020
ttgatttcta tgcataatgt agtttctcaa agattgtcat ttttcgaaaa tactttggat 1080
attgttcatt caatgcatat ttgtcaaat tggattccag atactatgtt ggaatttact 1140
ttgtatgata tttacagagt tttgagacca ggtgggtatt tctgggtgga tcattttctt 1200
tggttgggtt ctcaatttaa ttctacttac gttccaatgt tggatagagt tggttttaaa 1260
aagttgagat ggaatgctgg tatgaaattg gatagaggta ttgataaaaa tgaatggtac 1320
ttttctgctt tattggaaaa accagttaaa tcaataa 1356

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SEQ ID NO: 244      moltype = DNA length = 4236
FEATURE            Location/Qualifiers
source              1..4236
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 244

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atgtcatccg aatctccagg tcaatgggtg gaaaaagcat tattggattt gtgtaaaaag 60
attgaaactg gtttagattt ggatgggtgac attatttctg gtttagtttc ttattgtgaa 120
ttagcatctc cattagatgc aaaagaatac ttagataata ttattgggtc agaagcagggt 180
aaaaatgtta cagaagaata cttaaaaaga agaggtcatt ctgataatta caattctact 240
tctgatgctc caacttcaaa attgcatgca tatgttaaac caccatcaaa tgaagtttta 300
gtttctggta ctgaaaaacc atcaagagca ccaaaagaat tggctgcac ttcaaatcaa 360
gaaaagagag caccacatga aacagttgaa ttgagaaata tgaataaggg taatcaagggt 420
tcttcaagaa agaaaaaagc tggtaaaatt gtttcttttg ctgaagctgc taaaggttct 480
attgtttttc aacaaggtta accatgttca tgtcaagcaa gaaggcatag attggtttct 540
aattgtttat cttgtggtaa aattgtttgt gaacaagaag gtgaaggtcc atgtacattt 600
tgtgtgcat tagttttgaa agaaggttct tcttatgcag gtttagaaga aggtgcagtt 660
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agaaattctg cagctagaac tacagttatt gatgatcaat ctgattacta cgaatttgaa 780
ggtaattctt ggttgtcaat ggaagaaaaa gaattattga gaaagaaaaa agaagaaatt 840
gaagaagcag aagaactaa aagatcaaaa ttgattatga ctttgattt ggttggtaga 900
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tgtgaaaaac caggtttagc tgataatgca ggtagaaaac atggtcatca aaaagaagat 3240
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tctgaatcat cttggaactc acaaaactgca ttggttccat cattattgag aaaaatcatct 3360
caaaataaagc aaaaagagac tattggtaaa agatttttct tgggttttggt ttgttctggt 3420
ccatgtttgg ataaaaagtc tgtttacatt gatggttcag tttctcatgg tgttggtaaa 3480
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aattttgtga tggaaaattt ggcagttaaa aagcaattgg aagttgaaaa attggaagaa 3600
gaacaaagaa aatctttaga agtttttggt tctcatatga tgagaaaaag tgacgaaatt 3660
gctaaaaatt tggaaagaaa attatcaatg ttgacttggg atgcaattcc agcagctcaa 3720
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aatattggtg gtaaaagctcc aaaaggtttg ccatcaggtt tattgtcatg taaatctcaa 4080
aaagcagtta atgttgtctg aactgcagct tacagagcta atgaaaagac taaatcaatg 4140
tctttgggta aaccacaagc agactcaaaa agagttaaaag attttgaatt ttcaagagtt 4200
caacatccat ttgctgctag agttattgct ccaataa 4236

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SEQ ID NO: 245

moltype = DNA length = 1350

FEATURE

Location/Qualifiers

source

1..1350

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 245

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acaatttata	tttttactag	accatcttta	aatttgcac	caattttcaa	atacgattca	180
ttttctgatt	cttcattttt	attgcatgaa	ttaaatgcaa	ctatgaaaca	attagaaact	240
tctagatcac	aaattattgt	tttgaacaa	caattgaaat	ctactaatca	attggttaat	300
gaattgttag	ctcaattgaa	tattttgaat	caatctatta	atagaactcc	agattctact	360
ttattgaatt	ttaatgattt	gttagatgat	ttgtctaatt	aagcaaaatt	ggcttttagt	420
ccacataaat	taccattggg	ttactcacca	agatctggta	cagatgaagt	ttacgcatca	480
gttggtgggt	gttggttgat	gtacaaagaa	gatttggctc	aatatatgac	ttacagagtt	540
ggtaagaagt	gtccagtgtg	tgatgttttt	gctcaaagat	tgatgtttaa	aggtttgtgaa	600
ccattgccaa	gaagaagatg	tcattccaaa	gtccagttg	gttacgttga	accaactcca	660
ttgccaaaat	ctttgtgggc	aactccacca	gatacttcta	ttatttggga	cccatcacaca	720
tgtaaaaatt	acacttgttt	gattgaaaga	agaaaatttc	caggttttta	cgattgttaa	780
gattgttttg	atttgcgaat	cagagaaaag	acaagatggc	aatttgataa	tggttggttg	840
gatttttgta	tggatcaagt	tatggctact	agaaggcaag	gtacaattag	aatttggttg	900
gatattgggt	gtggtactgg	tacatttgct	gcaagaatga	aagaaagaaa	tgttactatt	960
gtttacaacat	caatgaattt	ggatgggtcca	ttcaattctt	ttattgcttc	tagaggtttg	1020
attttctatgc	atgttgggtg	ttctcaaaag	ttgccatttt	tcgaaaatac	tttggtatatt	1080
gttcattcta	tgcataattt	gtctaattgg	attccagatg	caatgtttaga	atttactttg	1140
tatgatattt	acagagtttt	gagaccaggt	gggtttatttt	gggtgggatca	tttcttttgt	1200
ttgggttcac	aattgaatgt	tacttacgtt	ccaatgttgg	aaagagttgg	ttttaaaaag	1260
ttgagatgga	atgctgggtat	gaatttggat	agaggtattg	ataaaaatga	atggtacttt	1320
tctgctttat	tggaaaaaac	aatttcataa				1350

SEQ ID NO: 246 moltype = DNA length = 1482
FEATURE Location/Qualifiers
source 1..1482
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 246

atgtcatccg	ctgctgcaaa	agggtcaagaa	agagaaatgc	aaaaatcata	ctggattgaa	60
catacaactg	atttgtcagt	tgaagctatg	atgttggatt	ctaaagcagc	tgatttggat	120
aaagaagaaa	gaccagaagt	tttgtcttta	ttgccaccat	acgaaggtaa	atctattttg	180
gaattaggtg	caggatattg	tagattcact	ggtgaattgg	ctaaaaaggg	tggtcaaatt	240
attgtcttgg	attttattga	agggtgctatt	aaaaagaatg	aaagaattaa	tggtcattgt	300
aaatcagcta	aattcatgtg	tgctgatgtt	acttctccag	aattagattt	ttcagcagaa	360
tctatggatt	tgtttttctc	aaattgggtta	ttgatgtact	tgtctgatga	agaagttgaa	420
gcattagcta	aaagaatggg	taaatgggtg	aaagttgggt	gtcatatttt	cttttagagaa	480
tcatgttttc	atcaatctgg	tgaccataaa	agaaagaata	atccaacaca	ttacagagaa	540
ccaagatttt	acactaaacc	ttttaagaa	tgctatttta	gagatgaatc	tggttaattct	600
tttgaattaa	cattgggttg	ttgtaaatgt	attggtgcat	acgttaaaaa	taagaaaaat	660
caaaatcaaa	tttgttggac	ttggaaaaag	gttagatctg	aagatgatag	agcatttcaa	720
agatttttgg	ataatgttca	atacaaatgt	actggtattt	tgagatacga	aagagttttt	780
gggtcaaggtt	atgtttctac	tggtgggtatt	gatactacaa	gagaatttgt	tgctaaattg	840
gatttgcac	cagggtcaaaa	gattttggat	gttggttgtg	gtattgtgtg	tggtgacttg	900
tacatggctg	aaaaatttga	tgttcatgtt	gttggatttg	atttgtctat	taatatgtgt	960
tcttttgctt	tggaaagagc	tattgggtta	caatgttcag	ttgaatttga	agttgcagat	1020
tgtactaaaa	agacttacc	agatgataca	tttgatgtta	tttactctag	agatactatt	1080
ttgcatattc	aagataaaac	agctttgttt	aaatttttct	ataaatgggt	aaaaccaggt	1140
ggtagagttt	taattttctga	ttactgtaaa	agagctgggt	caccatctga	agatttttga	1200
agatacat	aacaagaggt	ttacgatttg	catgatgttg	aagcatacgg	tcaaatgttg	1260
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gaagaagatt	atgatgat	tgttgggtgt	tggaaagcta	aattgattag	atcatcttct	1440
ggtagaacia	gatgggggtt	gtttattgct	aaaaagaatt	aa		1482

SEQ ID NO: 247 moltype = DNA length = 906
FEATURE Location/Qualifiers
source 1..906
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 247

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ttatttttgg	tatactttga	accaattttg	ttctttacac	aacatttctt	tccattcatt	180
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gtttgtttgt	ttttcgagc	tgcaaatttc	ttttactctt	cagtttcttt	acattggggt	300
atggctcaaa	gaattgttaa	tgtctgtcca	gattgggtcat	ctgttaaaaca	agcattggat	360
ttgggtgtgt	gtagaggtat	tttattgaat	gctgttgcaa	tgcaattgaa	aaagtgcagg	420
tcttcaggt	gagttgttgg	tttagatcca	actagagcac	atactaatgg	tccaactaca	480
ttatctactt	tgagaacagc	tggtttggaa	gggtgttcaa	aatatgttac	ttgtagagct	540
ggtagacc	gaaaattg	attttcagat	aattactttg	atgttgttgt	ttcatctgtt	600
tttgttcata	ctgttggtta	agaatttgg	caaaaatctg	ctgcagctgg	tcgagaaga	660

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atgagagttt tgggtgaagt tgttagagtt ttgaaatcag gtggtgttgg tgtgttttgg 720
gatttgggtc atgttccaga atacgttcaa agattgcaag aattgaaaat ggaagatatt 780
agagttttctg aaagagttac agctttttatg ttatcttcac atattgtttc ttttaaaaag 840
ccaaatcaac attttgttgg ttcaaatgaa gtttagagttg attggagatt caataatttg 900
tggttaa                                           906

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SEQ ID NO: 248      moltype = DNA length = 2766
FEATURE             Location/Qualifiers
source              1..2766
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 248
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ccaaatacat tgggtttacc aagaactaca acaagaactt ctgcaactac atcatctcaa 180
tcatcttcaa tgcagatttc acatgaaatt catccaacag aaccaatttc aaagaaatca 240
caaccagttt ttcaagattc tcaaggtgac ttgccagaag atgcaattca aacagatcat 300
gggtgctacac ataattgatca tactgatgaa tctccagaat taccaaatca atcttcaact 360
ggttctaaag ctcaagcaga agctgataat gtttctgaac aagattctgg tccagaaaat 420
aatggtgact ctgattcttc atctgataat aataatgaag ctgaaggtga cggtggtggt 480
gggtgacggtg gtaattgaaga agttgatcaa aaagattctg atgttgaagg taatccacaa 540
gaattgcatg gtaattgaaga agttgatcaa aaagattctg atgttgaagg taatccacaa 600
aattcagaag aattcttcaat tactcaaaat caagaagctg aaaagactag agaagatcaa 660
acaaaagcat cagctaataa ggattttgaa ggtaaagaaa atcaacatac tgcctgatgat 720
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caacaaaaga gattggaaac acatcaacaa caagaagatg atcaaatcca acatggtaaa 840
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gcagggtccag attacattcc atgtttggat aatgaagaag ctattgctaa attgcaatct 1260
agaaggcatt atgaagacat agaagacat tgtccaaaag aagctccaac ttgtttggtt 1320
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cattacattg atttcttggg agaagctggt ccagatattg cttggggtaa acattctaga 1560
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gttaaatatt gatccacaga tcttttgcca attatttatg aaagaggttt gtttggtatt 2460
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gaagttagaa atttgttaaa atctttgcat tgggaagttc atttgacttt ttctaaaaat 2700
caagaaggta ttttgtctgc tcaaaaatct gaatggagac cagattctta cgttgcatca 2760
tcataa                                           2766

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SEQ ID NO: 249      moltype = DNA length = 1032
FEATURE             Location/Qualifiers
source              1..1032
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 249
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ttagtttcat gttctggtat ttctggtgac agatctaaaa ttttggttcc aagatgttct 180
ttgtcagaat ctagaccaac atctcaacca agattcattc aacataaaaa ggaagcattt 240
tggttttata gatattttgc tattatttca gatcatgtta ttaatccagg tcattggact 300
gaagatatga gagatgaagc attggaacca gctgatttgt catctagaga tatgattgtt 360
gttgatgttg gtggtgttac aggtttttact acattgggta ttgttaaaac tgttgatgct 420
aaaaatgtta caatttttga tcaatctcca catcaattag ctaaagcaaa acaaaaagaa 480
ccattgaaa aattgtaaaa tattgaaggt gacgcagaag atttgccatt cagaactgat 540
tacgctgata gatacgtttc tgctggttct attgaatact ggccagatcc acaagaggtt 600

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attaagaag	catacagagt	tttaaaattg	ggtggtaaag	catgtttgat	tgttccagt	660
catccaactt	tttggttgtc	tagatttttc	gctgatattg	ggatgttggt	tccaaaagaa	720
gaagaataca	ttgaatgggt	tcaaaaagct	ggtttttaaag	atgttcaatt	gaaaagaatt	780
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ggtgtttaa	cagcttctgg	tgactctcca	ttacaattgg	gtccaaaagc	tgaagatgtt	900
tcaaaaccag	ttaatccaat	tacatttttg	ttaagatttg	ttttgggtgc	aatggctgca	960
acttactttg	ttttgggttc	aatttatatg	tggttgaaa	atcaaattgt	tccaaaaggt	1020
caaccaattt	aa					1032

SEQ ID NO: 250 moltype = DNA length = 1893
 FEATURE Location/Qualifiers
 source 1..1893
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 250

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tattttattg	gttcttacac	taattcaatt	tactcaattt	ctccatcacc	atctttgatt	180
agaagagata	caaattgtgg	tttgaagtt	aaaaataatt	tgaatactac	agtgaaaaat	240
gttcaagaac	cattgaattt	tgaaccattg	catacatgtg	ctttaccaat	ggaagatact	300
ccagaatctc	atcaaccatt	tccatattgt	ccaaagaatt	atacaaat	ctgtccatgt	360
caagatccaa	ataaggaaa	attgtttact	attgatagaa	gatttcata	agaaagacat	420
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acaaatcata	tctattgtgc	tcaaaaattg	aaaacttgga	aagctccaaa	attttgttct	1200
catgattttc	cagatcatgg	tttgtacaaa	aagatggaac	catgtattac	accattacca	1260
aaagttgaac	atgttagact	aacatctggt	ggttacttag	aaaaatggcc	aaaaagatta	1320
aaaactgtct	caccaagaat	tagatcagct	actgttgaag	gtgttactat	gagaactttt	1380
gatgaagata	ataagatttg	gaaaagaaga	gtttgttact	atggttctgt	tttgaaattt	1440
ttgtttaaag	gtaaatctat	aaatattatg	gatatgaact	ctggtttggg	tggttttgct	1500
gcagttttgt	ctgaatatcc	agcttgggtt	atgaatgttg	ttccatacga	tgctaaaaat	1560
aatacattgg	gtattgttta	cgaagagaggt	ttgatttgta	cttatatgga	ttgggtgtgaa	1620
ccattttcta	cttacccaag	aacttatgat	ttgattcatt	ctgatgggtt	gttttctttg	1680
tacatggaaa	aatgtgatat	tttagatatt	ttgtttgaaa	tgatagaat	tttgagacca	1740
gaaggtgcta	ttattatttag	agatcatgtt	gatgttattg	ttaaaattaa	agaaattaca	1800
gatagaatga	gatggaaact	aaaaatttct	cataatgaaa	tttcaccatt	tcattcagaa	1860
aagattttgt	ttattgataa	ttctccagaa	ttaa			1893

SEQ ID NO: 251 moltype = DNA length = 1161
 FEATURE Location/Qualifiers
 source 1..1161
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 251

atgtcatccg	attacaatat	tgttactatg	tctattgctc	caagagatat	tcatgaagct	60
caagttcaat	ttgctttgga	aagaggtgtt	ccagctttgt	taggtgtttt	gtctacttac	120
agattgcoat	ttccatctag	atgttttgat	atgattcatt	gttctagatg	tttggttcaa	180
tggaactgatt	ccggtgggtt	atacttgatg	gaaattgata	gattattgag	accaggtggg	240
tattgggttt	tgtctgtgtc	accaatttct	tggagaattt	cttacaaaag	ttgggaaaga	300
aatgcttctg	atttggaaat	tgaacaaaat	aatttggga	atttggctag	aaaattatgt	360
tggaaaaaagg	ttagagaatc	tgatccaatt	gctgtttggc	aaaaaccaac	aaatcatatt	420
cattgtgcac	aaaaattgaa	aacttggaaa	gctccaaaat	tttgttctca	tgattttcca	480
gatcatgggt	ggataaaaa	gatggaacca	tgtattactc	cattgccaaa	agttgaacat	540
gttagatcta	catcaggtgg	ttatttgga	aaatggccaa	aaagattgaa	aactgctcca	600
ccaagaattta	gatctgcaac	agttgaaggt	gttacaatga	gaacttttga	tgaagataat	660
aagatttgga	aaagaagagt	tttgatttac	ggttctgttt	tgaatttttt	gtttaaagg	720
aaatacagaa	atattatgga	tatgaactct	ggtttgggtg	gttttctgtc	agttttgtct	780
gaatatccag	cttgggttat	gaatgttgtt	ccatacga	ctaaaaataa	tactttgggt	840
attgtttacg	aaagaggttt	gatttggtact	tatatggatt	ggtgtgaacc	attttctact	900
taccaagaaa	ttacagattt	gattcattct	gatggtttgt	tttctttata	tatggaaaaa	960
tgtgatattt	tggatatttt	gtttgaaatg	tacagaattt	tgagaccaga	aggtgctatt	1020
attattagag	atcatgttga	gtttatttgt	aaaattaaag	aaattactga	tagaatgaga	1080
tggaaactcaa	aaatttctca	taataaaaat	tctccatttc	attctgaaaa	gattttgttt	1140
attgataatt	caccagaata	a				1161

SEQ ID NO: 252 moltype = DNA length = 1047

-continued

FEATURE
source

Location/Qualifiers
1..1047
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 252

atgtcatcct	ctaaccatc	agctttagat	ttggctacag	gtttgggtgg	taaaattgct	60
aaagatgatg	ttatttctgc	agttgataaa	tacgaaaaat	atcatgggtta	ctatgggtgg	120
gaagaagaag	aaagaaaagc	taattatact	gatatgggtta	ataagtaacta	tgatttgggt	180
acttcttttt	acgaatatgg	ttgggggtgaa	tcttttcatt	ttgcagctag	aatgaaaggt	240
gaatctttga	aagaatctat	taaaagacat	gaacatttct	tggttttgca	attaggtttg	300
aaaccaggtc	aaaaagtttt	ggatgttggg	tggtgtgttg	gtgggtccatt	aagagaaatt	360
gcacaatttt	cattgacttc	tgttacaggt	ttgaataata	atgaatatca	aatttctaga	420
ggtaaagcat	tgaatagagt	tgctgggtgt	gatagaactt	gtgattttgt	taaagctgat	480
tttatgaaaa	tgccattttc	agataatcaa	tttgatgctg	tttatgcagt	tgaagctact	540
tgatcatgctc	cagatgcagt	tggtttgttac	aaagaagttt	acagagtttt	aaaaccaggt	600
caatgttttg	ctgcatacga	atgggtgtatg	acagatttct	ttgatccaaa	taataaggaa	660
catcaaaaaga	ttaaagctga	aattgaattg	ggtaaatgggt	tgccagatat	tagattaact	720
gggtcaatgtt	tggaaagctgt	taaaaaggct	ggttttgaa	ttatttgggc	taaagatttg	780
gcagctgatt	ctccagttct	gtggtaactg	ccattggata	catctcattt	ttctttttca	840
tccttttagat	tgactgtctat	tggttagattg	attactaaaa	atttggttaa	aggtttggaa	900
tatattgggt	tggcaccatc	tggttcacaa	agagttcaag	atttcctgga	aaaagctgca	960
gaaggtttgg	tgtcaggtgc	taaaagggt	atttttactc	caatgtactt	ttcttgggt	1020
agaaaaccat	ttgtctgttc	agaataaa				1047

SEQ ID NO: 253

FEATURE
source

moltype = DNA length = 1014
Location/Qualifiers
1..1014
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 253

atgtcatccg	cttcttcatt	gttgtctggt	gcagaaaaat	tgaaacatat	tactggttct	60
cattttccag	gttctgaaca	tcaattgaaa	tactttccaa	agaaaggtcc	agtttcttca	120
gctaaatctt	caaaattttc	aactttggct	ccaagatgtt	ctttgtcatc	ttcaagacca	180
gtttcacaac	caagattctat	tcaacataaa	caagaagcat	tttggtttta	cagatttttg	240
tctattgttt	atgacatcat	tattaatcca	ggctattgga	ctgaagatat	gagagatgaa	300
gcattagaac	cagctgattt	gtactctaga	catttgactg	ttgttgatgt	tggttggtgt	360
actggtttta	caattgaata	tattgttcaa	catgttgatg	ctaaaaatgt	tacaattttg	420
gatcaatctc	cacatcaaat	ggctaaagca	aaacaaaaag	aaccattgaa	agaatgtaaa	480
attattgaag	gtgacgctga	agatttgcca	ttcagaactg	attacgctga	tagatacatt	540
tctgctgggt	caattgaaata	ctggccagat	ccacaaagag	gtattaaaga	agcatacaga	600
gttttgaaaa	ttggtggttaa	agcatgtgtt	attgggtccag	tttatccaac	attttgggtg	660
tctagatttt	tcgctgatgt	tgggatgttg	tttccaaaaa	aagaagaata	cattgaaatg	720
tttcaaaaag	ctggttttta	agatgttaaa	ttgaaaagaa	ttggtccaaa	atggatataga	780
gggtgttagaa	ggcatgggtt	gattatgggt	tggttcagtt	ctggtgttaa	accaatgtct	840
gggtgactct	cattacaatt	gggtccaaaa	caagaagatg	ttaaaaagcc	agttaataga	900
tttgtttttg	ttttgagatt	cattttgggt	gcaatggctg	ctgcatattt	tgttttgggt	960
ccaattttaca	tgtggttgaa	agatcaaat	attccaaaaa	gtatgccaat	ttaa	1014

SEQ ID NO: 254

FEATURE
source

moltype = DNA length = 1158
Location/Qualifiers
1..1158
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 254

atgtcatccg	ctttttgttc	acaatctaaa	ttgaataatt	tgaatttttg	gattaaaaaca	60
tctattgggt	catctacaaa	taatgttggg	catgattcaa	tgagaaaaag	aacttctaga	120
aatcatgcaa	atttgggtgt	tgttaaaatg	gcagcttcaa	catctggttc	aggttggtcaa	180
ggtagctgtg	cttttgaaga	aggtcaattg	gaaagaccac	aatggactgg	tgaacatttg	240
ttatctagat	tggttagagc	tttgatttct	tttaaaccat	tgttttctat	tatgaaattt	300
gggtctagac	aagttttgat	ttctacagca	gaaaagacaa	atattccgtg	gagagaaatg	360
acaagacaaa	ttttggaatc	tgatgtttat	aaagaaatgg	aatctattga	agataacttca	420
ttggtttacc	cagattatta	cttgaatcca	tttcatgctt	atgatgaagg	taatttgtct	480
tggttgccag	ctgctgaagc	agaagctgct	actatgtcta	tggtgagaag	agctattcca	540
gatgcttctt	cattggatga	agctaataca	attgttagag	gtaattgggt	gaaagctatt	600
gaagaacatc	ataataagta	ctctcaaaat	gttattatta	gagatatttt	ggatattggg	660
tggtctgttg	gtgtttctac	tagatatttg	gctgataaat	ttccagcagc	taaaattgaca	720
ggtttagatt	tgtctccata	ctttttggca	gttgctcaat	tcaagaaaaa	gaaatctaata	780
ccaagaaaaga	atactatttg	ttggattcat	gctaattggg	aatctacagg	tttgccatct	840
aaatcatacg	atattgtttc	aattgcttac	gtttttcatg	aatgtccagc	tggtgcaatt	900
aaaaatattg	ttaaagaagc	attcagattg	ttgagaccag	gtggtacatt	tgcattgact	960
gataattctc	caaaatcaaa	aattttacaa	gaattgtctc	cagttttgtt	tacattgatg	1020
aaatctacag	aaccattttt	ggatgaatat	tacttgacag	atattgaaaa	agttatgact	1080
gaagctgggt	ttattaatgt	tagaactgtt	ttgacagatc	caagacatag	aacagttact	1140
gctacagttc	catactaa					1158

SEQ ID NO: 255

moltype = DNA length = 2475

-continued

FEATURE
source

Location/Qualifiers
1..2475
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 255

atgtcatccg	cattggggtaa	atactcaaga	gttgatggta	gaaaaccatc	tacaaattat	60
tggtcaactg	ttacaattgt	tggttttgtt	gctttgtgtt	tagttgggtt	gtggatgatg	120
actttctcat	ctgttgttcc	agttcaaaat	tctgatgttg	cttctcaaca	agaatctaaa	180
aattgatgta	atgaaaatac	tagagtttct	aaatctaata	ctaataaaaa	tgattcagat	240
aattcttcaa	cagttaataa	gaatgataat	acttctgaat	catctatgaa	tgataatgaa	300
ggtaaatcta	aacaatttga	agataatcca	ggtgacttgc	cagaagatgc	tacaaaagggt	360
gacaatactg	tttcatctga	tcaagaagaa	aagaatccaa	atccacaaga	aaaaccaact	420
gaagaaaatc	aagaagaaaa	atcaggtgaa	gaaaatagag	aagaaggtgg	ttcaaaatct	480
gaagatgggt	ctaaaattga	aaaagaaaaa	gaagaaaatg	gtgaattgaa	atctgatgaa	540
aaagtttcag	aagctgggtg	agctaataga	tctaatagctt	ctggtgggtg	atctagagat	600
ggtaataaag	cagatgaaaa	ttttgatgaa	acaaaagatg	ctaataagga	agatgggtcaa	660
aaagttaaaa	ataagatgga	tactcaatct	gatcaagatt	caggtgttaa	aaaggaatca	720
gatcaaaacta	cagaagtttt	tccatctggt	gcacaatcag	aattgtttaa	tgaaccaact	780
acacaaaatg	gtgcattttc	tacacaagct	actgaatcta	aaaatgaaaa	agaagctcaa	840
caatcatctg	aattggaaaa	tcaatcaggt	acttcttggg	aattgtgtta	tgttactgca	900
aaatctgatt	atatctcatg	tttagataat	ttggaagcta	ttagatcttt	aagatcaact	960
aaacattacg	aacatagaga	aagacattgt	ccagataaat	caccaacttg	tttgggtgct	1020
tttccagaag	gttaccaaag	atctattgaa	tggccaactt	caagagaaaa	gatttgggtat	1080
cataatgttc	cacatacaaa	attggcagaa	gttaaaggctc	atcaaaattg	ggttaaagtt	1140
actggtgaat	acttgacatt	tccaggtggg	ggtaactaat	tcaaacatgg	tgctttacat	1200
tatatgtgatt	ttattcaaca	atctgttcca	gatattgctt	ggggtaaaag	atctagagtt	1260
gttttggtatg	ttggtgtgtg	tgtgtcttct	tttgggtggt	tcttatttga	tagagatggt	1320
ttgactatgt	ctttggcttc	aaaagatgaa	catgaagcac	aagttcaatt	tgctttggaa	1380
agaggtatct	cagctatttc	tgcagttatg	ggtaactaaa	gattgccata	tccagctggt	1440
gcttttgata	ttgttcatatg	tgctagatgt	agagttccgt	ggcatattga	aggttggtaaa	1500
ttattgttgg	aattaaatag	atgttgtaga	ccaggtgggt	actttgtttg	gtctgtctaca	1560
ccaattttat	aaaagatttc	agaagatgtt	gaaatttggg	aagctatgaa	aaagttaact	1620
aaatctatgt	gttgggaagt	tggttctatt	actaaagata	gagttaatgg	tggttggtatt	1680
gctgtttata	gaaaaccaat	tactaatgaa	tggtatgaac	aaagatctca	aatgaacca	1740
ccattgtgtc	agaatccaga	tgatccaaat	gctgcattgt	atgttccatt	gcaagtttgt	1800
atgcataaag	ttccaattgc	agaatcagaa	agaggttctc	aatggccaga	attgtggcca	1860
gctagattgg	ataaacacc	atactggttg	tcatcttctc	aagttgggtg	ttacggtaaa	1920
ccagctccag	aagattttgc	tactgatttt	caacattgga	gaagagttgt	tactaaatct	1980
tacgttaaatg	gttttaggtat	taattgggtc	gctgttagaa	atgttatgga	tatgagagca	2040
ttttacgggtg	gttttgcgc	agctttgaaa	gatatgaatg	tttgggttat	gaatattggt	2100
tcaattgatg	ctccagatac	tttgccaatt	atttatgaaa	gaggtttgtt	tggtattttac	2160
catgattgggt	gtgaatcatt	tctctcttac	ccaagatctt	acgatttggt	acatgctgat	2220
caattgtttt	caaaaattaa	aaagaaatgt	aattttaatg	cttttgttgc	agaagttgat	2280
agattattga	gaccagaagg	taaaattatt	gttagagata	atgttgaaac	aattaatgaa	2340
ttggaatcta	tttttaaatc	tttgcaattg	gatattagaa	tgacatatcc	taaaaataag	2400
gaaggtttgt	tatgtgtctc	aaagactatg	tggagaccaa	aagaatttga	aacagttact	2460
tatgcaattg	cttaa					2475

SEQ ID NO: 256

FEATURE
source

moltype = DNA length = 714
Location/Qualifiers
1..714
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 256

atgtcatccg	aagcatgttt	gactccatta	ccagaagttt	ctgatattaa	tgaaattgct	60
gggtgtccat	tagctaaatg	gccagaaaga	ttgactgctg	ttccaccaag	aattgtttct	120
gggtcattgg	atggtattac	tgctgcaaca	tttgaagaag	atactaattt	gtggaaaaag	180
agagttgctc	attataaagt	tttgatttca	caattggctg	aaccaggtag	atatagaaat	240
ttgttggtga	tgaatgctta	cttaggtggt	tttgetgcag	ctttgactaa	tgatccagtt	300
tggtgtatga	atgttgttcc	agttgaagct	aaattgaata	ctttgggtgc	tatttacgaa	360
agaggtttga	ttgttactta	ccaaacatgg	tgtagaagcta	tgtctactta	cccaagaact	420
tacgatttta	ttcatgcaga	tgctgttttc	tctttgtacc	aagatagatg	tgatatggaa	480
gatattttgt	tagaaatgga	tagaatttta	agaccacaag	gttctgttat	tattagagat	540
gatgttgata	tttttagtta	tgtaaaatct	gttgttgata	gattgcaatg	ggattctaga	600
atggttgatc	attgaagttg	tccacatgtt	agagaaaaat	tgtaatttgc	tgtaaaccaa	660
tattggactg	caccagctcc	atctgaagat	caagaacaat	ctagaactgc	ataa	714

SEQ ID NO: 257

FEATURE
source

moltype = DNA length = 1812
Location/Qualifiers
1..1812
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 257

atgtcatcca	aacataaaga	tggttaacca	tctacacaaa	ctggtaaatc	ttcaagaact	60
gttccaatgg	ctattatggt	tggtgtttgt	gtgtgtttgt	ctttttactt	aggttggtatt	120
ttctgttctg	aaaaagataa	atatgtctca	gcaaaagttg	aagatacagt	tgaatctcca	180

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aatcaacag cttctggttc attgcaagtt aaagatgttt ctttccaga atgtgggttg 240
gaataccaag attacacacc atgtactgat ccaagaagat ggaaaaagta cggtttacat 300
agattaacat ttttggaag acattgtcca ccagcttttg aaagaaaaga atgtttgggt 360
ccattgccag atggttataa aattccaatt agatggccaa aatcaagaga tcaatgttgg 420
tatagaaatg ttccatatga ttggattaat aagcaaaaat ctaatcaaca ttgggtgaaa 480
aaggaaagtg acaaatTTTT gtttccaggt ggtggtacta tgtttccaaa tgggtgttgg 540
agatacgttg atttgatgca agatttgatt ccaggtatga aagatggtac tattagaact 600
gctattgata caggttgttg tgttgcttct tggggtggtg acttggtaga tagaaatatt 660
ttgacagttt cattggcacc aagagataat catgaagctc aagttcaatt tgcattggaa 720
agaggtattc cagctattttt ggggtgttatt tctactcaa gattaccatt tccatctaatt 780
tcttttgata tggctcattg ttcaagatgt ttgattccgt ggactgaatt tgggtgttatt 840
tacttgttag aaattcatag aattttgaga ccaggtggtt tttgggtttt gtctggtcca 900
ccagtttaatt acgaaaaatag atggagaggt tggaaacta caattgaaga acaaaaatct 960
gattacgata aattacaaga ttgtttaaca tctttgtgtt ttaaatTTTA taaaaagaaa 1020
gatgataatt ctgttttgga aaaatctttg gataataatt gttacaaaaa gtttgattca 1080
tctgataaatt tcccaactaa atgtgatgat ggtactgaac cagattcagc ttggtacaca 1140
ccattgagac catgtgttgt tgttccaaat cataaataca aaaatttggg tttaaaatca 1200
ttgcaaaaat ggccacaaag attgcatgtt gctccagaaa gagttgctga tgttagaggt 1260
gggtctgaag gtgcttttaa tcatgatgat tctaagtga aaactagagt taaacattac 1320
aaaaagttat tgccagctgt tggtagagat aaaattagaa atgttatgga tatgaatact 1380
ttgtacggtg gttttgcagc tgctttaatt gattatccat tgtgggttat gaatgttgtt 1440
tcatcttaaca cttctaatac tttagctgtt gtttttgata gaggtttaat tggtagttat 1500
catgattggt gtgaagcatt tctactttat ccaagaacct atgatttgtt acatttggat 1560
ggtttgttta ctgctgaatc tcatagatgt gaaatTAAAT atgttttatt ggaattggat 1620
agaattttaa gaccaaatgg ttatgctatt attagagaat catcttatta cttggatgct 1680
gttgctactg ttgctaaaag aatgagatgg ggttgtagaa aagttgaaac tgaagataaa 1740
ttggataaag aaaagatttt gatttgtcaa aagaaattat ggtactctaa tactaaaaag 1800
tcatcaagat aa 1812

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SEQ ID NO: 258      moltype = DNA length = 1326
FEATURE
source              Location/Qualifiers
                    1..1326
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 258
atgtcatccg gtgaaaaata taaagaatac attactattt cttcttcatt attgcattca 60
tttccatctt ttccattttt gtcttttcca tttttgggtt tagttccaat ggcaacatgg 120
gctaatttgc catattactt aacattgaaa attaaaaaga ctaatttgta ttacatggct 180
atgattacag ttttatgttc aatgtcttac ttgatttgta tttggcaaca tgggtggttg 240
aaattagcag ctaatacagc aacttctaca ttaccacaat tgcattgtaa tccattgaaa 300
aagaaaattca caactatttg ttttccatta gattttactg ctcatcataa tgctgatgat 360
ttggttccat tgtcaacatc tgcgaagagt caacaatttc caccatgtaa accagaattg 420
tctgaataca ctccatgtga agatgctaaa agatcattga aatttgatag agataaaaaa 480
atttacagag aaagacattg tccagaaaaa gaagaattgt taaaatgtag aataccagca 540
ccataggtt acaaaactcc attcagatgg ccagcttcta gagataggtt ttggtacgct 600
aatgttccac ataaacattt gactgttgaa aaagctagac aaaattgggt tcaattcaaa 660
ggtagacagat tcaaatttcc aggtggtggt acaatgttcc caagaggtgc tgatgcata 720
attgatgata ttggtaaatt gattaaattg agagatggtt caattagaac tgctattgat 780
actggttggt gtggttgctc ttggggtgct tacttgttat ctagaaaatat tttgtctggt 840
tcatttgctc caagagatac tcatgttgct caagttcaat ttgcattgga aagaggtggt 900
ccagcttga ttggtgttat ggttcaatt agattgccat accatctag agcttttgat 960
attgctcatt gttctagatg ttgattccg tggggtaaa ttgatggtt atttttgatt 1020
gaagttgata gaattttgag accaggtggt tattgggatt tatctggtcc accaattaat 1080
tgggaaaaatc attggaagg ttgggtaga aaaccagaag atttgaaggc tgaacaagat 1140
ggatttgaaa atgttgcaaa aatgtgtgtg tggaaaaagt tgattcaaaa gaatgatatt 1200
gctatttggc aaaaaccaac aaatcataaa cattgtaaaa ttaatagaaa agtttttaaa 1260
aatccaccat ttgtagata taaaattcaa actaaacatg gtactccaga atggaaacca 1320
gcataa 1326

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SEQ ID NO: 259      moltype = DNA length = 1482
FEATURE
source              Location/Qualifiers
                    1..1482
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 259
atgtcatccg atttgaataa ggaacaagaa agagaaatgc aaaaatctta ttgggttgaa 60
catcacagctg atttgacagt tgaagctatg atgttggatt ctaaagctgc agatttggat 120
aaagaagaaa gaccagaagt tttgtcttta ttgccaccat atgatggtaa atctatttta 180
gaattgggtg ctggtatttg tagattcact ggtgaattgg ctaaaaaggc aggtcatttg 240
gttgctttgg attttattga atctgcaatt gaaaagaatg aatcttcaaa tggtcattac 300
aaaaatgc aaattcattt tgctgatgtt acttctccag aattgtcttt tccagttgaa 360
tctatggatt tgattttctc taattgggtt ttgatgtact tgtctgatga agaagttgaa 420
gcattggctg aaagaatggt taaatgggtg aaagtgtgtg gtcatatTTT ctttagagaa 480
tcttgttttc atcaatctgg tgaccataaa agaaagaata atccaacaca ttacagagtt 540
ccaagatttt atactaaagt ttttaagaa tgtcatttga gagatgggtt tggtaattct 600
tttgaattga cattggttgg ttgtaaatgt attggtgctt acgttagaaa taagaaaaat 660

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-continued

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caaaatcaaa tttgttgggt ttggcaaaaa gtttagatctc aagatgatag aggttttcaa 720
agatttttgg ataagtgttca atacaaatgt aatgggtattt tgagatatga aagagttttt 780
gggtcatgggt ttgttttctac aggttggtatt gaaacaacta gagaatttgt ttctaaattg 840
gatttgcac caggtcaaaa agttttggat gttggtttgtg gtattggtgg tggtagacttt 900
tatatggcag atgaatacga tgttcatggt gttggtattg atttgtctat taatatgatt 960
tcttttgggt ttgaaagagc tatgtgtttg aatgtttctg ttgaatttga agttgcagat 1020
tgtactaaaa agacttatcc agatggttct ttgatgtta ttattctag agatacaatt 1080
ttgcataatc aagataaaac agctttgttt aaatcttttt ataaatgggt aaaaccaggt 1140
ggtagagttt tgattttctga ttactgtaaa agatcagttt ctccatcaga agatttttgc 1200
aaatacatta aacaaagagg ttacgatttg catgatgttg aagcatacag acaaatgttg 1260
agagatgctg gtttttgatga agttattgct gaagatagaa cagatcaatt cattaaagt 1320
ttgcaaaagag aattggattc tgttgaaaaa gatagagaag cattcattaa agatttttct 1380
gaagaagatt acaatgatat ttgttggtgtg tggaaagcta aattgattag atctttctca 1440
ggtgacaaaa gatgggggtt gttttattgct aaaaagaatt aa 1482

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SEQ ID NO: 260      moltype = DNA length = 1062
FEATURE            Location/Qualifiers
source              1..1062
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 260
atgtcatccg gttctgtgta agaagaagca tcattgtttg ctatgcaatt ggcatctgct 60
tcagttttgc caatgggtttt gaaatctgct attgaattag atttgttggga attaattgct 120
aaagcagggtc caggtgtctta cgtttctcca tcaaaattgg ctgcacaatt gccaaactcat 180
aatccacaag ctcaggttat gttggataga attttgagat tggtagctac ttactctggt 240
ttgaattgta cttgcaaaaa ttgcccagat ggtggtgttg aaagattata cggtttggct 300
ccagtttgta aatttttgac taaaaatgct gatggtgttt ctatggctcc attattgttg 360
atgaatcaag ataaagtttt gatggaatct tggtaaccatt tgaaagatgc agttattgat 420
gggtggtatc cattcaataa ggcatatggt atgactgctt ttgaatatca tggtagacat 480
ccaagattca ataaggtttt taatcaaggt atgtcaaatc attctactat tacaatgaaa 540
aagatttttg aagttttcaa aggttttgaa ggtgttaaaa cagttgttga tgttggtggt 600
ggtagctggt caacattgaa tatgattatt gataaatacc caacaattaa aggtattaat 660
tttgaaatgc cacatgttgt tgaagatgct ccattttacc caggtgttga acatgttggt 720
ggtagacatg ttgtttctgt tccagaaggt gacgctattt ttatgaaatg gattttgtcat 780
gattgggtct atgacatttg tagaaaaatt gttaccatgc ttggccagaa 840
gggtgtaaa gttttttggc tgaatgtatt ttgccagatg ctccagatgc ttctttggct 900
actcaaaatg ttgttcatat tgatgttatt atgttggtgc ataaccagg tggtaaaaaa 960
agaactgaaa aagaatttga agcattgggt aaaggtgcag gttttaaaga atttagaaaa 1020
atttgttcag cagttaatac atgggattatg gaattatgta aa 1062

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SEQ ID NO: 261      moltype = DNA length = 1635
FEATURE            Location/Qualifiers
source              1..1635
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 261
atgtcatcca ttgatgatcc agaatctaatt gttaaagaaa ttaaaccatg tcatccaaga 60
tatacagatt acatctcatg tcaagatcaa aatagagcta tgaaatttcc aagagaaaaa 120
atgaattata gagaagaaga ttgtccacca caagaagaaa aattgcattg tttaattcca 180
gctccaaaag ttacgtttac accattcaga tggccaaaaa ctagagatta tgttccatgc 240
gctaattcac catcaaaatc ttgtactgtt gaaaaagcta ttcaaaattg gattcaatac 300
gaaggtaatg ttttttagatt tccaggtggt ggtacacaa ttccacaagg tgctgataaa 360
tacattgatc aattagcttc tgttattcca attgataatg gtattgttag aactgcttta 420
gatactgggt gtggtgttgc atcttgggtg gcatacttgt ggaaaagaaa agttattgca 480
atgtcttttg ctccaagaga ttctcatgaa gctcaagttc aatttgcatg ggaaagaggt 540
gttccagctg ttattgttgt tttgggtact attaaaatgc cataccatc tagagctttt 600
gatattggct attgttctag atgtttgatt ccgtgggggt ctaattgatg tatgtacatg 660
aaagaagtgt atagagtgtg gagaccaggt ggttattggg tttgtctggt tccaccaatt 720
aattggaaaa caaattacaa agcatggcaa agatcaaaa aagatttggg agaagaacaa 780
agaaaaattg aagatatgtc taaacaattg tgttgggaaa agaaatctga aaatgcagaa 840
attgctattt ggcaaaagag aattgatact gatgcttgta gagcagaaca agaaaattct 900
gggtgcagct tttgtcaagc agctgatcca gatgatgttt ggtacaaaaa gatggaagtt 960
tgtattactc catctgatag aaaaggtgac gaagcatagc gtgaaagaat taaaccattt 1020
ccagaagaat tatatgctat tccaccaaga attgcttcag gttctgttcc aggtttttct 1080
gcagaatctt acgaagaaga taataagaaa tggaaaaagc atgtttctgc ttacaaaaga 1140
attaataaga ttttggattc tggtagatat agaaatatta tggatatgaa tgctggttta 1200
gggtggtttg ctgcagcttt ggaatctcca aaattgtggg ttatgaatgt tgttccaaca 1260
attgctgaaa agaatacttt ggggtgttatt tacgaagag gtttgattgg tatttacct 1320
gattgggtgtg aaggtttttc tacataccca agaacatagc atttgattca tgctaagtgt 1380
ttgttttctt tgcacaaaga taaatgtgat ttgaaagata ttttattgga aatggataga 1440
attttgagac cagaaggtgc agttattatt agagatgaag ttgatgtttt gaatagagtt 1500
caaaaagattg ttgaggttat tagatgggat tttaaattga cagatcatga agatgggtcca 1560
ttagttccag aaaagatttt aattgcagtt aaaaagtact gggttgacag tgaataaat 1620
tcaacttcta cacaa 1635

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SEQ ID NO: 262      moltype = DNA length = 789

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-continued

FEATURE
source

Location/Qualifiers
1..789
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 262

atgtcatcct	cagggttatca	agctaagaa	tactctactg	gtagaccaac	ataccagaa	60
gaaatttttc	aatttgttgc	ttctaaaaca	ccatgtcatg	atttggcttg	ggatgttgg	120
actggtttcg	gtcaagcagc	tagatctttg	gctagaattt	acaaaaatgt	tggtgctact	180
gatgtttcac	cagaacaatt	ggaatgtgct	gttaaatg	caaatatata	atattactgt	240
acaccaccaa	aatgtctat	gactgaattg	gaacaaaatg	ttgcaactga	atcttcagtt	300
gatttggttt	ctattgtctac	agcaatgcat	tggtttgatt	tgccaacttt	ttaccaacaa	360
gttaaatggg	ttttgaaaaa	gccagatggg	gttattgctg	catggcttta	cacattgcc	420
caaattaata	agaatgttga	tactttattg	agaaaatgt	ttttgcaga	tgctgttcaa	480
tttttcggtc	cacaagctaa	attgattatt	gaagaaaatt	acagaactat	tgattttcca	540
tttgaaacag	ttgatgtgtt	ggaacatact	ggtccatttg	aatctaaaat	ggaaaaattg	600
attaatttgg	atggttattt	tacattgatt	aaatcatggg	ctgcttataa	aactgctaaa	660
gaaaaaggtg	ttgaattggt	acagaagat	ttgggttaag	aatttacttt	ggcttggaat	720
gaagatggta	aacaagaaaa	gattgtctact	tggccattgt	acatgagaat	tggtaaaagt	780
ggtaaacaa						789

SEQ ID NO: 263

FEATURE
source

Location/Qualifiers
1..714
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 263

atgtcatccg	atgttgatgt	tgaagcaaaa	atggatgctg	aaaaattgaa	aagaggatt	60
gctgaatttt	atgatgaatc	atctggtatt	tgggaagata	tttgggtga	ccatagcat	120
catggttttt	acgatccatc	ttcaactgtt	tcattattgg	atcatagatc	tgctcaaat	180
agaatgattg	aagaagcatt	gcaatttgct	tcagtttctg	atgctttgaa	tcaaccattt	240
ccagatggta	aatttgtatt	ggtttggctc	ttggaatcag	gtgaacatat	gccaaataag	300
gctaaatttg	ttaatgaatt	ggtagagtt	gcagctccag	gtgcaactat	tattattgtt	360
acttggtgtc	atagaaaatt	gtcatcttct	gaacaatctt	tacatccaga	tgaagagaaa	420
ttgttagata	aattgtgtga	tgcttactat	ttgccgcctg	ggtgttcaac	tgatgattac	480
gttaaaattg	tagaatcatt	ttctttgcaa	gatattaaag	cagctgattg	gactgaatac	540
gttgctccat	tttggccagc	tggtgtttct	tcagctttgt	cttggaagag	tattacttct	600
tttaatttgt	cttggttgaa	aactattaaa	ggtgcattgg	ctatgttatt	gatgattaaa	660
ggttataaaa	agggtttgat	taaatttgct	gttattactt	gtagaaaaac	agaa	714

SEQ ID NO: 264

FEATURE
source

Location/Qualifiers
1..1536
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 264

atgtcatccg	ctaaagatta	taatgggtct	ccaaaaccac	atcatttggg	aactcaaaga	60
aaaagattga	cttggatttt	agctgtttct	ggtttgtgtg	ttttgtttta	tgttttgggt	120
gcttggaat	ctacaaatcc	agctccatct	tcacaatctg	atgtttggga	aagagttggt	180
tgccaagatg	gttcaccaga	tcaaaaagtt	aataatccag	atttgtcaac	ttctacattg	240
ttctttatcat	gttcatggtc	atctacacca	ttggattttg	aatctcatca	tcaattggaa	300
gttaattctt	cagaagaatt	tagaaaaatt	ccaccatgtg	atacatctta	tactgaatac	360
acaccatgtc	aagatccacc	agaaggtaga	aaatttgata	gaaatatggt	gaaatacaga	420
gaaagacatt	gtccagaaaa	agaagaattg	ttacattggt	tgattccagc	accaccaaat	480
tacaaaactc	cattcaaatg	gccacaaaag	agagattacg	cttggtatgc	taatattcca	540
cataaacatt	tgtctgttga	aaaagccta	caaaattgga	ttcaatttga	aggtgacaga	600
ttcaaaattc	caggtggtgg	tacaatggtt	ccaagaggtg	cagatgctta	tattgatgat	660
attaatgaat	tgattccatt	gtctaattgt	actattagaa	cagctgttga	tactggttgt	720
gggtgtgcat	cttgggggtg	ttatttggtt	aaaagaaata	ttattgcaat	gtcttttgca	780
ccaagagata	ctcatgaagc	acaagtttgg	tttgccttgg	aaagaggtgt	ttccagctatg	840
attggtgtta	tgggtttctc	aagattgcca	tatccagcta	gagcatttga	tatgggtcat	900
tggtctagat	gttgatttcc	gtggcataaa	tatggtggtt	tgcatttgat	tgaagtgtat	960
agaattttta	gaccaggtgg	ttattggatt	ttgtctggtc	caccaattag	atggaaaaat	1020
tactggagag	gttgggaaag	aacacaagaa	gatttgaac	aagaacaaga	tgctattgaa	1080
gatgttgcta	aaaatatttg	ttggaaaaag	gttattgaaa	aagatgattt	ggcagtttgg	1140
caaaaaccaa	ttatcatcat	tgaatgtatt	aaatcaagag	caattcatca	tgctactcca	1200
atgtgtaaat	ctgataatgc	tgatgttgca	tgtgtataag	atatggaaac	ttgtatttca	1260
ccattaccag	aagtttcaaa	taaggatgaa	gttgctggtg	gttctttaga	aaaatggcca	1320
gaaagagttt	ttgcagttcc	accaagaatt	tcttcaggtt	ctggttgctg	tattactatg	1380
gaaaatttta	gagatgataa	tgatgtttgg	aaagaaagag	ttgctcatta	caaaagagtt	1440
gttggtccat	tggatcaagg	tcaatataga	aatgttatgg	atatgaatgc	atttttaggt	1500
ggtttgcaac	aattttgcga	aaatattcaa	tgtggt			1536

SEQ ID NO: 265

FEATURE
source

Location/Qualifiers
1..1875
mol_type = other DNA

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organism = synthetic construct

SEQUENCE: 265
atgtcatcca caagaggttag atctgatgggt gctcaaaaaga aaagattatt gacttcagtt 60
ggattatttg ctattttctt attgtttttg tacgtttatt ttggttcaaa atctaattgg 120
gaatctgctt tggaatatgg ttctagatct ttaagaaaat tgggttcac ttacttgggt 180
gggtgacgaag atgggtgactt gggttcaaaa caagatgatg cttcatctaa atttggcttg 240
gatgatgggtg aagatggttt ggttccaaaa tcatttttgg tttgtgatga tagacattca 300
gaattgattc catgttttaga tagacatttg atttaccaaa tgagattgaa attagatttg 360
tctttgatgg aacattacga aagacattgt ccattgccag aaagaagatt caattgtttg 420
attccaccac catctgggta caaagtcca attaaatggc caaaatcaag agatgaagtt 480
tggaaagcta atattccaca tacacatttg gcttcagaaa aatctgatca aaattggatg 540
gttgttaaa gtgaaaagat tatttttcca ggtgggtgta ctcattttca ttatgggtgct 600
gataaatata ttgcttcaat tgctaatag ttgaattttt ctgataataa tttgaataat 660
gaaggtagat tgagaactgt ttggatgtt gggttgggtg ttgcttcttt tgggtgcatac 720
ttgttatcat ctgatattat tgctatgtct ttggcaccaa atgatgttca tcaaaatcaa 780
attcaatttg cattagaaag aggtattcca gcatatttgg gtgttttagg tactatgaga 840
ttgcataacc catcaagatc ttttgaatta gctcattgtt ctagatgtag aattgatttg 900
ttgcaaaagag atggtattttt attggttagaa ttggatagat tgtaagacc aggtggttac 960
tttgccttatt cttcaccaga agcatacogt caagatgaag aagatttgag aatttggaga 1020
gaaatgtcag ctttgggtga aagaatgtgt tggaaaattg cagctaaaag aaatcaaact 1080
gttatttggg ttaaacattt gactaatgat tgttatatgg aaagagaacc aggtactcaa 1140
ccaccattgt gtcattctga taatgatcca gatgcagttt ggggtgttaa aatgggaagg 1200
tgtattacaa gatactctga tcatgatcat agaacaagag gttcaggttt agctccgttg 1260
ccagcaagat taacttcacc accaccaaga ttggcagatt ttggttattc ttcagaaatg 1320
tgggaaaaag atatgcaagt ttggagaaga agagttgaaa attattggaa tttgttatca 1380
ccaaaaattt cttctgattc ttttaaaaaa gttatggata tgaaagctaa tttgggttca 1440
tttgggtgct cattaaaaa aaaagatgtt tgggttatga atgttgttcc agaagatggt 1500
ccaaatacat taaaattggt ttatgataga ggtttgattg gtacaattca taattggtgt 1560
gaagcatttt ctacataacc aagaacatat gatttgttac atgcatggtc tgttttatca 1620
gatattgaaa agaaaagggt ttctgttgaa gatttgttat tggaaatgga tagaattttg 1680
agaccaacag gttttgttat tattagagat aaacaaccag ttattgattt tgttaaaaag 1740
tttttgggtg ctttaccatt ggaagctgtt ggtactgctg attcatcttc agatccagat 1800
caagaagggt acgaattggt ttttgttatt caaaagaaat tatggttgac ttcaggttct 1860
tttagagata cagaa 1875

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SEQ ID NO: 266      moltype = DNA length = 762
FEATURE            Location/Qualifiers
source              1..762
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 266
atgtcatccg ctactttgca accaccagtt tctcaattgc cacaaaaatc tccagctcca 60
aaattgttaa ataagccatt gtttaatat ttgttattga ctattttctg ttctttgtca 120
tattttattg gttcttacac taattctatt tactcaattt ctccatctcc atcattgatt 180
agaagagata ctaattgtgg ttttgaagtt aaaaataatt tgaatacaac tgttgaaaat 240
gttcaagaac cattgaattt tgaaccattg catactttgt ctttaccat ggaagatact 300
ccagaatctc atcaaccatt tccatactgt ccaaagaatt acactaatta ttgtccatgt 360
caagatccaa ataaggaaag attgtttact attgatagaa gatttcatag agaaagacat 420
tgtccagaaa aagggtgaag attgagatgt ttggttcaaa aaccagctgg ttacagaagg 480
ccatttccgt ggcctaaaatc tagagattac gcttgggtta aaaaatgttc attcaaaaaga 540
ttgactgaat acaaaaagtc tcaaaattgg gttagacaag aagggtgcaa attatttttc 600
ccaggtgggtg gtaactcttt tccaaaagggt gttaaagggt acattgattt gttaaaaaga 660
gttgttccat taaaattctg tgctattaga actgttttgg atgttgggtg tgggtgtttc 720
attaatgggt gttctaaaat ttttattggt ttgttagat tt 762

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SEQ ID NO: 267      moltype = DNA length = 1569
FEATURE            Location/Qualifiers
source              1..1569
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 267
atgtcatccg aaccatctac tggtaaacca tcatttttca gaaatatttt ggtagatta 60
tttttgtttt gtgtttttat ttgggtgtt agatttgctt atgttgttac tattagaggt 120
gaatcatgtg atttgggtga cttttgttct ttttctttgc cagataattt gaatacaatt 180
gtttctgggtg ttggtcaatt ggggtactaa tcattttctt catctgcaat tattgtgtt 240
ggtaaatctg ctccagcaaa accacataaa ttaccagatt tgtgggcaac tcaagggttt 300
caaaaagctg ctcaatttta ttcttcagtt ttccaagatt tgatttcaga aggtttcttg 360
actccaaatt caaaagcatt gtgtgttgat acaccaactg gtgctgatgt ttttgccttg 420
aaagaaattg gtgtttcaga ttctattggt atttttaaaa agtcttcaaa accattgggt 480
atttcaggtc aagcaattta acaaccattt ggtgacgata aatttgattt tgttttctca 540
gggtctggta tgattgaaa atctgttaaa gcagctgatt ttgcagctga agttttaga 600
acattaaaaa cagaagggtt ctttgttgtt catactgggt ctaagatata ttactcttac 660
aattcatttt tagatttggt taattgttgt aaattagtta aattcagaga tattgatggt 720
tttgattcaa aattaccatt cattagagaa attgttatgc aaaaagaaat caatgaaaat 780
gttgatgatg aaattcfaat gccaggtggt aattctgaag ataaatgttt tgttccaggt 840
tacaacaacg aattgattag aaatgctgaa ccattgattg ttgaagaacc aaagaaaccg 900

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-continued

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tggattactt tgaaaaagaa tattcaaaat attaaatatt tgccatctat ggctgatatt 960
tcttttaaac aaagatacgt ttatgttgat gttggtgcta gatcatacgg ttcttcaatt 1020
gtttcatggt ttaaaaagca atacccaaaa caaaatagaa catttgaat ttacgctatt 1080
gaagctgata aacattttca tgaagaatat aaatacaaaa agggtgttaa attgttacca 1140
tatgcagctt ggggttagaaa tgaaacattg tttttcgaaa ttaatgaaga tccaattgat 1200
aaagatgttg ttaaaagtag aggtatgggt agaatacaac cagttcaatc atcttcaggt 1260
gtttctgatg gtaatgttga tgaattcaa ggttttgatt ttgctgaatg gttgaaatct 1320
acagtttcag aaaaagattt tgtgttatg aaaatggatg ttgaaggtac agaatttgat 1380
ttgattccaa gattgattga aactggtaca atttgtttga ttgatgaagt tttcttgtaa 1440
tgtcattaca atagatggca aaaatgttgt ccaggtgaaa gaacttctaa ataccaaaag 1500
acttacgata aatgtttgga attgtttact tcattaagag aatctggtgt tttggttcat 1560
caatggtgg 1569

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SEQ ID NO: 268      moltype = DNA length = 2115
FEATURE            Location/Qualifiers
source              1..2115
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 268
atgtcatcct ctagaccatt gcatagaggt gtttctggtg gtggtagatt ttctccaaat 60
tcacatgatt tgtgggatga tgatgattca caaatgaaag ataaagcaga aaaagaagat 120
gggtacagaa aatctacaaa tgctgatcat acatacttgt caattagaca accattcaga 180
tttttgtag ctgataattc ttcatcaaaa catggtgtta ctaaagatgg tttgtttca 240
gatccatttt ctccatcacc aggtgtttct agaaatagac ataaattcat tttgttatta 300
ttgagatttg gtttagttgt tattgttatg ttagcttttg caggttcttt ttggtggact 360
atttcaatga caactctctc aagaggtcaa atttatagag gttacagaag attgcaacaa 420
aaattggttt cagatttggg tgatgttggg aaattgtcat taggttctac aagattgaaa 480
gatttggaat actgttcacc agaactcgaa cattacgttc catgttttaa tgttccagaa 540
aatttggttt tagttatttc aggtggtaat gaatacgata gacattgttg tccaggttct 600
agacaatctt gttttgtttt gccaccagtt aattacaaaa ctccattgag atggccatgg 660
ggtagagatt tgatttgggt gctaatattt aatattactg ctcaagaagt tagatcttca 720
gggtcattga ctaaaagaat tgatgtgtac gatgaagatc aaatttcttt tagatctgct 780
tcaccaggtt acgatgatat ggaagattac gctcatcaaa ttgcagaaat gattggtttg 840
agaaatgaat caaattttgt tcaatctggt gttagaacta ttttagatat tggttgtggt 900
tatggtaatt tgggtgtctc tttgttttca aatgaattgt taactatttg tattgtctaat 960
tacgaagcat ctggtttcaca agttcaattg acattagaaa gaggtttacc agctatgatt 1020
tcttctttta cttctaaaaa attgccatac ccattcttgt cattgatatt gattcattgt 1080
gctggttggt gtattgatgt ggaacaaaaa gatggtattt tattggttga agttgataga 1140
gttttgagac caggtgtgta tttgttttgg acatcacaaat tgacttctac tcaacaattt 1200
ttaagaaaaga atgaaaaatca aaaagaatgg aattttgttc atggttttgc taaatctttg 1260
tgttgggaaa tgttgtctca acaagttgat actgttgttt ggaaaaagac ttctaaaaag 1320
aattgttatt cagctagaaa atctggttca ggtccatctt tatgttcaag aggtcatgat 1380
gttgaatctc cacattacag accattgcaa ggttgatttg gtggtactca atcacaaaaga 1440
tggattccaa ttgaagaaag aactaaatgg ccagctagag ctgattgaa tgcaaaaaga 1500
ttagctattc atggtttaaa ttctgatgat tttgcagaag atacattgaa ttggaactct 1560
gcagttagaa attattgggtc attattgtct ccagttattt tctcagatca tccaaaaaga 1620
ccaggtgacg aagatccatc tccaccattc aatatgggta gaaatgtttt agatatgaat 1680
gcacattttg gtggttttaa tgctgcattg ttagaagcta aaaagtctgt ttgggttatg 1740
aatgttattc caactacagg taataatcat ttgccattga ttttagatag aggtttttgt 1800
gggtgtttgc atgattgggt tgaagcattt ccaacttacc caagaacata cgatttgggt 1860
catggtgaag gttctctgtc tttagaattt ggtgaagaac aaagatgtga aatgttggat 1920
gtttttactg aaattgatag aattttgaga ccagaaggtt ggtttatttt gagagatata 1980
actccattga ttgaatctgc tagaacattg actgcaagat tgaaatggga tgcaagaatg 2040
attgaaattg aaaataattc agatgaaaaga ttattgattt gtcaaaaacc atttgctaaa 2100
agacaagcaa tgtaa 2115

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SEQ ID NO: 269      moltype = DNA length = 306
FEATURE            Location/Qualifiers
source              1..306
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 269
atgtcatcca aatggatttt gcatgattgg gatgatggtc attgtttgaa attattgaaa 60
aattgttaca aagcattgtc tgataatggg aaagttattg ttgttgatgc tttggttcca 120
gttattccag atacttctcc atgtgtttaa actacatgtc aatctgattt gattatgatg 180
actcaaaatc caggtgtgaa agaaagatct gaaccagaat ttttggcatt agctactgca 240
gctggtttta aaggtatttag attggaatgt tacgttttga atttgtgggt tatggaattt 300
tacaaa 306

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SEQ ID NO: 270      moltype = DNA length = 420
FEATURE            Location/Qualifiers
source              1..420
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 270
atgtcatcca aaattcattc aagatctttg tgtccatcaa gatcttacgg ttgtcaagaa 60

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gttgaagttt ttcaaattgc tagaactatg atgcaatcta ttattttgta ctctattatt 120
tggtctttgc attgttttca tgcatttcca tcatctaatt catctactcc agcatacaat 180
gctttgtgga ataagtcttt aacttttgcca aatgttacta ataatagata ctttttgttt 240
atttttactc cattttctaa acatactaaa gttaaacatg ctactacaac tattgtttaa 300
ttgaataaatt acattttcaat tgttaatgat gttagaaatt tgtcagttat tgattttatt 360
tttgttattt acatttcattc atctagatgt cattacactt atagaatttc ttcaataaag 420

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SEQ ID NO: 271      moltype = DNA length = 1158
FEATURE            Location/Qualifiers
source              1..1158
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 271
atgtcatccc aaccacaaag aggtagaaaa agagaaagag aaagagatag agaagaaatg 60
gaatctgttc aatctaattc atcttcttca gatcaatttg ctatgaaagg tggtagacgat 120
gatttttcat atactaaaaa ttctacacat caaagagatg ctattcaagc tactaaaattt 180
ttcattcaag aatcaattgc tgaaaaaatt gatgttaata agttttgtgg taaagcattt 240
tgtgttgctg atttgggttg ttccagttgg ccaataacat tgattgctat gcaaaatatt 300
gttgaagcag ttgaattgaa attcaaaaaa agaaaagggt ttcattcacc aacaattcca 360
gaatttcaag tttctcttaa tgatcatact gttaatgatt ttaatacttt gtttagatct 420
ttgccaactg gtcatagata aagatattac ggtgttggtg ttccagggtc attttacggt 480
agattgttcc catgtgattc tattcatatt atgcatactt ctttttcaac tccatttttg 540
tctcaagttc caaaagaagt tatggataaa aattctgcag cttggaataa gggtagaata 600
catcataatt acgcaaaagc tgatgttttg aaagcatcac aagctcaaca tgctgaagat 660
attgattgtt tcttgactgc aagagctaaa gaattggttc atggtgggtt attgatggat 720
gttacttctt ttagaccaga tgggtttcca catactcatg ttttaactaa tattggtatg 780
gaagttttgg gttactgttt aatggatttg gctaaaaagg gtttaattga tgaagaaaat 840
gttgattctt ataattgttc agtttacttg caatctccag aagaattgaa acaagctggt 900
caaaagaaata agtacttttc tattgaaaag atggaaatcag ttccaatgat gattgattct 960
gatgtttcag caaaagctca acaatactct ttgggtatga gagctgttat gggtagcgtt 1020
attagagaac aatttgggtg tgaaattggt gataaattgt tgatttgggt taaaaagaaa 1080
ttggaagaac atccaaaatt tgcaaaagggt gttgttttgg atatgtttgt tttgttaaaa 1140
agaaatgctg aagattaa                                     1158

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```

SEQ ID NO: 272      moltype = DNA length = 1101
FEATURE            Location/Qualifiers
source              1..1101
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 272
atgtcatccg aatcttcaca atctaataat agattttgtg atcaatttgc tactaaagggt 60
ggtagacgat atttttctta tactaaaaat tcaattcatc aaagagatgc aattgaagct 120
actaaattat tgattcaaga agcaattgct gaaaaattgg atattaataa gttttgtggt 180
aaaacttttt gtattgtctc tttgggttgt tctacaggtc caactacatt gactgctatg 240
caaaatatta ttgaagctgt tgaattgaaa ttccaaaatc aaaaagggtt taattttcca 300
aattttccag aatttcaagt ttctttaat gatcatactg ttaatgattt taatgctttg 360
tttagatcat tgccaacagg tcatgataaa agattttacg gtgttggtgt tccaggttct 420
ttgtacggta gattgtttcc atgtaattct attcatatta cacatacttc tttttcaact 480
ccatttttgt ctttagttcc aaaaagaagt atggataaaa attcagcagc ttggaatgaa 540
ggtaaaatcc attacaatca tgctaagaaa gatgttttga aagcatacga aggtaaacat 600
gctgaagata tggattgttt ctttatggct agagcaaaag aattgggtca tgggtggttg 660
atgttaattg ttacatcttt gagaccagat ggtattccac atgttttgac taatattggt 720
atggaagttt taggttcttg ttgtatggat ttggctaaaa agggtttgat tgatgaagaa 780
aatgttgatt cttttaatgt tccagtttac ttggaatctc cagaagaatt gaaacaagct 840
attcaaagaa ataagtactt ttcaattgaa aaagtgaat ctgttccaat gattatggat 900
tctgatattt catgtagagc acaacaatat tctttggcta ttagagcagt tatggaagat 960
gttttttaga aacaatttga agctgaaatt ttagatcaat tgtttgattt gtttaaaaag 1020
aaattggaag gtcattccaa ttttgcactc gctgttggtt tagatttgggt tgttttggtt 1080
aaaagaaatg cagaagatta a                                     1101

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SEQ ID NO: 273      moltype = DNA length = 1107
FEATURE            Location/Qualifiers
source              1..1107
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 273
atgtcatccg aatcattgac aggtaataat agatcttacg atcaatttgc tatgaaagggt 60
ggtagacgat atttttctta cactaaaaat tctattcatc aaagagatgc tattgaagca 120
actaaattat tgattgaaga agcagttgct gaaaaattgg atatttcaaa tttttgtggt 180
aaagcatttt gtattgtctg tttgggttgt tctgttggtc caaatacttt aattgctatg 240
caaaatattg ttgaagaagt taaattgaaa ttcagaaata gaaaagggtt gaattttcca 300
aatattccag aatttcaagt tttctttaat gatcatactg ttaatgattt taatgctttg 360
tttagatcat tgccaactgg tcatgataaa tcttattacg gtgttggtgt tccaggttct 420
ttttacggta gattgtttcc atgtgatact attcatatta tgcatacttc tttttcaact 480
ccatttttgt cttcaagttc aaagaagtt atggaaaaga actctgcagc ttggaataag 540
ggtagaatca attacaatca tgctaagaaa gatgttttaa aagcatacga aggtaaacat 600

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gctgaagata tggattgttt ctttatggct agagcaaaag aattgggttca tgggtggtttg 660
atgttaattg ttactttctt gagaccagat ggtgttccac ataactagagt tttgactaat 720
attggtatgg aagttttggg ttcttgttta atggaattgg ctaaaaaggg tttgattgat 780
gaagaaaaag ttgattcttt taatgctcca gtttacttgg aatctccaga agaattaaaa 840
caagcaattc aaaagaataa gtacttttca attgaaaaga tggaaacagt tccaatgatt 900
atggattcag atattttcttg tagagcacaa caatactctt tggctattag agcagttatg 960
gaagatgttt ttgagaaca atttgaagct gaaatttttg atcaattatt tgatttgttt 1020
aaaaagaaat tgggaaggtc tccaaaattt gcttctgcag ttgttttaga tttgtttgtt 1080
ttgttaaaaa gaaatgcaga agattaa 1107

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SEQ ID NO: 274      moltype = DNA length = 1107
FEATURE            Location/Qualifiers
source             1..1107
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 274
atgtcatccg aatctgttca atcaaatctt tcatcttcag atcaatttgc tatgaaaggt 60
gggtgacgat atttttctta cactaaaaat tctactcatc aaagagatgc aattccaagct 120
actaaatttt tcattcaaga atctattgct gaaaaattgg atgttaataa gttttgtggt 180
aaagcatttt gtgttgctga tttgggttgt tctgttggtc caaatacttt gattgctatg 240
caaaatattg ttgaagctgt tgaattgaaa ttcaaaaata gaaaagggtt tcattctcca 300
actattccag aatttcaagt ttctttaat gatcatactg ttaatgattt taatactttg 360
tttagatctt tgccaactgg tcatgataaa agatattacg gtgttggtgt tccaggttct 420
ttttacggta gattgtttcc atgtgattca attcatatta tgcatacttc tttttcaaca 480
ccatttttgt ctcaagttcc aaaagaagtt atggataaaa attctgcagc ttggaataag 540
ggtagaatac atcataatta tgcaaaagct gatgttttga aagcatatga agctcaacat 600
gcagaagata ttgattgttt cttaactgct agagcaaaag aattgggttca tgggtggtta 660
ttgatggatg taacatcttt tagaccagat ggtgttccac atacacatgt tttgactaat 720
attggtatgg aagttttggg ttactgtttg atggatttgg ctaaaaaggg tttgattgat 780
gaagaaaatg ttgattctta caatgttcca gtttacttgc aatctccaga agaattgaaa 840
caagcagttc aaagaaataa gtacttttct attgaaaaga tggaaatcagt tccaatgatg 900
attgattctg atgtttctgc aaaagctcaa caatattctt tgggtatgag agcagttatg 960
gggtgacgta ttgagaaca atttgggtgt gaaattgttg ataaattgtt tgatttgttt 1020
aaaaagaaat tgggaagaca tccaaaattt gctaaagggt ttgttttgga tatgtttgtt 1080
ttattgaaaa gaaatgctga agattaa 1107

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SEQ ID NO: 275      moltype = DNA length = 1155
FEATURE            Location/Qualifiers
source             1..1155
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 275
atgtcatccc agcccccagg gggccgtaaa agggaacgtg aaagagacag ggaggagatg 60
gaaagccttc aaaacaacaa cagatactgt gaccaattcg ccatgatcgg aggggatgac 120
gattttagct atacaaaaaa ctcaatccac caaagagatg ctatcgaagc cacgaaattt 180
cttattcagg gatctattgc agaaaagtta gacatcaata atttttgcgg taaagcgttt 240
tgcgctgcgg atttgggggtg ctccgttggg cctaatacct tgatcgcgat gcagaatatt 300
gtggacgcgg tagaacttaa gtttaagaac aggaaagggt tccattcacc cacaattcca 360
gagttccaag tcttctttaa cgatcataca gttaatgact tcaatactct tttcagggtc 420
ttgcctacgg gacacgataa acgtttactac ggagtcggag ttcccggttt ttttcttctg 480
agtataatta gtctagattc aatcacacatt atgcacacga gcttctctac accgttccctg 540
agccaagtcc ctaaggaggt tatggataaa aactcagcag cctggaacaa agggcgtatc 600
caccacaatt acgctaaggc cgaatttcta aaggcctacg aggcgcaaca cgcgcaggac 660
atcgattgtt ttttaacagc aagggctaaa gagttagtcc atgggggttt acttatggat 720
gttacctcct tcaggccaga cggagtacct cacacccatg tgcttacaaa tatagggaatg 780
gaagttttag gttactgtct tatggacctg gctaaaaaag gccttataga cgaggagaat 840
gtagacagtt acaatgtccc agtttatcta cagagtccct aggagttgaa acaggctgtc 900
caaaagaaac agtatttctc aatcgaaaaa atggagagtg ttccaatgat gattgactct 960
gatgtatcag caaaagccca gcagttactcc ttgggaatga gggcggtaat gggggacgta 1020
atcagagagc agttcgtgtc ggaataatgt gacaaacttt tcatctgtgt caaaaagaag 1080
cttgaggagc accctaattt cgctaaggga gttgtcctag atatgttctg gctattaaag 1140
aggaacgcgg agggac 1155

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SEQ ID NO: 276      moltype = DNA length = 1146
FEATURE            Location/Qualifiers
source             1..1146
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 276
atgtcatccc agcccccagg aggtagaaaa cgtgaaaggg agagggacag agaggaaatg 60
tataaggcgc cctttgtgtt gtgcgctgcg gtccggttca taactgatca tatgaacttg 120
agtggcgcat gcaattatgg gagggatgca attgaggcaa caaaatttct aatacaggag 180
tctatagctg aaaagctgga catcaacaaa ttctgcggga aagccttctg tgttgccgat 240
ttgggatgca ggcgggccc taacaccctt atagcgtatg aaaaatatgt agaggcggtt 300
gaacttaaat tcaagaaccg taaaggcttt cattcaccca atataccga gtttcagggt 360
tttttcaatg accacaccgt aaacgatttt aacacattat tccgttctact acccaccggg 420

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cacgataaaa gatactatgg agtaggcgtc cccgtattgt tcaccgtgga ttattttttg 480
gttactttgt ttacacacat catgcacact tctttttcaa cgctttttct atctcaagta 540
cctaagaag taatggataa aaactctgcg gcgtggaata agggcgctat tcaccacaac 600
tatgcaaaagg ccgatgtctt aaaagcttat gaagcacaac acgcccaggga catagactgc 660
tttttaacgg cgagagctaa ggaattgggt catggcggtt tgtaaatgga tgtcacatca 720
tttagaccgg acggcgctacc gcacactcac gtacttacta atatagggtat ggagggtatta 780
ggctactgtt taatggactt agctaaaaag ggactaattg acgaggaaaa cgttgattcc 840
tataacgttc cagtttatct acagagtcca gaggagctga aacaggctgt tcaaaagaaac 900
aaatacttct ccatagaaaa gatggaatct gtoccaatga tgattgacag tgacgtctct 960
gcaaaagcac agcagtacag tttgggggatg agagccgtga tgggagatgt tatccgtgag 1020
cagttcgggc ccgagatagt ggacaagctg ttcgatctgt ttaaaaagaa acttgaggaa 1080
catcctaact tcgcgaaggg tgcgtactt gacatgtttg tcttattaaa gcgtaaatgct 1140
gaagac 1146

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SEQ ID NO: 277      moltype = DNA length = 873
FEATURE            Location/Qualifiers
source              1..873
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 277
atgtcatccc aaaatatagt agaagcagtg gagttaaat ttaagaatag aaaggggttt 60
cacagtccga cgatcccga gtttcaggta ttcttcaacg atcataccgt aaatgatttc 120
aatacattgt ttcgtagctt accgacagga cagcacaaga gatactatgg cgtgggagtg 180
ccaggtagtt tctacggtcg tttgttccca tgtgactcca tacatattat gcacactagc 240
tttagtactc ctttcttgag ccaagtaccg aaggagggtta tggacaaaaa ctacagctgcc 300
tggaataaag gtcgtattca tcataaattat gccaaaggcgg atgtattaaa agcttatgag 360
gcacaacacg cggaaagacat cgattgcttc ctaacagcac gtgcgaaggga actgggtccat 420
gggggattgt tgatggacgt taccagtttt agaccgatg gagttcccca cactcacgtc 480
ttaactaaca tttgggtagga ggtgctaggg tactgtctta tggacctggc taaaaagggt 540
ctaattgatg aggagaacgt cgactcctat aatgttccag tttatctgca gagcccagag 600
gaattaaaaa aagctgttca gaggaacaaa tatttttcaa tcgagaaaat ggagtcggtg 660
ccgatgatga tagatagtga cgttaagtgcg aaggcacaac aatactcatt gggaatgcgt 720
gctgttatgg gtgacgttat cagggaaacag tttggtgcag aaatcgtgga caagcttttc 780
gatttgttca agaaaaaatt ggaagagcac ccaaattttg cgaagggggg cgttctggat 840
atgttcgtcc tgctaaagag aaacgcagag gat 873

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SEQ ID NO: 278      moltype = DNA length = 1506
FEATURE            Location/Qualifiers
source              1..1506
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 278
atggcagcat tggcattggt gattttctct tctttgtttt tggctttggt gttgaaatta 60
ttgttgaaac ataagaacaa cagaaattct aaattaccac catgtccacc aaaattgcca 120
atcatcggtg atttgcattt gttgggtaat atgccacaag aatcattgca tcaattggct 180
aagaaatcac gtccagttat tttcttgcaa ttgggtgaag ttccaactat cgttatttct 240
tcagctagat tggcaaaaga agctatgaaa acattcgatt tggcattatc ttcaagacca 300
caaattttct ctgctaagca tttgttttat aattgtactg atgttgtttt ctctccatat 360
gggtcttact ggagaaacat tagaaaaaat tgtatcttgg aattgttatc tgcaaaagaga 420
gttcaatctt ttcatacgt tagagaagaa gaagtgtctc atttggttca aagagttgct 480
gaagcatatc caggtactac aaatttgaca aagatgttgg gtttgtacgc aaacaatttg 540
gtttgtagag ttgctttcgg taaagatttc tcacaagggtg gtgacaacca tagacataac 600
ttccaaaaat tgttagaaga ataccaagtt ttgttaggtg gtttttctat tggtgacttt 660
ttcccatcaa tggaaattgt tcataagttt actggtatgg aatctagatt aaaagatact 720
tttagaagat ttgatcaatt tttcgatgaa atcttgcaag aacatagaaa ccagaatca 780
agaaaagAAC ataaggattt ggttgatggt ttgttagaaa ttcaaaagaa tggtgaccca 840
gaaactccat taactacaga taacatcaag tctatcattt tggatatggt tgcgtcaggt 900
actgatacaa acttcatcac attggattgg gctatgactg aattaatgac aaaccaggt 960
gttttgaaaa gagcacaagc agaagttaga aatctcgttg gtgaaagaag atctgtttca 1020
gaaaacgatt tgcacaattt gccatatgtt aaggcagttg ttaaggaaac ttttagatta 1080
catccaccag ctccagtttt ggttccaaga gaatctatgg aagatgttac tattgatggt 1140
tacgatattc cagcaaaaac aagatttttc atcaacgctt gggcaattgg tagagatcca 1200
aaatcttggg aaatccaga tgtttttgaa ccagaaaagt tcatgaactc tcatatcgat 1260
ttcaagggtc aagattacga attgattcca tttggtgctg gtagaagatc ttgtccagca 1320
atcacttttt cagctgttac agttgaattg gctttagcac aattgttgca ttcttctgat 1380
tgggaattac caccaggtgt tgaagcaaaa gatttggata tgactgaagt ttttgggtatc 1440
acaatgcata agatctatcc attgatgggt gttgtctaac catacttttc acaagggttg 1500
aattaa 1506

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SEQ ID NO: 279      moltype = DNA length = 1506
FEATURE            Location/Qualifiers
source              1..1506
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 279
atggcagctt taccattggt gattttctct tctttgtttt tggctttggt gttgaaatta 60

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ttgttgaaac atactaaatt ggttaaatta ccaccaggtc caagaccatt accaatcacc 120
ggtaatttgc atcaattgaa cggttctttg ccacatagaa ttttgactga tttggctaag 180
aaatatggtc cattgatgca tttgcaattg ggtgaagttt caacagttat tgtttcttca 240
ccagctgttg caaaggaagt tatgaaggat catgatatta tttctgttta tagacctttg 300
ttattgtctc cagcaatctt gaactacggt ggtacagata ttgcttttgc accatatggt 360
gaatactgga gacaattgag aaaaatctgt actatcgaat tggtatcttc taaaagagtt 420
caaaactttta gatctattag agaacaagaa gtttcaaatt tggttagatc tattaaatca 480
gaagaaggtt ctggtgttaa tttgtctaag atgatctatt ctttgactta ctcaattaca 540
gctcaagctg catttggtta agaactaag taccaggata agtttatctc agttatgcaa 600
gatgctatta aattaatggg ttggtttaat atcgcagatg tttacccttc tattaaaatc 660
ttggaaaaga ttactgggat tagacataaa ttggaagaaa cacataacca aatcgatgaa 720
atcgttcaaa acattttgat tgaacataag gctaagagag cagaatcaaa attgggtaat 780
agaggtgcta aggaagatgt ggttgatgtt ttgttgaaca tccaatcttc aggcgaattt 840
gaaccaccat taactgatga taatttgaaa gctgttatat atgatatttt ctctgctggt 900
gggtgaaactt ctctactac agttattttg gcaatgtcag aatgatttag aaatccaaaa 960
gttatgaaga agctcaaga tgaagtttag gaagtttttg ctggtaatgg taatggtgat 1020
gaatctagat tgcattgaatt aaaatacttg caagctgtta ttaagaaaac tatgagatta 1080
aatccagcag ttccattatt gttaccaaga gaattgttctg aacattgtca aattaatggt 1140
ttcgatgttc cagctaaaac tagagttatt gttaatgctt gggcaatttg tagagatcca 1200
aaacaatgga cagaaccaga aaaattcaaa ccagatagat ttttgatttc tgaattcgat 1260
ttcagaggta aaactttttc ttttattcca ttggtgctg gtagaagaat ttgtccaggt 1320
atttcttttg ctttgccatt agttgaattg ccattagcac aattggtgta ccatttcgat 1380
tggaatttgc caggtcaatc aaaaccagat gaattagata tgactgaaat cttcggtttg 1440
acagttggta gaaagcatga tttgaatttg gttccaattc catcacagacc atgttctgtt 1500
ttgtaa 1506

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SEQ ID NO: 280      moltype = DNA length = 1506
FEATURE
source             Location/Qualifiers
                   1..1506
                   mol_type = other DNA
                   organism = synthetic construct

```

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SEQUENCE: 280
atggcagaat tggtgtttac ttgtgtgctt ttgtgggttt tcttttcttt cgcagttaag 60
atcgttagaa aaacaaaacc agttaaatta ccaccaggtc caagaccatt gccaatcacc 120
ggtaatttgc atcaattgaa cggttcttta ccacatagaa ttttaagaaa tttggctaag 180
aaatatggtc cattgatgca tttgcaattg ggtgaagttt caactgttat tatttcttca 240
ccagcttttg caaaggaagt catgaacgaa catgatattg tttctgttta cagaccattt 300
ttgttatctc catcaatctt gaactacggt ttgtctgata ttgcttttgc accatatggt 360
gaatactgga gacaattgag aaaaatctgt actatggaat tggttaacatc tagaagagtt 420
caaaactttta gatcaattag agaagttgaa gtttctaatt tgattagatc aatttcttca 480
gaagaaggtt ctggtgttaa tttgtctaag aaaatatatt ctttgactta ctcaattaca 540
gctcaagctg catttggtta aagatctaag taccatgata agtttatctc aatcatgcaa 600
gatgttgtta aattcgatgg ttggttttaa atcgcagata tttacccttc tttaaaaatc 660
ttggaaaagt tgactgggat tagacataaa ttgaaatgg ttccataacca agttgatgaa 720
gttattgaaa acattttgtc tgaacataag gttgaagag ctgattcaaa tttgggtaac 780
agagaagcaa aagaagattt ggttgatgtt ttgttgaaca tccaatcttc aggcgaattt 840
gggtgttccat taacagatca taacatcaag gctgttatat atgatgtttt ctctgctggt 900
gggtgaaactt ctctactac agttattttg acaatgtcag aatgatttag aaatccaaaa 960
gttatgaaga aagctcaaga tgaagtttag gaagtttttg ctggtaatgg taatggtgat 1020
gaatctagat tgcattgaatt aaaataattg caagctgtta ttaagaatc aatgagattg 1080
catccagctg caccattggt attgccaaga gaattgtctg aacattgtca aattaatggt 1140
tacgatattc cagctaaaac tagagttatt gttaatgctt gggcaatttg tagagattca 1200
aaagcatgga cagaaccaga aaagtttaat ccaaaccagat ttttagattc tgaattgat 1260
tttagaggta aaactttttc ttttattcca ttggtgctg gtagaagaat ttgtccaggt 1320
atttcttttg ctttaccatt ggttgaatta ccattggcac aattggtgta ccatttcgat 1380
tggaatttgc caggtcaatc aaaaccagaa gaattagata tgacagaaat cttcggtttg 1440
gcagttggta gaaagcatga tttgaatttg atcccaatcc catcacagacc atgttctaga 1500
gaataa 1506

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SEQ ID NO: 281      moltype = DNA length = 672
FEATURE
source             Location/Qualifiers
                   1..672
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 281
atggcaattg caaaagatgt ttgtgttact catcatccat cttttgcaga tagacaaaac 60
aatttggtca cagaatcatc cagatacagat caagaagata tgggtttttc accatatggt 120
gactactgga agcaaatgag aaagatctgt atgatgcaat tgttatctat gactaacgtt 180
gcatcatttt cttcaatcag acaagatgaa atttctaaat ttggtgaagc tattagatca 240
acaccaggtg ttccagttta ttgactgaa aagatttttc catgtacatc ttcaatcatt 300
tgtagagctg cactcggtaa agtttgtcaa gataaggaaa acatgattaa aaatttgaaa 360
atcgcttttg ttttgttagc aggtttttaa ttggctgatg tttttccatc tttgaagttt 420
attccaatca tcaactggtt taacacaaaa ttgattaaag ttcatcataa aatggatgaa 480
atcttgggtg acgttattaa agaacatctc ttgaacatg aaaaacgtta aaagtgtaac 540
gctgaatcag gtgacgaaga ttgatcgat gttttgttgc aacaacctaa aaatggtaat 600
ttgcatatcc caatcactgc aagaaatctc aaagcattga ttttagtttc tacaatttca 660

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-continued

ttgtgttttt aa 672

SEQ ID NO: 282 moltype = DNA length = 1341
 FEATURE Location/Qualifiers
 source 1..1341
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 282

atggcattgc	aattgggttc	tgttaaaaca	ttagttgttt	cttcagctga	agctgcacat	60
gaaatcatga	aaactcatga	tttgaatttc	ttgaacagac	cagaatcaaa	ggtaacaga	120
aaattgttgt	acgattttta	agatttgtct	gttgctccat	atgggtgaata	ctggagacaa	180
atgaaatcaa	tctgtgtttt	gcatttggtta	tctaataaga	gagttcaatc	aatcagatac	240
attagagaag	aagaaacttc	atttttgatc	actaagatta	gagaaggttc	tttagcttca	300
ccatcttcag	gtgcagttaa	tttgtctgaa	atggttcgtta	ctttgacaaa	tgatgttgtt	360
tcaagagctg	catttggttag	aaaatattct	gaggggtgaat	caggtagaaa	gtttagaaaa	420
ttgatgtctg	aatttggttc	agtttttaggt	ggtttcgatt	tcggtacagt	tttgccatgt	480
ttgggttggg	ttgatagatt	aaatggtttg	actgctagag	ttgaaagaat	cgcaaaggaa	540
atggataagt	tcttagaagg	tgttttgtaa	gaacatttgg	attctgatag	aaagaataac	600
gaatcaagag	ttgttatgac	tagaaaaaatt	catggtgaaa	acggtgaaga	ttttgctgat	660
gttttggtgg	gtatctataa	gaataaaggct	gcaggtattt	ctatcgatag	agattcaatt	720
aaaggtttgt	tattggatat	ttttgcaggt	ggtacagata	ctacatacac	tgttttagaa	780
tgggctatga	cagaaattatt	gagacatcca	aaagcattaa	aagaattgca	agatgaagtt	840
agaggtttgg	ttggtggtgg	tagagaaatc	acagaacaag	atttggaaaa	gatgaagtat	900
ttgaaagctg	ttattaaaga	aaactttaaga	ttgcatccac	caattccatt	attggttcca	960
agaaaagcta	ttgaattctg	aaagatcatg	gattacgata	tttctgcagg	tacaatggtt	1020
attactaacg	ctttcgcgat	tggtagagat	ccatctttgt	gggaagaacc	agatgagttt	1080
agaccagaaa	gattttttaa	ttcatgtatc	gatttcagag	gtcatgattt	ccaattgatt	1140
ccatttgggt	ctggtagaag	aggttgtcca	ggtatttcat	ttgctatggc	aacaaacgaa	1200
catgttttgg	ctaatttggg	tttgaatttc	gattgggttt	tgccagatgg	tgcaaaagggt	1260
actgatttgg	atatgactga	atgtacaggt	ttgactatcc	atagaaagtt	cccaatcttc	1320
gctgttgcaa	caccagttta	a				1341

SEQ ID NO: 283 moltype = DNA length = 1023
 FEATURE Location/Qualifiers
 source 1..1023
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 283

atggcattca	tgactttgac	aaacgatgtt	atgtctagag	ctgcatttgg	ttggatttgt	60
tcagaggggtg	aattctggtag	aaagttttaga	aaattgatgt	cagaatctgc	tagagttttt	120
gggtgtttttg	atttcggtag	ttatttgcca	tggttgggtc	aaatcgattc	agttaacggt	180
ttgatgtgtta	gagttgaaag	aatcgcaaag	gaaatggatg	aattttttaga	aagagttttg	240
gaagaacatt	tggattctga	tagaaaagaat	aacgaatcaa	gagttgttat	gacttctaag	300
attcatgggtg	aaaacggtga	agatttcgct	gatgttttat	ttggtatcta	taagaataag	360
gctgcaggtt	cttcaatcga	tagagattct	attaagcat	tggtgttgga	tatgtttgca	420
gggtgtactg	atacaaaacta	cacatttttg	gaatgggcta	tcactgaatt	attgagacat	480
ccaaaagcat	taaaagaatt	gcaagatgaa	gttagaggtt	tagttgatgg	tggttagagaa	540
atcacagaaac	aagattctga	aaagatgaag	tatttgaag	ctgttattaa	agaaaacttta	600
agattgcatc	ccaccattcc	attgtcagtt	agaaaagcta	ttaatgatgt	taagatcatg	660
gattacgata	tttctgcagg	tactacagtt	attacaaaacg	ctttcgcaat	tggttagagat	720
ccatcattgt	gggaagaacc	agatgagttt	agacctgaaa	gattttttaa	ctgttgtgtt	780
gatttcagag	gtcatgattt	ccaattgatt	ccatttgggt	caggtagaag	aggtttgtaca	840
gggtatttctt	ttgctatggc	aactaacgaa	catgttttgg	ctaatttggg	tttgaatttc	900
gattgggttt	tgccagatgg	tgcaaaagggt	acagatttga	acatgactaa	atctacaggt	960
ttgactatcc	ataggaaaaa	ttcttcttct	tggtgttgta	attcatcttt	attgactcaa	1020
taa						1023

SEQ ID NO: 284 moltype = DNA length = 1125
 FEATURE Location/Qualifiers
 source 1..1125
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 284

atggcagaat	tgttatctcc	aaagagagtt	caaactttta	gatcaattag	agaagggtgaa	60
gttttgaaca	tgtatcaagtc	tattgtctta	catatgggtt	cttcagttga	tttgtctaga	120
aaaattttct	ctttgccata	cgttattaca	gcacaagctg	cattcggttaa	aaagggttaa	180
taccaagaac	aatacatcat	gttgattgat	gaaatggtta	aattgtctgg	tggtttttca	240
tttagttgatg	tttcccatc	tattaaattg	ttagaagaag	tttcaggtac	tagaacaataa	300
ttagaaaaat	tgcataaaca	aattgatggt	attttacaaa	tcctcgttaa	cgaacataaa	360
aagaaattgg	aaaaagaaga	attaggttct	gggtgaaaaga	aagttgattt	gggttgatgtt	420
ttgttgaaca	tcggttaaaag	aggtgaattt	ggttcacatt	tggtcgataa	caacatcaag	480
gcaatcatct	atgatatttt	ctctgctggt	tcagaaaactt	cttcaactac	agttgaatgg	540
gcaattttctg	aaatgatcaa	gtcaagaaca	atcatgaaga	gagcacaaga	tgaagttaga	600
aatattttta	aggaacgtgg	taacgttgat	gaatctaatt	tgcatgaatt	gaaatactta	660
caagctgtta	ttaaagaaac	tatgagattg	catccaggtg	caccattggt	gttgccaaaag	720
gaatcaatcg	aaaagtgtga	agtttaagggt	tacgaatcc	cagctaaaac	aagagttatt	780

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attaatgctt	gggcattagg	tagagatcca	atgtactggt	ctgaagcaga	aaagtttaat	840
ccagaaagat	ttttggattc	agaattagat	ttcagaggta	accatttcga	atatattcca	900
tttggctgctg	gtagaagaat	ttgtccaggt	atttcttttg	ctttgccaaa	cactgaatta	960
ccattggcac	aattattggt	tcatttcgat	tggaaattgc	ctggtgactt	gaaacaagaa	1020
caattggata	tgactgaagt	ttttggtgtt	actgttagaa	gaaagaatga	tttgagattg	1080
attccagttc	catactctca	ttcttctttg	aataaggatt	tgtaa		1125

SEQ ID NO: 285 moltype = DNA length = 774
 FEATURE Location/Qualifiers
 source 1..774
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 285

atggcattgg	ctaagatcgt	tatgaaatct	aaaaacaaaa	attctggtaa	aaagttgcc	60
ccagggtcaa	gacaagttcc	aatcatcggt	aacatgcac	aattgtatgg	tccattacca	120
catcataaat	tgagagattt	ggctaagaaa	tacggtccat	tgatgcattt	gcaattgggt	180
gaaattttcta	ctatcgttgt	ttcttcacca	gaatttgcaa	aggaagttaa	cagaacacat	240
gatattatct	tcgctgaaag	agcagaattg	ttatctattt	caattgtttc	atacaatttc	300
ttgggtatcg	tttttgctcc	atatgcagat	tactggagac	aattaagaaa	aatctgtact	360
atggaattgt	tatctccaaa	gagagttcaa	acttttagat	caatctggga	aggtgaagtt	420
ttgaacatga	tcaagtctat	tgctttacaa	aagggttctt	cagttaattt	gtctagaaaa	480
attttctctt	taactttatg	tattacagca	caagctgcat	tcggtaaaaa	gggtaataac	540
caagaacaat	acgttatgtt	gggtgatgaa	atgggttaaa	tgtctggtgg	tttttctttg	600
gttgatttgt	accatcttat	taaattgtta	gaaagaattt	caggtactag	aacaaaattg	660
gaaatcttgc	ataagcaaat	tgatggtatt	ttgcaaaact	ttgttaatga	acataaaaaa	720
aaattggaaa	aagaagaatt	aggtctcggt	gaaaagaaat	ctttgtctgg	ttaa	774

SEQ ID NO: 286 moltype = DNA length = 1566
 FEATURE Location/Qualifiers
 source 1..1566
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 286

atggcagcaa	ttttctctaa	gacgcggtatc	gttgggtgctg	gtattttctgg	tattgctgca	60
gctaagcaat	tgagacatta	caagccagtt	gtttttgaag	caactgatcc	tttaggtggt	120
gtttggagac	attgttctta	cagatcaact	aaattgcaaa	caccaagatg	tgattacgaa	180
ttttcagatt	ttccttggac	agaaaagagat	aattcttcat	ttccatctta	catggaaatc	240
ttggattact	taaattctta	cgttccaaac	ttcgggtgtt	tgggaattgat	taagtttaat	300
tcaaaggttg	ttgaatttaa	gtttactggt	aacagagatc	aatctaacta	ctctgattca	360
ggttctaatt	caaaaggaata	cggttaattt	ttaggtggtc	aaccagtttg	ggaagttgca	420
gttcaaaact	ctgattcaga	tacaatcgaa	tggtacgctt	tcgaattttt	agttatttgt	480
actggtaaat	acggtgacat	tccaatcacc	ccaaagtcc	caaacataa	gggtccagaa	540
gtttttagag	gtcaagttat	gcatacattg	gattactcaa	aattggatga	agaagcatct	600
gttaatttgt	tgaagataa	gaaagttgtt	gttattgggt	acaagaaatc	tgctatcgat	660
ttggctgttg	aatgtgcaaa	agctaattgaa	gggtccagaag	gtaaacccatg	tactgtttgt	720
gttagaactt	tgacttggac	agttccacat	tactctatct	ggggtttgcc	attctatttg	780
ttttattcaa	caagagatcc	tcaattcttg	catgaaagac	caaaccaagg	tattttgaga	840
aatatcttgt	gtaaatgttt	atctccagtt	agaaaagcta	tgtcaaagat	catcgaattc	900
tatttggttt	ggaaattgccc	attgaagaaa	tacggtttga	aaccagatca	tccatttgaa	960
gaagattacg	gttcttctga	aatggcaatt	ttgccagaag	aatttttctgc	agaagctgat	1020
aagggtatga	tcgagtttaa	aagagcttca	caatgggtgt	tttgtgaagg	tggtgttcaa	1080
ttcgatgata	actcataaatt	agaagcagat	gttgttttgt	tagctactgg	tttcgatggt	1140
aaaagaaaaa	tgagatcaat	cttgccagaa	ccttttagat	ctttgattga	aactgatggt	1200
gttatgccat	tgtacagagg	tacaatccat	ccattgatcc	caaacatggc	attcgtttgt	1260
tacatcgaa	ctgtttcaaa	tttgcataca	gctgaaatca	gatgtatctg	gttgtcaaga	1320
ttagttgaaa	actacttcaa	attgccatct	ttagaaaaca	tgttggaaca	aactcaaaag	1380
gaaatggaaa	tcattgagaag	aactacaaga	tttttcaaaa	gatcttgtat	ctcaacattt	1440
tctattaatc	attctgatga	aatttgtgaa	gaaatgggtt	ggaaatcatg	gaggaaaaat	1500
tcttgggttg	cagaagcatt	ttctccatcc	aactctcaag	attaccaaga	aaagaaagtt	1560
aattaa						1566

SEQ ID NO: 287 moltype = DNA length = 990
 FEATURE Location/Qualifiers
 source 1..990
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 287

atggcagcaa	tgtcaaagg	tttgtctttg	ttttctctgc	aattcttgtg	tgttttgtgt	60
atttcagatt	tgtctaacgg	tcaaggtttg	gatgttaact	tctacaagaa	aacttgtcca	120
tctttagaag	atattgttaa	ggatactaca	gcaagattca	tttctagagc	accatctttg	180
gctgcacat	gtttaagaat	gcatttccat	gattgttttg	ttagaggttg	tgatggttca	240
gtttttgttaa	attctacttc	ttcaaaacca	tcagaaaagg	ctgcaatccc	aaaccaatct	300
ttaagaggtt	tccaagttaa	tgatgcagtt	aaatcagaat	tggaaaagaa	atgtccaggt	360
gttgtttctt	gtgctgatat	tgttgcctta	gttgcaagag	atgctattac	attgattaaa	420
gggtccatcat	ggcaagttga	attaggtaga	agagatggta	gagtttcaat	cgcttctcaa	480
gttttgaaata	agttgccatc	tccttttatg	aacatcgcac	aattgaaagc	atcttttcaa	540

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tctttgggtt tatcagttaa ggatttggca gttttgtctg gtggtcatat tattggtatt 600
tcacattgtg cttctgttaa caacagatta tacaacttca ctggtaaagg tgacacagat 660
ccaactttag atccaaagta cgttgcaact ttgaaaagaa agtgtcaacc aggtgacgtt 720
actacaattt tggaaatgga cccaggttct gctaaaacat tcgatactga ttacttcact 780
acagtttcaa aaagaagagg tttatttcaa tctgatgctg cattgttaga tgatactgaa 840
acaaatactt acgttcaaca acatttgtca tcttctggtt caggttcttt ctttgatgat 900
tttgggtgtc caatggttaa aatgggtaga attggtgttt tgacaggtaa ccaaggtgaa 960
attagaaaaa aatgtgcttt ggttaattaa

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SEQ ID NO: 288      moltype = DNA length = 990
FEATURE            Location/Qualifiers
source              1..990
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 288
atggcagcat ttccaagaca tttggctgtt tttatgttgt tgcaaatcat cttgtttcca 60
acagcatttc aacattgtta cgtcgaaggt ttgaaattag gtttctacaa gaaaacttgt 120
ccaggtgctg aagcaacagt taagcaaatc atgactggtg ttatttctag agcaccaaca 180
ttggctgcac caatgtttaag aatgcatttc catgatttgt ttgttagagg ttgtgatggt 240
tctgttttgt taaattcaac taaaaataat aagtctgaaa aggaagctat cccaaaccaa 300
tcattgagag gtttccaaat catcgataac atcaagtctg ctttggaata gaaatgtcca 360
ggtgttgttt catgtgctga tattttggct ttagttgcaa gagatgctgt ttctttgtta 420
aatggtccat ttgtgattgt tcatttaggt agaagagatg gtaaaagttc aatcatgaac 480
gaagcattga caaatttgcc accaccattc gctaactca ctaattgaa agcatttttc 540
tctgctaagg gtttaaatat aaaggatttg gcagttttat caggtggtca tactatttgt 600
acagctcatt ttgtgggttt tactaacaga ttatacaact ttactggtta aggtgacact 660
gatccatctt tagatccaaa ctacatccca ttgttgaaaa gaaagtgtaa gccaggtgac 720
gttactacat tagttgaaat gactccaggt acttttaaaa ctttcgatga agattactac 780
acattgatta ctaaaagaag aggttttatt caactctgat ctgcattgtt agatgattca 840
caacaagagc catacgttaa attgcaagct gcaactcatg gttctacttt ctttaaggat 900
ttcgtctgtt caatggttaa gttgggttaa gttggtgttt tgactggtta agcaggtgaa 960
attaggagac attgtgcttt agttaattaa

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SEQ ID NO: 289      moltype = DNA length = 951
FEATURE            Location/Qualifiers
source              1..951
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 289
atggcagctt ctccaattgt tccaactatt gatttggacc catttttgag agaaggtgac 60
gaagaagcta agaaaaatgc aatcgaagaa atctgtcagg catgttctga atacggtttc 120
tttcaagtgt ttaatcatgg tgttcatttg gatttgatga ctagagcaac agatatgttc 180
actgaatttt tcaattatcc aatcgaagaa aagttttaat ctatttcaaa gccaggtggt 240
tctccaccac caggttactt caactctcca gatcattcaa aggagggtta cgaacatttc 300
ttgatcttga caccatcatt gggtttaaat gtttaccag aaaatccacc agagttaga 360
aaggttttgg aagaatcttt cccaaatttg gctaaattag cattgttaat cgaagaat 420
gtttctgttt cattgggttt agcaccaaat ttcttgaaag aatacaacaa cgatagatct 480
tgtgattctt tgatgggttt acattacttt ccagctgaag aaactgaaaa taatggttaa 540
tcagcacatg aagatggtta ctgtatcaca ttcgtttttc aagatgaagt tgggtggttg 600
gaagttttga aaaatggtca atggattcca atcactccag ctcaaggtag attggttgtt 660
aacatcggtg acgtttgttc agttttatct aacaataagt ttaaatcagc aacacataga 720
gttgtagagc caaagggtta atctagatgc tcatatgttt tctttcataa ttgcatggt 780
gacaagtggg ttgaaccatt accacaattc actaaggata ttggtgaatc tgctaagtag 840
agaggtttcg tttacaagga atacatgcaa catagaatta gaaatagaac acatccacca 900
gcaagaccag aagatttgat ccatatcact cattactcta tttcaacata a 951

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SEQ ID NO: 290      moltype = DNA length = 954
FEATURE            Location/Qualifiers
source              1..954
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 290
atggcaccat tgttacaat gccaactgtt gatttgcatt ctttctttac aggtgacgat 60
gttggtcaaa agaattgttt gtctaaaatt tcacaagcat gttctgatta cgttttcttt 120
caaatcgtta atcatggttt gtgtttggat ttgatgaaca gagctttgga aatttctaga 180
gttttctttg aatcaccaac tgaagaaaaa ttgaaatgtt ctccaagacc aggtgcttca 240
ccaccagcag gttacaacag aaagccaaac ccagaatacg aattttctga atacttcac 300
atgtttccac caggttcaga ttccaacaga ttccaaccaa atttgccaca attgagagaa 360
gttatggaag aattattttc tcaatttttg aaaactgctt cagttatttc ttcaatcttg 420
atcgaatctt taggtttgcc acaatcaatc gatgaaggtt ttaatttcgc agatagatct 480
tcagatttgt tgattgcttt attttatttg tctgcaagg aaaaggaaaa catgggtgtt 540
aaccacata aggatgtttc ttgtttcaca atgggtttgc aagatgaagt tgggtgtttg 600
ggtgttgaaa aagatgttga atggttacca ttgggtccaa ctgaagggtc ttgatcgtt 660
aacatcgagc atgttttgca agttttgaca aacgataagt ttaaatctcc aaatcataga 720
gttttaagac caaaagctag atcaagatct tcatttgctt tctttcatga ttgcccaggt 780
aaaacttggt tagaaccatt gccagagttt actagagaaa ttggtgaatt accaaagtag 840

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agaaagtctt tgtacaagga ttacaaacaa atgagagtta gaaataagtc taaccacat 900
tcaagattag aagatgaaat cacaatttct cattaagcta tcgcatcaaa ataa 954

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SEQ ID NO: 291      moltype = DNA length = 1506
FEATURE            Location/Qualifiers
source              1..1506
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 291
atggcagctt tggcattggt aatttttgc tgcagttctt tgattgttgt tttgaaattt 60
gtttcaaac caaagaataa tagaaattct aaattgccac catgtccacc aaaattgcc 120
attattggta atttgcattt gttaggtaat atgccacaag aatctttgca tcaattggct 180
aaaaagtagc gtccagttat tttcttgcaa ttaggtgaag ttccaacaat tgttatttct 240
tcagcaagat tggctaaaga agcaatgaaa acttttgatt tggctttatc ttcaagacca 300
caaattttct cagcaaaaca ttgtgtttac aattgtacag atgtgtgttt ctctccatat 360
ggtgcttact ggagaaatat tagaaaaatt tgtattttag aattgtgtgc tgctaaaaga 420
gttcaatcat tttcttatgt tagagaagaa gaagttgctc atttgggtca aagagttgct 480
gaagcatatc caggtagaac taatttgaca aaaatgttag gtttatatgc taataatttg 540
gtttgtagag ttgcttttgg taaagatttt tctcaaggtg gtgacaatca tagacataat 600
tttcaaaaat ttttgggaaga ataccaagt ttgttaggtg gtttttctat tggtagcttt 660
ttcccatcta tggaaattgt tcataaatc actggtatgg aatctagatt aaaagatact 720
tttagaagat ttgatcaatt ttctgatgaa attttacaag aacatagaaa tccagaatct 780
agaaaagaac ataagattt ggttgatgtt ttattggaaa ttcaaaagaa tggtagacca 840
gaaactccat tgcaactga taatattaaa tctattattt tggatatggt tgcgtcaggt 900
acagatacta attttattac tttagattgg gctatgactg aattgatgac taatccaggt 960
gttttgaaaa gagcacaaagc tgaagttaga aatattgttg gtgaaagaag atcagtttct 1020
gaaaatgatt tgcacaaatt gccatattgt aaagcagttg ttaagaagaa ttttagattg 1080
catccaccag caccagtttt agttccaaga gaatctatgg aagatgttac tattgatggt 1140
tacgatattc cagctaaaac aagatttttc attaatgcat gggctattgg tagagatcca 1200
aaatcatggg aaaatccaga tgtttttgaa ccagaaagat tcatgaactc acatattgat 1260
tttaaaggtc aagattttga attgattcca ttggtgctg gtagaagatc ttgtccagca 1320
attacttttt ctgctgtttc tgttgaatta gctttggcac aattgttaca ttcttttgat 1380
tggaatttac caccaggtgt tgaagctaaa gatttggata tgactgaagt ttttggattt 1440
actatgcata aaatttacc attgatgggt gttgctaaac catacttttc acaaggtttg 1500
aattaa 1506

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SEQ ID NO: 292      moltype = DNA length = 1506
FEATURE            Location/Qualifiers
source              1..1506
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 292
atggcagcat taccattggt tatttttgca tctatttctt tggttccagt tttgggtatt 60
ttattgatga aaaaggataa aagatcaggt aaattgccac catctccacc aaaattgcc 120
attattggta atttgcattc aattggtaat atgccacatg aatcatttca taatttggct 180
aaaaagtttg gtccaattat gtttttggg ttaggtcaag ttccaactgt tgttatttct 240
tcagcaagat tggctaaaga agcattgaaa acatctgatt tggcattgtc atctagacca 300
gaattgtttt ctgctaaca ttgtcattac aattgtacag atttggcatt ttctccatac 360
ggtacttatt ggagaaatat tagaaaaatt acaactttac aattgttatt tgctaaaaga 420
gttcaatcat tttcttttgt tagaaaagat gaagttgcac atttgggtta aagaattact 480
gaatcttatt catcaactat taatttgcct catatgttgg aagtttatgc taataataat 540
atttgtagag ttgttttggg taaagatttt tctcaaggtg gtgacaatgt tagacataat 600
tttcaaaaag ttttgggaaga ataccaaatg ttgttaggtg gtttttcaat tggtagctac 660
tttccatcta tggaaatttt gcataaatc actggtatgg aatcaagatt gaaagatact 720
tttagaagat ttgatcatat ttgtgataaa attttggaag aacatagaaa aacagaaaat 780
agaaaagatc aaaaagattt ggttgatgtt ttattggaaa ttcaacataa tggtagactg 840
gaaacaccat taactacaga taatattaaa gctattatta ttgatattg ttgttgggtg 900
actgatacaa attttattac attagattgg acaatgactg aattgattat taatccaaaa 960
gttatgcaaa gagcacaaagc tgaagttaga tctattgttg gtgaaagaag atttgtttca 1020
gaaaatgatt tgtctcattt gggttacatg aaagctgttg ttaagaagaa ttttagattg 1080
catccaccag ctccattatt ggcaccaaga gaatctattc aagatgttac aattgatggt 1140
tacaatattc cagctaaaac tagatttttc attaatgcat gggctattgg tagagattct 1200
gaattgtggg aaaatccaga aatgtttgaa ccagaaagat tcatgaactc acatattgat 1260
tttaaaggtc aagattttga attgattcca ttggtgcag gtagaagaat ttgtccagct 1320
atttctttta ctgttgctac tgttgaatta gctttggctc aattattgca tggttttgat 1380
tggaatttgc caccaggtgt tgaagctaaa gatttggata tgactgaagc atttggattt 1440
acaatgcata gaactttctc attgatgggt gttgcaaaac catacttttc tgatcatttg 1500
aattaa 1506

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SEQ ID NO: 293      moltype = DNA length = 1506
FEATURE            Location/Qualifiers
source              1..1506
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 293
atggcagcat taccattggt tatttttgca tcaatttcat tggttttagt tttgggtatt 60

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ttatcaatga aaaaggataa aagatctgtt aaattaccac catcaccacc aaaattgcc 120
attattggta atttgcata aattggtaaat atgccacatg aatcttttca taatttggct 180
aaaaagtttg gtccaattat gtttttatgg ttgggtcaag ttccaactgt tgttatttca 240
tctgctagat tggctaaaga agcattgaaa acatttgatt tggcattatc taatagacca 300
gaattgtttt cagctaaaca ttgcatatg aattgtactg atttggcatt ttctccatat 360
ggtagcttat ggagaaatat tagaaaaatt acaactttac aattgttatc tgctaaaaga 420
gttcaatcat ttcttttgt tagaaaagat gaagtgttgc atttgggtta aagaattact 480
gaatcttacc catctactat taatttgtct catgttttgg aagttttacgc taataataat 540
atattgtccag ttgcttttgg taaagatttt tctcaaggtg gtgacaatgt tagacataat 600
tttcaaaaag ttttagaaga ataccaaatg ttgttaggtg gtttttctat tggtgactac 660
tttccatcta tggatttttt gcataaaatc actggtagtg aatcaagatt aaaagatact 720
ttgagaagat ttgatcatat ttgtgataaa attttggaag aacatagaaa aactgaaaat 780
agaaaagatc aaaaagattt ggttgatgtt ttgttagaaa ttcaacataa tggtgacttg 840
gaaacaccat tgactacaga taatattaaa gctattatta ttgatattgt tgttgggtgg 900
actgatacaa attctattac ttggattggg acaatgactg aattgattat gaatccaaaa 960
gttatgcaaa ggcacaaagc tgaagttaga tcaattgttg gtgaaaagag atttgtttct 1020
gaaaatgatt tgcctactat gggttacatg aaagctgttg ttaaaagaaac ttttagatta 1080
catccaccag caccattatt ggtcccaaga gaattctatg aagattgtac tattgatggg 1140
tataatattc cagctaaaac aagatttttc attaatgcat gggctattgg tagagattct 1200
gaattgtggg aaaaacgaga aatgtttgaa ccagaaaagt tcatgaactc tcatattgat 1260
tttaaggttc aagattttga attgattcca ttgggtgctg gtagaagaat ttgtccagct 1320
atttctttta ctgttgcac agtgaaatg gctttagcac aattgttaca ttcttttgat 1380
tgggaattgc caccaggtgt tgaagcaaaa gatttggata tgacagaagc atttgggtatt 1440
actatgcata gaacttctcc attgatgggt gttgcaaaac catacttttc tgatcatatg 1500
aattaa 1506

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SEQ ID NO: 294      moltype = DNA length = 1494
FEATURE            Location/Qualifiers
source              1..1494
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 294
atggcagctt taccattggt tatttttgc tctatttcat tggttccagt tttgggtatt 60
ttattgatga aaaaggataa aagatctgtt aaattgccac catctccacc aaaattgcc 120
attattggta atttgcata aattggtaaat atgccacatg aatcttttca taatttggct 180
aaaaagtttg gtccaattat gtttttggg ttagggtcaag ttccaacagt tgttatttca 240
tctgcaagat tggctaaaga agcattgaaa acttctgatt tggctttatc atctagacca 300
gaattgtttt ctgcaaaaaca ttgcatatg aattgtacag atttggcatt ttctccatat 360
ggtagcttat ggagaaatat tagaaaaatt acaactttgc aattattgtc tgctaaaaga 420
gttcaatcat ttcttttgt tagaaaagat gaagtgtgctc atttgggtta aagaattact 480
gaatcttacc catcaacaat taatttgc catatgttgg aagttttacgc taataataat 540
atattgtagag ttgcttttgg taaagatttt tctcaaggtg gtgacaatgt tagacataat 600
tttcaaaaag ttttgaaga ataccaaatg ttgttaggtg gtttttctat tggtgactac 660
tttccatcaa tggatttttt gcataaaatc acaggtatgg aatctagatt gaaagatact 720
tttagaagat ttgatcatat ttgtgataaa attttagaag aacatagaaa aacagaaaat 780
agaaaagatc aaaaagattt ggttgatgtt ttattggaaa ttcaacataa tggtgacttg 840
gaaactccat tgacaactga taatattaaa gatattgttg ttggtggtag tgatacaaat 900
tttattactt tggattggac aatgactgaa ttaattatta atccaaaagt tatgcaaaaga 960
gcacaagctg aagttagatg aattgttggg gaaagaaagt ttgtttctga aaatgatttg 1020
tctcatattg gttacatgaa agctgttgtt aaagaaaact ttagattgca tccaccagct 1080
ccattattgg ctccaagaga atctattcaa gatgttacta ttgatgggta taatattcca 1140
gctaaaaacta gtttttctat taatgcatgg gctattggta gagattctga attgtgggaa 1200
aatccagaaa tgtttgaacc agaaaagatc atgaactctc atattgattt taaaggtcaa 1260
gattttgaat tgattccatt tgggtgctgt agaagaattt gtccagctat ttcttttact 1320
gttgctacag ttgaattggc tttagcacaa ttattgcatg gttttgattg ggaattgcc 1380
ccaggtgttg aagcaaaaaga tttggatatg actgaagcat ttgggtattc tatgcataga 1440
acttctccat tgatgggtgt tgcataacca tatttttctg atcatttgaa ttaa 1494

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SEQ ID NO: 295      moltype = DNA length = 1506
FEATURE            Location/Qualifiers
source              1..1506
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 295
atggcagctt tgccattatt gattttctca tctttgtttt tagctttgtt attgaaatta 60
ttgttaaaac ataaaaataa tagaaattct aaattgccac catgtccacc aaaattgcc 120
attattggta atttgcata aattggtaaat atgccacatg aatcttttca taatttggct 180
aaaaagtttg gtccaattat gtttttggg ttagggtcaag ttccaactgt tgttatttca 240
tctgcaagat tggctaaaga agcattgaaa acttttgatt tggcattatc ttcaagacca 300
gaattgtttt cagcaaaaaca ttgcatatg aattgtacag atttggcttt ttctccatat 360
ggtagcttat ggagaaatat tagaaaaatt actacattgc aattattgtc tgctaaaaga 420
gttcaatctt ttcttttgt tagaaaagat gaagtgtgac atttgggtta aagaattact 480
gaatcatacc catctactat taatttgc catatgttag aagttttacgc taataataat 540
atattgtagag ttgcttttgg taaagatttt tctcaaggtg gtgacaatgt tagacataat 600
tttcaaaaag ttttgaaga ataccaaatg ttattgggtg gtttttcaat tggtgactac 660
tttccatcta tggatttttt gcataaaatc acaggtatgg aatctagatt gaaagatact 720

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tttagatctt	ttgatcatat	ttgtgataaa	atthttggaag	aacatagaaa	aactgaaaat	780
agaaaagatc	aaaaagattt	gggtgatgtt	ttgttggaag	ttcaacataa	tggtgacttg	840
gaaactccat	tgacaactga	taatatataa	gctattatta	ttgatattgt	tggtggtggt	900
actgatacta	atttgattac	attagattgg	acaatgactg	aattgattat	taatccaaaa	960
gttatgcaaa	gagcacaagc	tgaaagttaga	tctattgttg	gtgaaagaag	atttgtttct	1020
gaaaatgatt	tgtctcattt	gggttacatg	aaagctgttg	ttaagaagac	ttttagattg	1080
catccaccag	caccattatt	gggtccaaga	gaatcaattc	aagatgttac	tattgatggt	1140
tattacattc	cagcaaaaac	aagatttttc	attaatgctt	gggctattgg	tagagatcca	1200
gaattgtggg	aaaatccaga	aatgtttgat	ccagaaagat	tcatgaactc	tcatattgat	1260
tttaaaggtc	aagattttga	attgattcca	tttgggtcag	gtagaagaat	ttgtccagct	1320
attacatttt	caattgtcta	tgttgaatta	gctttggcac	aattattgca	ttcatttgat	1380
tggaatttac	caccagggtg	tgaagctaaa	gatttggata	tgacagaagc	atttgggtatt	1440
acaatgcata	gaattttctc	attgatgggt	gttgctaaac	catactgtgc	agattacttg	1500
aattaa						1506

SEQ ID NO: 296 moltype = DNA length = 1494
 FEATURE Location/Qualifiers
 source 1..1494
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 296

atggcagcat	taccattggt	tatttttgc	tcaatttctt	tggttttagt	tttgggtatt	60
ttgtctatga	aaaaggataa	aagatctgtt	aaattgccac	catctccacc	aaaattgcc	120
attattggta	atttgcata	aattggta	atgccacatg	aatcatttca	taatttggct	180
aaaaagttg	gtccaattat	gtttttgtg	ttgggtcaag	ttccaaactg	tggtatttct	240
tcagcaagat	tagetaaaga	agcattgaaa	acttttgatt	tggtttgtgc	taatagacca	300
gaattatttt	ctgcaaaaac	tttgatttac	aattgtacag	atctggcttt	ttctccatac	360
ggtagcttatt	ggagaaaat	tagaaaaatt	actacattac	aattgtgtgc	agcaaaaaga	420
gttcaatctt	tttcatctgt	tagaaaagat	gaagtgtgtc	atttgggtta	aagaattact	480
gaattctacc	ctcaactat	taatttgc	catgttttgg	aagtttacgc	taataataat	540
atttgtccag	tatcatcat	ttaagatttt	tctcaagggtg	gtgacaatgt	tagacataat	600
tttcaaaaag	ttttggaaga	atatcaaatg	ttattgggtg	gtttttcaat	tggtgactat	660
tttccatcta	tggaaatttt	acataaatc	actggtatgg	aatcaagatt	gaaagataca	720
ttagaagat	ttgatcatat	ttgtgataaa	atthttggaag	aacatagaaa	aactgaaaat	780
agaaaagatc	aaaaagattt	gggtgatgtt	ttgttagaaa	ttcaacataa	tggtgacttg	840
gaaactccat	taacaactga	taatatataa	gatattgttg	ttgttggtac	agataactat	900
tctattactt	tagattggac	tatgacagaa	ttgattatga	atccaaaagt	tatgcaaga	960
gcacaagctg	aagttagatc	aattgttgg	gaaagaagat	ttgtttctga	aaatgatttg	1020
tctcatttgg	gttaccatga	agctgttgtt	aaagaaactt	ttagattgca	tccaccagct	1080
ccattatttg	caccaagaga	tactatgcaa	gatgttacta	ttgatgggtta	caatattcca	1140
gctaaaaacta	gatttttcat	taatgcttgg	gcaattggta	gagattcaga	attgtgggaa	1200
aatccagaaa	tggttgaacc	agaagaattc	atgaactcac	atattgattt	ttaagggtcaa	1260
gattttgaa	tggttctcgt	gggtgcaggt	agaagaattt	gtccagctat	ttcttttaca	1320
gttgcaacag	ttgaattggc	tttagctcaa	ttgttacatt	cttttgattg	ggaattgcc	1380
ccaggtgttg	aagctaaaga	tttgatatg	acagaagcat	ttggtattac	tatgcataga	1440
acatctccat	taattggtgt	tgcaaaaacca	tacttttccag	atcatatgaa	ttaa	1494

SEQ ID NO: 297 moltype = DNA length = 1506
 FEATURE Location/Qualifiers
 source 1..1506
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 297

atggcagcat	tgccattatt	gatttttgc	tcaattgctt	tgattgttgt	tttgaatttt	60
ttattgaaac	caaagaataa	tagaaaattc	aaattgccac	catgtccacc	aaaattgcc	120
attattggta	atttgcattt	gttaggta	atgccacaag	aatctttgca	tcaattggct	180
aaaaagtatg	gtccaattat	ttcttgc	ttaggtgaag	ttccaacagt	tggtatttct	240
tcagctagat	tggtcaaat	agctatgaaa	acttttgatt	tggtttatc	ttcaagacca	300
caaattttct	ctgctaaaca	ttgtttttat	aattgtactg	atgttgtttt	ctcaccatac	360
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gttcaatcat	tttcttctgt	tagagaagaa	gaagtgtgtc	atttgggtca	aagagttgct	480
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gtttgtagag	ttgcatttgg	ttaagatttt	tctcaagggtg	gtgacaatca	tagacataat	600
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tttagaagat	ttgatcaatt	ttcogatgaa	atthttgcaag	aacatagaaa	tccagaatct	780
agaaaagaac	ataaagattt	gggtgatgtt	ttgttagaaa	ttcaaaaaga	tggtgaccca	840
gaaactccat	tgactacaga	taatatataa	gctattattt	tagatatggt	tgctgctggt	900
actgatacaa	attttattac	tttgatttgg	gctatgactg	aattgatgac	taatccaggt	960
gttttgaaaa	gagcacaagc	agaagttaga	aatattgttg	gtgaaagaag	atcagtttct	1020
gaaaatgatt	gtccacaatt	accatattgt	aaagcagttg	ttaagaagac	ttttagattg	1080
catccaccag	ctccagtttt	gggtccaaga	gaatctatgg	aagatgttac	tattgatggt	1140
tacgatattc	cagctaaaac	tagatttttc	attaatgctt	gggcaatttg	tagagatcca	1200
aaatcttggg	aaaatccaga	tgtttttgaa	ccagaaagat	tcatgaactc	tcatattgat	1260
tttaaaggtc	aagattacga	attgattcca	tttgggtgctg	gtagaagatc	ttgtccagct	1320
attacttttt	ctgcagttac	tgttgaatta	gcattggctc	aattgttaca	ttcttttgat	1380

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tggaattac	caccaggtgt	tgaagctaaa	gatttggata	tgactgaagt	tttgggtatt	1440
actatgcata	aaatttatcc	attaatgggt	gttgctaaac	catatttttc	acaaggttta	1500
aattaa						1506

SEQ ID NO: 298 moltype = DNA length = 1554
 FEATURE Location/Qualifiers
 source 1..1554
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 298

atggcagcat	taattgaaga	tcattctaca	caagcatcag	tttttatttt	aatggaattg	60
tttttaactt	tgtgtgcttt	attggcattt	ttctcaattg	ctttgaaaat	tgtagaaaa	120
actaaattgg	ttaaattacc	accaggtcca	agaccattac	caattattgg	taatttgcac	180
caattaaatg	gttctttgcc	acatagaatt	tgactgatt	tggtaaaaa	gtacgggtcca	240
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gcaattttga	attacgggtg	tactgatatt	gcatttgctc	catacgggtg	atattggaga	420
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attttgattg	aacataaagc	aaaaagagct	gaatctaaat	taggttaaatg	aggtgctaaa	840
gaagatttgg	ttgatgtttt	attgaatatt	caatcatctg	gtgaatttga	accaccattg	900
actgatgata	atttgaaagc	tgttattttac	gatattttct	ctgcaggtgg	tgaaacatct	960
tcaactacag	ttattttgggc	tatgtctgaa	atgattagaa	atccaaaagt	tatgaaaaag	1020
gctcaagatg	agtttagaga	agtttttctg	ggtaatggta	atgttgatga	atctagattg	1080
catgaattaa	aattatttgc	agctgtttat	aaagaaacta	tgagattaaa	tccagctgtt	1140
ccattattgt	taccaagaga	atgttcagaa	cattgtcaaa	ttaattggta	cgatgttcca	1200
gctaaaacaa	gagttattgt	taatgcatgg	gctattggta	gagatccaaa	acaatggact	1260
gaaccagaaa	aattcaaaac	agatagattt	ttggattctg	aaattgattt	tagaggtaaa	1320
actttttctt	ttattccatt	tggtgcaggt	agaagaattt	gtccaggtat	ttcttttctg	1380
ttgccattag	ttgaattacc	attggcacaa	ttattgtacc	attttgattg	gaaattgcca	1440
ggccaatcta	aaccagatga	attggatatg	actgaaattt	ttggtttgac	tggtggtaga	1500
aaacatgatt	tgaatttggt	tccaattcca	tatagaccat	gttcagtttt	gtaa	1554

SEQ ID NO: 299 moltype = DNA length = 1524
 FEATURE Location/Qualifiers
 source 1..1524
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 299

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gcttcattat	ctttgattgt	tggtttgaaa	tttttgtaa	aaacaaagaa	taatagaaat	120
tcataaattgc	caccatgtcc	accaaaattg	ccaattattg	gtaatttgca	ttgttgggt	180
aatatgccac	aagaatcttt	gcatcaattg	gctaaaaagt	acgggtccaa	tattttctta	240
caattgggtg	aagttccaac	tggtgttatt	tcttcagcta	gattagctaa	agaagcaatg	300
aaaacttttg	atttggcatt	atcttcaaga	ccacaaattt	tctctgctaa	acatttgttt	360
tacaattgta	ctgatgttgt	tttctacca	tatggtgctt	actggagaaa	tattagaaaa	420
atttgtattt	tagaattggt	gtcagctaaa	agagttcaat	ctttttcata	cgttagagaa	480
gaagaagtgg	cacatttggt	tcaaaagagt	gcagaagcat	accaggttac	tacaaaattg	540
actaaaaatg	tggttttata	cgcaaaata	ttggtttgta	gagttgcttt	tggtaaagat	600
ttttctcaag	gtggttgacaa	tcatagacat	aattttcaaa	aattgttaga	agaataccaa	660
gttttggttag	gtggtttttc	tattggtgac	tttttcccat	ctatggaatt	tgttcataaa	720
ttcactggta	tggaatcaag	attgaaagat	acttttagaa	gatttgatca	atttttcgat	780
gaaattttgc	aagaacatag	aatccagaa	tctagaaaa	aacataaaga	tttgggtgat	840
gttttattgg	aaattcaaaa	gaatggtgac	ccagaaactc	cattgactac	agataatatt	900
aaagctatta	ttttggatat	gtttgcagct	ggtactgata	ctaattttat	tactttggat	960
tggtgcaatga	ctgaattgat	gactaatcca	ggtgttttga	aaagagcaca	agctgaagtt	1020
agaaatattg	ttggtgaaag	aagatcagtt	tctgaaaaatg	atttgccaca	attgccatac	1080
gttaaagcag	ttgttaaaga	aactttttaga	ttgcatccac	cagcaccagt	tttggttcca	1140
agagaatcaa	tgaagatgt	tactattgat	ggttacgata	ttccagctaa	aacaagattt	1200
ttcattaatg	catgggcaat	tggttagagat	ccaaaatctt	gggaaaaatc	agatgttttt	1260
gaaccagaaa	gattcatgaa	ctctcatatt	gattttaaag	gtcaagatta	cgaattaatt	1320
ccatttggtg	caggtagaag	atcttgtcca	gctattactt	tttctgtgtg	tactgttgaa	1380
ttggcattgg	ctcaattatt	gcattctttt	gattgggaat	tgccaccagg	tggtgaagca	1440
aaagatttgg	atatgcacaga	agtttttgggt	attacaatgc	ataaaaattta	ccattgatag	1500
gttgttgcta	aaccatattt	ttca				1524

SEQ ID NO: 300 moltype = DNA length = 963
 FEATURE Location/Qualifiers
 source 1..963
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 300

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atggcagctt tgattgaaga tcattctaca caagcatctg tttttatatt gatggaattg 60
tttttaactt tgtgtgcttt attggcattt ttctcaattg ctttgaaaat tgtagaaaaa 120
acaaaatttag ttaaatattacc accagggtcca agaccattgc caattatttg taatttgcat 180
caattaaatg gttcttttgc acatagaatt ttgactgatt tggctaaaaa gtacgggtcca 240
ttaatgcatt tgcatttagg tgaagtttct actgttattg tttcttcacc agctgttgca 300
aaagaagtta tgaagaatca tgatattatt ttctgttaca gaccattggt attgtctcca 360
gctattttga attacgggtg tactgatatt gcttttgac catacgggtga atattggaga 420
caattaagaa aaattttgtac aattgaattg ttatcttcaa aaagagtcca aactttttaga 480
tcaatttagag aacaagaagt ttctaatttg tttagatcta ttaaatctgc tgaaggttct 540
gttgtaattt tgtctaaaat gatttactct ttgacttatt ctattactgc tcaagcagct 600
tttggtaaaag aatctaaata ccaagataaa ttcatctctg ttatgcaaga tgctattaaa 660
ttgatgggtg gttttaatat tgcgtgatgt taccatcta ttaaaatttt agaaaagatt 720
acaggatta gacataaatt ggaagaaca cataatcaaa ttgatgaat tgttcaaat 780
attttaattg aacataaagc taaaagagca gaatctaaat tgggtaaat aggtgctaaa 840
gaagatttgg ttgatgtttt attgaatatt caatcatctg gtgaatttga accaccattg 900
actgatgata atttgaagc tgttatttac gtttcattgt ttactcacc atctcatatt 960
aaa

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SEQ ID NO: 301      moltype = DNA length = 1395
FEATURE            Location/Qualifiers
source              1..1395
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 301
atggcagctt tggttagatcc gtggcaaaaa gaatcatttt tgcattcatt catttttgct 60
tctatttcat tgggtccagt tttgggtatt ttattgatga aaaaggataa aagatctgtt 120
aaattgccac catctccacc aaaattgcca attatttgga atttgcacca aattggtaat 180
atgccacatg aatcttttca taatttggct aaaaagtgtg gtccaattat gtttttggtg 240
ttaggtcaag ttccaactgt tgttatttca tcagctagat tagcaaaaaga agcattgaaa 300
acttctgatt tggctttatc atctagacca gaattgtttt ctgctaaaca tctgcattac 360
aattgtacag atttggcttt ttaccatat ggtacatat ggagaaaat tagaaaaatt 420
acaaacattgc aattattgtc tgcataaaga gttcaatctt ttctatttgt tagaaaagat 480
gaagtgtctc atttgggtta aagaattact gaatcatatc catcaactat taatttgtca 540
catatgttgg aagtttatgc aaataataat attttagag ttgctatttg taaagatttt 600
tcacaagggt gtgacaatgt tagacataat tttcaaaaag ttttagaaga atatacaatg 660
ttgttaggtg gtttttctat tgggtgactat ttccatcta tggaaatttt acataaattc 720
actggtatgg aatcttagtc aaaaagatac tttagaagat ttgatcatat ttgtgataaa 780
attttagaag aacatagaaa aactgaaaat agaaaagatc aaaaagattt ggttgatgtt 840
ttattggaat ttcaacataa tgggtgactg gaaacaccat tgactacaga taatattaaa 900
gctattatta ttgatattgt tgttgggtgt actgatacta attttattac tttggatttg 960
actatgacag aattgattat taatccaaaa gttatgcaaa gagcacaagc tgaagttaga 1020
tctattgttg gtgaagaagc atttgttctt gaaaatgatt tgcctcattt gggtttatatg 1080
aaagctgttg ttaagaagaa ttttagattg catccaccag ctccattatt ggctccaaga 1140
gaatctattc aagatgttac tattgatggt tataaatatt catttggtgc tggtagaaga 1200
atttgtccag ctatttcttt tacagttgct actgttgaat tggcattagc tcaattgttg 1260
cattcttttg attgggaatt gccaccaggt gttagaagca aagatttgga tatgactgaa 1320
gcatttggtt ttactatgca tagaacttct ccattaatgg ttgttgctaa accatacttt 1380
tctgatcatt tgaat

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SEQ ID NO: 302      moltype = DNA length = 1608
FEATURE            Location/Qualifiers
source              1..1608
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 302
atggcagcta atgttaaagc taataagcat aatattgttt gtattttgtg tttcttattt 60
ttgacaatgg ctaaacaaat ttttgtttta gatttgccat ttctatcttt gtttccattg 120
gctttgttat tgttgacttt agttattttg ttttacgctt cttcaagacc aaaaagaga 180
ttgccaccat caccacaaaa attgccaatt gttggtcatt tgttttagatt gggattttat 240
ccacatagaa ctttgcaaaag attatctaaa aagtacggtc cattgatgat gattcatttg 300
gggtctaaac caatttttgg tgcctcttca gcagatgttg cttgtgaaat tatgaaaaat 360
catgatattg tttttgcaaa tagaccaaaa agatctatta attcaaaatt attgtacgat 420
tgtaaagata ttgcaactgc tcaatatggt gaatactgga gacaaattag atctatttgt 480
gttttgcaat tattgtcaaa taagagagtt caatctttta gacatgttag agaagaagaa 540
acttcaatga tgattcaaaa gattagagaa agatgttcat cttctcttct ttcatgtatt 600
aatttgtcag atgctttgat tactttgaca aatgatgtta tttgtagagt tgccttgggt 660
agaaaatatt caaatgatga aaatggtaga aaatctatga aaattatgaa agaatttgg 720
gaattgttag gtactttttc tgttggtagc tacttgccgt ggtagcttg ggtaaataga 780
ttcaatggtt tggatgctaa attagaaaaa gttgctaaac atattgatca atttttgtaa 840
gggtgtattg aagaacataa attcagaaaag aaaggtgaag attacgttga ttctactgaa 900
aagaaaagcat tggatttggg tgatatattt ttggatattc aaaaagataa ttctgttgg 960
ttagctttgg aattagattc tgttaaagca attatttttg atatgtttgc aggtggtact 1020
gatacatcat acactgcaat ggaatgggca atgtctgaat tattgagaga tccaaaaact 1080
ttggaaaaat tacaagttga agttagaaaa gttgctcaag gtaaatctga aattactgaa 1140
gatgatttgg ataaaaatga ttacttaaaa gcagttatta aagaaacttt gagattacat 1200
actccaattc cattgttagt tccaagagaa tcaactcaac atgttaaaat ttaggttat 1260

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-continued

gatattccag	ctggtactga	agttatgggt	aatgcacggg	ctattgctag	agatccagtt	1320
caatggaaaa	agccagaaga	atTTTTgcca	gaaagatttt	taaattctac	tattgatttt	1380
agagggtttta	atTTTgaatt	tattccattt	ggtgctggta	gaagagggtg	tccagggtgca	1440
acttttgcta	ttgctgttaa	tgaattggct	ttggcaaaat	tggtacatga	atttgatttt	1500
gcattggcag	atgggtgctaa	accagaagat	ttggatatgt	ctgatgctac	agggttttact	1560
gttcatagaa	aatttccatt	gttagcagtt	gctattccac	attcttgt		1608

SEQ ID NO: 303 moltype = DNA length = 1617
 FEATURE Location/Qualifiers
 source 1..1617
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 303

atggcagctg	ttagaatttt	gaaaaagaaa	aagtttcaat	tcacaatttt	gaataactaa	60
aaggaaatgt	ctatttttca	atttgaagaa	attacagctg	gtttatctaa	acatttttcca	120
tctttgtgtg	tattgttatt	gatttcttta	ttggtgttga	gatggaaaaa	ttctacatca	180
gatacaactc	aaaatttgcc	accatctcca	ccacaattac	caattattgg	taatttgcac	240
caattggggt	ctattccaca	aagatctttg	aatcattgg	cacaaaaata	cggttctttg	300
atggtattga	gattgggttc	taaacaggtt	ttgattattt	cttcagcaga	tgctgctgaa	360
gaagttttga	aaactcatga	tttgattttt	gcaaatagac	caaaaccaga	ttttgttggg	420
agattattgt	acaattttta	agatattgct	ttttctccat	acggtgggta	ttggagacat	480
attagatcta	tttgtgtttt	gcaattattg	actaataaga	gagttcaatc	ttttagaaat	540
attagagaag	aagaagttga	attaatgttg	gaaaagatta	gagaatcatg	tgcttcttca	600
tctattattt	tgattgttga	aattttggct	actttgacaa	ataatattgt	ttctagagtt	660
gcaactggta	aaagatacac	tgaagaagaa	tctggttcaa	aattcaaaga	tttgtttgaa	720
gaattttcaa	tggtgttagg	tgtttttaat	gttgggtgact	acattccgtg	gttggcattg	780
attaatcata	ttaattgttt	ggaagcaaaa	gttaaaagag	ttggtaaaga	ttttgatgaa	840
tacattggata	aaattgttca	agaaggtatt	attagacaag	aaaagagagg	ttctaattgt	900
ggtgaccatg	aaaaagaaca	acaaaatttg	gttgatgttt	tgttagaatt	gcaaaagaact	960
aattctgctg	gttttgcatt	tggttagagat	tctattaaag	ctattattac	aaatatgttt	1020
gttgggtgta	ctgatacaac	ttttcttta	ttggaatggg	ctatttctga	attattgaga	1080
aatacaaatg	ctatgcataa	attgcaaaaa	gaagttagac	aatttttggg	tagaaaaatc	1140
agaattactg	aagatgattt	ggaataattg	cattacttga	aagcagttat	taaaagaagca	1200
ttgagattgc	atccaccagc	tccattgttg	attccaagag	aatcatctaa	agctgttaaa	1260
ttaattgggt	acaattatgc	tcaggttaca	caagttatta	ttaatgcacg	ggcaattggg	1320
agagatccaa	aattgtggga	tgatgctggg	gaatttagac	cagaagaatt	tttaaatcca	1380
tctgttgata	ttaattgttca	acattttgaa	ttaattccat	ttggttcagg	tagaagatct	1440
tgctccagga	ctgcatttgc	tatgattact	gctgaattgg	cattagctaa	tttgatttgg	1500
aattttgatt	ttgaattggc	agggtgggtg	agaccagaag	atttggtatg	gacagaagca	1560
ccaggtatta	ttactccaag	aaaagttcca	ttaagattgg	ttgcttcttt	gccaaat	1617

SEQ ID NO: 304 moltype = DNA length = 1545
 FEATURE Location/Qualifiers
 source 1..1545
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 304

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gttgctgttt	ctttattatt	gccatctttt	atgattttta	ttttgttttt	atctaaatgg	120
tggtttttga	caacttctaa	aaagaaaaat	ttgccaccat	caccaccaag	attgccaat	180
attggttaatt	tgcatcaaat	tggttctgtt	gcacatagat	cttttaaatc	tttggtgtaa	240
aaacatggtc	cattgatgtt	gttacatgtt	ggtagacaaa	aattggtagt	tgtttcttca	300
gctgaaactg	caagagaagt	tttgaaaaac	catgatgttg	cattttgttg	tagaccagat	360
tctgaagtta	caagaagaat	tttctataat	ttgaaaaata	tttcttgtgc	accatacggg	420
gactactgga	aaattgttag	atctattgct	gttaatcaaa	ttttgaatca	aaagagagtt	480
caatctttta	gatctgttag	agaagaagaa	atttctttgt	tagttgaaag	aattaaagaa	540
tcttgtgcat	catcttctgt	tatttgtatg	aatacattat	tgacaacttt	agttaatgat	600
attgtttcta	gaattacaat	tggtaaaaag	tcatctggta	aagggtgggtc	tagatttggc	660
gaattgttat	ttgcatctac	agaattgttt	gggtctttct	tagctgggtga	ctttttccca	720
gggtttgggt	gggttaggtta	tattactggg	tttgaagcta	aaactaaaaa	gatttgcaaaa	780
gatttggatc	aatgttttaga	aaatttgatt	gaagaacaaa	ttaatagaaa	taagagaggt	840
gacaatcaag	gtaaaagata	tcaaaatttc	ttgcaagttt	tattggaaat	tcaaaagaact	900
gattcatctg	gtcatgcttt	ggatagagaa	tctattaaag	ctattattat	ggatattggt	960
ctgggtgggt	ttgatataca	ctctttattg	gaatgggcta	catctttatt	ggttaaacat	1020
ccagatgtta	tgaaaaaggt	gcaaaatgaa	gttaaaagag	ttgctgggtc	aaaacttttt	1080
atttcagaag	atgatttggc	taaattacaa	tacttaaaag	ctgttatttaa	agaaactttt	1140
agatttgtgc	caggtgcttt	tattgttaga	gtttctacaa	aagatgttaa	aattatgggt	1200
tatgatattg	ctgcaggtac	acaagttttg	gttaatgcac	tggtcatttg	tagagatcca	1260
actttatggg	aaaaaccaga	agaatttaga	ccagaaagat	ttttaaattc	ttcaattgat	1320
ctgaaaggtc	atcattttga	attgttacca	tttggttctg	gtagaagatc	atgtccagggt	1380
tcaacatttg	ctttgggttat	tgatgaattg	gttttagcaa	atttgggttg	taattttta	1440
tatgctttgc	caggtgggtg	aaaagctgaa	gatttggata	tgacagaagc	tccaggtgct	1500
atgccaagaa	gaagaactcc	attgttattg	gttccatctt	tatgt		1545

SEQ ID NO: 305 moltype = DNA length = 1536
 FEATURE Location/Qualifiers

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source 1..1536
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 305

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gcgtccgtct	ccttaactcg	cgtcctaaag	ttcgatatcaa	aaccaaagaa	caatcgtaac	120
tcaaaattac	ctccgtgtcc	tctaagttta	ccgatcattg	ggaacttaca	cttgctaggg	180
aacatgccac	aggagagtct	acatcagtta	gcaaagaaat	acgggcccg	tatcttcttg	240
caattgggag	aagtggcgac	aatagtcac	agcagtgcaa	gattagcgaa	agaggccatg	300
aagacattcg	atctggccct	atcaagcaga	ccacaaatat	ttccgcccaa	gcaccttttc	360
tataattgca	ctgacgtggt	gtttagtcgg	tacggagctt	attggcgtaa	tataagaaaa	420
atatgtatat	tggagctttt	atctgccaa	agagttcaaa	gtttctccta	tgtacgtgaa	480
gaggaggtag	ctcaccttgt	gcaacgtgta	gctgaagcct	accctggtag	caccaatctg	540
accaagatgt	tggggctata	tgctaataac	ctagtgtgca	gagtcgcgtt	tggaaaagac	600
ttctcacaa	ggggagacaa	tcatagggac	aacttccaaa	agctactgga	agagtatcaa	660
gtccttctag	gaggttttag	tatagtgac	ttctttccat	ccatggagtt	cgtacacaaa	720
tttacgggca	tggaaagtgc	tctgaaggac	accttcagaa	ggtttgatca	gtttttcgat	780
gaaatcctac	aagagcacag	gaacccggaa	tcaagggaag	agcacaagga	tctggtggat	840
gtacttctg	agatccagaa	gaacggtgac	cccgaactc	ccttaactac	agacaacata	900
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tgggcccata	ctgagttgat	gacaaacct	ggtgtcctga	aaagagcaca	ggcagaagt	1020
cgtaacatag	tgggggagcg	tagaagcgtg	agcgagaatg	atcttcccca	actgccctat	1080
gttaaggcag	tgggtgaagga	gaacttcaga	ctacatcctc	ctgcgcag	tctgtttcca	1140
agggagtcac	tggaggatgt	gacgatcgat	ggttacgaca	ttcccgcaaa	gaccagattt	1200
ttcataaacg	cgtgggagcg	cggtagagat	cctaagagct	gggagaatcc	tgatgtgttc	1260
gagccagaaa	gatttatgaa	ctcccacata	gacttcaagg	gacaagacta	tgaattgac	1320
cccttcgggg	caggagagcg	tagctgtcct	gcgattacct	tctccgcgt	tacagtagag	1380
ttagcattag	cacagctttg	acactcattc	gattgggaat	tgcaccggg	agttgaagcg	1440
aaggatttgg	atatgcacaga	ggtttttggg	atcaccatgc	ataagattta	cctctaatg	1500
gtggttgcca	aaccatactt	cagtcagggc	ttaaat			1536

SEQ ID NO: 306 moltype = DNA length = 1503
FEATURE Location/Qualifiers
source 1..1503
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 306

atggcagagt	tattttcttac	actttgtgcg	ctactagett	tcttcagcat	agcccttaaa	60
atcgtagaaa	aaacaaagtt	agtcacaaata	ccgccaggac	cacgtctctt	accgatcatc	120
ggcaacttgc	atcaattaaa	tggatcactt	ccacatcgta	tcttaaccga	cctagcgaa	180
aaatatggac	cattgatgca	cctgcaatta	ggcgagggtg	gcaccgttat	cgtgagttct	240
ccggcggttg	ccaaagaggt	catgaaggat	cacgacatta	tcttttgcta	tagacctttg	300
ctattatcac	ctgcgatact	aaactatggt	ggcacggaca	tagccttcgc	tccgtacgga	360
gagtagctgc	gtcagttgag	gaaaaatatgt	accatcgagc	ttctttctct	caaaagggtta	420
caaacattta	ggagtatccg	tgagcaagag	gtatcaaatc	tgtttcgttc	tataaagagc	480
gcggaaggaa	gcgtagtgtg	tttagcaag	atgatctatt	cattaacata	cagtattact	540
gctcaggcgg	cttttggaag	ggagtccaag	tatcaggata	agttcatctc	tgttatgcag	600
gatgcgataa	aactaatggg	tggcttcaac	attgccgatg	tttaccgcgc	catcaaaatt	660
ttagaaaaga	ttaccggaat	caggcacaa	ctagaggaga	cacacaatca	aatcgatgag	720
attgtgcaga	acatactgat	agagcacaa	gctaaacgtg	cggaaaagta	actaggcaat	780
cgtggagcta	aagaagacct	ggttgacgtg	cttcttaata	tccagtcagg	tggcgaattt	840
gagccgcgcg	tgacagatga	caacctgaag	gccgtgattt	atgacatctt	cagtcgcggt	900
ggggaaacaa	gctctacaac	ggttatctgg	gcgatgtctg	agatgattag	gaatcccaaa	960
gtaattgaaa	aggctcagga	cgaagtgaga	gaagtgtttg	cagggaacgg	aaacgtggac	1020
gaaagcagac	ttcacgagct	taaataacct	caagcgggta	taaaagagac	tatgcgtttg	1080
aacctgtctg	taccgctgct	tcttccgagg	gagtgctctg	agcactgcca	aataaacgga	1140
tatgatgtac	cagccaagac	gagggttata	gtcaacgctt	gggctatcgg	cagggaccgg	1200
aaacagtgga	cagaacccga	gaaattcaaa	ccggaccggt	tctggacag	cgaaatcgat	1260
tttagaggca	agactttctc	ctttatacca	tttggcgccg	gtagaaggat	atgtccgggt	1320
atcagttttg	ccttgccctc	ggtagaatta	cctctagcac	agcttttgta	tacttttgat	1380
tggaaaactc	ctggacaatc	aaagcccgat	gaacttgata	tgacagaaat	ttttgggcta	1440
acgggttgaa	ggaacacaga	tctgaatctg	gttccgatcc	cttacaggcc	gtgctctgtg	1500
cta						1503

SEQ ID NO: 307 moltype = DNA length = 1503
FEATURE Location/Qualifiers
source 1..1503
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 307

atggcagagt	tatttctaac	actgtgcgca	ctgctggcct	tcttttcaat	cgcaattaag	60
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gggaacctac	accaacttaa	cggctcacta	ccacacagga	tactaaactga	tttggctaaa	180
aaatatgggc	cgtaatagca	tctacaattg	ggagaagtga	gtacggtaat	agtttctctc	240
cctgcggtg	ccaagggaat	catgaaggat	cacgacatta	ttttctgcta	caggcctctg	300
ctactatcac	cagctatact	taactacgga	ggaactgaca	tagccttcgc	accatatgga	360

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gagtactgga gacaacttag aaagatctgc acgatcgagc tattgtcctc aaagagggtt 420
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gaagagggga gtgttgtaaa tttatccaaa aggatatact ctttgacata ttctataacc 540
gcgcaggcag cctttggaaa agagtccaaa taccaagaca aattcatatc cgtgatgcaa 600
gatgccatga tgctaattggg tggattcaac atagccgatg tttatccctc aattaaaaatc 660
ttggagaaga ttacagggaat aagacataag ctggagggaag tacataatca ggttgatgaa 720
attgttcaga acatcttaat cgagcacaaa gctaaaaggg ctgaaagcaa gttgggtaat 780
aggggagcta aagaggactt agtggatgtt ttgttgaata tccagagcag cggagaattt 840
gaggcgcttt taacagacga taatctaaaa gccgtaatTT acgacatatt ttccagctggt 900
ggggagacca gtacgcagac tgtcatttgg gcgatgagcg agatgatcag gaacccgaaa 960
gtgatgaaaa agggccagga cgaagttaga gaggtcttcg ctggtaacgg taacgcttgac 1020
gagtcagcat tacatgagct taaatatTTg caggcagtaa tcaagaaac aatgagatta 1080
aaccttgcag ttccactact gcttccgctg gagtgctcag aacactgtca gattaatggg 1140
ttcgacgtcg ctgtaagac gagagtgaac gtcaatgcat gggccatcgg tcgtgatccg 1200
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atcagtttcc gctacacgta agttgagcta ccccttgcgc agctattata tcatttcgac 1380
tggaacctac cagggcagtt aaaaccagac gagcttgata tgaccgagat cttcggactt 1440
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ctt 1503

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SEQ ID NO: 308      moltype = DNA length = 1605
FEATURE
source              Location/Qualifiers
                    1..1605
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 308
atggcaataa gttctagtat caagagtata ttacttaagt taacaatcgc tcagttggca 60
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tatttgatgg tggccaagat tcacaggaaa tctaccggcg aaaacagcaa gttaaatttg 180
cttccaggtc cgaaccggtt accgataaatt ggtaatatgc atcagttgca cggtccccc 240
attcatcacc ttcttaggga cttggcgaag aagtatggtc cgctgatgca tttaaagctg 300
ggcgaaatat cagctattat agttacaagt ccggaaatgg caaaggagat ctacaaaacg 360
aacgatattg ctaattgcag cagaccaggt acccacctgg ctttaagat cattagtta 420
aacttcacgg gtatcatata tagcgcgtat ggaaatcact ggagacaact gcgtaagatt 480
tgtacagcag agttgttgtc accaaaaaga gtgcagactt tcaagtctat tagagaggac 540
gaagttttta atttaattca gttgatctac agccagaaag gctccataat taactctgagc 600
aggtgtatct tcttactatc atatagcacc actagcagaa ttgcccctcg taaaagaaac 660
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tctcttgca ccatgtatcc ttctgtgaag ctattacagg ttctatcaat catgctgtat 780
aaacttgaga tagctcatcg tcaggtagac caaatactgg aaaacatcat caacgaccac 840
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aggaatgttt atgacgagcg tggcaatgtt gacgaatcct cctacatga actgaaatat 1140
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agggaaatgt gggagcagtg cgagattaat ggcacgtga tccctgcaaa agcaagaata 1260
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gaggagctgg atatgactga agatttttgt gtaactacca agagggtctaa tgacttacac 1560
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SEQ ID NO: 309      moltype = DNA length = 1509
FEATURE
source              Location/Qualifiers
                    1..1509
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 309
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atgcatcagc tgcattgtag ccaactacat catctattaa gggatctggc taaaatttat 180
gggtccctaa tgcacctaaa acttggtaga atatccacga ttatcgtgac gagccccgag 240
atggctagag aaatttataa aaccaatgac ctgcttttct catctaggcc gagccattac 300
cttgctttca aaatctcttc ctataacttt cgggatatag cattcagccc ctatggtaac 360
tattggaggc aaatgagaaa gatattgacc ttggaacttt tgagcaccaa aagggttcaa 420
acatttcaga gcatccgtga agatgaggtc ttcaatctga tgaagagtat aagtactcag 480
aaagggtcta ttattaactt atcaagggtc atattcactg tgatttacac tataacttcc 540
cgtgcagcgt tcgggaagag aaataaagat actgaaaggt atatcaggtt ggtggaggag 600
gtcacgaaagc tagccagctg gttctcactt gccgacgtat acccgagcgt gaaattattc 660
cagttgatgt ctactgttag gcacaaagct gccatagtcg ataacaagt cgatgaaatc 720
attgaaaaca tccctaataa ccacaaggag aagatcgaag aagctaaaca ggcgtccggc 780
gaagccaagg aggatttagt tgatgtactt ctatccattc aaaatagtg agatttcgag 840
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gaaacctcta	gtacagtgtg	ggaatgggca	atctccgaaa	tgataaaaaa	tcccagagact	960
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tctaaacttc	acgaacttaa	atatttgaat	gctatcatta	aagaacacct	aagattacat	1080
ccttctgctc	cactgctagt	tccaagggag	tgtactgagc	agtgtgagat	caatggctac	1140
gtaattcccg	ccaaagccag	attgtttgtc	aatgcctggg	ccatagggag	ggacccgaaa	1200
tactggatcg	atgcggaaaa	ttttaatccc	gagcgtttcc	ttgactctaa	ggtagactat	1260
aggggcaatg	atttcgagta	catccctttt	ggtgctggaa	gacgtatatg	ccctggcatt	1320
acattcagta	ggcctggaa	cgaactggca	ttagctcagc	ttctgttcca	ttttgactgg	1380
aaagtcccg	gtgagcttaa	gcaagaggaa	cttgatatga	cagagaactt	tggaatgact	1440
gtaaggcgta	aaaacgactt	gaacttaatt	cccgttcttt	acaatagatc	ttgcctgagc	1500
aaggacaac						1509

SEQ ID NO: 310 moltype = DNA length = 1503
 FEATURE Location/Qualifiers
 source 1..1503
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 310

atggcagagt	tggtccctaac	cttatgcgct	ttactagctt	tcttttccat	tgccctaaag	60
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gggaaccttc	atcagctgaa	cgggtctcta	ccccatcgta	ttttaacaga	ccttgccgaag	180
aaatatggtc	cggttaatgca	tcttcaatta	ggcagggttt	ctactgtaat	cgttagtcca	240
cccgagttg	ctaaggaggt	gatgaaagat	catgataatta	tcttttgcta	tcgtccgctg	300
cttttgagtc	ccgcgatact	gaactatggt	ggaaccgcga	tagccttcgc	tccgtatggc	360
gagtatggga	gacaactacg	taaaatctgt	accatcgaa	ttcttagtag	taaaagagtg	420
cagacattca	ggagtatcag	agagcaagag	gtctcaaaact	tatttaggtc	catcaagtca	480
gcggaaggga	gtgttgtaaa	tttaagcaag	atgatctact	cccttaccta	ttcaataaact	540
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ttagaaaaga	tacgggtgat	caggcataag	ttggaagaaa	cacacaatca	aatagacgag	720
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aggggtgcga	aggaagacct	tgtggacgtt	ctattaaata	tccagagttc	tgccgagttc	840
gaacctccgt	tgacagacga	taattctaaaa	gcggtagatct	atgacatttt	ttcagccgga	900
ggagagacaa	gtagtacgac	ggtgatattg	gcgatgtcag	aaatgattag	aaaccccaag	960
gtcatgaaaa	aagctcagga	cgaagtgaga	gaggtgtttg	caggtaacgg	taatgttgac	1020
gaatccaggg	ttcttagagct	taagtatctg	caagcgggtg	tcaagagacg	tatgaggttg	1080
aatcccgctg	ttctttgtct	tctgccgagg	gagtgttctg	aacactgcc	aatcaacgga	1140
tacgacgtgc	cagcaaaagac	tagagtgata	gtaaatgcct	gggctatttg	aagagatcct	1200
aaacagtgga	cggagccaga	gaagttaag	cccgatagat	ttttggacag	cgaatccgac	1260
ttcaggggga	aaacggtctc	attcatccct	tttggggcag	gccgtagaat	ctgccagga	1320
atctcattcg	ctctgccatt	ggtagaacta	cctttagcac	agctacttta	ccacttcgat	1380
tggaagctgc	cgggacagag	caaaccagat	gagctggaca	tgaccgagat	cttcggactt	1440
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tta						1503

SEQ ID NO: 311 moltype = DNA length = 1503
 FEATURE Location/Qualifiers
 source 1..1503
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 311

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atgggtgaaga	actcaaaacc	agacaatctg	ccaccaggtc	cgatgcaact	accaattatt	120
ggtaacatac	atcagttatt	cggtagttta	gcatacagga	ttcttacggg	acttgccgaag	180
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gcagaggtgg	cgaagaagat	aatgaatgaa	catgacataa	ttttcgcgctc	acgtcctaac	300
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gagaccttta	ggccaatcag	ggaggaggca	gtgcttgaca	tgataaaaaa	tatagcgctg	480
cagaggggct	ctgacgtcaa	tctgagtaaa	gagatcttta	gcttgacctta	taggattacg	540
gcgcaagcgg	cgttccggaaa	gagaaaataag	caccaagaag	agtttatttc	tgttatgggtg	600
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ggcgatagta	aagaagactt	ggtagatgtg	ttgttgaaa	tgacagaaatc	aggagagttc	840
ggggcgccac	tatcaataaa	taacctaaag	gccgtgattc	ttgacatatt	ctccgctggc	900
gacgaatcaa	gtagtacaac	tgcgcaatgg	acgatgttgg	aaatgatgaa	aaatcccggt	960
accattaaaa	ggtcacaaga	ggaggtgaga	gaagtgtttg	gtgacaaggg	caatgtagac	1020
gagtcacagg	tacatgagct	taagtactta	caagcggtaa	ttaaggatgc	gctgaggtta	1080
aggcctccag	ttccgctgct	attgccccag	gagtgccgag	agccatgtga	gatccatggt	1140
tacgaaatcc	ccgccaaagc	tcgtatgata	gtaaaacgct	gggcaatttg	aagagattca	1200
aggaggtgga	ctgaacttga	aaaaatttatc	tccgaaagg	ttcttgacag	tgactatgac	1260
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tggaacttcc	caggggagat	aaaggaaaca	cagctggata	tgaccgaaac	cttcgggcta	1440
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aac 1503

SEQ ID NO: 312 moltype = DNA length = 1527
 FEATURE Location/Qualifiers
 source 1..1527
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 312

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ttacccataa	taggtaacat	gcaccaatta	catgggagtc	ccatccacca	ccttatgcgt	180
gatcttgcta	agaaatatgg	tccgcttatg	catcttaagc	tgggagagat	atccgctata	240
attgtcacct	cacctgaaat	ggcgaaagaa	atctacaaaa	caaacgacct	attatttgct	300
tccagacctt	catctcatct	ggcgctttaa	attatttccct	ataacttcgc	ggacatcata	360
ttttcagcgt	atggtaatca	ttggcgctcag	ttgagaaaaa	tttgcactgt	tgaattgctg	420
tcacctaaag	gtgtccagac	tttcaagtca	gtacgtgaat	acgaggtttt	taaattgata	480
caatcaatca	gttcaaaagag	gggaagtata	ataaatcttt	ctagaagcat	cttccatta	540
tcatatagca	tcacttcacg	tatcgcttcc	gggaaaaagga	ataagtcac	agagcgcttc	600
atacagatag	ttgaggaatt	taataagttg	gtctcaggtt	tctcacttgc	agatatgtac	660
ccgagtgtea	aacttctacg	tggtatgtct	ataatgagat	ataaacttgt	tatcgctcac	720
aaacaggtcg	atcaaatatt	ggagaatatt	cttaatgatc	acaagcagaa	tgccacagaa	780
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gagacgctta	ggctacaccc	ctcagcacca	cttctacttc	ctagggagtg	tagtgaacaa	1140
tgtgagataa	atggttacgt	catcccagct	aaagccagaa	ttatagttaa	tgcttgggct	1200
ataggtcgtg	accaggtatg	ctgggacagaa	gcagaaaaat	ttaatccaga	aagggtttct	1260
gacagtgaat	ttgattttaa	aggaaacaat	cttgaatata	ttccctttgg	tgagggaaga	1320
agaatatgtc	cggggatcag	tttttcacaa	ccgcacatgg	agcttgtgct	ggcgcagctg	1380
cttttttact	tgatttgga	acttctctgc	gaactgaaac	aggaagagct	agatatgact	1440
gaagattttg	gcgtcactat	taggaggacg	aatgatcttc	atctaattcc	gattccttat	1500
aatcgtagtt	gcttacgtat	ggacgga				1527

SEQ ID NO: 313 moltype = DNA length = 1503
 FEATURE Location/Qualifiers
 source 1..1503
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 313

atggcagaac	tattttcttac	tctgtgtgcg	ctgctggcat	tctttagtat	tgctataaaa	60
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ctaccaatta	ttgggaatat	acaccagtta	tttggttcac	ttccacacag	aatactgacg	180
gatcaagcaa	aaaagtatgg	tccacttatg	cacttacagt	tgggcgagat	cagcacgttt	240
gtagtatcct	ctgccgaggt	cgctaaagaa	tgatgaacg	aacacgatat	aacctttgcg	300
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tcagtcattg	tcgatattgt	tgagctagtg	gctggtttta	atatagccga	tatgtatccc	660
agcgtcaaga	tcctagagcg	tatcagtggt	atgagtcaga	aactagagag	aacgcacaaa	720
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tacgaaatac	ccgctaagc	tcgtgttctg	gtgaacgctt	gggctatagg	tcgtgatccc	1200
aggcattgga	ccgaacctga	caaattcata	ccggagagat	tcctggattc	cgactatgac	1260
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tggaaccttc	caggagacat	cggcgagcaa	caagtggata	tgactgagac	gtttgggtct	1440
gcggtccgtc	ctaaacacgg	agtcacactg	gtgcccttca	gcttttctaa	ttcttctgaa	1500
aat						1503

SEQ ID NO: 314 moltype = DNA length = 2064
 FEATURE Location/Qualifiers
 source 1..2064
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 314

atgactacgc	tgctaaaatc	ttatttcgga	gagtttgagg	gaatgtacgt	gcctcaaatc	60
ctaattgccg	ccctaaggca	acttgaggaa	gcattcgtca	gcgcacagaa	agaccccgag	120

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ttccaggcac agttcaatga cctattgaag aattacgccc gtagacctac tgcattaact 180
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atgggcaaaa cagaatttat cgcgagagact ggggcaggcc aacacggcgt tgccagcgct 360
ctagcatcag cgttactggg actgaaatgt agaatttata tgggcgcgaa ggatgtggaa 420
cgtcagtcgc ccaatgtctt taggatgagg ctgatgggtg ccgaggttat tcctgttcat 480
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tacgagacgg cccattacat gctagggacc gccgcaggtc cccaccgta tccgacaata 600
gttagagaat ttcaagaat gataggagag gaaaccaagg ctcatgtttt agagagggaa 660
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ttcgccgatt ttattaatga gacgaatgtc ggcctaattg gtgtggaacc aggcggacac 780
ggaatcgaaa cgggtgaaca tgggtcacca cttaagcatg gaagggtggg catctatttc 840
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agcgcccgac tggacttccc aagcgtaggt ccgcaacatg catacctaaa cagtacgggg 960
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tctggtgagg gacgtgggag cctattgact tgcgcgatg tcgaggagaa tccggggcca 1260
atggaacggt acgagtcact ttttgctcaa ctgaaggaga gaaaggagggt ggccctcgtg 1320
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gtgggcgtcg attcagtgct ggtcgcgagc gtacccttag aggaatccgc tcccttcaga 1680
caggccgcgc ttcgtcataa cgttgacccg atatttatat gcccccaaa cgcagacgac 1740
gatttgttac gtcaaatcgc ctcttaccgg cgtggataca cttacttgc aagtagggct 1800
ggggttaactg gggtgagaa ccgtgctgct ctacccttga accatttagt cgctaaactg 1860
aaagagtata tcgtgtctcc acccctgcag ggccttggga taagtgcacc agatcaggta 1920
aaggcggcta atgatgcagg agcggcgggc gcgatatcag gcagcgcgat agtaaaagatt 1980
atagaacaac acataaatga gccgaaaag atgctggccg cgttgaagggt attcgttcag 2040
cccatgaagg ctgccactcg ttca 2064

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SEQ ID NO: 315      moltype = DNA length = 396
FEATURE            Location/Qualifiers
source              1..396
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 315
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cctgggggag aagaggtggt attggatcat aaaggaaaag acggtctctga agatttcgac 180
gatgtgggccc attccaaaga cgcaatggaa ctaatgaaga cctaccagggt gggcgctactt 240
gtagaaagtg agcgtagtaa taagccgcgc aaaaagggtt gggtcgcagg atataacaca 300
aagcagagtg aaaaagtatgt ccaaccaggc gttccttttt atctgctgat tgggggtatc 360
gcgttggtag tcgcgctttt cttctatgct tactaa 396

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SEQ ID NO: 316      moltype = DNA length = 408
FEATURE            Location/Qualifiers
source              1..408
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 316
atggcagcaa gtgaccagaa gttgcacaaa ttcgacgagg tcagtaagca caataagacc 60
aaagactggt ggctaattat caacggtaag gtttacgacg ttacccctt tatggatgat 120
caccocggcg gggatgaagt gctgttgagt gcaacaggca aggatgctac aaatgatctc 180
gaggacgttg gccatagcga tagtgcgaga gagatgatgg ataagtacta tataggcgaa 240
atggatatgg caaccgtgcc actgaagaga acttacatac ctcccagca agctcagtat 300
aatcctgaca agactccaga atttgtgac aaaatcctgc aatccttgt tcccttggtt 360
atcttagggc ttgcgttcgc ggtacgtcat tatactaagg aaaagtaa 408

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SEQ ID NO: 317      moltype = DNA length = 2040
FEATURE            Location/Qualifiers
source              1..2040
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 317
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tccggcagga gcctggtagt tttctacggt agccaaaccc gcaccgcaga agagtgttgc 300
ggaagacttg ccaaggaagg agcagagatat agacttaaa gaaatgttgc tgacctggag 360
gactacgaaa tggaggagct agtgaatcta aagaatatcc cgaactcatt ggctgtgttc 420
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SEQ ID NO: 318      moltype = DNA length = 2049
FEATURE
source             Location/Qualifiers
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                   mol_type = other DNA
                   organism = synthetic construct
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SEQ ID NO: 319      moltype = DNA length = 2148
FEATURE
source             Location/Qualifiers
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                   mol_type = other DNA
                   organism = synthetic construct
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SEQUENCE: 319
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SEQ ID NO: 320      moltype = DNA length = 2094
FEATURE             Location/Qualifiers
source              1..2094
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 321 moltype = DNA length = 1767
 FEATURE Location/Qualifiers
 source 1..1767
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 321

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SEQ ID NO: 322 moltype = DNA length = 1791
 FEATURE Location/Qualifiers
 source 1..1791
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 322

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SEQ ID NO: 323      moltype = DNA length = 1770
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                   organism = synthetic construct

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gaagatgaat ttaattcaga tgggtcagtt aaaaatccgt ggagaatttg tactattcaa 960
caagttgaag aggttaaaat tttgttgaaa ttgattccag tttggatttc ttgtattatt 1020
tactatattt cagttgttca acaacaaaat tacgtttgtt ttcaagcatt gcaatctgat 1080
agaagattat ttggtacatc taaattcaaa attccagctg catcttacct tgttattgct 1140
atgtttggct ttgactattt gattccaaat ttgtatagaa ttattgttcc atgtttgcaa 1200
aagattactg gtaaaagaag ttgtattact attttcaaaa gaattgggtat tggtagtggt 1260
ttgtcagttt tgacaatgat tacatcagct ttggttgaat ctaaaagaag aaaatttgct 1320
ttgaatagac caacattggg ttttgaacca agaaaagggt caattttctc aatgtctggt 1380
gttttggttaa ttccacaatt ggccttatca ggtttgtctg aagcattcac tattattggt 1440
gaaaatgaat tgttttataa agaatttcca gaaaatatga gatcaattgc tatggcattg 1500
gtttttgttg gttttgtctg ttcttcatac ttgtcatctt ttttatcttc aattgttcaa 1560
aagacaactg gtgtgaatga atcttggtta gctgaagatt tgaatagagg tagattggat 1620
tacttttact attgattggg tgctttagaa acattaaatt tgatttacct tttagtttgt 1680
tctaaatggt ataaatcaaa acaagttgaa aagattccac aagttgctat ggaaaaatta 1740
aataatgttg aagaaaaatc attagtttaa 1770

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SEQ ID NO: 324      moltype = DNA length = 1785
FEATURE            Location/Qualifiers
source             1..1785
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 324
atggcagaaa agaatgaaaa attggttcca aatgggtacta aagaagatgg taaagttgca 60
gctaaaaatg aaccaaatta tagaggattt aaagcaatgc catttgtttt gggtaatgaa 120
acatgtgaaa aattgggtac aattgggtact tcttcaaat ttgtctgttta cttgtcaaca 180
gtttttcata tgaatactat gtctgcaaat aatgttatta atatttttaa tggtagatgt 240
tacttgggtt ctttgggttg tgcatttttg tgtgatacat attttggtag atacaaaact 300
ttagggtttg gttctgttgc ttcatttttg ggtatgttga tgttaacatt gactgctgca 360
atttcaactt tacatccaac taattgtgat aattctggtg actgtaaagg tccatctgct 420
gggtcaaatg catttttagt ttgtgttttt gttttgatgg ttattgtgtc ttcaggattt 480
agaccatgta attggccttt tgggtcgtgat caattcaatc caaatacaga atctggtaga 540
agaggattta attcatTTTT caattgggtat tacttttact atacatttgc tgttatgggt 600
tctgtacaaa ttattgtttt tgttcaatca gatattttct gggtcaattg ttttctattt 660
ccaacatcat tgatgttttt atcttgttct ttgtttttcg ctggtagatg gttgtatggt 720
aaagttttgc cacaagggtc accaatgact tctttagttc aagttattgt tgtttctggt 780
aaaaagagaa gattaaaatt gccagggtgaa ccatctttat tgtttaatca tattccatct 840
aatgcaatta attcctaatt accaagaact aataatttta gattttttaga taaagctgca 900
attttgacac cagaagataa agttaatcca gatggttctg cagctaataa gtggaaatta 960
tgttcaattc aacaagttga agaagttaaa tgtgttttga gagttttacc aatttggatt 1020
gctgggtgta tttttcaatt ttctgttact caacaatcaa attatgttcc atttcaagca 1080
tttcaatcag atagacattt gattaaaggt aataagtttc aaattccagc aggtacatat 1140
aatgtttctc caatgttagc ttgtactatt tgggtgccaa tttacgtag aattattatt 1200
ccattgttaa gaagatggac taaaaaggaa gatggtatta cattgttaca aagaatgggt 1260
gttggatttg ttttatcaat ttgtcaatg ttggtttcag ctttgggtga atctagaaga 1320
agaaatttgg ctttgactag accaacttta ggtattcaag aaggtcaagg ttctatttct 1380
tcaatgtcag ctttatgggt tgttccacaa ttggctttta tgggtttggc tgaaggtttt 1440
acttttatgt gtgaaaatga attgttttat aaagaatttc cagaaaaatg gagatctatt 1500
gcagcttctt tggttttgtg ttgtgttctg gttgcttcat atttgacatc ttttattgct 1560
gaaattgttc ataaaaaac tagatcaggt actagagatt ggttggctca agatttgaat 1620
aaggctagat tggattactt ttgtacatg ttgtctgttt tagaagcatt aaatttgggt 1680
tactttttag tttgtgctaa atgggtataga tacaagggtta ctggtgactt gggtaaatta 1740
gaagttgcta tggaaaaatt ggaaacagaa aaacatttgg ttttaa 1785

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SEQ ID NO: 325      moltype = DNA length = 2052
FEATURE            Location/Qualifiers

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source          1..2052
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 325
atgttgaatc atccatctca aggttcagat gatgctcaag atgaaaaaca aggtgacttt 60
ccagttattg aagaagaaaa gacacaagct gttactttga aagattcata cgtttctgat 120
gatgttgcta attctactga aagatacaat ttgtctccat ctccagaaga tgaagatttt 180
gaagctccaa ctgaagaaga aatgcaaaaca ttgagacatg ttggtggtaa aaatgaacca 240
aattacagag gtattaaagc aatgccattt gttttgggta atgaaacttg tgaaaaattg 300
ggtaacaatt gtacttcacg taatttgta gtttacttat caactgtttt tcatatgaat 360
acaatgtctg ctaataatgt tattaatatt tttaatggta catgttactt gggttctttg 420
gttgggtgat ttttatgtga tacatacttt ggtagatata aaactttggg tttaggttct 480
gttgcatcat ttttgggtat gttgatgta actttgacag cagctatttc aactttgcat 540
ccaacaaatt gtgataattc tggtagctgt aaaggtccat cagcagtgta aatggcattt 600
ttggttggtg gttttgtttt aatggttatt ggtgcttctg gtattagacc atgtaatttg 660
gcatttggtg ctgacacaat caatccaaat acagaatcag gtagaagagg tattaattca 720
tttttcaatt ggtattactt tacatactat tttgctgta tggtttctgc tactattatt 780
gtttatgttc aatctgatat ttcttggta attggttttg ctattccaac ttcatgtatg 840
tttttatctt tgcttttatt ttctgctggt acatgggtat acgttaaagt ttaccacaa 900
ggttctccaa tgacatcatt ggttcaagtt attgttgggt ctgttaaaaa gagaagattg 960
aaattgccag gtgaaccatc attattgttt aatcatattc catctaagtc tattaattct 1020
aaattaccaa gaacaaataa ttttagattt ttggataaag cagctatttt gaccacagaa 1080
gataaagtta atccagatgg ttctgtgca aataagtgga aattatgttc aattcaacaa 1140
gttgaaagag ttaaatgtgt ttttaagagt ttgccaatth ggattgcagg tgttattttt 1200
caaatttctg ttcacaaca atcaaatcac gttccatttc aagcatttca atctgataga 1260
catttgatta aaggaataaa gtttcaaat ccagctggta cttacaatgt tttctctatg 1320
ttagcattga ctatttgggt gccaatttat gatagaatta ttattccatt gtttaagaaga 1380
tggactaaaa aggaagatgg tattacatta ttgcaaaaga tgggtgttgg tattgttttg 1440
tcaattttat ctatgttggg ttccagcttg gttgaatcta gaagaagaaa tttggcttta 1500
actagaccac cattgggtat tcaagaaggt caaggttcta tttcatcaat gtcgtcttgg 1560
tgggttggtc cacaattagc ttgatgggt ttggcagaag gttttacttt tattgggtgaa 1620
aatgaattgt tttacaagaa atttccagaa aatatgagat ctattgcagc ttcatagtt 1680
ttgtgtgggt ttgcagttgc ttcatacttg acttctttta ttgcagaagt tgttcataaa 1740
acaactagat ctggtactag agattgggtg gcacaagatt tgaataaggc aagattggat 1800
tactttttct acatgggtgc tgttttagaa gcattgaatc tgggttactt tttgggttgt 1860
gctaaatggg acagataata aggtacaggt gacttgggta aattggaagt tgcattggaa 1920
aaattggaaa ctgaaaaaca ttggttgtaa gaagaagatg aatttgattt gaatccaatt 1980
tcagacccaa aagctaataa tattgaaatt ttggaaccaa tggattcttt aagatcaaca 2040
gctaataatt aa                                     2052

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SEQ ID NO: 326      moltype = DNA length = 1950
FEATURE             Location/Qualifiers
source              1..1950
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 326
atgttaaatc atccatctca aggttcagat gatgctcaag atgaaaaaca aggtgacttt 60
ccagttattg aagaagaaaa gactcaagca gttacattga aagattcata tgtttcagat 120
gatgttgcta attcaactga aagatacaat ttgtcaccat ctccagaaga tgaagatttt 180
gaagcaccac ctgaagaaga aatgcaaaact ttaagacatg ttggtggtaa aaatgaacca 240
aattacagag gtattaaagc tatgccattt gttttgggta atgaaacatg tgaaaaattg 300
ggtaactatt gtacatctac taatttgta gtttacttat ctactgtttt tcatatgaat 360
acaatgtctg ctaataatgt tattaatatt tttaatggta cttgttactt gggttcttta 420
gttgggtgat ttttgggtga tacatacttt ggtagatata aaactttggg tttaggttct 480
gttgcttcac ttttgggtat gttaatgttg actttaacag cagctatttc tacattgcat 540
ccaactaatt gtgataattc tggtagctgt aaaggtccat ctgctggta aatggcattt 600
ttagttgggt gttttgtttt aatggttatt ggtgcttctg gtattagacc atgtaatttg 660
gcttttgggt cagatcaatt caatccaaat acagaatcag gtagaagagg tattaattca 720
tttttcaatt ggtactactt tacatacaca tttgctgta tggtttctgc tactattatt 780
gtttatgttc aatcagatat ttcatggctt attggttttg ctattccaac atcatgtatg 840
ttttgtcatg gtgcattggt ttctgctggt acttgggtgt acgttaaagt ttaccacaa 900
ggttctccaa tgacatcttt ggttcaagtt attgttgggt cagttaaaaa gagaagatta 960
aaattgccag gtgaaccatc attgttgggt aatcatattc catctaagtc aatttaattca 1020
aaattaccaa gaactaataa ttttagattt ttagataaag ctgcaatttt aactccagaa 1080
gataaagtta atccagatgg ttctgcagct aataagtgga aattgtgttc tattcaacaa 1140
gttgaaagag ttaaatgtgt ttgagaggt ttgccaatth ggattgcagg tgttattttt 1200
caaatttcag ttactcaaca atctaattat gttccatttc aagcatttca atctgataga 1260
catttgatta aaggtaataa gtttcaaat ccagctggta cttataatgt tttctctatg 1320
ttagctttga caatttgggt accaattttac gatagaatta ttattccatt gtttaagaaga 1380
tggactaaaa aggaagatgg tattacatta ttgcaaaaga tgggtgttgg tattgttttg 1440
tctattttgt caatgttggg ttcaagcttg gttgaatcaa gaagaagaaa tttggcatta 1500
acaagaccac cattaggtat tcaagaaggt caaggttcaa tttcatcaat gtcagcttta 1560
tgggttggtc cacaattagc tttaatgggt ttggcagaag gttttacttt tattgttgaa 1620
aatgaattgt tttacaagaa atttccagaa aatatgagat caattgtctg atctttgggt 1680
ttgtgtgggt ttgctgttgc atcttactta ttgctgaaat tgttcataaa 1740
acaactagat caggtactag agattgggtg gctcaagatt tgaataaggc aagattggat 1800

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tactttttct atatggttgc tgttttggaa gcattaaatt tggtttactt tttggtttgt 1860
gctaaatggt acagatacaa aggtactggt gacttgggta aattggaagt tgctatggaa 1920
aaattggaag cagaaaaaca tttggtttaa 1950
```

```
SEQ ID NO: 327      moltype = DNA length = 1275
FEATURE            Location/Qualifiers
source             1..1275
                   mol_type = other DNA
                   organism = synthetic construct
```

```
SEQUENCE: 327
atgggaggtg cgagtgcag cgggttagac aataaaattha ataaggaaag ggtgatcgct 60
attggtgaga taatgagatt accaaatctt aattctttac aggtcgctgc attcatcaat 120
tcactaaggg atgatccgag ccaaagcgcc aacctacttg ccgaagcgaa gaaattaaac 180
gacgcccagg ctcccaaagg aggtagcagc gcaagtcttg caggagcgag tagtgttgac 240
aacaatatta acaaggaagc gcaaaactgc ggtgtggaaa tcatggaact gccaaacctg 300
aatacacggc agctcctggc ttcatccag tcattacgtg atgatccgtc ccagtcagcg 360
aatttgctgg cagaggctaa aaagctcaac gatgcgcaag ctctaaaaac cagtggaggc 420
agttccgcct ccagcgcagg aggggtcaag gttagacaaca aattcaacaa agaaagagtg 480
attcgatag gtgagatcat gcggttacct aacttaacaa gtctccaagt ggtggccttc 540
atcaatagtt tacgggatga cccaagccaa agcgctaact tactagcaga agctaaaaag 600
ctaaatgatg ctacggcacc gaaaggaggt tctagtgcgt ccagtgcagg agggagcagt 660
gttgataaca agttcaacaa ggaggctcag accgcagggg tggagattat ggagctacct 720
aacctaaaca ctagacaact ttggcggtt atccagtccc ttaggggacga cccatcccaa 780
tccgcaaatc tattggcgga ggctaagaag ctgaacgatg cacaagcccc caagacaagt 840
ggtagcggtg catctgtag ttgtgcaggt ggaagcagcg tagacaataa gtttaataag 900
gagcgtgtta tagccatcgg agaaattatg cgtttgccaa atctgaactc cctacagggt 960
gtggctttca taaatagttt acgtgatgac cctcacaat ccgcgaactt gtttagctgag 1020
gcaaaaaagt taaacgacgc tcaggctcca aaaggttccg gtgcctccgg aagcggagca 1080
ggtaggatca gcgttgacaa taaatttaac aaagagggcg agactgcagg cgtggaaatt 1140
atggaattgc cgaacttgaa caccgcgcag cttttggcct tcatacagag cttacgtgat 1200
gacctatctc aatcagctaa cctattagca gaggctaaga agttaaacga tgctcaggcc 1260
ccaaagacaa gtttaa 1275
```

```
SEQ ID NO: 328      moltype = DNA length = 210
FEATURE            Location/Qualifiers
source             1..210
                   mol_type = other DNA
                   organism = synthetic construct
```

```
SEQUENCE: 328
atggtagaca acaaatccaa caaagaactg ggttgggcga cctgggagat ctcaactta 60
cctaacttaa acggtgtgca agtgaaggcc ttcacgata gtttacggga tgaccaaacg 120
caaagcgcta acttgctggc agaagctaaa aagctaaatg atgctcaggc gccgaaaggt 180
tcaagcagct caggttcgag ttcttcggc 210
```

```
SEQ ID NO: 329      moltype = DNA length = 210
FEATURE            Location/Qualifiers
source             1..210
                   mol_type = other DNA
                   organism = synthetic construct
```

```
SEQUENCE: 329
atggttagaca acaaatccaa caaagaacaa atacaagcga gtcaagagat cagactatta 60
cctaacttaa acggttagaca aaagctcgcc ttcacccaca gtttacttga tgaccaaacg 120
caaagcgcta acttgctggc agaagctaaa aagctaaatg atgctcaggc gccgaaaggt 180
tcaagcagct caggttcgag ttcttcggc 210
```

```
SEQ ID NO: 330      moltype = DNA length = 168
FEATURE            Location/Qualifiers
source             1..168
                   mol_type = other DNA
                   organism = synthetic construct
```

```
SEQUENCE: 330
atgcaactgc tgagatgttt tagcatcttt tccgttattg cgagcgtggt agctcaggaa 60
cttactacta tctgcgaaca aatcccaagc cccacgctgg aatcaacccc ttactcactg 120
agtacaacaa cgatacttgc gaacggcaaa cttgacaaga gaggaagt 168
```

```
SEQ ID NO: 331      moltype = DNA length = 216
FEATURE            Location/Qualifiers
source             1..216
                   mol_type = other DNA
                   organism = synthetic construct
```

```
SEQUENCE: 331
atgaagggtc ttgatttgct aaccgtgctg agtgettccct cactactgag caccttcgca 60
gcggcggagt caacggctac tgctgatagc acaactgcag ccagtagtac ggcatcttgt 120
aatcttttaa aaactactgg atgcacacct gacaccgcgt tagccaactag cttcagcgaa 180
gactttctcat cttccagctc tgacaagcgt gggtcc 216
```

-continued

SEQ ID NO: 332 moltype = DNA length = 153
 FEATURE Location/Qualifiers
 source 1..153
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 332
 atgttatcac tgaagacttt gttgtgtact cttctaacag tctcaagcgt gttggcgacg 60
 ccagttcctg ccagggaccc ctctccatt caattcgcc atgaagaaaa caagaagagg 120
 tactatgact acgaccacgg gtccttggc gaa 153

SEQ ID NO: 333 moltype = DNA length = 99
 FEATURE Location/Qualifiers
 source 1..99
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 333
 atgaagtttg cgtatagttt actgcttccc cttgccggtg tttcagcctc cgtcatcaac 60
 tataaacgtg acggcgactc attagacaaa cgtggaagt 99

SEQ ID NO: 334 moltype = DNA length = 198
 FEATURE Location/Qualifiers
 source 1..198
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 334
 atgaagctac agtctcttct agtgtctgca gcggtactta catccctaac agaaaatgta 60
 aatgcatgga gtcctaataa ttctatgtc cgggctaacg tgacctgcga cgacgacata 120
 aatctagtcg gtgaggcatc tggccttagc gacaacgaaa ctgagtggtt gaagaaacgt 180
 gacgttata cgaaagag 198

SEQ ID NO: 335 moltype = DNA length = 60
 FEATURE Location/Qualifiers
 source 1..60
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 335
 atgctattgc aggctttctt gttcttgctg gcaggattcg cggcaaaaat cagcgcatct 60

SEQ ID NO: 336 moltype = DNA length = 297
 FEATURE Location/Qualifiers
 source 1..297
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 336
 atgaactgt cgcgtacgac tctaactcgc gcaagcctaa taggctattc tacgattgtg 60
 tctgctcttc cgtatgcggc ggacattgac acaggatgta cgacgactgc gcacggttcc 120
 catcagcaca agagagctgt ggcgtgaacc tacgtgtacg agacagttac cgtggataaa 180
 aatggtcaaa cagtaacgcc gaccagtacc gaagccagta gtaccgtcgc cagcaccaca 240
 acgcttatta gtgaatcttc cgtcacaaa gtcagctccc tggataaaa gggttct 297

SEQ ID NO: 337 moltype = DNA length = 276
 FEATURE Location/Qualifiers
 source 1..276
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 337
 atgcgttttc cttccatctt tacggacgtg ttgttcgctg cgtcaagcgc cttagccacc 60
 cctgttaaca cgaccacaga agacgagaca gctcagatcc ctgccgaggc ggttataggt 120
 tacagtgacc ttgaagggga cttcgatgtt gccgttttgc cgtttttctaa ttccaccaat 180
 aacgggttgc tattcataaa taccacaatc gcgtcaatcg cagctaagga agaaggagta 240
 agtttagaga agaggggaagc aacgaatttc ggatca 276

SEQ ID NO: 338 moltype = DNA length = 300
 FEATURE Location/Qualifiers
 source 1..300
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 338
 atggaagagc aaagagagat attagaacag ttgaagaaaa ctttacagat gctgactgtt 60
 gagccgtcca agaataatca aatcgccaac gaagaaaaag agaagaaaga aaacgaaaat 120
 agctgggtgca tcttggaaca caattacgag gatattgcac aagagtttat cgactttatt 180
 tataagaacc ccacaacctc tcacgttgtt tctttcttcg cgggaattgct ggacaagcac 240
 aacttcaaat acttatcaga gaaaagcaac tggcaggatt ccatcgagga agatggcgga 300

SEQ ID NO: 339 moltype = DNA length = 225
 FEATURE Location/Qualifiers

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source                1..225
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 339
atgttcagaa tacaattacg taccatgtca tctaagacgt gcaagagtga ctaccctaag 60
gagtttgcct cattcctgaa tagcagtcac agtccatata ataccgttca caacattaag 120
aaacatttag taagtaatgg attcaaggag ctaagcgagc gtgactcatg ggctggacat 180
gtggcgagaa aaggcaagta ttctgttaca cgtaacggga gtagt 225

SEQ ID NO: 340        moltype = DNA length = 891
FEATURE               Location/Qualifiers
source                1..891
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 340
atgtctgaca gcagtcgcagc tattaacttt attaacttta accaaaccgg tacatgtatc 60
agtttaggaa cgtccaaagg cttcaagata ttcaactgtg aacctttcgg caagttctat 120
tcagaagaca gtggcgggata tgccattgta gagatgctat tcagtaccag tcttctagcc 180
ttagttggca taggtgatca acccgctctt agccctaggg gtctaaggat catcaatact 240
aagaaacact ctatcatctg tgaggtgacg ttcccgactt ccattttgtc cgttaagatg 300
aataagtcac gtcttgtcgt acttttgcaa gagcagatat atatctatga cataaacaca 360
atgaggcttc tacacactat tgagaccaat ccgaacccaa gaggtttgat ggcgatgtca 420
ccaagtgtag ccaacagcta tctggtgtac ccttctcttc ctaaaagtaat taacagcgag 480
attaaggctc acgcaacaac gaataatata acactgtctg tcggtggaaa cactgagacg 540
agttttaaac gtgaccagca agacgcgggc catagcgaca tctctgatct ggaccaatac 600
tctagcttca ctaagaggga cgatgctgac cccacgtctt ctaacggttg caatagcagt 660
attattaaga atggcgacgt tatagtttcc aacctggaga ctcttcagcc cactgatggtc 720
atcgaggcac ctaagggtga gatcgcgagg atggcaattt ccttgacgg aacacttatg 780
gcaacagcgt ccgacaaggg gactataatt agagtatttg atatagaaac gggagacaaa 840
atctatcagt tccgtagggg cacatacgcg acaagaattt atagcataag c 891

SEQ ID NO: 341        moltype = DNA length = 171
FEATURE               Location/Qualifiers
source                1..171
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 341
atgattgccc ttctgttaga aaaagcacca aggaagtctc tttggcagag acacagggcc 60
ttcatctcgc ggatcgctgc tctaattatt ataggtacgt tttcttaaac ttctggttg 120
catctgctc caccgcatga agcaaaacgt ccccatcatg ggaaaggacc c 171

SEQ ID NO: 342        moltype = DNA length = 105
FEATURE               Location/Qualifiers
source                1..105
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 342
atgttttagcc tgaaggcatt attgccgctg gcgttggtac ttgtctctgc caatcaggtt 60
gccgccaagg tccacaaggc gaaaatatac aaacacgagt tgtcc 105

SEQ ID NO: 343        moltype = DNA length = 297
FEATURE               Location/Qualifiers
source                1..297
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 343
atgaaattgg agaacacttt atttacactt ggtgcactgg gttccatttc tgctgcgtta 60
gtaattccga atctggagaa tgcagccgat cccatgaat tgataaataa agaggaccac 120
catgagaggg ccagggaagg tgaatttaca aaggatgacg acgaggaacc ttctgacagc 180
gaggacaagg aacacgggaa attccacaag aagggtagga aaggtcagga caaagagtct 240
cccgaattta atggaagag ggccctcagg agccacgggt ccgcgcatga aggagga 297

SEQ ID NO: 344        moltype = DNA length = 492
FEATURE               Location/Qualifiers
source                1..492
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 344
atgaaagcgt tcacatctct tctgtgcggg ttgggtttgt caacgacctt ggccaaagcc 60
atctccctac aaagggcggt aggcctggac aaagacgtgc ttcttcaagc agcggaagaa 120
ttcgggttag acttggatct ggatcatttg ctaaaaggaa tggacagcaa cgtcttggat 180
gcgtgggctc aaatcgaaac tctatacccg aatcaagtaa tgtctctgga aacgtcaacc 240
aaaccgaaat tccagagggc aatcaagacc aaaaaggact gggacttcgt agttaagaac 300
gacgcaatcg agaactatca actgcgtgtc aacaaaatta aagatccaaa gattttggga 360
atcgacccca acgttaactca atacacgggc tacttggatg tcgaagacga ggataagcac 420
ttcttctctt ggacttttga gtcaagaaat gacctgcca aagatccagt gatcttggg 480

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-continued

ctgaatggtg ga 492

SEQ ID NO: 345 moltype = DNA length = 225
 FEATURE Location/Qualifiers
 source 1..225
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 345
 atgacccaaa actttattgt tacgttgaag aaaaatacgc cggacgtgga agcaaagaaa 60
 ttcttgata gcgtccacca tgctggaggg tcaatcgtec ataagttcga tataatcaaa 120
 ggctacacaa taaaagtgcg cgaatgctcg cacctgaata aactaaagga aaaacacaat 180
 gacgtgatag aaaatgtcga agaagacaaa gaggtacata cgaac 225

SEQ ID NO: 346 moltype = DNA length = 234
 FEATURE Location/Qualifiers
 source 1..234
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 346
 atgaaatctt ttattactag aaacaaaact gccatttttg cgacggttgc cgccactggt 60
 actgctattg gtgcctatta ctactacaac caattacagc agcagcaaca gcgtggtaag 120
 aaaaatacaa ttaacaagga tgagaagaag gacacaaaagg attcacagaa ggaacagaa 180
 ggcgcaaaaga agtctactgc accttctaac ccacgatat acccagttag ctct 234

SEQ ID NO: 347 moltype = DNA length = 351
 FEATURE Location/Qualifiers
 source 1..351
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 347
 atggcgatgt tctgtacctt ctttgagaaa catcatagaa aatgggacat cttactagaa 60
 aagagcacgc gagtgatgga ggcgatgaaa gtaacttcag aagagaaaga gcagttgtct 120
 acagctatcg atagaatgaa tgaaggtctg gacgcattta ttcaactata taacgaatcc 180
 gagatcgatg aacctttaat ccagttggat gacgatacag cagaactaat gaaacaggct 240
 agggacatgt acggccaaga gaaacttaac gagaaattaa acacaataat caaacaaatc 300
 ctgtcaattt ctgtctccga agagggtagg aaagaaggaa gcgcatcagg c 351

SEQ ID NO: 348 moltype = DNA length = 513
 FEATURE Location/Qualifiers
 source 1..513
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 348
 atggctgaca gggacaggag tggcatttac ggaggtacct atggccaaca ccagcaggga 60
 ggaggtagac ccatggggcga acaggttaaa ggaatgattc aggacaaggg tcccactgcc 120
 tctcaggcgc ttaactgtgc aaccttattc cctctgggag gtctactttt ggtcctatct 180
 ggattggcct tagccgcaag cgttgtcggt ttagctctgg cgacccccgt atttctacta 240
 ttttctccg ttctgtgcc ggctgcttta ctaatcggtta ccgctgttac ggggtttctg 300
 acctcagggt tcttaggact gggaggactt ttagcctta cttgcttggc caataccgct 360
 agacaggcgt ttcagaggac tctgactat gtagaggaag cccgtctgag aatggcgga 420
 gccgctgcac atgctggcca aaagacggct caggcaggac acgccaatcca gtcaagggct 480
 caagaggctg catctggagg cgcaggtgct gga 513

SEQ ID NO: 349 moltype = DNA length = 279
 FEATURE Location/Qualifiers
 source 1..279
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 349
 atgtctacgc tacagagaag gcgtgttaac agagcagact ctggtgacac gtccagcatc 60
 cacagctccg caaataatgc aaaaagtgat aagattgcaa acatcgctgt cgatggcgat 120
 gacgataacg gtacaaaaca gaaaatagca tacgatcccg aagagagtaa gctaagagat 180
 aacataaaca ttctactttt gactttaatg gaggagggtc ttttgatggg ttacgtgac 240
 agagagggat acctatcctt ctggaatgat tccatcagt 279

SEQ ID NO: 350 moltype = DNA length = 264
 FEATURE Location/Qualifiers
 source 1..264
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 350
 atgtcacttc ccttgagaac gatagtacat ttggtaaagc cctttgcttg cactgctagg 60
 tttagtgcga gataccaaat ccacgtoatt gttgttgctg ttttattgag tgccgctgct 120
 tatctatccg tgacacaatc ttaccttaac gaatggaagc tggactctaa tcagtattct 180
 acatacttaa acataaagcc ggaatgagttg tttgagaaat gcacacacta ctataggctc 240
 ggtggaaacg ctgcaattag aagc 264

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SEQ ID NO: 351 moltype = DNA length = 87
FEATURE Location/Qualifiers
source 1..87
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 351
atggaatagca aagggtcctc ccagaaggga tcaagattgc ttctgctgct agtatccaat 60
cttttattgt gccaggggggt tgtatca 87

SEQ ID NO: 352 moltype = DNA length = 69
FEATURE Location/Qualifiers
source 1..69
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 352
atgaagcgt tcacgtcctt attatgtggc ttgggtttga gcacgacact ggctaaggcc 60
atctcactg 69

SEQ ID NO: 353 moltype = DNA length = 219
FEATURE Location/Qualifiers
source 1..219
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 353
atgtccgcga ccaaaagcat cgtaggggag gctctggaat acgtcaatat cgggctttca 60
cacttccttg ctcttccctt agcccaaaga ataagtctga tcattattat tccattcactc 120
tataaatattg tatggcagct tctatatctt ttgcgtaagg acaggccccc gcttggttttc 180
tactggattc cctgggtcgg gtccgcagtt gtgtacggg 219

SEQ ID NO: 354 moltype = DNA length = 138
FEATURE Location/Qualifiers
source 1..138
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 354
atgcctttcg gtattgataa tactgatttc acggtcctag cgggtttggt attagcagtt 60
ctattatatg taaagcgtaa ctccatcaag gagttgctta tgtctgatga cggggatata 120
accgccgtgt cctctggt 138

SEQ ID NO: 355 moltype = DNA length = 99
FEATURE Location/Qualifiers
source 1..99
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 355
atggagtga ccatggcctc tacgatgagt ctggccttac tgggtgtgag cgcggcctac 60
gttctagtag cactgagaag aagtagatct tccagcagc 99

SEQ ID NO: 356 moltype = DNA length = 210
FEATURE Location/Qualifiers
source 1..210
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 356
atgtcatctg tggcagaaaa cataattcag cagcgaacac acaacagtac gctgcaccaa 60
cttgcaagg accaaccttc cgtcggcggt acgaccgcat ttagcatctt agacactcta 120
aagagcatga gctatctgaa gatattcgct actcttattt gtatcttgct tgtatgggat 180
caagttgect accagattaa gaaaggctca 210

SEQ ID NO: 357 moltype = DNA length = 75
FEATURE Location/Qualifiers
source 1..75
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 357
atgaagaat tattaaaaca agcactacaa caagcacagc aattattgca acaggcccaa 60
gagttggcga agaag 75

SEQ ID NO: 358 moltype = DNA length = 753
FEATURE Location/Qualifiers
source 1..753
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 358
atggtatcta aaggagaaga attgtttaca ggcgtgggtac caattctggt tgaattggac 60

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ggtgacgtga acggacacaa attcagcgtg agtgggaga gcgagggaga tgctacctat 120
ggcaagtgtga cgcttaaaact gatctgcaca acgggcaaat taccagtgcc ctggccgacg 180
cttgttaacaa ctcttggtata cgggttacag tgctttgccc gttatccaga ccatatgaaa 240
cagcatgact tcttcaaatc tgcgatgccg gagggatatg tacaggaacg tacgattttc 300
tttaaggacg atgggaaacta caagactcgt gctgagggtta agtttgaagg cgacactcta 360
gtcaatagga tagaattaaa ggggtattgat ttttaaggagg atgggaacat cctgggccc 420
aaactagagt acaactacaa ttacataat gtctacatca ccgctgataa acagaagaac 480
gggatcaaaag ctaatttcaa gatacgtcat aatatcgaag atgggtggcgt ccagcttgct 540
gaccactacc agcagaacac gcctataggc gacgggccgg tggtgtacc tgacaatcat 600
tatctgtcct atcagtcocg cctttcaaaa gaccctaag agaagaggga tcatatgggtg 660
cttttagaat ttgtaacocg ggcagggatc acacttggga tggatgagct gtataaaggga 720
ggtagtggaa gtgcgggatc tggcggtagt agc 753

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SEQ ID NO: 359      moltype = DNA length = 1161
FEATURE            Location/Qualifiers
source              1..1161
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 359
atgaagatcg aagaagggaaa gttagtgtac tggataaaatg gtgataaaagg ctacaatggg 60
ttggcgggaag taggaaaaaaa gttcgagaaa gacacaggaa tcaaagttag ggctcgagcac 120
cccataaac tagagggaaaa gttccacag gtagctgcta cgggggacgg accagacatt 180
atcttttggg ccacagatag attcgggggt tatgctcagt ccggaactct ggccgagatt 240
actccagaca aggccttcca agacaaactt taccggtcca catgggacgc agtcagggtac 300
aatggaaaagc tgattgata tccgatagct gtggaggcac ttagcctaatt ttacaacaag 360
gatctactac ctaacctccc caagacttgg gaagaaattc cagctctgga caaggagtta 420
aaagcaaaagg gtaagagtgc acttatgttc aatctacaag agccttattt cacatggccc 480
ctaatagcgc cgcacggagg ctatgccttt aagtaacgaaa acggcaagta tgacataaag 540
gatgtttgggg tagacaacgc gggagccaaag gctggattaa ctttccctgg ggatttaatt 600
aagaacaac acatgaacgc agacactgac tactctatcg cagaagcagc gttcaataaa 660
ggcgaaacgg cgatgacaat taacgggccc tgggcttgggt caaacattga cacgagtaaa 720
gttaactatg gtgtaacggt attgcccaca tttaagggac aaccagtaaa acctttcgta 780
ggagtcttgt cagccgggat caatgcagct tcccgaata aagagcttgc taaggaaatt 840
cttgaaaatt atcttttaac cgatgaggga ttggaggcgg ttaacaagga caagcctctt 900
ggtgtctgtg cctgaaatc ctatgaagaa gagttagcta aggacccaag aatcgccgca 960
acaatggaga atgctcagaa gggagaaatt atgccaaata taccacaaat gagtgccttc 1020
tggtatgcgg taaggacggc agttattaat gccgcttcag gtagacaaac agtcgatgag 1080
gctttgaaaag atgcacagac taacagttca tccaacaata ataacaataa caataacaat 1140
aacctgggta tcgagggcgc t 1161

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SEQ ID NO: 360      moltype = DNA length = 1473
FEATURE            Location/Qualifiers
source              1..1473
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 360
atgacatcac tatccaagtc attcatgcag tctggttaga tttgcgcgcg ctgcttctac 60
cttcttttca ctctgcttag catccctata tcttttaaag taggcggtct tgaatgcgga 120
ctttctttta cgtttacatt gtttacgcta tacttcatta caacaacact taatgtactg 180
gcgagaaggc atggaggggcg tttgtacatc ttcttccacta attgctcata ctacagccaa 240
cacttcatca tcgcgagttt gctttattta tttctttccg gcttctccaa cgacgagtta 300
ggcaatgtgt tgaagaataa gtacaacgag agtgagtctt ttttagaagc acttaaaaac 360
agtttgaact caaaccaaat gaattacgtc ctgtactatt actactaccg tttcgttgtg 420
caaccttggc aattcgtaact aacaaaatct actccattct tcacactatc tgaaggattt 480
ttactattt tagctattca agcagtcggc gagaccaata ggtggctatc taacgacttg 540
aactccaaca cctggatcat ctctcatta ctaacatccg ggggcgtgat taccgcttcc 600
ttgtattatc tgtataggat atatgtcact cctatatggc ctcttagtat tcagactgca 660
tctttgttgg gtttggcttt gagtatggta tgggactag gactgtacgg aattgttagc 720
caaaaggggt cagtgtattga gtcttctttg ttcttcgcct acattgttgc ttgtatatat 780
gaaatcagtc cgaagctggc gacaacagct actgatgaga ttttaaaact gttcaaaagat 840
gtatggcaga agcaccagc taactcgcg accgcagaca accttttggt ctattttcac 900
aatgttatct tgaagaacgc ggaggtactt tggggtagct tcataccaag aggtagaaaag 960
aaaacaggag attttcacga caagctaatt tccattctat cattcgaaaa agtaagttaa 1020
atctctaagc cactctggaa attcttcaaa aacttcacgt tcagtgtacc gtttaagtac 1080
aatgaatttt gccaaagtgc catcaagatg gcgagtgaag gcgtttctcc cgcaatagta 1140
ataaacctgt gttttagggt tctgatgttt tacagcgaga ccagaataat tccagctttg 1200
caaaagaaaa acgacaacaa gttgaggaag agcaggcgta tcatgaaagg attgtactgg 1260
tacagtccat gcattctgat cgccatgtac actcacttaa ttcttcaata ctcaggcgaa 1320
ctgaagaaaag atttatgtat ttggggatgc agcgaaaagt ggtttggagt cgaccaaccc 1380
gaaattatag tagactcatg gggcttttgg aactgggtgca acatattctg tactattttg 1440
gtgtacgcaa ctgactcatg tgggagcggga agt 1473

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SEQ ID NO: 361      moltype = DNA length = 912
FEATURE            Location/Qualifiers
source              1..912
                    mol_type = other DNA

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organism = synthetic construct

SEQUENCE: 361
atgcaacaag cgactggcaa tgaattgtta ggtatattgg atttagataa tgacatcgac 60
tttgagacgg cgtaccaaat gttatcaagt aacttcgatg atcaaatgtc cgcacatata 120
cacgaaaaata ccttttagcgc cacctcacct cctttactta ctcacgaact aggcatcata 180
ccgaacgtgg caaccgtaca accaagtcac gtggagacca ttccggcgga caatcaaact 240
caccatgctc cgttacatcac tcatgcccac tacttgaacc acaatccaca ccagccaagc 300
atgggcttcg atcaggcgct tggattaaaag ctgtcccca gcagcagcgg cttgtgtgca 360
accaacgaaa gtaatgcgat tgagcagttc ttagacaacc ttatttcaca agacatgatg 420
agttccaatg cgtctatgaa tagtgagagt cacctacaca tcagatcacc aaaaaaacag 480
caccgttaca ccgaattgaa ccaacgttac ccgagacgc acccccactc caataccggt 540
gagctaccga ctaatacggc tgatgtcccg actgaattca caaccctgga gggcccgcac 600
cagcccatcg gaaatgacca ctataatcct cccccctttt ctgttctgga gataaggatt 660
ccagatagcg atattctctg gaattattgag gatgaccccg taaaagttag gaagtggaaa 720
catgtgcaaa tggaaaagat ccgtagaatt aatacaaaaagg ggcgtttgga gcgtttgatc 780
aagtctgtga gaactccgcc gaaggagaat ggcaaaaagga ttccgaaca tatcctgcta 840
acctgtgtta tgaacgacat aaaaagtatc agaagcgcca atgaagcatt gcagcacatt 900
ctggacgatt ca                                     912

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SEQ ID NO: 362      moltype = DNA length = 1152
FEATURE            Location/Qualifiers
source             1..1152
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 362
atgagtaaaa gcaagacatt cctgttcacg agcgaatcag ttggcgaggg gcaccccgat 60
aagatatgcg accagggttag tgatgcgatt ctggacgcat gtctggaaca ggatcctttt 120
tcaaaaagtgg cgtgtgagag ggcgcggaag acgggaatga tcatggtgtt cggagagatc 180
accactaaaag ccggtttaga ttatcaacaa attgtacgtg ataccataaa gaagataggt 240
tatgatgact cagcaaaagg gtttgattat aaaacttgta acgtcttggt cgctatcgag 300
caacagtcctc ctgatatgtc acaaggttta cattatgaga agtcccttga agacttagga 360
gcaggcgatc aaggtattat gtttgggtac gcaaccgacg agaccccgga aggactgcct 420
ttaacaatac ttctgtctca taagctgaac atggcaatgg cagacgcgag gagggatggt 480
agcctgcctc ggttgcgtcc agatactaaa actcaagtga cggttgaata tgaagatgat 540
aacgggaggtt ggttgccgaa gaggattgac acggttggtt tctccgcaca gcacgccgat 600
gagatctcca cggccgacct acgtacacaa ctacaaaaag atatttggtga aaaagtaatt 660
ccgaaggata gtgtggatga aaatacaaaag tattttatcc aacctctgga aaggttgcgc 720
ataggggggtc cccaagggga cgtcggccta actggcagga agattattgt agatgcgtat 780
ggcgggagcaa gtacgcttgg agggggggct ttcagcgcca aagactatcc caaggtagat 840
aggagcgccg cgtacgcggc gagatgggtg gcaaaatcct tagtggctgc aggactttgt 900
aagcgtgttc aagtgcaatt ctcatcgcg atcggtatag cggagccctc ttccttacat 960
gttgatacgt atggcacagc aactaaaaag gacgatgaga taatagaaat cattaagaaa 1020
aacttcgatt tgagacctgg cgtactagtc aaagagttgg acttagcaag accaatctac 1080
cttcccacgg cttcttatgg acattttacc aaccaggagt actcttggga gaagccgaaa 1140
aagctggagt tc                                     1152

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SEQ ID NO: 363      moltype = DNA length = 666
FEATURE            Location/Qualifiers
source             1..666
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 363
atgtcctcca gcgaatccat ccgtatggtg ctgattggtc ccccaggggc ggggaagggt 60
acgcaagccc caaatctgca agaaagattt cacgctgctc acctagctac aggagatgat 120
ttgcgtagtc agattgcaaa aggtacacag ctaggcctgg aagctaaaaa aattatggac 180
caaggcggcc ttgttagtga cgtattatag gtaaacatga taaaagatga gcttacgaat 240
aaccgcgctt gcaagaatgg attcatctta gatggtttcc cgcgtacaat tccgcaggcc 300
gaaaagtgtg atcaaatgct gaaagaacaa ggcacacctt tagagaaggc cattgagctt 360
aaagttgatg acgagctact agtggcccgat atcacgggga gacttatcca ccccgcagc 420
ggaagaagtt accacaagat attcaaccct ccgaagaaga acatgaaaga cgatgtcacc 480
ggagaagcat tagttcaaa gctctgatgc aacgcggatg cgctaaaaaa gagattggct 540
gcgtaccacg cacagacgga accgattgta gatttctata agaaaacagg gatttgggcc 600
ggagtcgatg cttcccaacc cctgccact gtatgggctg atatccttaa taaactaggt 660
aaggat                                     666

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SEQ ID NO: 364      moltype = DNA length = 1761
FEATURE            Location/Qualifiers
source             1..1761
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 364
atggacatac ttaaaagagg aaatgagagc gacaaattta ctaaaataga gacggaatct 60
acaactatac ccaacgactc agacagatcc ggatctctga ttagacgtat gaaagactca 120
tttaaacaaat caaacctgca tgttattcca gaggatttgg aaaaacagtga acagacggaa 180
caggagaaaaa tacagtggaa actggctagt caaccgtacc agaaggtttt gtctcagaga 240
cacttgacca tgatcgctat cgggggcaca ctaggaaacag gactatttat tggccttggt 300

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tactctttgg cctctgggccc agcagcacta ctaattgggtt tccctcttgt tgggacctcc 360
atgttttttg tagttcagtc cgcgcagaa ttgtcctgtc agtttccgt ttctggtagc 420
tacgcccacac acgtgtctag gtccatagac gaatctgtgg gtttcacagt agcaacaaac 480
tatgccttgg cgtggctgat ctcccttccct agtgaactga tcggatgtgc gctaacgatt 540
tcttactgga accaaacgggt aaatccggcg gtgtgggtag caatatttta tgtattttatc 600
atggtgtttaa atttggtttg tggtcgtggc ttcgcagaga cagaattcgc gcttagtata 660
attaaggtaa tagcaatctt tatatttatt atcattggga tcgtgttaat agcaggagg 720
ggccctaatt caacaggtata tataggggcg aagtactggc acgacctgg agcggtcgct 780
aagccggtgt tcaagaatth atgcaataca ttgtctcag cggcggtctc ctttgggtgg 840
agtgaactag cctctttgac ctctactgaa agcaagaaca ttccagcgat aagtaggcg 900
gcaaagggtta cctctcgtag gatgtctatc ttctatatca ctaccgtagt cataatcggc 960
tgcttagtac catataacga cccaaggctg ttaagtggct ctaactctga agacgtatcc 1020
gcctcccggt ttgtcatcgc gctatccaac accgggagca tgggtgctaa ggtgtcaaat 1080
tttatgaacg tcgtaattct atagcagta gtatccgtgt gcaattcatg tgtttacgca 1140
agctcaaggc tgattcaagc gcttggggcc tctggtcaac taccatcagt gtgtagttat 1200
atggatagaa aaggagagac gttagtgggg atagggatta gcggagcttt cggcttgta 1260
ggattctctg tggcaagttaa aaaggaagac gaagtgttta cctggctttt tgcctgtgc 1320
tcaataagca gcttttttac ctggttttgt atatgtatgt cccagatcag gtttaggatg 1380
gctctaaagg cgcagggcag gagtaacgac gagattgcgt ataagagcat ccttggggt 1440
tacggaggaa tattaggttg cgtgctgaat gctttgctga tagcggggga gatctatgtc 1500
agcgcgggcg ctgtcggatc cccaagtagt gcggaggcat tttttagta ttgtctatcc 1560
atccccatca tgatagtcgt ttattttgct catcgttttt atagacgtga ctggaacat 1620
ttctacatta aaagggtccga gatagatttg gacacaggat gctccgtcga aaatttggag 1680
ttgtttaagg cccagaaaga agctgaagag caattgatag ccagcaaac tttctactat 1740
aagatctata gattttggtg t 1761

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SEQ ID NO: 365      moltype = DNA length = 1722
FEATURE
source             Location/Qualifiers
                   1..1722
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 365
atgagcgaag gaagaacgtt tttatctcag ctaaatgttt tcaataagga aaattaccaa 60
ttctcttcta gcaccacaaa aaaagaagta tcaaaactcta cagtggacgc tgacaacgga 120
gcttctgatt ttgaggccgg gcagcagttc gccaccgaat tagaccaggg ggaaaagcag 180
ctaggggatt taagttgcatt tgggttgata tgaataagga tgttggggac ggggggtgtt 240
gcagtgaagt caacgatata tacgttatgc gggagcgtgg ggctggctct tatcatgtgg 300
gcagttggag ctattattgc aatctcaggt ttgtacgtct acatggagtt tgggaccgct 360
atacctaaaa atggcggcga gaagaattac ttggaagcaa tcttcagaaa acctaaattc 420
tttataacct gtagtgcgc cgcatatatt tttttcttag gatgggctgc cggtaacagt 480
ataaacacag caattatgtt cctaaccggt gctgacacgg aggtgactaa gtggaatcag 540
agaggaaatt gtagtagcgt ggtgttttcc gcatctctga taaattcctt gaattgtaa 600
attggtctat atttacaata catactagga attttcaaga tcggaatagt attattcatc 660
tctatcactg gttgggtggc gctaggggga ggtctaaaaa acggatacca gagccacaat 720
tttagaaatg cgttcgaggg aacggagacc gcgaccgctc atggcattgt taatgcatta 780
tactcagtaa tttggagttt tgcggatac tccaacgtca attatgcgt tggtagggtg 840
aaaaaccocg ttaggacatt gaagatcgct gggcctacat ccatggtttt cttggccatc 900
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SEQ ID NO: 366      moltype = DNA length = 1776
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source             Location/Qualifiers
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                   mol_type = other DNA
                   organism = synthetic construct

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SEQ ID NO: 367      moltype = DNA length = 1800
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source             Location/Qualifiers
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                   mol_type = other DNA
                   organism = synthetic construct

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aataaaattcc atgctttcta ttacacttta aaaagttcca gctcttctct cgaagaaaag 360
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ttactggggg gtgtaaattg ggatgtgaac gtatcaaac agggcttgcc gcgtgaaacg 480
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SEQ ID NO: 368      moltype = DNA length = 1527
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source             Location/Qualifiers
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                   mol_type = other DNA
                   organism = synthetic construct

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tggccaaata gatggactac tgctggtaat ggtaaaactt acacttcttt gtataatgtt 240
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acttcagttt tggattttta aaagggtgaa tcatctccgt ggactccatt gtctaatatg 420
caaatttcta gaggttacca atcatcttgt acaacttctg aaggtaaaaa ttttgttatt 480
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SEQ ID NO: 369      moltype = DNA length = 984
FEATURE            Location/Qualifiers
source              1..984
                    mol_type = other DNA
                    organism = synthetic construct

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agagtttggt ctgctgttaa ttaa 984

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1. A polynucleotide comprising a sequence as disclosed in a Sequence Listing filed herewith, or a sequence having greater or equal to about 90% sequence identity to any one of said sequences.

2. The polynucleotide according to claim 1, having greater or equal to about 95% sequence identity to any of said sequences.

3. An expression vector comprising one or more polynucleotide sequences according to claim 1 and an operatively linked promoter sequence that allows for expression of the one or more polynucleotide sequences in a cell.

4. A recombinant cell comprising a polynucleotide according to claim 1.

5. The recombinant cell according to claim 4, wherein the cell comprises a bacterial cell, a fungal cell, or a yeast cell.

6. The recombinant cell according to claim 5, wherein the cell is a species of yeast that includes *Saccharomyces*, *Candida*, *Pichia*, *Schizosaccharomyces*, *Scheffersomyces*, *Blakeslea*, *Rhodotorula*, or *Yarrowia*.

7. The recombinant cell according to claim 5, wherein the cell is a species of a filamentous fungus that includes *Aspergillus* or *Penicillium*.

8. The recombinant cell according to claim 3, wherein the cell is a species of bacteria that includes *Escherichia*, *Corynebacterium*, *Caulobacter*, *Pseudomonas*, *Streptomyces*, *Bacillus*, or *Lactobacillus*.

9. The recombinant cell according to claim 4, wherein at least one copy of the polynucleotide sequence or polynucleotide sequences is located on at least one plasmid.

10. The recombinant cell according to claim 4, wherein at least one copy of the polynucleotide sequence or polynucleotide sequences is stably integrated into the genome of the cell.

11. The recombinant cell according to claim 9, wherein a plurality of copies of the polynucleotide sequence or polynucleotide sequences are stably integrated a plasmid or into the genome of the cell.

12. The recombinant cell according to claim 4, wherein the polynucleotide sequence or polynucleotide sequences are the same sequence.

13. The recombinant cell according to claim 4, wherein the polynucleotide sequence or polynucleotide sequences are different sequences.

14. The recombinant cell according to claim 4, wherein the polynucleotide sequence or polynucleotide sequences encode the same class of enzyme or amino acid sequence.

15. The recombinant cell according to claim 4, wherein the polynucleotide sequence or polynucleotide sequences encode a different class of enzyme or amino acid sequence.

16. A method for the biosynthesis of a downstream monoterpene indole alkaloid compound, or a precursor or metabolite thereof, the method comprising culturing a recombinant cell of claim 4 under conditions that allow for the biosynthesis of the compound.

17. The method according to claim 16, wherein the method further comprises contacting the recombinant cell with a feed comprising one or more exogenous substrate compounds, wherein the one or more exogenous substrate compounds comprise strictosidine, a strictosidine

analog, corynantheidine, or a corynantheidine analog, or the feed comprises a plant extract that includes the one or more of the exogenous substrate compounds.

18. The method according to claim 16, wherein the one or more exogenous substrate compounds are produced by a recombinant cell.

19. The method according to claim 16, wherein the method generates one or more strictosidine, hydroxystictosidine, methoxystictosidine, strictosidine aglycone, hydroxystictosidine aglycone, methoxystictosidine aglycone, demethylcorynantheidine (20R, 20S), 9-hydroxydemethylcorynantheidine, 9-methoxydemethylcorynantheidine, corynantheidine, dihydrocorynantheine, 9-hydroxycorynantheidine, hydroxydihydrocorynantheine, speciogynine, and/or mitragynine, among others.

* * * * *